

JEE-ADVANCED QUESTION BANK

PART - I (XI TOPIC)

PHYSICS

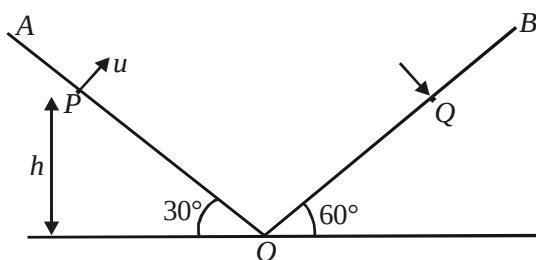
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KINEMATICS

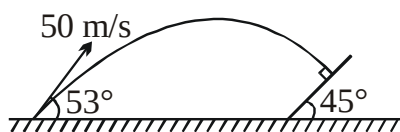
EXERCISE (S-2)

(Not to be discussed in class)

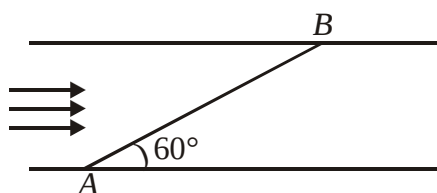
1. A particle goes from A to B with a speed of 40km/h and B to C with a speed of 60km/h. If $AB = 6BC$, the average speed in km/h between A and C is.
2. Two inclined planes OA and OB having inclinations 30° and 60° respectively, intersect each other at O as shown in figure. A particle is projected from point P with velocity $u = 10\sqrt{3}$ m/s along a direction perpendicular to plane OA. If the particle strikes the plane OB perpendicularly at Q. Calculate
 - (i) Time of flight.
 - (ii) Velocity with which particle strikes the plane OB.
 - (iii) Vertical height of P from O.
 - (iv) Maximum height from O attained by the particle.
 - (v) Distance PQ.



3. A projectile is thrown with velocity of 50 m/s towards an inclined plane from ground such that it strikes the inclined plane perpendicularly. The angle of projection of the projectile is 53° with the horizontal and the inclined plane is inclined at an angle of 45° to the horizontal.
 - (i) Find the time of flight.
 - (ii) Find the distance between the point of projection and the foot of inclined plane.



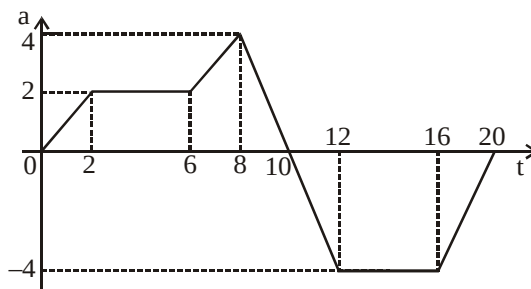
4. A swimmer starts to swim from point A to cross a river. He wants to reach point B on the opposite side of the river. The line AB makes an angle 60° with the river flow as shown. The velocity of the swimmer in still water is same as that of the water.
 - (i) In what direction should he try to direct his velocity? Calculate angle between his velocity and river velocity.
 - (ii) Find the ratio of the time taken to cross the river in this situation to the minimum time in which he can cross this river.



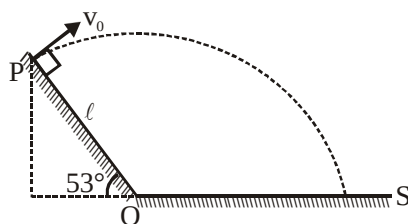
EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. A parachutist jumps out of an airplane and accelerates with gravity for 6 seconds. He then pulls the parachute cord and after a 4 s deceleration period, descends at 10 m/s for 60 seconds, reaching the ground. From what height did the parachutist jump? Assume acceleration due to gravity to be 10 m/s^2 throughout the motion.
 (A) 840 m (B) 920 m (C) 980 m (D) 1020 m
2. If initial velocity of particle is 2 m/s , the maximum velocity of particle from $t = 0$ to $t = 20 \text{ sec}$ is :

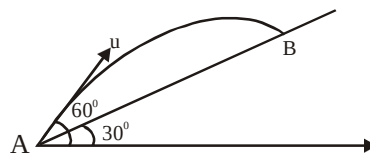


- (A) 20 m/s (B) 18 m/s (C) 22 m/s (D) 24 m/s
3. A particle moving on the x-axis with constant acceleration has displacements of 6 m from $t = 4 \text{ s}$ to $t = 7 \text{ s}$ and 3 m from $t = 5 \text{ s}$ to $t = 8 \text{ s}$. The distance covered from $t = 6 \text{ s}$ to $t = 9 \text{ s}$ is :-
 (A) 1.75 m (B) 2.25 m (C) 3.0 m (D) 0
4. A particle leaves the origin with an initial velocity $\vec{v} = (3\hat{i} + 4\hat{j}) \text{ ms}^{-1}$ and a constant acceleration $\vec{a} = (-\hat{i} - 0.5\hat{j}) \text{ ms}^{-2}$. When the particle reaches its maximum x-coordinate, what is the y-coordinate?
 (A) $\frac{27}{4} \text{ m}$ (B) $\frac{37}{4} \text{ m}$ (C) $\frac{29}{4} \text{ m}$ (D) $\frac{39}{4} \text{ m}$
5. A stone is projected from point P on the inclined plane with velocity $v_0 = 10 \text{ m/s}$ directed perpendicular to the plane. The time taken by the stone to strike the horizontal ground S is (Given $PO = \ell = 10 \text{ meter}$)



- (A) 1.5 sec (B) 1.4 sec (C) 2 sec (D) 2.3 sec

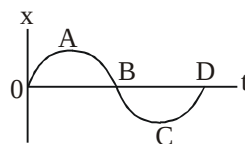
6. Time taken by the projectile to reach from A to B is t . Then the distance AB is equal to :-



- (A) $\frac{ut}{\sqrt{3}}$ (B) $\frac{\sqrt{3}ut}{2}$ (C) $\sqrt{3} ut$ (D) $2 ut$
7. Man A sitting in a car moving at 54 km/hr observes a man B in front of the car crossing perpendicularly the road of width 15 m in three seconds. Then the velocity of man B will be
 (A) $5\sqrt{10}$ towards the car (B) $5\sqrt{10}$ away from the car
 (C) 5 m/s perpendicular to the road (D) None
8. A particle is projected with a velocity of $\sqrt{20}$ m/s such that it strikes on the same level as the point of projection at a distance of $\sqrt{3}$ m. Which of the following options is/are incorrect (mass = 1kg) :
 (A) The maximum height reached by the projectile can be 0.25 m.
 (B) The minimum velocity during its motion can be $\sqrt{5}$ m/s
 (C) The time taken for the flight can be $\sqrt{3/5}$ s.
 (D) Minimum kinetic energy during its motion can be 6 J.

MULTIPLE CORRECT TYPE QUESTIONS

9. A particle moving along a straight line with uniform acceleration has velocities 7m/s at A and 17 m/s at C. B is the mid point of AC. Then
 (A) The velocity at B is 12 m/s.
 (B) The average velocity between A and B is 10 m/s.
 (C) The ratio of the time to go from A to B to that from B to C is 3 : 2.
 (D) The average velocity between B and C is 15 m/s.
10. A particle has a rectilinear motion and the figure gives its displacement as a function of time. Which of the following statements are true with respect to the motion



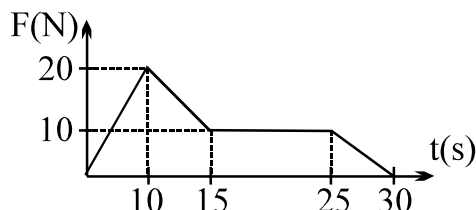
- (A) in the motion between O and A the velocity is positive and acceleration is negative
 (B) between A and B the velocity and acceleration are positive
 (C) between B and C the velocity is negative and acceleration is positive
 (D) between C and D the acceleration is positive

- 11.** A point mass is moving in the x-y plane. Its acceleration is a constant vector perpendicular to the x-axis. Which of the following do/does not change with time?
- (A) only y-component of its velocity vector
 (B) only x-component of its velocity vector
 (C) only y-component of its acceleration vector
 (D) only x-component of its acceleration vector
- 12.** A ball is thrown from ground such that it just crosses two poles of equal height kept 80 m apart. The maximum height attained by the ball is 80 m. When the ball passes the first pole, its velocity makes 45° with horizontal. The correct alternatives is/are :- ($g = 10 \text{ m/s}^2$)
- (A) Time interval between the two poles is 4 s.
 (B) Height of the pole is 60 m.
 (C) Range of the ball is 160 m.
 (D) Angle of projection is $\tan^{-1}(2)$ with horizontal.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 13 to 16

In figure shown, the graph shows the variation of a unidirectional force F acting on a body of mass 10 kg (in gravity free space), with time t . The velocity of the body at $t = 0$ is zero. (Area under F - t curve gives change in momentum)



- 13.** The velocity of the body at $t = 30 \text{ s}$ is
 (A) 30 m/s (B) 20 m/s (C) 40 m/s (D) none
- 14.** The power of the force at $t = 12 \text{ s}$ is (Power = force $\times$ velocity)
 (A) 225.0 W (B) 217.6 W (C) 226.7 W (D) none
- 15.** The average acceleration of the body from $t = 0$ to $t = 15 \text{ s}$ is :-
 (A) 1.25 m/s^2 (B) $4/7 \text{ m/s}^2$ (C) $5/6 \text{ m/s}^2$ (D) $7/6 \text{ m/s}^2$
- 16.** The change in momentum of the body between the time $t = 10 \text{ s}$ to 15 s is :-
 (A) 100 kg.m/s (B) 75 kg.m/s (C) 125 kg.m/s (D) none

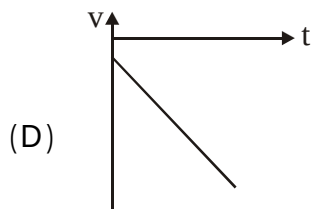
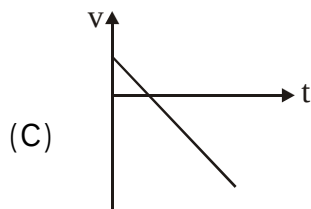
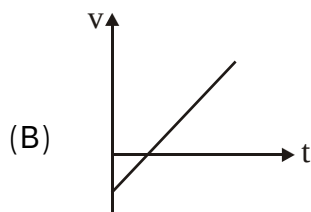
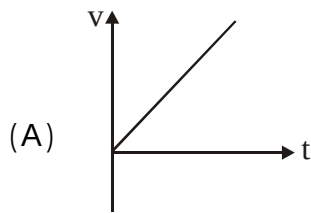
MATRIX MATCH TYPE QUESTION

- 17.** A balloon rises up with constant net acceleration of 10 m/s^2 . After 2 s a particle drops from the balloon. After further 2 s match the following : ($g = 10 \text{ m/s}^2$)

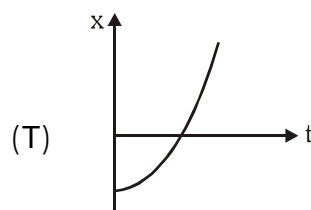
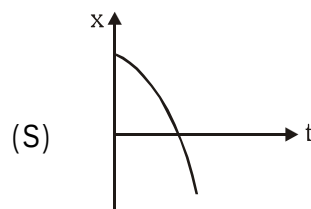
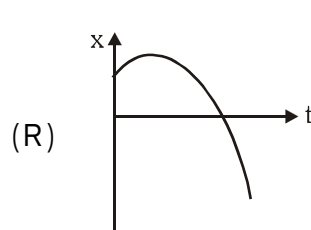
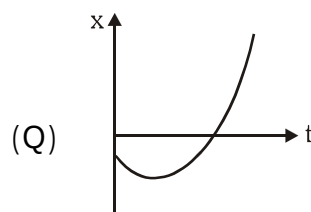
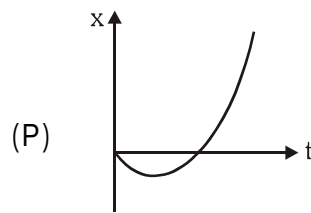
Column-I	Column-II
(A) Height of particle from ground	(P) Zero
(B) Speed of particle	(Q) 10 SI units
(C) Displacement of Particle	(R) 40 SI units
(D) Acceleration of particle	(S) 20 SI units

18. In the first column of the given table, some velocity-time ($v-t$) graphs and in the second column some position-time ($x-t$) graphs are shown. Suggest suitable match or matches.

Column-I



Column-II



EXERCISE (JA)

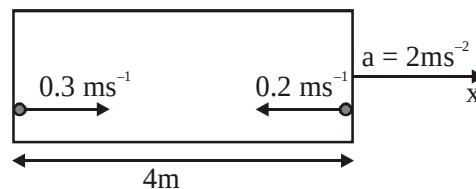
(Not to be discussed in class)

1. A train is moving along a straight line with a constant acceleration 'a'. A boy standing in the train throws a ball forward with a speed of 10 m/s , at an angle of 60° to the horizontal. The boy has to move forward by 1.15 m inside the train to catch the ball back at the initial height. The acceleration of the train, in m/s^2 , is

[IIT-JEE 2011]

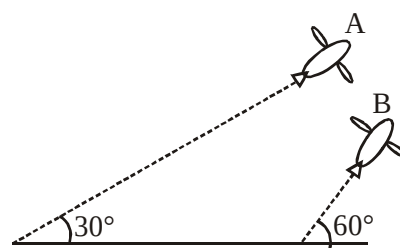
2. A rocket is moving in a gravity free space with a constant acceleration of 2 ms^{-2} along $+x$ direction (see figure). The length of a chamber inside the rocket is 4 m . A ball is thrown from the left end of the chamber in $+x$ direction with a speed of 0.3 ms^{-1} relative to the rocket. At the same time, another ball is thrown in $-x$ direction with a speed of 0.2 ms^{-1} from its right end relative to the rocket. The time in seconds when the two balls hit each other is

[JEE Advanced 2014]



3. Airplanes A and B are flying with constant velocity in the same vertical plane at angles 30° and 60° with respect to the horizontal respectively as shown in figure. The speed of A is $100\sqrt{3} \text{ ms}^{-1}$. At time $t = 0 \text{ s}$, an observer in A finds B at a distance of 500 m . This observer sees B moving with a constant velocity perpendicular to the line of motion of A. If at $t = t_0$, A just escapes being hit by B, t_0 in seconds is

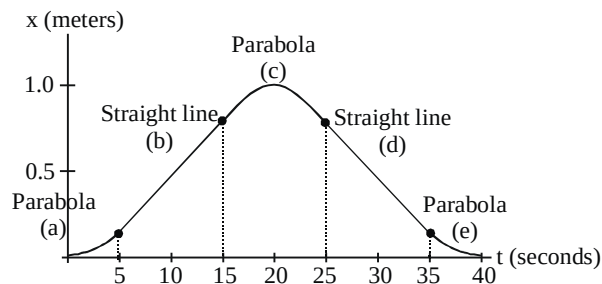
[JEE Advanced-2014]



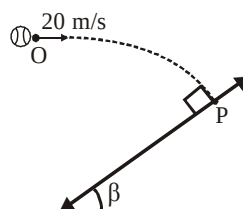
REVIEW TEST

(Not to be discussed in class)

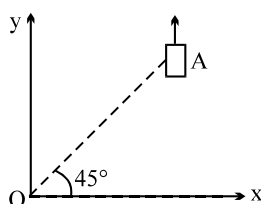
- The figure below is a displacement vs time plot for the motion of an object, answer questions (i) & (ii) with the letter of appropriate section of the graph.
(i) Which section represents motion in the forward direction with positive acceleration?
(ii) Which section represents uniform motion backwards ($-x$ direction)?



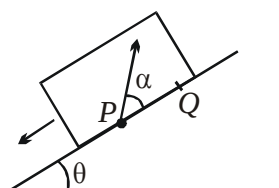
- A particle is projected in the x - y plane with y -axis along vertical. Two second after projection the velocity of the particle makes an angle 45° with the X -axis. Four second after projection, it moves horizontally. Find the velocity of projection.
- A particle is projected upwards with a velocity of 100 m/s at an angle of 60° with the vertical. Find the time when the particle will move perpendicular to its initial direction, taking $g = 10 \text{ m/s}^2$.
- A Bomber flying upward at an angle of 53° with the vertical releases a bomb at an altitude of 800 m . The bomb strikes the ground 20 s after its release. Find: [Given $\sin 53^\circ = 0.8$; $g = 10 \text{ m/s}^2$]
(i) The velocity of the bomber at the time of release of the bomb .
(ii) The maximum height attained by the bomb .
(iii) The horizontal distance travelled by the bomb before it strikes the ground
(iv) The velocity (magnitude & direction) of the bomb just when it strikes the ground .
- A ball is thrown horizontally from a point O with speed 20 m/s as shown. Ball strikes the incline plane along the normal to it after two seconds. Find value of x , if $\beta = \pi/x$ (where β is the angle of incline in degree).



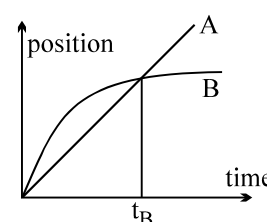
6. On a frictionless horizontal surface, assumed to be the x-y plane, a small trolley, A is moving along a straight line parallel to the y-axis (see figure) with a constant velocity of $(\sqrt{3} - 1)$ m/s. At a particular instant, when the line OA makes an angle of 45° with the x-axis, a ball is thrown along the surface from the origin O. Its velocity makes an angle ϕ with the x-axis and it hits the trolley.
- (a) The motion of the ball is observed from the frame of trolley. Calculate the angle θ made by the velocity vector of the ball with the x-axis in this frame.
- (b) Find the speed of the ball with respect to the surface, if $\phi = \frac{4\theta}{3}$.



7. A helicopter takes off along the vertical with an acceleration of 3 m/sec^2 & zero initial velocity. In a certain time, the pilot switches off the engine. At the point of takeoff, the sound dies away in 30 sec. Determine the velocity of the helicopter at the moment when its engine is switched off, assuming the velocity of sound is 320 m/sec .
8. A large heavy box is sliding without friction down a smooth plane of inclination θ . From a point P on the bottom of the box, a particle is projected inside the box. The initial speed of the particle with respect to box is u and the direction of projection makes an angle α with the bottom as shown in figure. **[JEE]**
- (i) Find the distance along the bottom of the box between the point of projection P and the point Q where the particle lands. (Assume that the particle does not hit any other surface of the box. Neglect air resistance).
- (ii) If the horizontal displacement of the particle as seen by an observer on the ground is zero, find the speed of the box with respect to the ground at the instant when the particle was projected.



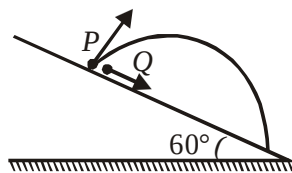
9. The graph shows position as a function of time for two trains running on parallel tracks. Which statement is true?
- (A) At time t_B , both trains have the same velocity.
- (B) Both trains have the same velocity at some time after t_B .
- (C) Both trains have the same velocity at some time before t_B .
- (D) Somewhere on the graph, both trains have the same acceleration.



10. A particle is projected from a horizontal plane (x - z plane) such that its velocity vector at time t is given by $\vec{V} = a\hat{i} + (b - ct)\hat{j}$. Its range on the horizontal plane is given by

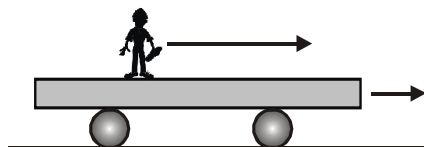
(A) $\frac{ba}{c}$ (B) $\frac{2ba}{c}$ (C) $\frac{3ba}{c}$ (D) None

11. A particle P is projected from a point on the surface of smooth inclined plane (see figure). Simultaneously another particle Q is released on the smooth inclined plane from the same position. P and Q collide after $t = 4$ s. The speed of projection of P is :-



(A) 5 m/s (B) 10 m/s (C) 15 m/s (D) 20 m/s

12. A trolley is moving horizontally with a constant velocity of v with respect to earth. A man starts running from one end of the trolley with a velocity $1.5v$ with respect to trolley. After reaching the opposite end, the man returns back and continues running with a velocity of $1.5v$ w.r.t. the trolley in the backward direction. If the length of the trolley is L then the displacement of the man with respect to earth during the process will be



(A) $2.5L$ (B) $1.5L$ (C) $\frac{5L}{3}$ (D) $\frac{4L}{3}$

13. An elevator car (lift) is moving upward with uniform acceleration of 2 m/s^2 . At the instant, when its velocity is 2 m/s upwards a ball is thrown upward from its floor. The ball strikes back the floor 2 s after its projection. Find the velocity of projection of the ball relative to the lift.

(A) $10 \text{ m/s} \uparrow$ (B) $10 \text{ m/s} \downarrow$ (C) $12 \text{ m/s} \uparrow$ (D) $12 \text{ m/s} \downarrow$

14. A motor boat is to reach at a point 30° upstream (w.r.t. normal) on other side of a river flowing with velocity 5 m/s . The angle 30° is measured from a direction perpendicular to river flow. Velocity of motorboat with respect to water is $5\sqrt{3} \text{ m/s}$. The driver should steer the boat at an angle

(A) 120° with respect to stream direction.
(B) 30° with respect to the perpendicular to the bank.
(C) 30° with respect to the line of destination from starting point.
(D) None of these.

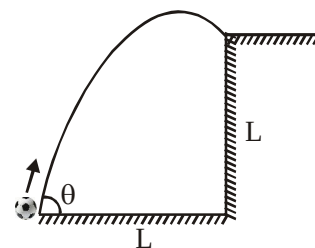
15. A particle moves in the x - y plane with a constant acceleration g in the negative y -direction. Its equation of motion is $y = ax - bx^2$, where a and b are constants. Which of the following is/are correct?
- (A) The x -component of its velocity is constant.
- (B) At the origin, the y -component of its velocity is $a\sqrt{\frac{g}{2b}}$.
- (C) At the origin, its velocity makes an angle $\tan^{-1}(a)$ with the x -axis.
- (D) The particle moves exactly like a projectile.
16. Two particles A and B projected along different directions from the same point P on the ground with the same velocity of 70 m/s in the same vertical plane. They hit the ground at the same point Q such that $PQ = 480$ m. Then :- [$g = 9.8$ m/s²]
- (A) Ratio of their times of flight is $4 : 5$
- (B) Ratio of their maximum heights is $9 : 16$
- (C) Ratio of their minimum speeds during flights is $4 : 3$
- (D) The bisector of the angle between their directions of projection makes 45° with horizontal
17. The position vector of a particle is determined by $\vec{r} = 3t^2\hat{i} + 4t^2\hat{j} + 7\hat{k}$. The distance travelled in first 10 sec is :-
- (A) 100 m (B) 150 m (C) 500 m (D) 300 m
18. A particle is thrown from a stationary platform with velocity v at an angle of 60° with the horizontal. The range obtained is R . If the platform moves horizontally in the direction of target with velocity v , the range will increase to :
- (A) $\frac{3R}{2}$ (B) $\frac{5R}{2}$ (C) $2R$ (D) $3R$
19. A swimmer swims in still water at a speed = 5 km/hr. He enters a 200 m wide river, having river flow speed = 4 km/hr at point A and proceeds to swim at an angle of 127° with the river flow direction. Another point B is located directly across A on the other side. The swimmer lands on the other bank at a point C , from which he walks the distance CB with a speed = 3 km/hr. The total time in which he reaches from A to B is
- (A) 5 minutes (B) 4 minutes (C) 3 minutes (D) None
20. A ball is thrown at an angle θ up to the top of a cliff of height L , from a point at a distance L from the base, as shown in figure. Assuming that one of the following quantities is the initial speed required to make the ball hit right at the edge of the cliff, which one is it :-

(A) $\sqrt{\frac{gL}{2(\tan \theta - 1)}}$

(B) $\frac{1}{\cos \theta} \sqrt{\frac{gL}{2(\tan \theta - 1)}}$

(C) $\frac{1}{\cos \theta} \sqrt{\frac{gL}{2(\tan \theta + 1)}}$

(D) $\sqrt{\frac{gL \tan \theta}{2(\tan \theta + 1)}}$



- 21.** A projectile is thrown with speed u into air from a point on the horizontal ground at an angle θ with horizontal. If the air exerts a constant horizontal resistive force on the projectile then select correct alternative(s).

- (A) At the farthest point, the velocity is horizontal.
(B) The time for ascent equals the time for descent.
(C) The path of the projectile may be parabolic.
(D) The path of the projectile may be a straight line.

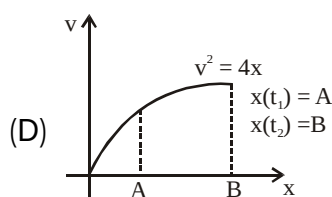
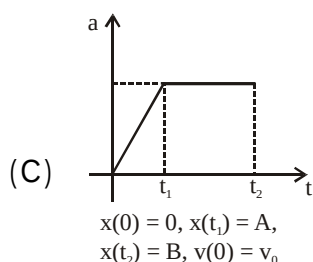
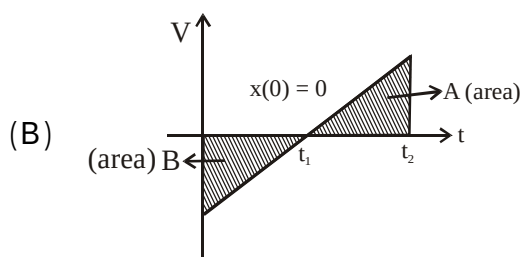
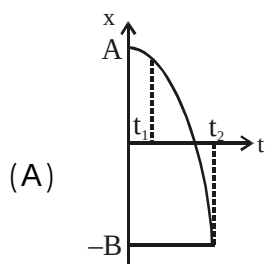
22. Match the column :-

Column-I : Shows graph of One Dimension motion of a particle. Symbols have their usual meaning such as $x(0)$ = initial position, $x(t_1)$ = position at $t = t_1$, $v(0)$ = initial velocity.

Column-II : Shows physical quantities. Displacement and distance are asked for $0 < t < t_2$, and average values are asked for $0 < t < t_2$

V_0 , A & B are positive constant

Column-I



Column-II

(P) |Displacement| = Distance

(Q) |Instantaneous velocity| = |Instantaneous speed|

(R) |Average velocity| ≤ Average speed

(S) Instantaneous acceleration = Average acceleration

(T) Displacement = $A - B$ and Distance = $A + B$

23.

Column-I

Column-II

- (A) Time for a boat to cross a river of width ℓ by the shortest distance ($\vec{v}$ - velocity of boat with respect to water; $\vec{u}$ - velocity of water)

(P) $\frac{\ell}{|\vec{v} + \vec{u}|}$

- (B) Time for two particles moving with velocities $\vec{v}$ and $\vec{u}$ in opposite directions to meet each other. (initial separation of particles is ℓ)

(Q) $\frac{\ell}{\sqrt{v^2 - u^2}}$

- (C) Time for a boat to cross a river of width ℓ in the shortest time ($\vec{v}$ - velocity of boat with respect to water; $\vec{u}$ - velocity of water)

(R) $\frac{\ell}{|\vec{v}| + |\vec{u}|}$

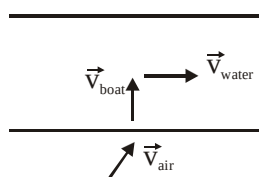
- (D) Time for a boat to travel a distance ℓ downstream ($\vec{v}$ - velocity of boat with respect to water;

(S) $\frac{\ell}{|\vec{v}|}$

$\vec{u}$ - velocity of water)

(T) $\frac{\ell}{\sqrt{u^2 + v^2}}$

24. A boat is being rowed in a river. Air is also blowing. Direction of velocity vectors of boat, water and air in ground frame are as shown in diagram.



Column-I

- (A) Direction in which boat is being steered
 (B) Direction in which a flag on the boat may flutter
 (C) Direction of velocity of water relative to boat
 (D) Direction of velocity of air relative to a piece of wood floating on river.

Column-II

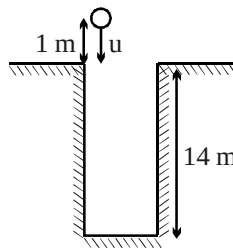
(possible directions)

- (P) 
 (Q) 
 (R) 
 (S) 

25. Assuming right side to be positive x , Figure shows the motion diagram of a drag racer. The camera took one frame every 2 s. Comment about the acceleration of drag racer.



26. A boy throws a ball with speed u in a well of depth 14 m as shown. On bounce with bottom of the well the speed of the ball gets halved. What should be the minimum value of u (in m/s) such that the ball may be able to reach his hand again? It is given that his hands are at 1 m height from top of the well while throwing and catching.



27. A flower pot falls off a window sill and falls past the window below. It takes 0.30 s to pass a window 3.45 m high. How far is the top of the window below upper window sill?
28. Two guns, situated at the top of a hill of height 10 m, fire one shot each with the same speed $5\sqrt{3}$ m/s at some interval of time. One gun fires horizontally and other fires at an angle of 60° up the horizontal. The shots collide in air at a point P . Find
(i) the time interval between the firings, and (ii) the coordinates of the point P .
(Take origin of the coordinates system at the foot of the hill vertically below the muzzle and trajectories in X - Y plane)
29. A ball is dropped from a building. Somewhere down it crosses a window of length 4 m in 0.5 sec. Speed of ball at top of window is v_1 and at bottom v_2 , then choose the **CORRECT** option(s) :-
($g = 10 \text{ m/s}^2$) :-

- (A) $v_2 - v_1 = 5 \text{ m/s}$ (B) $v_2 + v_1 = 16 \text{ m/s}$ (C) $\frac{v_2}{v_1} = 9$ (D) $\frac{v_2}{v_1} = \frac{21}{11}$

ANSWER KEY**EXERCISE (S-2)**

1. Ans. 42km/hr 2. Ans. (i) 2 s (ii) 10 m/s (iii) 5m (iv) 16.25 m (v) 20 m
 3. Ans. (i) 7 s, (ii) 175 m 4. Ans. (i) 120° (ii) $2/\sqrt{3}$

EXERCISE (O-2)**SINGLE CORRECT TYPE QUESTIONS**

1. Ans. (B) 2. Ans. (C) 3. Ans. (B) 4. Ans. (D) 5. Ans. (C)
 6. Ans. (A) 7. Ans. (B) 8. Ans. (D)

MULTIPLE CORRECT TYPE QUESTIONS

9. Ans. (B,C,D) 10. Ans. (A,C,D) 11. Ans. (B, C) 12. Ans. (A, B,C,D)

COMPREHENSION TYPE QUESTIONS

13. Ans. (A) 14. Ans. (B) 15. Ans. (D) 16. Ans. (B)

MATRIX MATCH TYPE QUESTION

17. Ans. (A) - (R); (B) - (P) ; (C) - (S) ; (D) - (Q)
 18. Ans. (A) - (T); (B) - (P,Q); (C) - (R); (D) - (S)

EXERCISE (JA)

1. Ans. 5 2. Ans. 8 or 2 3. Ans. 5

REVIEW TEST

1. Ans. (i) section (a) as slope = $v = \frac{dx}{dt}$ is positive and increasing.

- (ii) section (d) as slope = $v = \frac{dx}{dt}$ is negative and constant.

2. Ans. $20\sqrt{5}$ m/s 3. Ans. 20 s

4. Ans. (i) 100 m/s (ii) 980 m (iii) 1600 m (iv) $(80\hat{i} - 140\hat{j})$

5. Ans. 4

6. Ans. (a) 45°, (b) 2 m/sec 7. Ans. 80 m/sec

8. Ans. (i) $\frac{u^2 \sin 2\alpha}{g \cos \theta}$ (ii) $v = \frac{u \cos(\alpha + \theta)}{\cos \theta}$

9. Ans. (C) 10. Ans. (B) 11. Ans. (B) 12. Ans. (D) 13. Ans. (C) 14. Ans. (C)

15. Ans. (A, B, C, D) 16. Ans. (B, C, D) 17. Ans. (C) 18. Ans. (D)

19. Ans. (B) 20. Ans. (B) 21. Ans. (B,C,D)

22. Ans. (A)-P,Q,R; (B)-Q,R,S,T; (C)-P,Q,R; (D)-P,Q,R,S

23. Ans. (A) - (P,Q); (B) - (R); (C) - (S); (D) - (P,R) 24. Ans. (A)-P; (B)-Q, S; (C)-S; (D)-P,R

25. Ans. Initially acceleration is positive then zero & then negative

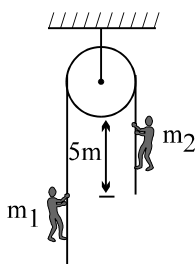
26. Ans. 30 27. Ans. 5m 28. Ans. (i) 1 s, (ii) $(5\sqrt{3}$ m, 5 m) 29. Ans. (A, B, D)

NEWTON'S LAWS OF MOTION & FRICTION

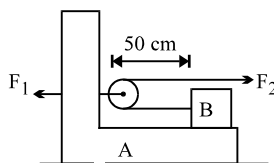
EXERCISE (S-2)

(Not to be discussed in class)

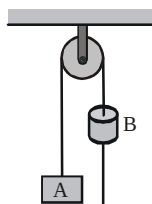
1. Two men of masses m_1 and m_2 hold on the opposite ends of a rope passing over a frictionless pulley. The man m_1 climbs up the rope with an acceleration of 1.2 m/s^2 relative to the rope. The man m_2 climbs up the rope with an acceleration of 2 m/s^2 relative to the rope. Find the tension in the rope if $m_1 = 40 \text{ kg}$ and $m_2 = 60 \text{ kg}$. Also find the time after which they will be at same horizontal level if they start from rest and are initially separated by 5 m .



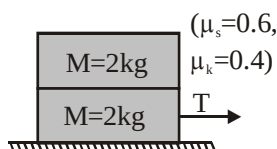
2. A 1 kg block B rests as shown on a bracket A of same mass. Constant forces $F_1 = 20 \text{ N}$ and $F_2 = 8 \text{ N}$ start to act at time $t = 0$ when the distance of block B from pulley is 50 cm . Time when block B reaches the pulley is _____.



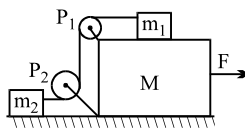
3. A 2 kg block A is attached to one end of a light string that passes over an ideal pulley and a 1 kg sleeve B slides down the other part of the string with an acceleration of 5 m/s^2 with respect to the string. Find the acceleration of the block, acceleration of sleeve and tension in the string. [$g = 10 \text{ m/s}^2$]



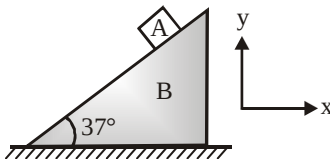
4. The coefficient of static and kinetic friction between the two blocks and also between the lower block and the ground are $\mu_s = 0.6$ and $\mu_k = 0.4$. Find the value of tension T applied on the lower block at which the upper block begins to slip relative to lower block.



5. In the figure masses m_1 , m_2 and M are 20 kg, 5 kg and 50 kg respectively. The co-efficient of friction between M and ground is zero. The co-efficient of friction between m_1 and M and that between m_2 and ground is 0.3. The pulleys and the string are massless. The string is perfectly horizontal between P_1 and m_1 and also between P_2 and m_2 . The string is perfectly vertical between P_1 and P_2 . An external horizontal force F is applied to the mass M . Take $g = 10 \text{ m/s}^2$. [IIT-JEE 2000]

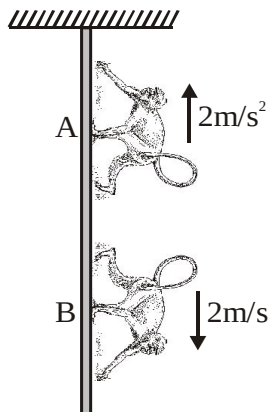


- (i) Draw a free-body diagram for mass M , clearly showing all the forces.
- (ii) Let the magnitude of the force of friction between m_1 and M be f_1 and that between m_2 and ground be f_2 . For a particular F it is found that $f_1 = 2 f_2$. Find f_1 and f_2 . Write down equations of motion of all the masses. Find F , tension in the string and accelerations of the masses.
6. In the figure shown the acceleration of A is, $\vec{a}_A = (15\hat{i} + 15\hat{j}) \text{ m/s}^2$. If A is sliding on B then the acceleration of B is



EXERCISE (O-2)

1. Two monkeys of masses 10 kg and 8 kg are moving along a vertical light rope, the former climbing up with an acceleration of 2 m/s^2 , while the latter coming down with a uniform velocity of 2 m/s . Find tension in the rope at the fixed support.



- (A) 180 N (B) 200 N (C) 80 N (D) 216 N
2. The rear side of a truck is open and a box of mass 20 kg is placed on the truck 4 m away from the open end, $\mu = 0.15$ and $g = 10 \text{ m/s}^2$. The truck starts from rest with an acceleration of 2 m/s^2 on a straight road. The distance moved by the truck when box starts fall down is
- (A) 4 m (B) 8 m (C) 16 m (D) 32 m

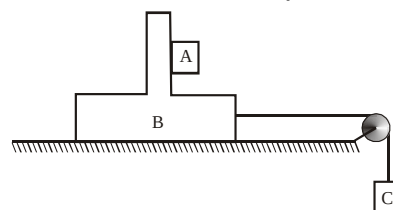
3. In the arrangement shown in the figure, mass of the block B and A is $2m$ and m respectively. Surface between B and floor is smooth. The block B is connected to the block C by means of a string-pulley system. If the whole system is released, then find the minimum value of mass of block C so that A remains stationary w.r.t. B. Coefficient of friction between A and B is μ .

(A) $\frac{m}{\mu}$

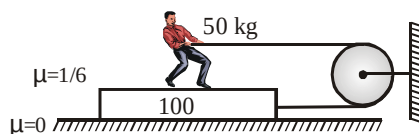
(B) $\frac{2m+1}{\mu+1}$

(C) $\frac{3m}{\mu-1}$

(D) $\frac{6m}{\mu+1}$



4. A man of mass 50 kg is pulling on a plank of mass 100 kg kept on a smooth floor as shown with force of 100 N. If both man & plank move together, find force of friction acting on man.



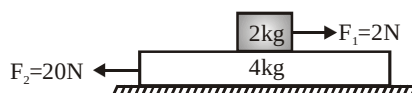
(A) $\frac{100}{3}$ N towards left

(B) $\frac{100}{3}$ N towards right

(C) $\frac{250}{3}$ N towards left

(D) $\frac{250}{3}$ N towards right

5. In the arrangement shown in figure, coefficient of friction between the two blocks is $\mu = 1/2$. The force of friction acting between the two blocks is



(A) 8 N

(B) 10 N

(C) 6 N

(D) 4 N

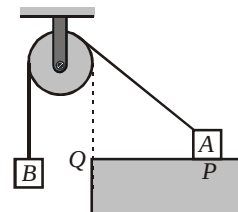
6. A flexible chain of weight W hangs between two fixed points A & B which are at the same horizontal level. The inclination of the chain with the horizontal at both the points of support is θ . What is the tension of the chain at the mid point?

(A) $\frac{W}{2} \cdot \operatorname{cosec} \theta$ (B) $\frac{W}{2} \cdot \tan \theta$ (C) $\frac{W}{2} \cot \theta$ (D) none

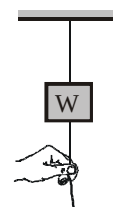
MULTIPLE CORRECT TYPE QUESTIONS

7. Two small identical blocks are connected to the ends of a string passing over pulley as shown when the system is released from rest.

(A) block A and B do not move
 (B) block A accelerates towards pulley along the string.
 (C) block A does not leave contact with table till it reaches to the edge Q of the table
 (D) Normal reaction of table on block A is less than weight of block A between P and Q and at Q it vanishes



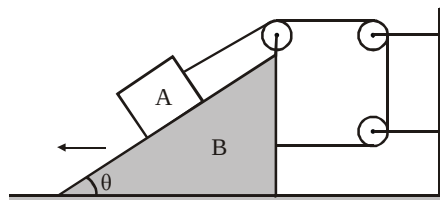
8. A block of mass m is suspended from a fixed support with the help of a cord. Another identical cord is attached to the bottom of the block. Which of the following statement is /are true?
- (A) If the lower cord is pulled suddenly, only the upper cord will break.
 (B) If the lower cord is pulled suddenly, only the lower cord will break.
 (C) If pull on the lower cord is increased gradually, only the lower cord will break.
 (D) If pull on the lower cord is increased gradually, only the upper cord will break.



9. A carpenter of mass 50 kg is standing on a weighing machine placed in a lift of mass 20 kg . A light string is attached to the lift. The string passes over a smooth pulley and the other end is held by the carpenter as shown. When carpenter keeps the lift moving upward with constant velocity ($g = 10 \text{ m/s}^2$)
- (A) the reading of weighing machine is 15 kg
 (B) the man applies a force of 350 N on the string
 (C) net force on the man is 150 N
 (D) Net force on the weighing machine is 150 N



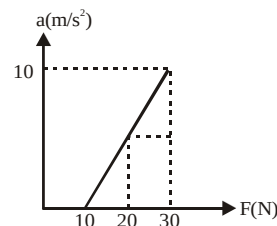
10. A block A and wedge B connected through a string as shown. The wedge B is moving away from the wall with acceleration 2 m/s^2 horizontally and acceleration of block A is vertical upwards. Then



(A) Acceleration of A with respect to B is 4 m/s^2 .
 (B) Acceleration of A with respect to B is $2\sqrt{3} \text{ m/s}^2$.
 (C) Angle θ is 60° .
 (D) Acceleration of A is $2\sqrt{3} \text{ m/s}^2$.

- 11.** A block placed on a rough horizontal surface is pushed with a force F acting horizontally on the block. The magnitude of F is increased and acceleration produced is plotted in the graph shown.

- (A) Mass of the block is 2 kg.
(B) Coefficient of friction between block and surface is 0.5.
(C) Limiting friction between block and surface is 10 N.
(D) When $F = 8$ N, friction between block and surface is 10 N.

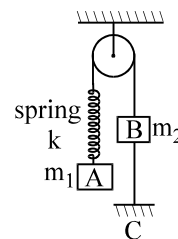


- 12.** A block is released from rest from a point on a rough inclined plane of inclination 37° . The coefficient of friction is 0.5.

- (A) The time taken to slide down 9 m on the plane is 3 s.
(B) The velocity of block after moving 4 m is 4 m/s.
(C) The block travels equal distances in equal intervals of time.
(D) The velocity of block increases linearly.

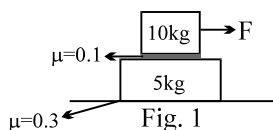
- 13.** In the system shown in the figure $m_1 > m_2$. System is held at rest by thread BC. Just after the thread BC is burnt :

- (A) initial acceleration of m_2 will be upwards
(B) magnitude of initial acceleration of both blocks will be equal to $\left(\frac{m_1 - m_2}{m_1 + m_2}\right)g$.
(C) initial acceleration of m_1 will be equal to zero
(D) magnitude of initial acceleration of two blocks will be non-zero and unequal.



COMPREHENSION TYPE QUESTIONS

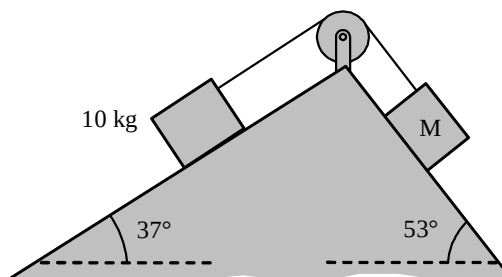
Paragraph for Question No. 14 to 18



- 14.** When $F = 2$ N, the frictional force between 5 kg block and ground is
(A) 2 N (B) 0 (C) 8 N (D) 10 N
- 15.** When $F = 2$ N, the frictional force between 10 kg block and 5 kg block is
(A) 2 N (B) 15 N (C) 10 N (D) None
- 16.** The maximum F which will not cause motion of any of the blocks is
(A) 10 N (B) 15 N (C) data insufficient (D) None
- 17.** The maximum acceleration of 5 kg block is :-
(A) 1 m/s^2 (B) 3 m/s^2 (C) 0 (D) None
- 18.** The acceleration of 10 kg block when $F = 30$ N is
(A) 2 m/s^2 (B) 3 m/s^2 (C) 1 m/s^2 (D) None

Paragraph for Question No. 19 to 21

The blocks are on frictionless inclined ramp and connected by a massless cord. The cord passes over an ideal pulley. [$g = 10 \text{ m/s}^2$]



19. When set free, the 10 kg block slides down the ramp with acceleration of 2 m/s^2 . Mass M is closest to
 (A) 13 kg (B) 8 kg (C) 5 kg (D) 4 kg
20. When the 10 kg block slides down the ramp with acceleration of 2 m/s^2 , tension in the cord is closest to
 (A) 80 N (B) 60 N (C) 40 N (D) 30 N
21. What value of M would keep the system at rest.
 (A) 10 kg (B) 8 kg (C) 7.5 kg (D) 6 kg

MATRIX MATCH TYPE QUESTION

22. Velocity of three particles A, B and C varies with time t as, $\vec{v}_A = (2t\hat{i} + 6\hat{j}) \text{ m/s}$; $\vec{v}_B = (3\hat{i} + 4\hat{j}) \text{ m/s}$ and $\vec{v}_C = (6\hat{i} - 4\hat{j}) \text{ m/s}$. Regarding the pseudo force match the following table

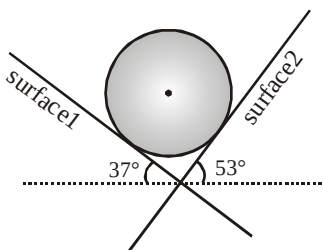
Column I

- (A) On A as observed by B
 (B) On B as observed by C
 (C) On A as observed by C
 (D) On C as observed by A

Column II

- (P) Along positive x -direction
 (Q) Along negative x -direction
 (R) Along positive y -direction
 (S) Along negative y -direction
 (T) Zero

23. A sphere of mass 10 kg is placed in equilibrium in a V shaped groove plane made of two smooth surfaces 1 and 2 as shown in figure. ($g = 10 \text{ ms}^{-2}$)


Column I

- (A) Normal reaction by Surface 1
 (B) Normal reaction by surface 2
 (C) Force on sphere by Earth
 (D) Net force on sphere

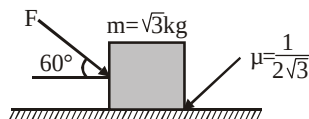
Column II

- (P) Zero
 (Q) 60 N
 (R) 80 N
 (S) 100 N
 (T) 120 N

EXERCISE (JA)

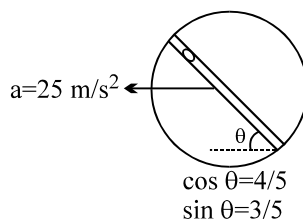
(Not to be discussed in class)

1. What is the maximum value of the force F such that the block shown in the arrangement, does not move? [2003, 2M]

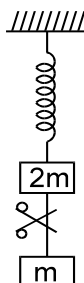


- (A) 20 N (B) 10 N (C) 12 N (D) 15 N
2. A circular disc with a groove along its diameter is placed horizontally. A block of mass 1 kg is placed as shown. The co-efficient of friction between the block and all surfaces of groove in contact is $\mu = 2/5$. The disc has an acceleration of 25 m/s^2 . Find the acceleration of the block with respect to disc.

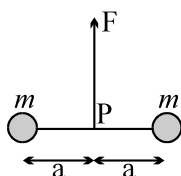
[IIT-JEE 2006]



3. System shown in figure is in equilibrium and at rest. The spring and string are massless. Now the string is cut. The acceleration of mass $2m$ and m just after the string is cut will be :



- (A) $g/2$ upwards, g downwards (B) g upwards, $g/2$ downwards
(C) g upwards, $2g$ downwards (D) $2g$ upwards, g downwards
4. Two particles of mass m each are tied at the ends of a light string of length $2a$. The whole system is kept on a frictionless horizontal surface with the string held tight so that each mass is at a distance ' a ' from the center P (as shown in the figure). Now, the mid-point of the string is pulled vertically upwards with a small but constant force F . As a result, the particles move towards each other on the surfaces. The magnitude of acceleration, when the separation between them becomes $2x$, is [IIT-JEE 2007]



(A) $\frac{F}{2m} \frac{a}{\sqrt{a^2 - x^2}}$ (B) $\frac{F}{2m} \frac{x}{\sqrt{a^2 - x^2}}$ (C) $\frac{F}{2m} \frac{x}{a}$ (D) $\frac{F}{2m} \frac{\sqrt{a^2 - x^2}}{x}$

5. **Statement-1** : A cloth covers a table. Some dishes are kept on it. The cloth can be pulled out without dislodging the dishes from the table. **[IIT-JEE 2007]**

and

Statement-2 : For every action there is an equal and opposite reaction

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

6. **Statement-1** : It is easier to pull a heavy object that to push it on a level ground.

and

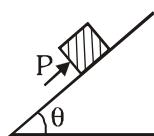
[IIT-JEE 2008]

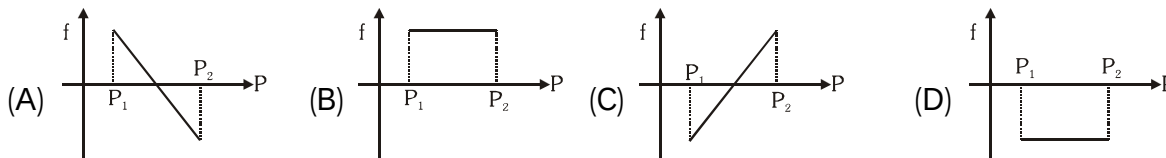
Statement-2 : The magnitude of frictional force depends on the nature of the two surfaces in contact.

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True
7. A piece of wire is bent in the shape of a parabola $y = kx^2$ (y-axis vertical) with a bead of mass m on it. The bead can slide on the wire without friction. It stays at the lowest point of the parabola when the wire is at rest. The wire is now accelerated parallel to the x-axis with a constant acceleration a . The distance of the new equilibrium position of the bead, where the bead can stay at rest with respect to the wire, from the y-axis is **[IIT-JEE-2009]**

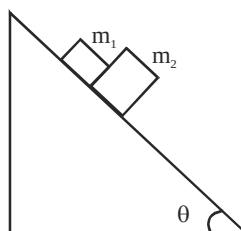
(A) $\frac{a}{gk}$ (B) $\frac{a}{2gk}$ (C) $\frac{2a}{gk}$ (D) $\frac{a}{4gk}$

8. A block of mass m is on an inclined plane of angle θ . The coefficient of friction between the block and the plane is μ and $\tan\theta > \mu$. The block is held stationary by applying a force P parallel to the plane. The direction of force pointing up the plane is taken to be positive. As P is varied from $P_1 = mg(\sin\theta - \mu\cos\theta)$ to $P_2 = mg(\sin\theta + \mu\cos\theta)$, the frictional force f versus P graph will look like **[IIT-JEE-2010]**





9. A block is moving on an inclined plane making an angle 45° with the horizontal and the coefficient of friction is μ . The force required to just push it up the inclined plane is 3 times the force required to just prevent it from sliding down. If we define $N = 10\mu$, then N is **[IIT-JEE-2011]**
10. A block of mass $m_1 = 1$ kg another mass $m_2 = 2$ kg, are placed together (see figure) on an inclined plane with angle of inclination θ . Various values of θ are given in List I. The coefficient of friction between the block m_1 and the plane is always zero. The coefficient of static and dynamic friction between the block m_2 and the plane are equal to $\mu = 0.3$. In List II expressions for the friction on block m_2 are given. Match the correct expression of the friction in List II with the angles given in List I, and choose the correct option. The acceleration due to gravity is denoted by g .
[useful information : $\tan(5.5^\circ) \approx 0.1$; $\tan(11.5^\circ) \approx 0.2$; $\tan(16.5^\circ) \approx 0.3$] **[IIT-JEE-2014]**



List-I

- (P) $\theta = 5^\circ$
(Q) $\theta = 10^\circ$
(R) $\theta = 15^\circ$
(S) $\theta = 20^\circ$

Code :

- (A) P-1, Q-1, R-1, S-3
(C) P-2, Q-2, R-2, S-4

List-II

- (1) $m_2 g \sin \theta$
(2) $(m_1 + m_2) g \sin \theta$
(3) $\mu m_2 g \cos \theta$
(4) $\mu(m_1 + m_2) g \cos \theta$

- (B) P-2, Q-2, R-2, S-3
(D) P-2, Q-2, R-3, S-3

ANSWER KEY

EXERCISE (S-2)

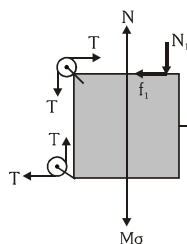
1. Ans. 556.8 N, 1.47 s

2. Ans. 0.5 s

3. Ans. 5 m/s² downwards, 0 m/s², 10 N

4. Ans. 40 N

5. Ans. (i)

(ii) $a = 3/5 \text{ m/s}^2$, $T = 18 \text{ N}$, $F = 60 \text{ N}$ 6. Ans. $-5 \hat{i} \text{ m/s}^2$

EXERCISE (O-2)

SINGLE CORRECT ANSWER TYPE QUESTION

1. Ans. (B)

2. Ans. (C)

3. Ans. (C)

4. Ans. (A)

5. Ans. (A)

6. Ans. (C)

MULTIPLE CORRECT TYPE QUESTIONS

7. Ans. (C, D)

8. Ans. (B, D)

9. Ans. (A, B)

10. Ans. (A, C, D)

11. Ans. (A, B, C)

12. Ans. (A, B, D)

13. Ans. (A, C)

COMPREHENSION TYPE QUESTIONS

14. Ans. (A)

15. Ans. (A)

16. Ans. (A)

17. Ans. (C)

18. Ans. (A)

19. Ans. (D)

20. Ans. (C)

21. Ans. (C)

MATRIX MATCH TYPE QUESTION

22. Ans. (A) - (T) ; (B) - (R) ; (C) - (R) ; (D) - (Q) 23. Ans. (A) - (R) ; (B) - (Q) ; (C) - (S) ; (D) - (P)

EXERCISE (JA)

1. Ans. (A)

2. Ans. 10 m/s²

3. Ans. (A)

4. Ans. (B)

5. Ans. (B)

6. Ans. (B)

7. Ans. (B)

8. Ans. (A)

9. Ans. 5

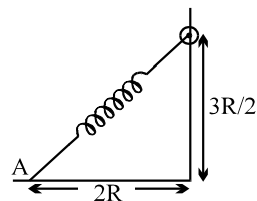
10. Ans. (D)

CIRCULAR MOTION & WORK-POWER-ENERGY

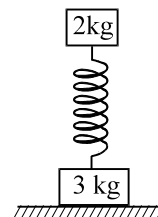
EXERCISE (S-2)

(Not to be discussed in class)

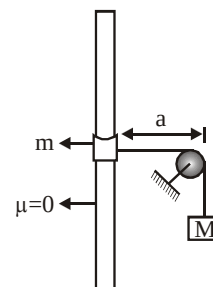
1. A ring of mass m can slide over a smooth vertical rod. The ring is connected to a spring of force constant $K = \frac{4mg}{R}$ where $2R$ is the natural length of the spring. The other end of the spring is fixed to the ground at a horizontal distance $2R$ from the base of the rod. The mass is released at a height of $1.5R$ from ground
- calculate the work done by the spring.
 - calculate the velocity of the ring as it reaches the ground.



2. The ends of spring are attached to blocks of mass 3 kg and 2 kg . The 3 kg block rests on a horizontal surface and the 2 kg block which is vertically above it is in equilibrium producing a compression of 1 cm of the spring. The 2 kg mass must be compressed further by at least _____, so that when it is released, the 3 kg block may be lifted off the ground.



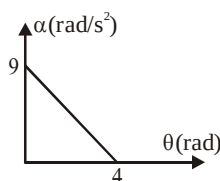
3. A ring of mass m slides on a smooth vertical rod. A light string is attached to the ring and is passing over a smooth peg distant a from the rod, and at the other end of the string is a mass M ($M > m$). The ring is held on a level with the peg and released. Show that it first comes to rest after falling a distance



$$\frac{2mMa}{M^2 - m^2}.$$

4. A particle is moving with kinetic energy E , straight up an inclined plane with angle α , the coefficient of friction being μ . The work done against friction before the particle comes down to rest is
5. A stone is launched upward at 45° with speed v_0 . A bee follows the trajectory of the stone at a constant speed equal to the initial speed of the stone.
- Find the radius of curvature at the top point of the trajectory.
 - What is the acceleration of the bee at the top point of the trajectory? For the stone, neglect the air resistance.

6. A particle starts moving in a non-uniform circular motion, has angular acceleration as shown in figure. The angular velocity at the end of 4 radian is given by ω rad/s then find the value of ω .

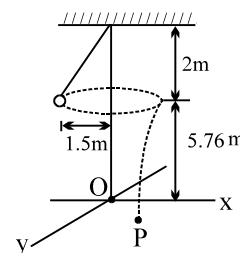


7. A particle is moving along a circular path of radius R in such a way that at any instant magnitude of radial acceleration & tangential acceleration are equal. If at $t = 0$ velocity of particle is V_0 . Find the

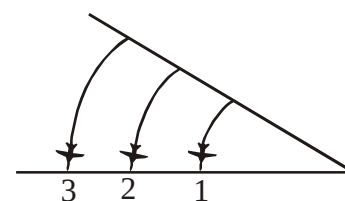
speed of the particle after time $t = \frac{R}{2V_0}$

8. A boy standing on a tower is swinging a particle in a horizontal circle of radius 1m at a height of 10 m with angular velocity $2 \frac{\text{rad}}{\text{sec}}$ above the ground. The tower is just below the centre of the horizontal circle. When boy releases the string, how far (in m) from the foot of the tower the particle will hit the ground ?

9. A particle suspended from the ceiling by inextensible light string is moving along a horizontal circle of radius 1.5 m as shown. The string traces a cone of height 2 m. The string breaks and the particle finally hits the floor (which is xy plane 5.76 m below the circle) at point P. Find the distance OP in cm.



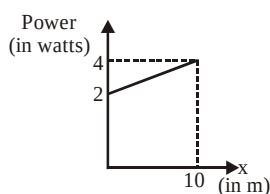
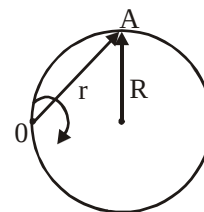
10. Three aircraft make a turn in the horizontal plane at same uniform speed, moving along concentric circular trajectories that are shown in figure. The aircraft move such that they are at constant distance of 600 m from each other at any time. The aircraft closest to the center moves in a circle of radius $R = 600$ m. The aircraft 2 is moving at a speed of $v_2 = 720$ km/h. Find the acceleration of third aircraft (in m/s^2)



11. The speed of an object undergoing uniform circular motion is 4 m/s. The magnitude of the change in the velocity during 0.5 sec is also 4 m/s. Find the minimum possible centripetal acceleration (in m/s^2) of the object. [Take $\pi = 25/8$]

EXERCISE (O-2)

- A ring of radius r and mass per unit length m rotates with an angular velocity ω in free space. The tension in the ring is :
 (A) zero (B) $\frac{1}{2}m\omega^2 r^2$ (C) $m\omega^2 r^2$ (D) $mr\omega^2$
- A uniform rod of mass m and length ℓ rotates in a horizontal plane with an angular velocity ω about a vertical axis passing through one end. The tension in the rod at a distance x from the axis is :
 (A) $\frac{1}{2}m\omega^2 x$ (B) $\frac{1}{2}m\omega^2 \frac{x^2}{\ell}$ (C) $\frac{1}{2}m\omega^2 \ell \left(1 - \frac{x}{\ell}\right)$ (D) $\frac{1}{2} \frac{m\omega^2}{\ell} [\ell^2 - x^2]$
- A particle is moving in a circular path. The acceleration and momentum of the particle at a certain moment are $\vec{a} = (4\hat{i} + 3\hat{j}) \text{ m/s}^2$ and $\vec{p} = (8\hat{i} - 6\hat{j}) \text{ kg-m/s}$. The motion of the particle is
 (A) uniform circular motion (B) accelerated circular motion
 (C) de-accelerated circular motion (D) we can not say anything with $\vec{a}$ and $\vec{p}$ only
- A particle A moves along a circle of radius $R=50 \text{ cm}$ so that its radius vector r relative to the point O (figure) rotates with the constant angular velocity $\omega=0.40 \text{ rad/s}$. Then modulus of the velocity of the particle, and the modulus of its total acceleration will be
 (A) $v=0.4 \text{ m/s}$, $a=0.4 \text{ m/s}^2$
 (B) $v=0.32 \text{ m/s}$, $a=0.32 \text{ m/s}^2$
 (C) $v=0.32 \text{ m/s}$, $a=0.4 \text{ m/s}^2$
 (D) $v=0.4 \text{ m/s}$, $a=0.32 \text{ m/s}^2$
- A rod of length L is pivoted at one end and is rotated with a uniform angular velocity in a horizontal plane. Let T_1 and T_2 be the tensions at the points $L/4$ and $3L/4$ away from the pivoted ends.
 (A) $T_1 > T_2$
 (B) $T_2 > T_1$
 (C) $T_1 = T_2$
 (D) The relation between T_1 & T_2 depends on whether the rod rotates clockwise or anticlockwise
- A particle 'A' of mass $\frac{10}{7} \text{ kg}$ is moving in the positive x -direction. Its initial position is $x = 0$ & initial velocity is 1 m/s . The velocity at $x = 10 \text{ m}$ is : (use the graph given)

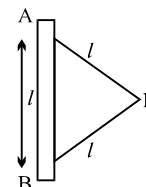


- (A) 4 m/s (B) 2 m/s (C) $3\sqrt{2} \text{ m/s}$ (D) $100/3 \text{ m/s}$

7. A light spring of length 20 cm and force constant 2 N/cm is placed vertically on a table. A small block of mass 1 kg falls on it. The length h from the surface of the table at which the block will have the maximum velocity is :
 (A) 20 cm (B) 15 cm (C) 10 cm (D) 5 cm

8. A particle P of mass m is attached to a vertical axis by two strings AP and BP of length l each. The separation $AB = l$. P rotates around the axis with an angular velocity ω . The tensions in the two strings are T_1 and T_2

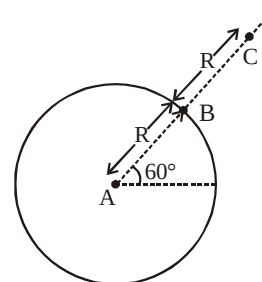
- (A) $T_1 = T_2$
 (B) $T_1 + T_2 = m\omega^2 l$
 (C) $T_1 - T_2 = 2mg$



- (D) BP will remain taut only if $\omega \geq \sqrt{\frac{2g}{l}}$

9. Diagram shows three particles A, B & C. Particle B is moving in a circle of radius R with constant angular velocity ω clockwise. A & C are at rest. Choose the correct statement(s)

- (A) Angular velocity of B wrt A is ω clockwise
 (B) Angular velocity of B wrt A is ω anticlockwise
 (C) Angular velocity of B wrt C is constant & equal to ω
 (D) Angular velocity of B wrt C is not constant and is $\leq \omega$



10. A particle is projected from ground at an angle θ . At a certain instant the velocity vector $\vec{v}$ of particle makes an angle α with horizontal.

- (A) $\frac{d\vec{v}}{dt} = \vec{g}$ (B) Modulus of $\frac{d|\vec{v}|}{dt} = g \sin \alpha$
 (C) Radius of curvature is $\frac{v^2 \cos^2 \alpha}{g \sin \alpha}$ (D) Magnitude of radial acceleration is $g \cos \alpha$

11. A car is moving with constant speed on a rough banked road.

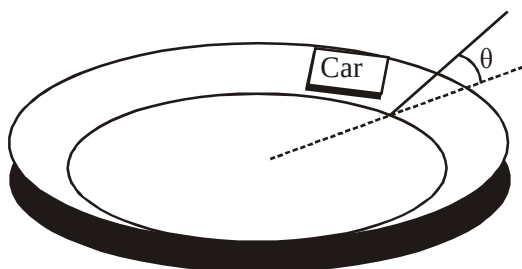
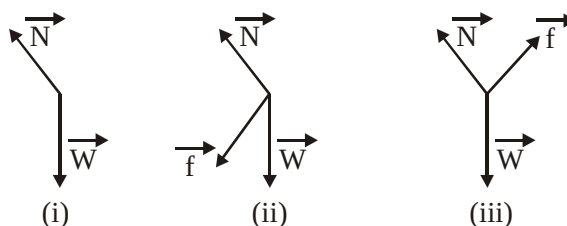


Figure (i), (ii) and (iii) show the free body diagram of car A, B & C respectively:-



- (A) Car A has more speed than car C (B) Car A has less speed than car B
 (C) FBD for car A is not possible (D) If $\mu > \tan \theta$ the FBD for car C is not possible

- 12.** A heavy particle is tied to the end A of a string of length 1.6 m. Its other end O is fixed. It revolves as a conical pendulum with the string making 60° with the vertical. Then

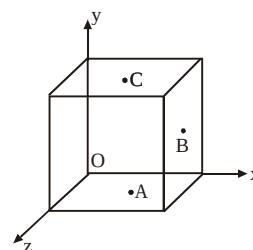
- (A) its period of revolution is $\frac{4\pi}{7}$ sec.
(B) the tension in the string is double the weight of the particle
(C) the velocity of the particle = $2.8\sqrt{3}$ m/s
(D) the centripetal acceleration of the particle is $9.8\sqrt{3}$ m/s².

- 13.** A particle of mass 2 kg is projected with an initial speed $u = 10$ m/sec at an angle $\theta = 30^\circ$ with the horizontal

- (A) The magnitude of work done on the particle during the first half of the total time of flight of the particle is 25J.
(B) The magnitude of work done on the particle during the total time of flight of the particle is 0 J.
(C) The magnitude of average power delivered to the particle during the first half of the flight is 50watt.
(D) The radius of curvature of the trajectory of the particle at the highest point of the projectile is 7.5m.

- 14.** A particle is shifted from A to B and then from B to C where A, B and C are the midpoints of the corresponding faces of a cube of side 2m. If a force $\vec{F} = (3\hat{i} + 4\hat{j} - 5\hat{k})$ N is continuously acting on the particle, then select correct alternative is

- (A) work done from A to B is 7 J
(B) work done from B to C is 1 J
(C) work done A to C is 8 J
(D) force $\vec{F}$, is conservative force



- 15.** Which of the following is/are conservative force(s)?

- (A) $\vec{F} = 2r^3\hat{r}$ (B) $\vec{F} = -\frac{5}{r}\hat{r}$ (C) $\vec{F} = \frac{3(xi+yj)}{(x^2+y^2)^{3/2}}$ (D) $\vec{F} = \frac{3(yi+xj)}{(x^2+y^2)^{3/2}}$

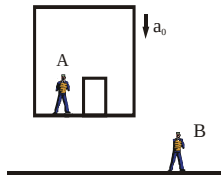
- 16.** If one of the forces acting on a particle is conservative then :

- (A) Its work is zero when the particle moves exactly once around any closed path.
(B) Its work equals the change in the kinetic energy of the particle.
(C) It obeys Newton's second law.
(D) Its work depends on the end points of the motion, not on the path between.

COMPREHENSION TYPE QUESTIONS

Comprehension for Q. 17 to 21

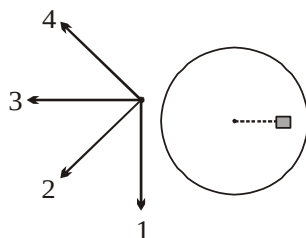
A block of mass m is kept in an elevator which starts moving downward with an acceleration a_0 as shown in figure. The block is observed by two observers A and B for a time interval t_0 .



17. The observer B finds that the work done by gravity is
 (A) $\frac{1}{2} mg^2 t_0^2$ (B) $-\frac{1}{2} mg^2 t_0^2$ (C) $\frac{1}{2} mg a_0 t_0^2$ (D) $-\frac{1}{2} mg a_0 t_0^2$
18. The observer B finds that work done by normal reaction N is
 (A) zero (B) $-N a_0 t_0^2$ (C) $+\frac{N a_0 t_0^2}{2}$ (D) None of these
19. The observer B finds that work done by pseudo force is
 (A) zero (B) $-m a_0^2 t_0$ (C) $+m a_0^2 t_0$ (D) $-mg a_0 t_0$
20. According to observer B, the net work done on the block is
 (A) $-\frac{1}{2} m a_0^2 t_0^2$ (B) $\frac{1}{2} m a_0^2 t_0^2$ (C) $\frac{1}{2} mg a_0 t_0^2$ (D) $-\frac{1}{2} mg a_0 t_0^2$
21. According to the observer A
 (A) the work done by gravity is zero (B) the work done by normal reaction is zero
 (C) the work done by pseudo force is zero (D) all the above

MATRIX MATCH TYPE QUESTIONS

22. A block is placed on a horizontal table which can rotate about its axis. The block is placed at a certain distance from centre as shown in figure. Table rotates such that particle does not slide. Select possible direction of net acceleration of block at the instant shown in figure.



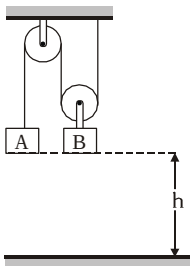
Column-I

- (A) When rotation is clockwise with constant ω
 (B) When rotation is clockwise with decreasing ω
 (C) When rotation is clockwise with increasing ω
 (D) Just after clockwise rotation begins from rest

Column-II

- (P) 1
 (Q) 2
 (R) 3
 (S) 4

23. In the figure shown are two blocks A and B of same mass connected with pulley and string to each other. Initially both of them are at a height of $h = 0.5$ m from ground. After they are released they move in either direction and one of them strike the ground. For, the interval from releasing to when one of them strike, some physical quantities are in column I and their modulus values in SI units are in column II.



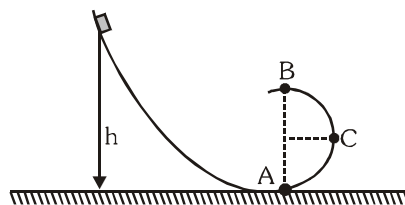
Column I

- (A) Velocity of A immediately before any one of them strike ground.
(B) Velocity of B immediately before any one of them strike ground.
(C) Ratio of work done by gravity on A to workdone by gravity on B.
(D) Acceleration of block A before any one of them strike ground.

Column II

- (P) 1
(Q) 2
(R) 3
(S) 4
(T) 5

24. A block of mass m is released from top of a smooth track as shown in the figure. The end part of the track is a circle in vertical plane of radius R . N is normal reaction of the track at any point of the track. Match the entries of column I with entries of column-II.



Column-I

- (A) $h = \frac{5}{2} R$
(B) $h = \frac{9}{2} R$
(C) $h = R$
(D) $h = 2R$

Column-II

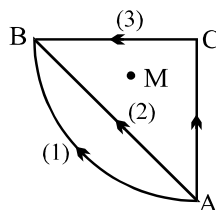
- (P) Net force on the block at C is mg
(Q) $N_A - N_B = 6 mg$
(R) Block leaves contact before B
(S) Block will keep contact with the track is region between A & B.
(T) $N_C > mg$

EXERCISE (JA)

(Not to be discussed in class)

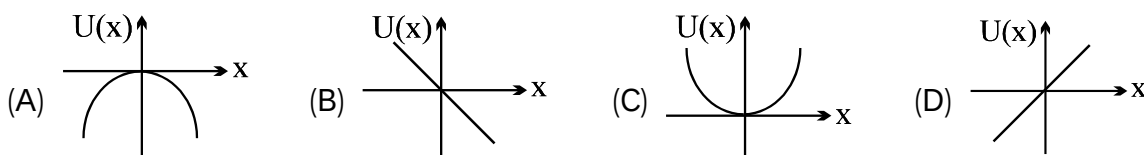
1. In a region of only gravitational field of mass 'M' a particle is shifted from A to B via three different paths in the figure. The work done in different paths are W_1 , W_2 , W_3 respectively then

[IIT-JEE (Scr.)'2003]



- (A) $W_1 = W_2 = W_3$ (B) $W_1 = W_2 > W_3$
 (C) $W_1 > W_2 > W_3$ (D) $W_1 < W_2 < W_3$
2. A particle is placed at the origin and a force $F = kx$ is acting on it (where k is a positive constant). If $U(0) = 0$, the graph of $U(x)$ versus x will be (where U is the potential energy function)

[IIT-JEE' 2004(Scr)]



3. STATEMENT-1

A block of mass m starts moving on a rough horizontal surface with a velocity v . It stops due to friction between the block and the surface after moving through a certain distance. The surface is now tilted to an angle of 30° with the horizontal and the same block is made to go up on the surface with the same initial velocity v . The decrease in the mechanical energy in the second situation is smaller than that in the first situation.

[IIT-JEE 2007]

because

STATEMENT-2

The coefficient of friction between the block and the surface decreases with the increase in the angle of inclination.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

4. **Statement-1** : For an observer looking out through the window of a fast moving train, the nearby objects appear to move in the opposite direction to the train, while the distant objects appear to be stationary.

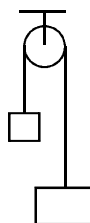
[IIT-JEE 2008]

and

Statement-2 : If the observer and the object are moving at velocities $\vec{V}_1$ and $\vec{V}_2$ respectively with reference to a laboratory frame, the velocity of the object with respect to the observer is $\vec{V}_2 - \vec{V}_1$.

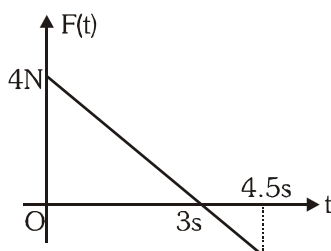
- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True
5. A light inextensible string that goes over a smooth fixed pulley as shown in the figure connects two blocks of masses 0.36 kg and 0.72 kg. Taking $g = 10 \text{ m/s}^2$, find the work done (in **joules**) by the string on the block of mass 0.36 kg during the first second after the system is released from rest.

[IIT-JEE-2009]



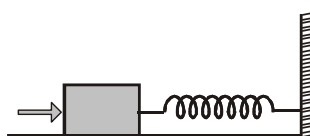
6. A block of mass 2 kg is free to move along the x-axis. It is at rest and from $t=0$ onwards it is subjected to a time-dependent force $F(t)$ in the x-direction. The force $F(t)$ varies with t as shown in the figure. The kinetic energy of the block after 4.5 second is

[IIT-JEE-2010]

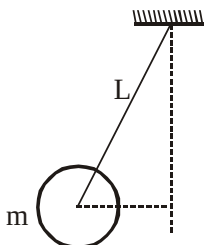


- (A) 4.50 J (B) 7.50 J (C) 5.06 J (D) 14.06 J
7. A block of mass 0.18 kg is attached to a spring of force-constant 2 N/m. The coefficient of friction between the block and the floor is 0.1. Initially the block is at rest and the spring is un-stretched. An impulse is given to the block as shown in the figure. The block slides a distance of 0.06 m and comes to rest for the first time. The initial velocity of the block in m/s is $V = N/10$. Then N is

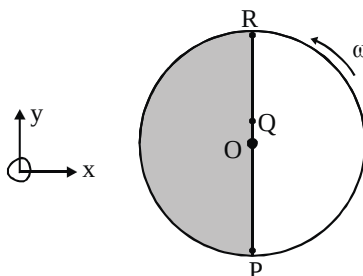
[IIT-JEE-2011]



8. A ball of mass (m) 0.5 kg is attached to the end of a string having length (L) 0.5 m. The ball is rotated on a horizontal circular path about vertical axis. The maximum tension that the string can bear is 324 N. The maximum possible value of angular velocity of ball (in radian/s) is **[IIT-JEE-2011]**



- (A) 9 (B) 18 (C) 27 (D) 36
9. Consider a disc rotating in the horizontal plane with a constant angular speed ω about its centre O. The disc has a shaded region on one side of the diameter and an unshaded region on the other side as shown in the figure. When the disc is in the orientation as shown, two pebbles P and Q are simultaneously projected at an angle towards R. The velocity of projection is in the y-z plane and is same for both pebbles with respect to the disc. Assume that (i) they land back on the disc before the disc has completed $\frac{1}{8}$ rotation, (ii) their range is less than half the disc radius, and (iii) ω remains constant throughout. Then **[IIT-JEE-2012]**



- (A) P lands in the shaded region and Q in the unshaded region.
 (B) P lands in the unshaded region and Q in the shaded region.
 (C) Both P and Q land in the unshaded region.
 (D) Both P and Q land in the shaded region.
10. The work done on a particle of mass m by a force, $K \left[\frac{x}{(x^2 + y^2)^{3/2}} \hat{i} + \frac{y}{(x^2 + y^2)^{3/2}} \hat{j} \right]$ (K being a constant of appropriate dimensions), when the particle is taken from the point $(a, 0)$ to the point $(0, a)$ along a circular path of radius a about the origin in the x-y plane is :- **[IIT-JEE-2013]**

- (A) $\frac{2K\pi}{a}$ (B) $\frac{K\pi}{a}$ (C) $\frac{K\pi}{2a}$ (D) 0

11. A particle of mass 0.2 kg is moving in one dimension under a force that delivers a constant power 0.5W to the particle. If the initial speed (in ms^{-1}) of the particle is zero, the speed (in ms^{-1}) after 5 s is.

[IIT-JEE-2013]

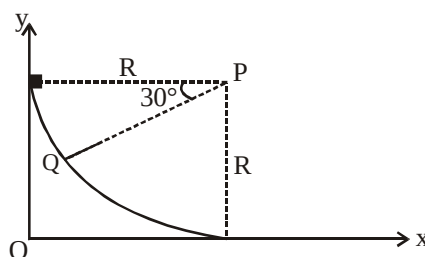
12. A bob of mass m , suspended by a string of length ℓ_1 is given a minimum velocity required to complete a full circle in the vertical plane. At the highest point, it collides elastically with another bob of mass m suspended by a string of length ℓ_2 , which is initially at rest. Both the strings are mass-less and inextensible. If the second bob, after collision acquires the minimum speed required to complete a full circle in the vertical plane, the ratio $\frac{\ell_1}{\ell_2}$ is.

[IIT-JEE-2013]

Paragraph for Questions 13 and 14

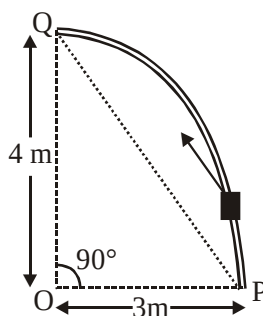
A small block of mass 1 kg is released from rest at the top of a rough track. The track is a circular arc of radius 40 m. The block slides along the track without toppling and a frictional force acts on it in the direction opposite to the instantaneous velocity. The work done in overcoming the friction up to the point Q, as shown in the figure below, is 150 J. (Take the acceleration due to gravity, $g = 10 \text{ m s}^{-2}$)

[IIT-JEE-2013]



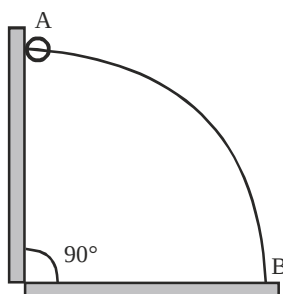
13. The magnitude of the normal reaction that acts on the block at the point Q is
(A) 7.5 N (B) 8.6 N (C) 11.5 N (D) 22.5 N
14. The speed of the block when it reaches the point Q is
(A) 5 ms^{-1} (B) 10 ms^{-1} (C) $10\sqrt{3} \text{ ms}^{-1}$ (D) 20 ms^{-1}
15. Consider an elliptically shaped rail PQ in the vertical plane with $OP = 3 \text{ m}$ and $OQ = 4 \text{ m}$. A block of mass 1 kg is pulled along the rail from P to Q with a force of 18 N, which is always parallel to line PQ (see the figure given). Assuming no frictional losses, the kinetic energy of the block when it reaches Q is $(n \times 10)$ Joules. The value of n is (take acceleration due to gravity = 10 ms^{-2})

[JEE-Advance-2014]



16. A wire, which passes through the hole in a small bead, is bent in the form of quarter of a circle. The wire is fixed vertically on ground as shown in the figure. The bead is released from near the top of the wire and it slides along the wire without friction. As the bead moves from A to B, the force it applies on the wire is :-

[JEE-Advance-2014]

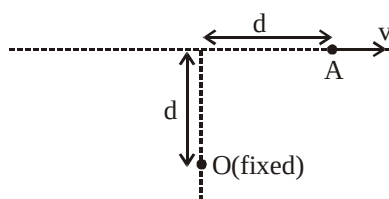


- (A) Always radially outwards
- (B) Always radially inwards
- (C) Radially outwards initially and radially inwards later.
- (D) Radially inwards initially and radially outwards later.

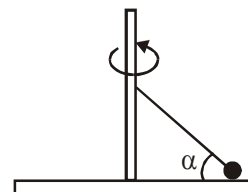
REVIEW TEST

(Not to be discussed in class)

1. Find angular velocity of A with respect to O at the instant shown in the figure.



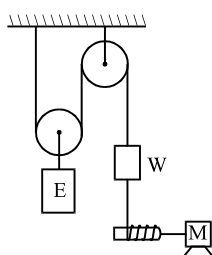
2. A circular platform rotates around a vertical axis with angular velocity $\omega = 10 \text{ rad/s}$. On the platform is a ball of mass 1 kg, attached to the long axis of the platform by a thin rod of length 10 cm ($\alpha = 30^\circ$). Find normal force exerted by the ball on the platform (in newton). Friction is absent.



3. A block of mass $m = 20 \text{ kg}$ is kept at a distance $R = 1 \text{ m}$ from central axis of rotation of a round turn table (A table whose surface can rotate about central axis). Table starts from rest and rotates with constant angular acceleration, $\alpha = 3 \text{ rad/sec}^2$. The friction coefficient between block and table is

$\mu = 0.5$. At time $t = \sqrt{\frac{x}{3}}$ sec from starting of motion (i.e. $t = 0 \text{ sec}$) the block is just about to slip. Find the value of x .

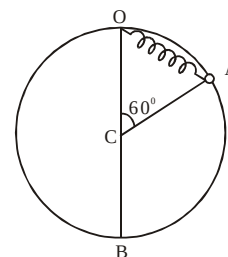
4. The elevator E has a mass of 3000 kg when fully loaded and is connected as shown to a counterweight W of mass 1000 kg. Determine the power in kilowatts delivered by the motor
- (a) when the elevator is moving down at a constant speed of 3 m/s,
- (b) when it has an upward velocity of 3 m/s and a deceleration of 0.5 m/s^2 .



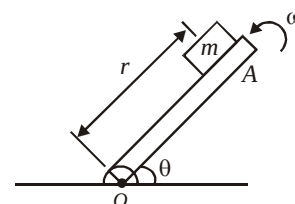
5. Two blocks of mass $m_1 = 10 \text{ kg}$ and $m_2 = 5 \text{ kg}$ connected to each other by a massless inextensible string of length 0.3 m are placed along a diameter of a turn table. The coefficient of friction between the table and m_1 is 0.5 while there is no friction between m_2 and the table. The table is rotating with an angular velocity of 10 rad/sec about a vertical axis passing through its centre. The masses are placed along the diameter of the table on either side of the centre O such that m_1 is at a distance of 0.124 m from O. The masses are observed to be at rest with respect to an observer on the turn table.

- (i) Calculate the frictional force on m_1
- (ii) What should be the minimum angular speed of the turn table so that the masses will slip from this position.
- (iii) How should the masses be placed with the string remaining taut, so that there is no frictional force acting on the mass m_1 .

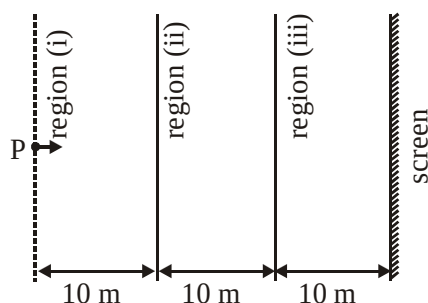
6. A particle of mass 5 kg is free to slide on a smooth ring of radius $r = 20$ cm fixed in a vertical plane. The particle is attached to one end of a spring whose other end is fixed to the top point O of the ring. Initially the particle is at rest at a point A of the ring such that $\angle OCA = 60^\circ$, C being the centre of the ring. The natural length of the spring is also equal to $r = 20$ cm. After the particle is released and slides down the ring the contact force between the particle & the ring becomes zero when it reaches the lowest position B. Determine the force constant of the spring.



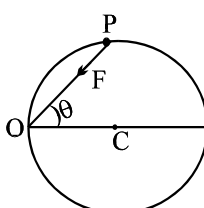
7. The member OA rotates in vertical plane about a horizontal axis through O with a constant counter clockwise velocity $\omega = 3$ rad/s. As it passes the position $\theta = 0$, a small mass m is placed upon it at a radial distance $r = 0.5m$. If the mass is observed to slip at $\theta = 37^\circ$, find the coefficient of friction between the mass & the member.



8. A small particle initially at point p starts moving from rest. The whole space where particle will move is divided into three regions as shown in figure. In region (i) particle accelerates through (5 m/s^2) where direction of acceleration is along the normal of the screen while in region (ii) the acceleration acts in such a way that it is always perpendicular to the direction of motion resulting the particle to move on a circular track having radius $\frac{20}{\sqrt{3}}$ m. There is uniform acceleration in region (iii) in such a manner that velocity of particle become thrice (without change in direction) when it just reach the screen. Find the average speed of a particle (in m/s).



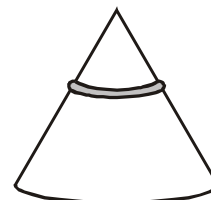
9. A particle P is moving on a circle under the action of only one force acting always towards fixed point O on the circumference. Find ratio of $\frac{d^2\theta}{dt^2}$ & $\left(\frac{d\theta}{dt}\right)^2$.



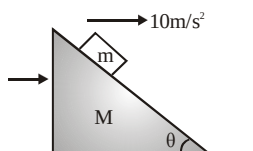
- 10.** A block of mass m moves on a horizontal circle against the wall of a cylindrical room of radius R . The floor of the room on which the block moves is smooth but the friction coefficient between the wall and the block is μ . The block is given an initial speed v_0 . As a function of the speed v write (a) the normal force by the wall on the block, (b) the frictional force by the wall and (c) the tangential acceleration of the block. (d)

Integrate the tangential acceleration $\left(\frac{dv}{dt} = v \frac{dv}{ds} \right)$ to obtain the speed of the block after one revolution.

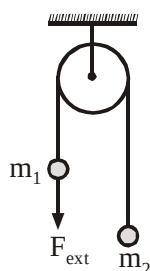
- 11.** A smooth right circular cone of semi vertical angle $\alpha = \tan^{-1}(5/12)$ is at rest on a horizontal plane. A rubber ring of mass 2.5 kg which under a tension of 15 N has an extension of 10 cm is placed on the cone as shown. Find the increase in the radius of the ring in equilibrium.



- 12.** In the figure shown all the surfaces are frictionless, and mass of the block, $m = 1 \text{ kg}$. The block and wedge are held initially at rest. Now wedge is given a horizontal acceleration of 10 m/s^2 by applying a force on the wedge, so that the block does not slip on the wedge. Then work done by the normal force in ground frame on the block in $\sqrt{3}$ seconds is :



- (A) 30 J (B) 60 J (C) 150 J (D) $100 \sqrt{3} \text{ J}$
- 13.** Two bodies of mass m_1 and m_2 ($m_2 > m_1$) are connected by a light inextensible string which passes through a smooth fixed pulley. The instantaneous power delivered by an external agent to pull m_1 with constant velocity v is :



- (A) $(m_2 - m_1) g/v$ (B) $(m_2 - m_1) v/g$ (C) $(m_2 - m_1) gv$ (D) $(m_1 - m_2) gv$
- 14.** Assume the aerodynamic drag force on a car is proportional to its speed. If the power output from the engine is doubled, then the maximum speed of the car.
- (A) is unchanged (B) increases by a factor of $\sqrt{2}$
(C) is also doubled (D) increases by a factor of four.

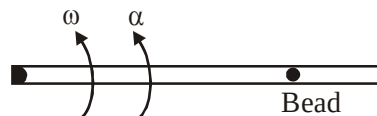
- 15.** A bead is constrained to move on rod in gravity free space as shown in figure. The rod is rotating with angular velocity ω and angular acceleration α about its end. If μ is coefficient of friction. Mark the correct option. (Rod rotates in the plane of paper.)

(A) If $\mu = \frac{\omega^2}{\alpha}$ friction on bead is static in nature

(B) If $\mu > \frac{\omega^2}{\alpha}$ friction on bead is kinetic in nature

(C) If $\mu < \frac{\omega^2}{\alpha}$ friction is static

(D) If bead does not slide relative to rod. Friction will not exist between bead and rod.



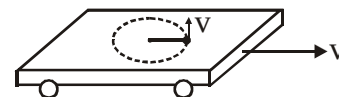
- 16.** On a train moving along east with a constant speed v , a boy revolves a bob with string of length ℓ on smooth surface of a train, with equal constant speed v relative to train. Mark the correct option(s).

(A) Maximum speed of bob is $2v$ in ground frame.

(B) Tension in string connecting bob is $\frac{4mv^2}{\ell}$ at an instant.

(C) Tension in string is $\frac{mv^2}{\ell}$ at all the moments.

(D) Minimum speed of bob is zero in ground frame.



- 17.** An ant travels along a long rod with a constant velocity $\vec{u}$ relative to the rod starting from the origin. The rod is kept initially along the positive x-axis. At $t = 0$, the rod also starts rotating with an angular velocity ω (anticlockwise) in x-y plane about origin. Then

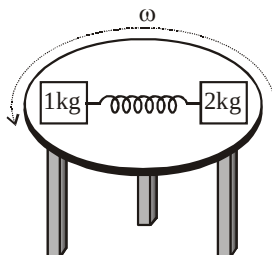
(A) the position of the ant at any time t is $\vec{r} = ut[\cos \omega t \hat{i} + \sin \omega t \hat{j}]$

(B) the speed of the ant at any time t is $u\sqrt{1 + \omega^2 t^2}$

(C) the magnitude of the tangential acceleration of the ant at any time t is $\frac{\omega^2 tu}{\sqrt{1 + \omega^2 t^2}}$

(D) the speed of the ant at any time t is $\sqrt{1 + 2\omega^2 t^2}u$

- 18.** On a circular turn table rotating about its center horizontally with uniform angular velocity ω rad/s placed two blocks of mass 1 kg and 2 kg, on a diameter symmetrically about center. Their separation is 1m and friction is sufficient to avoid slipping. The spring between them as shown is stretched and applied force of 5N. If f_1 and f_2 are values of friction on 1 kg & 2kg block respectively:-



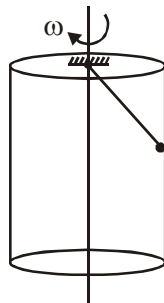
(A) For $\omega = 2$ rad/s, $f_1 = 3$ N & $f_2 = 1$ N

(B) For $\omega = 3$ rad/s, $f_1 = 0.5$ N & $f_2 = 4$ N

(C) For $\omega = \sqrt{10}$ rad/s, $f_1 = 0$ & $f_2 = 5$ N

(D) For $\omega = \sqrt{10}$ rad/s, $f_1 = 0$ & $f_2 = 0$ N

19. In the shown figure inside a fixed hollow cylinder with vertical axis a pendulum is moving conically with its axis same as that of the cylinder with uniform angular velocity. Radius of cylinder is 30 cm, length of string is 50 cm and mass of bob is 400 gm. The bob makes contact with the inner frictionless wall of the cylinder while moving :-



- (A) The minimum value of angular velocity of the bob so that it does not leave contact is 5 rad/s
(B) Tension in the string is 5N for all values of angular velocity
(C) For angular velocity of 10 rad/s the bob pushes the cylinder with a force of 9N
(D) For angular velocity of 10 rad/s, tension in the string is 20N

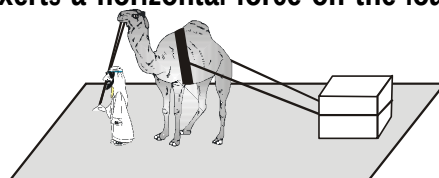
Paragraph for Question no. 20 to 24

Ram and Shyam are two fast friends since childhood. Shyam neglected studies and now has no means to earn money other than a camel whereas Ram becomes an engineer. Now both are working in the same factory. Shyam uses camel to transport the load within the factory.

Due to low salary & degradation in health of camel, Shyam becomes worried and meet his friend Ram and discusses his problem. Ram collected some data & with some assumptions concluded the following:

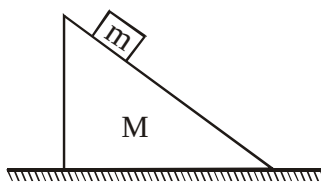
- The load used in each trip is 1000 kg and has friction coefficient $\mu_k = 0.1$ and $\mu_s = 0.2$.
- Mass of camel is 500 kg.
- Load is accelerated for first 50 m with constant acceleration, then it is pulled at a constant speed of 5m/s for 2 km and at last stopped with constant retardation in 50 m.
- From biological data, the rate of consumption of energy of camel can be expressed as $P = 18 \times 10^3 v + 10^4$ J/s where P is the power and v is the velocity of the camel. After calculations on different issues Ram suggested proper food, speed of camel etc. to his friend. For the welfare of Shyam, Ram wrote a letter to the management to increase his salary.

(Assuming that the camel exerts a horizontal force on the load) :



20. Sign of work done by the camel on the load during parts of motion : accelerated motion, uniform motion and retarded motion respectively are :
- (A) +ve, +ve, +ve (B) +ve, +ve, -ve
(C) +ve, zero, -ve (D) +ve, zero, +ve

21. The ratio of magnitude of work done by camel on the load during accelerated motion to retarded motion is
 (A) 3 : 5 (B) 2.2 : 1 (C) 1 : 1 (D) 5 : 3
22. Maximum power transmitted by the camel to load is
 (A) 6250 J/s (B) 5000 J/s (C) 10^5 J/s (D) 1250 J/s
23. The ratio of the energy consumed of the camel during uniform motion for the two cases when it moves with speed 5 m/s to the case when it moves with 10 m/s.
 (A) $\frac{19}{20}$ (B) $\frac{19}{10}$ (C) $\frac{10}{19}$ (D) $\frac{20}{19}$
24. The total energy consumed of the camel during the trip of 2100 m is
 (A) 2.1×10^6 J (B) 4.22×10^7 J (C) 2.22×10^4 J (D) 4.22×10^6 J
25. A block of mass m lies on wedge of mass M . The wedge in turn lies on smooth horizontal surface. Friction is absent everywhere. The wedge block system is released from rest. All situation given in column-I are to be estimated in duration the block undergoes a vertical displacement 'h' starting from rest (assume the block to be still on the wedge, g is acceleration due to gravity).

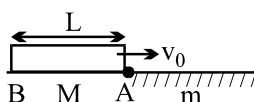
**Column-I**

- (A) Work done by normal reaction acting on the block is
- (B) Work done by normal reaction (exerted by block) acting on wedge is
- (C) The sum of work done by normal reaction on block and work done by normal reaction (exerted by block) on wedge is
- (D) Net work done by all forces on block is

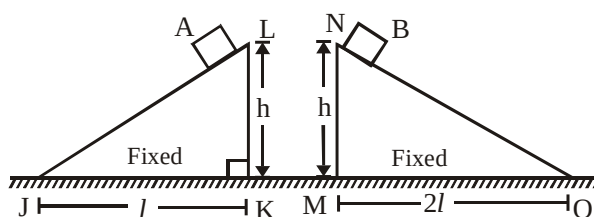
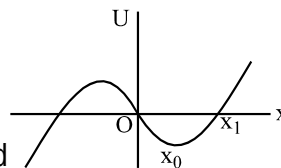
Column-II

- (P) Positive
- (Q) Negative
- (R) Zero
- (S) Less than mgh in magnitude

26. A uniform rod of mass m length L is sliding along its length on a horizontal table whose top is partly smooth & rest rough with friction coefficient μ . If the rod after moving through smooth part, enters the rough with velocity v_0 .
- (i) What will be the magnitude of the friction force when its x length ($< L$) lies in the rough part during sliding.
- (ii) Determine the minimum velocity v_0 with which it must enter so that it lies completely in rough region before coming to rest.



27. A particle is confined to move along the +x axis under the action of a force $F(x)$ that is derivable from the potential $U(x) = ax^3 - bx$.
- (i) Find the expression for $F(x)$
- (ii) When the total energy of the particle is zero, the particle can be trapped within the interval $x=0$ to $x=x_1$. For this case find the values of x_1 .
- (iii) Determine the maximum kinetic energy that the trapped particle has in its motion. Express all answers in terms a and b. At what value of x will the kinetic energy be maximum ?
28. A 650-kg elevator starts from rest. It moves upward for 3.00 s with constant acceleration until it reaches its cruising speed of 1.75 m/s.
- (i) What is the average power of the elevator motor during this period?
- (ii) How does this power compare with its power when it moves at its cruising speed?
29. A thin circular loop of radius R rotates about its vertical diameter with an angular frequency ω . Show that a small bead on the wire loop remains at its lowermost point for $\omega \leq \sqrt{g/R}$. What is the angle made by the radius vector joining the centre to the bead with the vertical downward direction for $\omega = \sqrt{2g/R}$? Neglect friction.
30. Two identical blocks A and B are placed on two inclined planes as shown in diagram. Neglect air resistance and other friction. Choose the correct statement :
- Statement I :** Kinetic energy of 'A' on sliding to J will be greater than the kinetic energy of B on falling to M.
- Statement II :** Acceleration of 'A' will be greater than acceleration of 'B' when both are released to slide on inclined plane.
- Statement III :** Work done by external agent to move block slowly from position B to O is negative



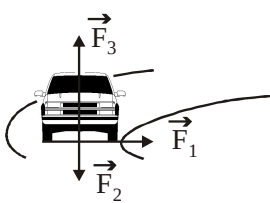
- (A) statement I is true
- (B) statement II is true
- (C) statement I and III are true
- (D) statement II and III are true

31. Column-I shows certain situations and column-2 shows information about forces.

Column-I

Column-II

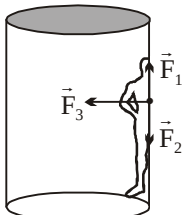
(A) Situation



Front view of a car rounding a curve with constant speed.

(P) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3$ is centripetal force.

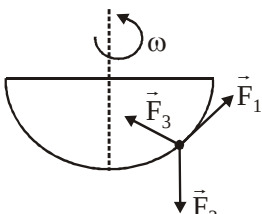
(B)



Passengers in a rotor not sliding relative to rotor wall cylindrical rotor is rotating with constant angular velocity about its symmetry axis.

(Q) $\vec{F}_1$ is static friction.

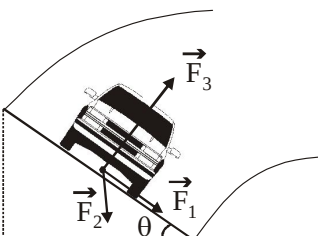
(C)



Particle kept on rough surface of a bowl, no relative motion of particle in bowl, bowl has constant angular velocity

(R) $\vec{F}_1$ can be in direction opposite to that shown in figure.

(D)



Car moving on a banked road with constant speed, no sideways skidding

(S) $\vec{F}_1 + \vec{F}_2 = \vec{0}$

(T) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$

ANSWER KEY

EXERCISE (S-2)

 1. Ans. (i) $mgR/2$, (ii) $2\sqrt{gR}$

2. Ans. 2.5cm

 4. Ans. $\frac{E\mu \cos \alpha}{\sin \alpha + \mu \cos \alpha}$

 5. Ans. (i) $\frac{V_0^2}{2g}$, (ii) $2g$

6. Ans. 6

 7. Ans. $2V_0$

8. Ans. 3

9. Ans. 0390 cm

10. Ans. 50

11. Ans. 0008.33

EXERCISE (O-2)

1. Ans. (C) 2. Ans. (D)

3. Ans. (B)

4. Ans. (D)

5. Ans. (A)

6. Ans. (A) 7. Ans. (B)

8. Ans. (B, C, D)

9. Ans. (A, D)

10. Ans. (A,B,D)

11. Ans. (A, B)

12. Ans. (A, B, C, D)

13. Ans. (ABCD)

14. Ans. (A,B,C,D)

15. Ans. (A,B,C)

16. Ans. (A,C,D)

17. Ans. (C)

18. Ans. (D)

19. Ans. (A)

20. Ans. (B)

21. Ans. (D)

22. Ans. (A)-R ; (B)-S ; (C)-Q ; (D)-P

23. Ans. (A)→(Q); (B)→(P); (C)→(Q); (D)→(S)

24. Ans. (A) → (Q,S,T); (B) → (Q,S,T); (C) → (P); (D) → (R,T)

EXERCISE (JA)

1. Ans. (A) 2. Ans. (A)

3. Ans. (C)

4. Ans. (B)

5. Ans. 8

6. Ans. (C)

7. Ans. 4

8. Ans. (D)

9. Ans. C OR D

10. Ans. (D)

11. Ans. 5

12. Ans. 5

13. Ans. (A)

14. Ans. (B)

15. Ans. 5

16. Ans. (D)

REVIEW TEST

 1. Ans. $\frac{v}{2d}$

2. Ans. 5

3. Ans. 5

4. Ans. (a) -30 kW, 19.5 kW

5. Ans. (i) 36N, (ii) 11.66rad/sec, (iii) 0.1m, 0.2m

6. Ans. 1000 N/m

 7. Ans. $\mu = \frac{3}{16}$

8. Ans. 10

9. Ans. $2 \tan \theta$ 10. Ans. (a) $\frac{mv^2}{R}$, (b) $\frac{\mu mv^2}{R}$, (c) $-\frac{\mu v^2}{R}$, (d) $v_0 e^{-2\pi\mu}$ 11. Ans. $\Delta r = \frac{mg \cot \alpha}{4\pi^2 k}$, 1cm

12. Ans. (C)

13. Ans. (C) 14. Ans. (B)

15. Ans. (A)

16. Ans. (A, C, D)

17. Ans. (A, B, C)

18. Ans. (A, B, C)

19. Ans. (A, B, C)

20. Ans. (A)

21. Ans. (D)

22. Ans. (A) 23. Ans. (D)

24. Ans. (B)

25. Ans. (A) Q,S (B) P (C) R (D) P,S

26. Ans. (i) $f = -\frac{\mu m}{\ell} xg$; (ii) $\sqrt{\mu g \ell}$ 27. Ans. $F = -3ax^2 + b$, $x = \sqrt{\frac{b}{a}}$ 28. Ans. (i) $6 \times 10^3 \text{ W}$ (ii) $1 \times 10^4 \text{ W}$

30. Ans. (D)

31. Ans. (A) P,Q (B) P,Q,S (C) P,Q,R (D) P,Q,R

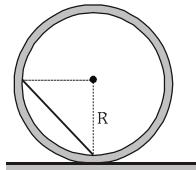
Important Notes

Important Notes

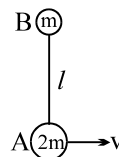
CENTRE OF MASS, MOMENTUM & COLLISION

EXERCISE (S-2)

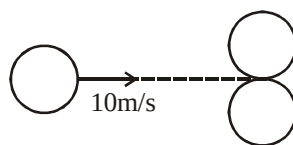
1. Inside a hollow uniform sphere of inner radius R a uniform rod of length $R\sqrt{2}$ is released from the state of rest as shown. The mass of the rod is same as that of the sphere. Assume friction to be absent everywhere. Horizontal displacement of sphere with respect to earth in the time in which the rod becomes horizontal, is



2. Two masses A and B connected with an inextensible string of length ℓ lie on a smooth horizontal plane. A is given a velocity of v m/s along the ground perpendicular to line AB as shown in figure. Find the tension in string during their subsequent motion.



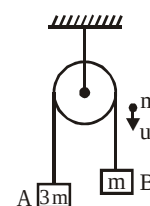
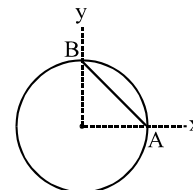
3. A massive vertical wall is approaching a man at a speed u . When it is at a distance of 10 m, the man throws a ball with speed 10 m/s at an angle of 37° which after completely elastic rebound reaches back directly into his hands. Find the velocity u of the wall.
4. A ball with initial speed of 10 m/s collides elastically with two other identical ball whose centres are on a line perpendicular to the initial velocity and which are initially in contact with each other. All the three ball are lying on a smooth horizontal table. The first ball is aimed directly at the contact point of the other two balls. All the balls are smooth. Find the velocities of the three balls after the collision.



5. A sphere of mass m is moving with a velocity $4\hat{i} - \hat{j}$ when it hits a smooth wall and rebounds with velocity $\hat{i} + 3\hat{j}$. Find the impulse it receives. Find also the coefficient of restitution between the sphere and the wall.
6. Two bodies of same mass tied with an inelastic string of length ℓ lie together. One of them is projected vertically upwards with velocity $\sqrt{6g\ell}$. Find the maximum height up to which the centre of mass of system of the two masses rises.
7. Two particles A and B of mass $2m$ and m respectively are attached to the ends of a light inextensible string of length $4a$ which passes over a small smooth peg at a height $3a$ from an inelastic table. The system is released from rest with each particle at a height a from the table. Find
- The speed of B when A strikes the table.
 - The time that elapses before A first hits the table.
 - The time for which A is resting on the table after the first collision & before it is first jerked off.

EXERCISE (O-2)

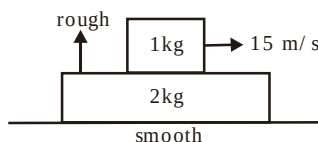
1. An object comprises of a uniform ring of radius R and its uniform chord AB (not necessarily made of the same material) as shown. Which of the following can not be the centre of mass of the object
- (A) $(R/3, R/3)$ (B) $(R/3, R/2)$
 (C) $(R/4, R/4)$ (D) $(R/\sqrt{2}, R/\sqrt{2})$
2. A system of two blocks A and B are connected by an inextensible massless strings as shown. The pulley is massless and frictionless. Initially the system is at rest when, a bullet of mass ' m ' moving with a velocity ' u ' as shown hits the block ' B ' and gets embedded into it. The impulse imparted by tension force to the block of mass $3m$ is

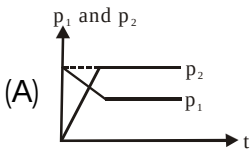
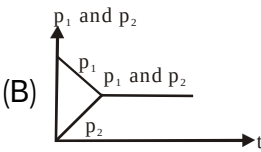
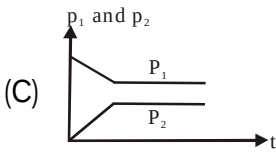
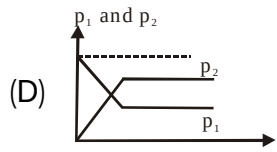


- (A) $\frac{5mu}{4}$ (B) $\frac{4mu}{5}$ (C) $\frac{2mu}{5}$ (D) $\frac{3mu}{5}$

Question No. 3 to 5 (3 questions)

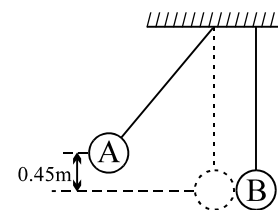
A 1 kg block is given a velocity of 15 m/s towards right over a very long rough plank of mass 2 kg as shown in figure.



3. The correct graph showing linear momentum of 1 kg (i.e. p_1) and of 2 kg (i.e. p_2) versus time is :
- (A)  (B)  (C)  (D) 
4. If coefficient of friction between the two blocks is equal to 0.4, then magnitude of initial slope of p_1 versus t and p_2 versus t (in SI unit) will be :-
- (A) 4 and 2 (B) 2 and 4 (C) 4 and 4 (D) 2 and 2
5. Momentum of both the blocks are equal at time $t = \dots\dots\dots$ seconds :
- (A) 1.75 (B) 1.875 (C) 2.5 (D) 1.25
6. A cannon of mass $5m$ (including a shell of mass m) at rest on a smooth horizontal ground, fires the shell at an angle θ with the horizontal at a velocity u relative to cannon. Find the horizontal distance of the point where shell strikes the ground from the initial position of the cannon:

- (A) $\frac{4u^2 \sin 2\theta}{5g}$ (B) $\frac{u^2 \sin 2\theta}{5g}$ (C) $\frac{3u^2 \sin 2\theta}{5g}$ (D) $\frac{8u^2 \sin 2\theta}{5g}$

7. Two balls A and B having masses 1 kg and 2 kg, moving with speeds 21 m/s and 4 m/s respectively in opposite direction, collide head on. After collision A moves with a speed of 1 m/s in the same direction, then correct statements is :
- (A) The velocity of B after collision is 6 m/s opposite to its direction of motion before collision.
(B) The coefficient of restitution is 0.2.
(C) The loss of kinetic energy due to collision is 200 J.
(D) The impulse of the force between the two balls is 40 Ns.
8. A particle moving with kinetic energy = 3 joule makes an elastic head on collision with a stationary particle. Mass of the stationary particle is double of the moving particle.
- (A) The minimum kinetic energy of the system is 1 joule.
(B) The maximum elastic potential energy of the system is 2 joule.
(C) Momentum and total kinetic energy of the system are conserved at every instant.
(D) The ratio of kinetic energy to potential energy of the system first decreases and then increases.
9. Two massless string of length 5 m hang from the ceiling very near to each other as shown in the figure. Two balls A and B of masses 0.25 kg and 0.5 kg are attached to the string. The ball A is released from rest at a height 0.45 m as shown in the figure. The collision between two balls is completely elastic. Immediately after the collision, the kinetic energy of ball B is 1 J. The velocity of ball A just after the collision is
- (A) 5 m/s to the right (B) 5 m/s to the left (C) 1 m/s to the right (D) 1 m/s to the left
10. A particle of mass m , kinetic energy K and momentum p collides head on elastically with another particle of mass $2m$ at rest. After collision :

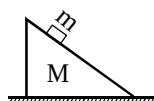


Column I	Column II
(A) Momentum of first particle	(P) $\frac{3}{4}p$
(B) Momentum of second particle	(Q) $-K/9$
(C) Kinetic energy of first particle	(R) $-p/3$
(D) Kinetic energy of second particle	(S) $\frac{8K}{9}$
	(T) None

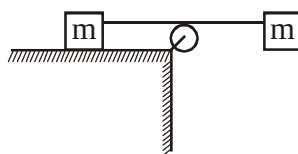
- 11.** In each situation of column-I, a system involving two bodies is given. All strings and pulleys are light and friction is absent everywhere. Initially each body of every system is at rest. Consider the system in all situation of column I from rest till any collision occurs. Then match the statements in column – I with the corresponding results in column-II

Column I

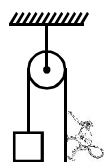
- (A) The block plus wedge system is placed over smooth horizontal surface. After the system is released from rest, the centre of mass of system



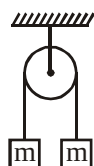
- (B) The string connecting both the blocks of mass m is horizontal. Left block is placed over smooth horizontal table as shown. After the two block system is released from rest, the centre of mass of system



- (C) The block and monkey have same mass. The monkey starts climbing up the rope. After the monkey starts climbing up, the centre of mass of monkey+block system



- (D) Both block of mass m are initially at rest. The left block is given initial velocity u downwards. Then, the centre of mass of two block system afterwards

**Column II**

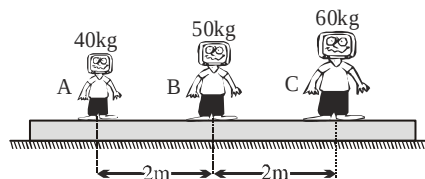
- (P) Shifts towards right

- (Q) Shifts downwards

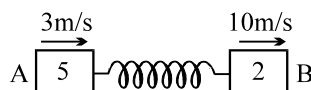
- (R) Shifts upwards

- (S) Does not shift

12. Three man A, B & C of mass 40 kg, 50 kg & 60 kg are standing on a plank of mass 90 kg, which is kept on a smooth horizontal plane. If A & C exchange their positions then man B will shift



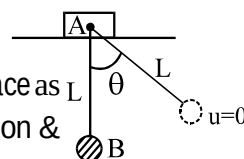
- (A) $1/3$ m towards left w.r.t. ground
(B) $1/3$ m towards right w.r.t. ground
(C) will not move w.r.t. ground
(D) $5/3$ m towards left w.r.t. ground
13. Two blocks A (5kg) and B(2kg) attached to the ends of a spring constant 1120N/m are placed on a smooth horizontal plane with the spring undeformed. Simultaneously velocities of 3m/s and 10m/s along the line of the spring in the same direction are imparted to A and B then



- (A) when the extension of the spring is maximum the velocities of A and B are zero.
(B) the maximum extension of the spring is 25 cm.
(C) maximum extension and maximum compression occur alternately.
(D) the minimum speed of B is 0.

Question No. 14 to 16 (3 questions)

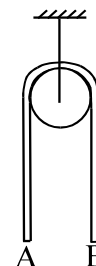
A small ball B of mass m is suspended with light inelastic string of length L from a block A of same mass m which can move on smooth horizontal surface as shown in the figure. The ball is displaced by angle θ from equilibrium position & then released.



14. The displacement of block when ball reaches the equilibrium position is
- (A) $\frac{L \sin \theta}{2}$ (B) $L \sin \theta$ (C) L (D) none of these
15. Maximum velocity of block during subsequent motion of the system after release of ball is
- (A) $[g(1 - \cos \theta)]^{1/2}$ (B) $[2g(1 - \cos \theta)]^{1/2}$
(C) $[gl \cos \theta]^{1/2}$ (D) informations are insufficient to decide
16. The displacement of centre of mass of A + B system till the string becomes vertical is
- (A) zero (B) $\frac{L}{2}(1 - \cos \theta)$ (C) $\frac{L}{2}(1 - \sin \theta)$ (D) none of these

Question No. 17 and 18 (2 questions)

A uniform chain of length $2L$ is hanging in equilibrium position, if end B is given a slightly downward displacement the imbalance causes an acceleration. Here pulley is small and smooth & string is inextensible

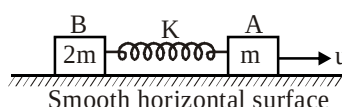


17. The acceleration of end B when it has been displaced by distance x , is
- (A) $\frac{x}{L} g$ (B) $\frac{2x}{L} g$ (C) $\frac{x}{2} g$ (D) g

18. The velocity v of the string when it slips out of the pulley (height of pulley from floor $> 2L$)

- (A) $\sqrt{\frac{gL}{2}}$ (B) $\sqrt{2gL}$ (C) $\sqrt{gL}$ (D) none of these

19. Two blocks A and B of mass m and $2m$ respectively are connected by a massless spring of spring constant K . This system lies over a smooth horizontal surface. At $t = 0$ the block A has velocity u towards right as shown while the speed of block B is zero, and the length of spring is equal to its natural length.



Column-I

- (A) The velocity of block A
 (B) The velocity of block B
 (C) The kinetic energy of system of two block
 (D) The potential energy of spring

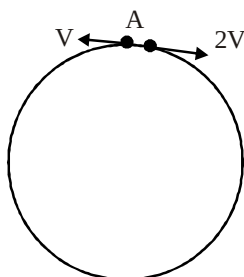
Column-II

- (P) can never be zero
 (Q) may be zero at certain instants of time
 (R) is minimum at maximum compression of spring
 (S) is maximum at maximum extension of spring

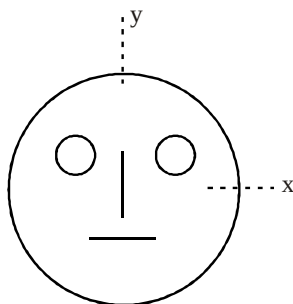
EXERCISE (J-A)

Straight Objective Type

- Two blocks of masses 10 kg and 4 kg are connected by a spring of negligible mass and placed on a frictionless horizontal surface. An impulse gives a velocity of 14 m/s to the heavier block in the direction of the lighter block. The velocity of the centre of mass is : **[IIT-JEE(Scr) 2002]**
(A) 30 m/s (B) 20 m/s (C) 10 m/s (D) 5 m/s
- Two small particles of equal masses start moving in opposite directions from a point A in a horizontal circular orbit. Their tangential velocities are v and $2v$, respectively, as shown in the figure. Between collisions, the particles move with constant speeds. After making how many elastic collisions, other than that at A, these two particles will again reach the point A? **[IIT-JEE-2009]**



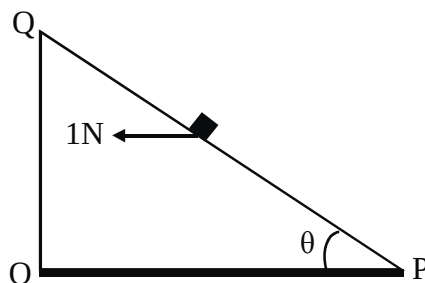
- (A) 4 (B) 3 (C) 2 (D) 1
- Look at the drawing given in the figure which has been drawn with ink of uniform line-thickness. The mass of ink used to draw each of the two inner circles, and each of the two line segments is m . The mass of the ink used to draw the outer circle is $6m$. The coordinates of the centres of the different parts are: outer circle $(0, 0)$, left inner circle $(-a, a)$, right inner circle (a, a) , vertical line $(0, 0)$ and horizontal line $(0, -a)$. The y -coordinate of the centre of mass of the ink in this drawing is **[IIT-JEE-2009]**



- (A) $\frac{a}{10}$ (B) $\frac{a}{8}$ (C) $\frac{a}{12}$ (D) $\frac{a}{3}$

Multiple Correct Type Questions

4. Two balls, having linear momenta $\vec{p}_1 = p\hat{i}$ and $\vec{p}_2 = -p\hat{i}$, undergo a collision in free space. There is no external force acting on the balls. Let $\vec{p}'_1$ and $\vec{p}'_2$ be their final momenta. The following option(s) is(are) NOT ALLOWED for any non-zero value of p , a_1 , a_2 , b_1 , b_2 , c_1 and c_2 . [IIT-JEE 2008]
- (A) $\vec{p}'_1 = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$; $\vec{p}'_2 = a_2\hat{i} + b_2\hat{j}$ (B) $\vec{p}'_1 = c_1\hat{k}$; $\vec{p}'_2 = c_2\hat{k}$
- (C) $\vec{p}'_1 = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$, $\vec{p}'_2 = a_2\hat{i} + b_2\hat{j} - c_1\hat{k}$ (D) $\vec{p}'_1 = a_1\hat{i} + b_1\hat{j}$; $\vec{p}'_2 = a_2\hat{i} + b_1\hat{j}$
5. A point mass of 1 kg collides elastically with a stationary point mass of 5 kg. After their collision, the 1 kg mass reverses its direction and moves with a speed of 2 m/s. Which of the following statement(s) is(are) correct for the system of these two masses? [IIT-JEE 2010]
- (A) Total momentum of the system is 3 kg m/s.
 (B) Momentum of 5 kg mass after collision is 4 kg m/s.
 (C) Kinetic energy of the centre of mass is 0.75 J.
 (D) Total kinetic energy of the system is 4 J.
6. A small block of mass of 0.1 kg lies on a fixed inclined plane PQ which makes an angle θ with the horizontal. A horizontal force of 1 N acts on the block through its center of mass as shown in the figure. The block remains stationary if (take $g = 10 \text{ m/s}^2$) [IIT-JEE 2012]



- (A) $\theta = 45^\circ$
 (B) $\theta > 45^\circ$ and a frictional force acts on the block towards P
 (C) $\theta > 45^\circ$ and a frictional force acts on the block towards Q
 (D) $\theta < 45^\circ$ and a frictional force acts on the block towards Q
7. A particle of mass m is projected from the ground with an initial speed u_0 at an angle α with the horizontal. At the highest point of its trajectory, it makes a completely inelastic collision with another identical particle, which was thrown vertically upward from the ground with the same initial speed u_0 . The angle that the composite system makes with the horizontal immediately after the collision is :- [IIT-JEE 2013]

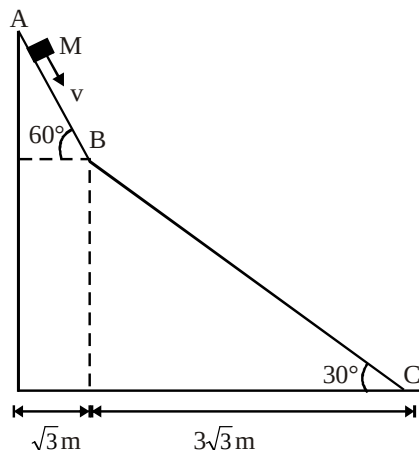
- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{4} + \alpha$ (C) $\frac{\pi}{2} - \alpha$ (D) $\frac{\pi}{2}$

Linked Comprehension Type Questions

Comprehension Q.8 to Q.10 (3 Questions)

[IIT-JEE 2008]

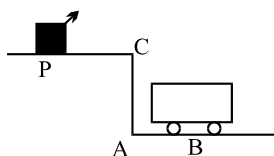
A small block of mass M moves on a frictionless surface of an inclined plane, as shown in figure. The angle of the incline suddenly changes from 60° to 30° at point B . The block is initially at rest at A . Assume that collisions between the block and the incline are totally inelastic ($g = 10 \text{ m/s}^2$)



8. The speed of the block at point B immediately after it strikes the second incline is :
 (A) $\sqrt{60} \text{ m/s}$ (B) $\sqrt{45} \text{ m/s}$ (C) $\sqrt{30} \text{ m/s}$ (D) $\sqrt{15} \text{ m/s}$
9. The speed of the block at point C , immediately before it leaves the second incline is :
 (A) $\sqrt{120} \text{ m/s}$ (B) $\sqrt{105} \text{ m/s}$ (C) $\sqrt{90} \text{ m/s}$ (D) $\sqrt{75} \text{ m/s}$
10. If collision between the block and the incline is completely elastic, then the vertical (upward) component of the velocity of the block at point B , immediately after it strikes the second incline is :
 (A) $\sqrt{30} \text{ m/s}$ (B) $\sqrt{15} \text{ m/s}$ (C) 0 (D) $-\sqrt{15} \text{ m/s}$

Subjective Type Questions

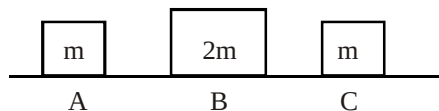
11. A car P is moving with a uniform speed of $5(3^{1/2}) \text{ m/s}$ towards a carriage of mass 9 Kg at rest kept on the rails at a point B as shown in figure.



The height AC is 120 m . Cannon balls of 1 Kg are fired from the car with an initial velocity 100 m/s at an angle 30° with the horizontal. The first canon ball hits the stationary carriage after a time t_0 and sticks to it. Determine t_0 . At t_0 , the second cannon ball is fired. Ignore the resistive force between the rails and the carriage. If the second ball also hits and sticks to the carriage. What will be the horizontal velocity of the carriage just after the second impact?

[IIT-JEE 2001]

- 12.** Three objects A, B and C are kept in a straight line on a frictionless horizontal surface. These have masses m , $2m$ and m , respectively. The object A moves towards B with a speed 9 m/s and makes an elastic collision with it. Thereafter, B makes completely inelastic collision with C. All motions occur on the same straight line. Find the final speed (in m/s) of the object C. **[IIT-JEE-2009]**



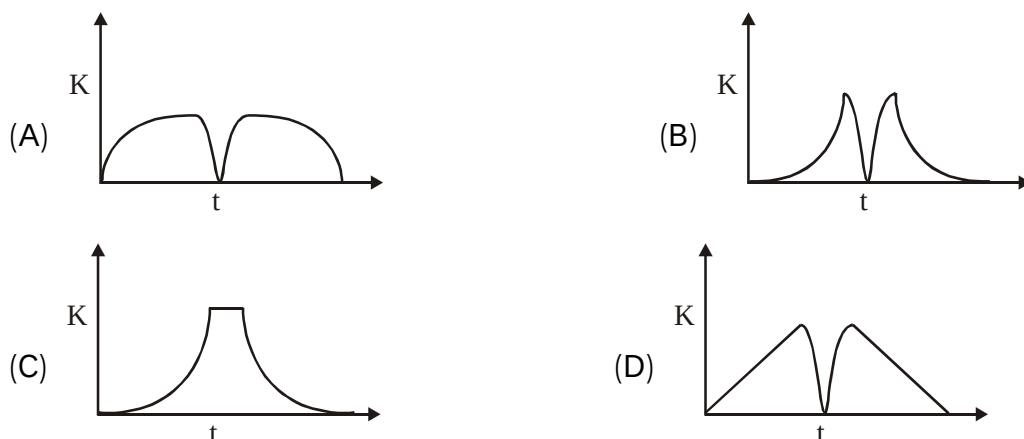
- 13.** A bob of mass m , suspended by a string of length ℓ_1 is given a minimum velocity required to complete a full circle in the vertical plane. At the highest point, it collides elastically with another bob of mass m suspended by a string of length ℓ_2 , which is initially at rest. Both the strings are massless and inextensible. If the second bob, after collision acquires the minimum speed required to complete

a full circle in the vertical plane, the ratio $\frac{\ell_1}{\ell_2}$ is.

[JEE Advance 2013]

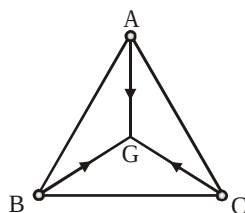
- 14.** A tennis ball is dropped on a horizontal smooth surface. It bounces back to its original position after hitting the surface. The force on the ball during the collision is proportional to the length of compression of the ball. Which one of the following sketches describes the variation of its kinetic energy K with time t most appropriately? The figures are only illustrative and not to the scale.

[JEE Advance 2014]

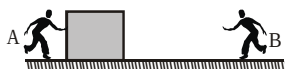


REVIEW TEST

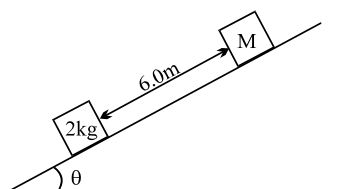
1. A particle A of mass 2kg lies on the edge of a table of height 1m. It is connected by a light inelastic string of length 0.7m to a second particle B of mass 3kg which is lying on the table 0.25m from the edge (line joining A & B is perpendicular to the edge). If A is pushed gently so that it start falling from table then, find the speed of B when it starts to move. Also find the impulsive tension in the string at that moment.
2. Three particles A, B and C of equal mass move with equal speed v along the medians of an equilateral triangle as shown in fig. They collide at the centroid G of the triangle. After the collision, A comes to rest, B retraces its path with the speed v . What is the velocity of C?



3. A body is thrown vertically upwards from ground with a speed of 10 m/s. If coefficient of restitution of ground, $e = 1/2$. Find (i) the total distance travelled by the time it almost stops. (ii) time elapsed (after the ball has been thrown) when it is at its subsequent maximum height for the third time.
4. Two persons A and B each of mass 100 kg are on a frictionless horizontal surface. Person A at rest is holding a block of mass 50 kg, suddenly pushes the block with some velocity (v) towards the person B approaching at a velocity of 5 m/s. B catches the block & slows down. Now the separation between A and B becomes constant. Find the speed v (in m/s).



5. Two blocks of mass 2kg and M are at rest on an inclined plane and are separated by a distance of 6 m as shown. The coefficient of friction between each of the blocks and the inclined plane is 0.25. The 2kg block is given a velocity of 10 m/s up the inclined plane. It collides with M , comes back and has a velocity of 1 m/s when it reaches its initial position. The other block M after the collision moves 0.5 m up and comes to rest. Calculate the coefficient of restitution between the blocks and the mass of the block M . [Take $\sin\theta \approx \tan\theta = 0.05$ and $g = 10 \text{ m/s}^2$]



6. Two interacting particles form a closed system whose centre of mass is at rest.



Figure illustrates the positions of both particles at a certain moment and the trajectory of the particle of mass m_1 . Select the trajectory of the particle of mass m_2 if $m_2 = m_1$.



7. Two particles bearing mass ratio $n : 1$ are interconnected by a light inextensible string that passes over a smooth pulley. If the system is released, then the acceleration of the centre of mass of the system is

(A) $2(n-1)g$ (B) $\left(\frac{n+1}{n-1}\right)^2 g$ (C) $\left(\frac{n-1}{n+1}\right)^2 g$ (D) $\left(\frac{n+1}{n-1}\right) g$

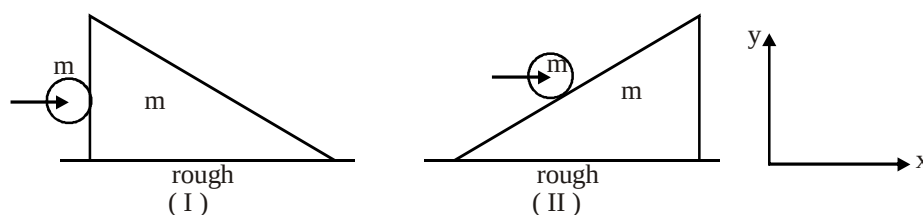
8. A system of particles consists of several particles. Total mass of all the particles is 10 kg. To apply Newton's laws of motion to one of the particles of mass 2 kg in centre of mass frame, you have to assume a pseudo force of $(4\hat{i} - 2\hat{j})$ N acting on it. What is the net external force $\Sigma \vec{F}$ acting on the whole system?

(A) $\Sigma \vec{F} = (10\hat{i} - 20\hat{j})$ N (B) $\Sigma \vec{F} = (-10\hat{i} + 20\hat{j})$ N
 (C) $\Sigma \vec{F} = (20\hat{i} - 10\hat{j})$ N (D) $\Sigma \vec{F} = (-20\hat{i} + 10\hat{j})$ N

9. A force exerts an impulse I on a particle changing its speed from u to $2u$. The applied force and the initial velocity are oppositely directed along the same line. The work done by the force is

(A) $\frac{3}{2} I u$ (B) $\frac{1}{2} I u$ (C) $I u$ (D) $2 I u$

10. A ball of mass m collides horizontally with a stationary wedge on a rough horizontal surface, in the two orientations as shown. Neglect friction between ball and wedge. Two student comment on system of ball and wedge in these situations

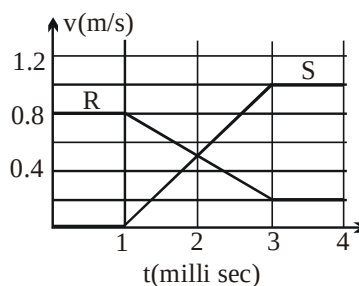


Saurav : Momentum of system in x-direction will change by significant amount in both cases.

Rahul : There are no impulsive external forces in y-direction in both cases hence the total momentum of system in y-direction can be treated as conserved in both cases.

- (A) Saurav is wrong and Rahul is correct (B) Saurav is correct and Rahul is wrong
 (C) Both are correct (D) Both are wrong

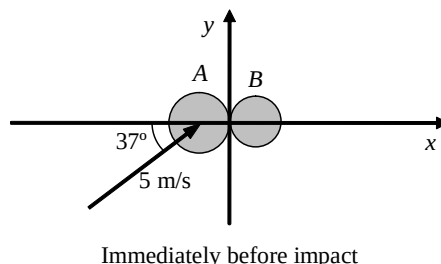
- 11.** Two particles approach each other with different velocities. After collision, one of the particles has a momentum $\vec{p}$ in their centre of mass frame. In the same frame, the momentum of the other one is
 (A) zero (B) $-\vec{p}$ (C) $-\vec{p}/2$ (D) more information required
- 12.** A system of N particles is free from any external forces.
 (i) Which of the following is true for the magnitude of the total momentum of the system?
 (A) It must be zero
 (B) It could be non-zero, but it must be constant
 (C) It could be non-zero, and it might not be constant
 (D) The answer depends on the nature of the internal forces in the system
 (ii) Which of the following must be true for the sum of the magnitudes of the momenta of the individual particles in the system?
 (A) It must be zero
 (B) It could be non-zero, but it must be constant
 (C) It could be non-zero, and it might not be constant
 (D) It could be zero, even if the magnitude of the total momentum is not zero
- 13.** The diagram to the right shows the velocity-time graph for two masses R and S that collided elastically.



Which of the following statements is true?

- (I) R and S moved in the same direction after the collision.
 (II) Kinetic energy of the system (R & S) is minimum at $t = 2$ milli sec.
 (III) The mass of R was greater than mass of S.
 (A) I only (B) II only (C) I and II only (D) I, II and III
- 14.** In a one-dimensional collision between two particles, their relative velocity is $\vec{v}_1$ before the collision and $\vec{v}_2$ after the collision
 (A) $\vec{v}_1 = \vec{v}_2$ if the collision is elastic (B) $\vec{v}_1 = -\vec{v}_2$ if the collision is inelastic
 (C) $|\vec{v}_2| = |\vec{v}_1|$ in all cases (D) $\vec{v}_1 = -k\vec{v}_2$ in all cases, where $k \geq 1$

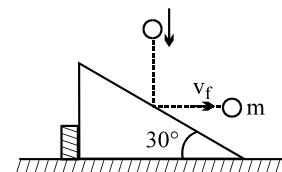
15. Two disks A of mass 3 kg and B of mass 1 kg can slide on a frictionless horizontal surface. Disk A collides with disk B, which is at rest before the collision.



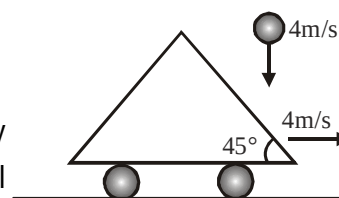
Immediately before impact

If coefficient of restitution is 0.5, velocity vectors $\vec{v}_A$ and $\vec{v}_B$ of the disks after the impact are

- (A) $\vec{v}_A = (2.5\hat{i} + 3\hat{j})$ m/s and $\vec{v}_B = 4.5\hat{i}$ m/s (B) $\vec{v}_A = (-2.5\hat{i} + 3\hat{j})$ m/s and $\vec{v}_B = 4.5\hat{i}$ m/s
 (C) $\vec{v}_A = (3.5\hat{i} + 3\hat{j})$ m/s and $\vec{v}_B = 1.5\hat{i}$ m/s (D) $\vec{v}_A = (-3.5\hat{i} + 3\hat{j})$ m/s and $\vec{v}_B = 1.5\hat{i}$ m/s
16. As shown in the figure a body of mass m moving vertically with speed 3 m/s hits a smooth fixed inclined plane and rebounds with a velocity v_f in the horizontal direction. If $\angle$ of inclined is 30° , the velocity v_f will be
- (A) 3 m/s (B) $\sqrt{3}$ m/s
 (C) $1/\sqrt{3}$ m/s (D) this is not possible

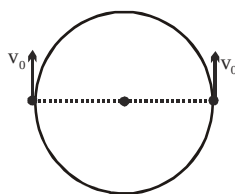


17. A small ball falling vertically downward with constant velocity 4 m/s strikes elastically a massive inclined cart moving with velocity 4 m/s horizontally as shown. The velocity of the rebound of the ball is



- (A) $4\sqrt{2}$ m/s (B) $4\sqrt{3}$ m/s
 (C) 4 m/s (D) $4\sqrt{5}$ m/s
18. Few boys are standing on a flatcar that being pulled at constant speed. They start walking on the flat car. Which of the following statements is/are correct?
- (A) If net momentum of the boys relative to the flat car is in the direction of motion of the flat car, the total kinetic energy of the system consisting of the flat car and the boys increases.
 (B) If net momentum of the boys relative to the flat car is in opposite direction of motion of the flat car, the total kinetic energy of the system consisting of the flat car and the boys decreases.
 (C) If net momentum of the boys relative to the flat car is zero, the total kinetic energy of the system consisting of the flat car and the boys does not change.
 (D) Irrespective of amount of net momentum relative to the car, the total kinetic energy of the system consisting of the flat car and the boys increases.

19. An experiment involves an elastic collision between two particles. In a reference frame fixed with the laboratory, one of the particles was observed at rest before the collision. Which of the following statement is correct?
- (A) Total kinetic energy is conserved in laboratory as well as center of mass frame of reference, but it is observed more in the former one.
- (B) Total kinetic energy is conserved in laboratory as well as center of mass frame of reference, but it is observed more in the latter one.
- (C) Total kinetic energy is conserved in laboratory as well as center of mass frame of reference, and it is observed equal in both of them.
- (D) Total kinetic energy is conserved in laboratory frame but not in the center of mass frame of reference.
20. Two particles of mass m , constrained to move along the circumference of a smooth circular hoop of equal mass m , are initially located at opposite ends of a diameter and given equal velocities v_0 shown in the figure. The entire arrangement is located in gravity free space. Their speed just before collision is



- (A) $\frac{1}{\sqrt{3}} v_0$ (B) $\frac{\sqrt{3}}{2} v_0$ (C) $\frac{2}{\sqrt{3}} v_0$ (D) $\frac{\sqrt{7}}{3} v_0$
21. The bob a simple pendulum of thread pulled aside and held in level with the point of suspension O with its thread slacked as shown in the figure. When the bob is released, it first falls vertically under gravity until the thread becomes straight, thereafter it starts swinging on a circular arc of radius equal to thread length. To analyze subsequent motion, you should apply



- (A) work kinetic energy principle to find speed immediately before the thread becomes straight after the vertical fall of the bob.
- (B) impulse momentum principle to find change in speed in time interval from the instant the thread becomes straight and when the bob starts swinging on circular path.
- (C) work kinetic energy principle to find change in speed between positions where it starts swinging on circular path and its lowest position.
- (D) newtons laws to find tension in the thread at any point on circular arc while the bob is swinging.

- 22.** In a one dimensional collision between two identical particles A and B, B is stationary and A has momentum p before impact. During impact, B gives impulse J to A.
- (A) The total momentum of the 'A plus B' system is p before and after the impact, and $(p-J)$ during the impact.
- (B) During the impact A gives impulse of magnitude J to B
- (C) The coefficient of restitution is $\frac{2J}{p} - 1$
- (D) The coefficient of restitution is $\frac{J}{p} + 1$
- 23.** A shell is fired from the ground with velocity u at an angle θ with the horizontal. Before it reaches the highest point, it explodes into two fragments A and B. Fragment A immediately comes to an instantaneous rest and then falls vertically down. Where would the mass center of the two pieces be, when the fragment A hits the ground?
- (A) On the ground at distance $\frac{u^2 \sin(2\theta)}{g}$ away from the point of projection.
- (B) Somewhere in the air at horizontal distance $\frac{u^2 \sin(2\theta)}{g}$ from the point of projection.
- (C) Somewhere in the air at horizontal distance less than $\frac{u^2 \sin(2\theta)}{g}$ from the point of projection.
- (D) Somewhere in the air at horizontal distance more than $\frac{u^2 \sin(2\theta)}{g}$ from the point of projection.

Question No. 24 to 26 (3 questions)

One particle of mass 1 kg is moving along positive x -axis with velocity 3 m/s. Another particle of mass 2 kg is moving along y -axis with 6 m/s. At time $t = 0$, 1 kg mass is at $(3m, 0)$ and 2 kg at $(0, 9m)$, x - y plane is the horizontal plane. (Surface is smooth for question 1 and rough for question 2 and 3)

- 24.** The centre of mass of the two particles is moving in a straight line which equation is :
- (A) $y = x + 2$ (B) $y = 4x + 2$ (C) $y = 2x - 4$ (D) $y = 2x + 4$
- 25.** If both the particles have the same value of coefficient of friction $\mu = 0.2$. The centre of mass will stop at time $t = \dots\dots$ s :
- (A) 1.5 (B) 4.5 (C) 3.0 (D) 2.0
- 26.** Co-ordinates of centre of mass where it will stop finally are
- (A) (2.0 m, 14.25 m) (B) (2.25 m, 10 m) (C) (3.75 m, 9 m) (D) (1.75 m, 12 m)

ANSWER KEY

EXERCISE (S-2)

1. Ans. $\frac{R}{4}$

2. Ans. $\frac{2mv^2}{3\ell}$

3. Ans. 13/3 m/s

4. Ans. -2m/s, 6.93 m/s $\angle 30^\circ$

5. Ans. $m(-3\hat{i} + 4\hat{j})$, $e = \frac{9}{16}$

6. Ans. ℓ

7. Ans. (i) $\sqrt{\frac{2ag}{3}}$ (ii) $\frac{3v}{g}$ (iii) $\frac{2v}{g}$

EXERCISE (O-2)

- | | | | | | | |
|-----------------------------------|---------|--------------------------------|-------------------------------------|---------|--------|--------------|
| 1. (B,D) | 2. (D) | 3. (D) | 4. (C) | 5. (B) | 6. (A) | 7. (A, B, C) |
| 8. (A, B, D) | 9. (D) | 10. (A)-R; (B)-T; (C)-T; (D)-S | | | | |
| 11. (A)-Q; (B)-P, Q; (C)-R; (D)-S | 12. (B) | 13. (B,C,D) | 14. (A) | 15. (A) | | |
| 16. (B) | 17. (A) | 18. (C) | 19. (A)-Q; (B)-Q; (C)-P,R; (D)-Q, S | | | |

EXERCISE (J-A)

- | | | | |
|-----------|------------|---|----------------|
| 1. C | 2. C | 3. A | 4. A, D |
| 5. A, C | | 6. A, C | 7. A 8. B |
| 9. B | 10. C | 11. $t = 12 \text{ sec}$, $v = \frac{100\sqrt{3}}{11} \text{ m/s}$ | |
| 12. 4 m/s | 13. Ans. 5 | 14. Ans. (B) | |

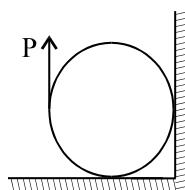
REVIEW TEST

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|---|------------------|--|-----------|
| 1. 1.2 m/s, 3.6 N-s | 2. v | 3. $\frac{40}{3} \text{ m}$, $\frac{13}{4} \text{ sec}$ | 4. Ans. 4 |
| 5. Ans. $e = (5 + \sqrt{3})/8$, $M = 26/\sqrt{3} \text{ kg}$ | | 6. (B) | 7. (C) |
| 8. (D) | 9. (B) | 10. (D) | 11. (B) |
| 12. (i) (B) (ii) (C) | 13. (D) | 14. (D) | 15. (A) |
| 16. (B) | 17. (D) | 18. (D) | 19. (A) |
| 20. (A) | 21. (A, B, C, D) | 22. (B, C) | 23. (C) |
| 24. (B) | 25. (C) | 26. (D) | |

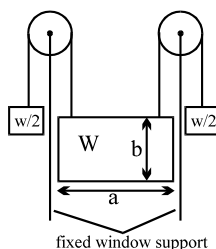
ROTATIONAL DYNAMICS

EXERCISE (S-2)

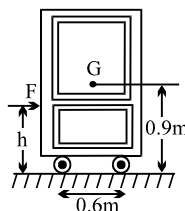
- Find the MI of a rod about an axis through its centre of mass and perpendicular to the length whose linear density varies as $\lambda = ax$ where a is a constant and x is the position of an element of the rod relative to its left end. The length of the rod is ℓ .
- Figure shows a vertical force applied tangentially to a uniform cylinder of weight F_g . The coefficient of static friction between the cylinder and both surfaces is 0.500. In terms of F_g , find the maximum force P that can be applied that does not cause the cylinder to rotate.



- A slight loosely fit window is balanced by two strings which are connected to weights $w/2$ each. The strings pass over the frictionless pulleys as shown in the figure. The strings are tied almost at the corner of the window. The string on the right is cut and then the window accelerates downwards. If the coefficients of friction between the window and the side supports is μ then calculate the acceleration of the window in terms of μ , a , b and g , where a is width and b is the length of the window.

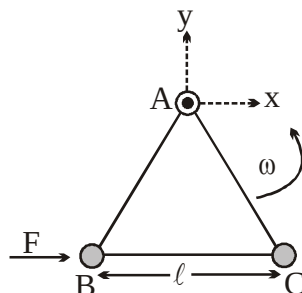


- A 20 kg cabinet is mounted on small casters that allow it to move freely ($\mu = 0$) on the floor. If a 100 N force is applied as shown, determine

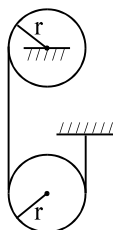


- the acceleration of the cabinet,
- the range of values of h for which the cabinet will not tip.

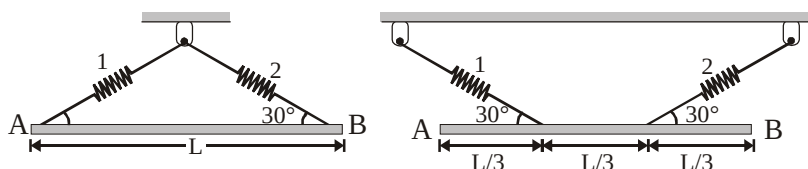
5. Three particles A, B, and C each of mass m , are connected to each other by three massless rigid rods to form a rigid, equilateral triangular body of side ℓ . This body is placed on a horizontal frictionless table (x - y plane) and is hinged to it at the point A, so that it can move without friction about the vertical axis through A (see figure). The body is set into rotational motion on the table about A with a constant angular velocity ω .



- (a) Find the magnitude of the horizontal force exerted by the hinge on the body.
 (b) At time T , when the side BC is parallel to the x -axis, a force F is applied on B along BC (as shown). Obtain the x -component and the y -component of the force exerted by the hinge on the body, immediately after time T . **[IIT-JEE' 2001]**
6. The door of an automobile is open and perpendicular to the body. The automobile starts with an acceleration of 2 ft/sec^2 , and the width of the door is 30 inches. Treat the door as a uniform rectangle, and neglect friction to find the speed of its outside edge as seen by the driver when the door closes.
7. Two uniform cylinders, each of mass $m = 10 \text{ kg}$ and radius $r = 150 \text{ mm}$, are connected by a rough belt as shown. If the system is released from rest, determine

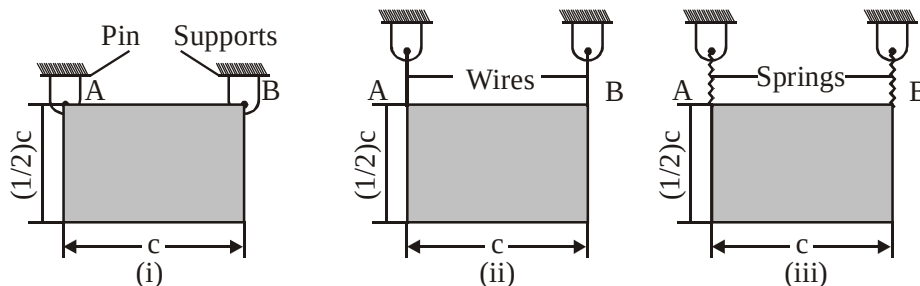


- (a) the velocity of the centre of cylinder A after it has moved through 1.2 m &
 (b) the tension in the portion of the belt connecting the two cylinders.
8. A uniform slender bar AB of mass m is suspended from two springs as shown. If spring 2 breaks, determine at that instant ;

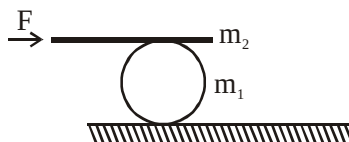
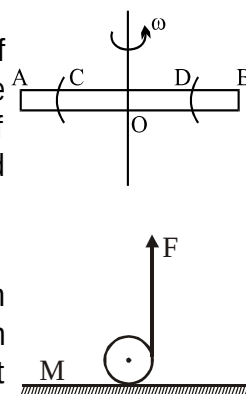


- (a) the angular acceleration of the bar .
 (b) the acceleration of point A .
 (c) the acceleration of point B .

9. A uniform plate of mass m is suspended in each of the ways shown. For each case determine immediately after the connection at B has been released ;



- (a) the angular acceleration of the plate.
 (b) the acceleration of its mass center .
10. A rigid horizontal smooth rod AB of mass 0.75 kg and length 40 cm can rotate freely about a fixed vertical axis through its mid point O. Two rings each of mass 1 kg are initially at rest at a distance of 10 cm from O on either side of the rod. The rod is set in rotation with an angular velocity of 30 radians per second. Find the velocity of each ring along the length of the rod in m/s when they reach the ends of the rod
11. The free end of the string wound on the surface of a solid cylinder of mass M & radius r is pulled up by a force F as shown. If there is sufficient friction between cylinder & floor so that the cylinder is able to roll without slipping, find the maximum angular acceleration that the cylinder can have (without cylinder leaving contact with surface).
12. A man pushes a cylinder of mass m_1 with help of a plank of mass m_2 as shown. There is no slipping at any contact. the horizontal component of the force applied by the man is F . Find:

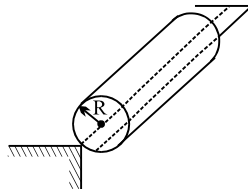


- (a) the accelerations of the plank and the centre of mass of the cylinder, and
 (b) the magnitudes and directions of frictional forces at contact points. [JEE '99 , 6 + 4]
13. A 160 mm diameter pipe of mass 6 kg rests on a 1.5 kg plate. The pipe and plate are initially at rest when a force P of magnitude 25 N is applied for 0.75 s. Knowing that $\mu_s = 0.25$ & $\mu_k = 0.20$ between the plate and both the pipe and the floor, determine;

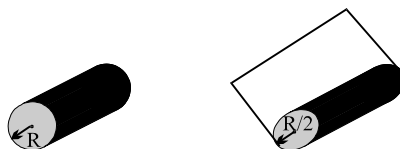


- (a) whether the pipe slides with respect to the plate .
 (b) the resulting velocities of the pipe and of the plate .

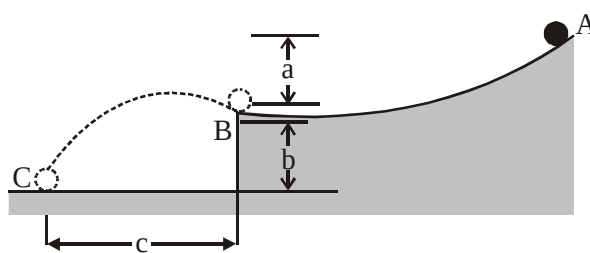
- 14.** A rectangular rigid fixed block has a long horizontal edge. A solid homogeneous cylinder of radius R is placed horizontally at rest with its length parallel to the edge such that the axis of the cylinder and the edge of the block are in the same vertical plane as shown in figure. There is sufficient friction present at the edge so that a very small displacement cause the cylinder to roll off the edge without slipping. Determine



- (a) the angle θ_c through which the cylinder rotates before it leaves contact with the edge,
 (b) the speed of the centre of mass of the cylinder before leaving contact with the edge, and
 (c) the ratio of the translational to rotational kinetic energies of the cylinder when its centre of mass is in horizontal line with the edge.
- 15.** A carpet of mass ' M ' made of inextensible material is rolled along its length in the form of a cylinder of radius ' R ' and is kept on a rough floor. The carpet starts unrolling without sliding on the floor when a negligibly small push is given to it. The horizontal velocity of the axis of the cylindrical part of the carpet when its radius reduces to $R/2$ will be:



- 16.** A small sphere of mass m and radius r is released from rest at A and rolls without sliding on the curved surface to point B where it leaves the surface with a horizontal velocity. Knowing that $a = 1.5 \text{ m}$ & $b = 1.2 \text{ m}$, determine;
 (a) the speed of the sphere as it strikes the ground at C .
 (b) the corresponding distance c .

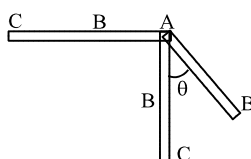


- 17.** A uniform disk of mass m and radius R is projected horizontally with velocity v_0 on a rough horizontal floor so that it starts off with a purely sliding motion at $t = 0$. After t_0 seconds it acquires a purely rolling motion as shown in figure.



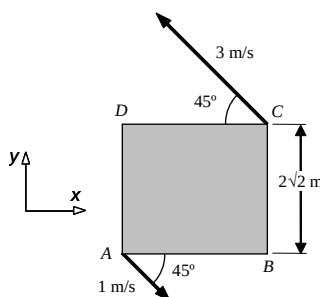
- (i) Calculate the velocity of the centre of mass of the disk at t_0 .
 (ii) Assuming the coefficient of friction to be μ calculate t_0 . Also calculate the work done by the frictional force as a function of time and the total work done by it over a time t much longer than t_0 .

- 18.** A uniform disc of mass m and radius R rotates about a fixed vertical axis passing through its centre with angular velocity ω . A particle of same mass m and having velocity $2\omega R$ towards centre of the disc collides with the disc moving horizontally and sticks to its rim. Find
- the angular velocity of the disc.
 - the impulse on the particle due to disc.
 - the impulse on the disc due to hinge.
- 19.** A rod hinged at one end is released from the horizontal position as shown in the figure. When it becomes vertical its lower half separates without exerting any reaction at the breaking point. Then find the maximum angle ' θ ' made by the hinged upper half with the vertical.



EXERCISE (O-2)

- A ring rolls without slipping on the ground. Its centre C moves with a constant speed u . P is any point on the ring. The speed of P with respect to the ground is v .
 (A) $0 \leq v \leq 2u$
 (B) $v = u$, if CP is horizontal
 (C) $v = u$, if CP makes an angle of 30° with the horizontal and P is below the horizontal level of C.
 (D) $v = \sqrt{2}u$, if CP is horizontal
- A wheel of radius r rolling on a straight line, the velocity of its centre being v . At a certain instant the point of contact of the wheel with the ground is M and N is the highest point on the wheel (diametrically opposite to M). The incorrect statement is:
 (A) The velocity of any point P of the wheel is proportional to MP.
 (B) Points of the wheel moving with velocity greater than v form a larger area of the wheel than points moving with velocity less than v .
 (C) The point of contact M is instantaneously at rest.
 (D) The velocities of any two parts of the wheel which are equidistant from centre are equal.
- A square plate is rotating about an axis that is normal to the plane of the plate. At an instant velocities of two of its opposite corners are shown in the figure.



- Angular velocity of the plate is 1 rad/s.
- The instantaneous axis of rotation passes through a point that is on diagonal AC and 1 cm away from A.

(C) Velocity of corner D is $\vec{v}_D = -\frac{1}{\sqrt{2}}(3\hat{i} + \hat{j})$ m/s.

(D) Velocity of corner A relative to B is in the positive y-direction.

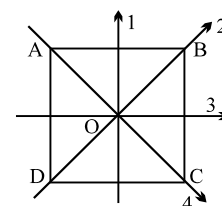
- ABCD is a square plate with centre O. The moments of inertia of the plate about the perpendicular axis through O is I and about the axes 1, 2, 3 & 4 are I_1, I_2, I_3 & I_4 respectively. It follows that :

(A) $I_2 = I_3$

(B) $I = I_1 + I_4$

(C) $I = I_2 + I_4$

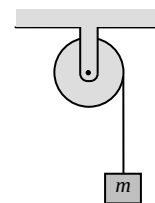
(D) $I_1 = I_3$



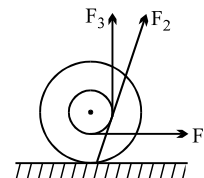
5. A body is in equilibrium under the influence of a number of forces. Each force has a different line of action. The minimum number of forces required is
 (A) 2, if their lines of action pass through the centre of mass of the body.
 (B) 3, if their lines of action are not parallel.
 (C) 3, if their lines of action are parallel.
 (D) 4, if their lines of action are parallel and all the forces have the same magnitude.
6. A rod of weight w is supported by two parallel knife edges A and B and is in equilibrium in a horizontal position. The knives are at a distance d from each other. The centre of mass of the rod is at a distance x from A.

- (A) the normal reaction at A is $\frac{wx}{d}$ (B) the normal reaction at A is $\frac{w(d-x)}{d}$
 (C) the normal reaction at B is $\frac{wx}{d}$ (D) the normal reaction at B is $\frac{w(d-x)}{d}$

7. A block of mass m is suspended with the help of a light cord wrapped over a cylindrical pulley of mass M and radius R as shown in the figure. The system is released from rest. Magnitude of torque on the pulley
 (A) decreases on increasing mass of the pulley.
 (B) increases on increasing radius of the pulley.
 (C) is proportional to the mass of the hanging block.
 (D) is independent of mass and radius of the pulley and depends on mass of the hanging block.



8. A yo-yo is resting on a perfectly rough horizontal table. Forces F_1 , F_2 and F_3 are applied separately as shown. The correct statement is
 (A) when F_3 is applied the centre of mass will move to the right.
 (B) when F_2 is applied the centre of mass will move to the left.
 (C) when F_1 is applied the centre of mass will move to the right.
 (D) when F_2 is applied the centre of mass will move to the right.



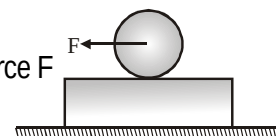
9. If a cylinder is rolling down the incline with sliding.
 (A) after some time it may start pure rolling
 (B) after sometime it will start pure rolling
 (C) it may be possible that it will never start pure rolling
 (D) none of these

Question No. 10 and 11 (2 questions)

A disc of mass m and radius R is placed over a plank of same mass m . There is sufficient friction between disc and plank to prevent slipping. A force F is applied at the centre of the disc.

10. Acceleration of the plank is :-

- (A) $\frac{F}{2m}$ (B) $\frac{3F}{4m}$ (C) $\frac{F}{4m}$ (D) $\frac{3F}{2m}$



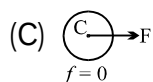
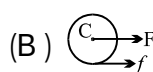
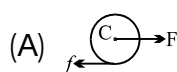
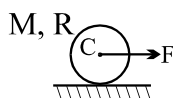
11. Force of friction between the disc and the plank is :-

(A) $\frac{F}{2}$ (B) $\frac{F}{4}$ (C) $\frac{F}{3}$ (D) $\frac{2F}{3}$

Question No. 12 and 13 (2 questions)

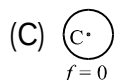
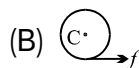
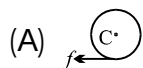
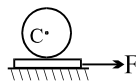
In the following problems, indicate the correct direction of friction force acting on the cylinder, which is pulled on a rough surface by a constant force F .

12. A cylinder of mass M and radius R is pulled horizontally by a force F . The friction force can be given by which of the following diagrams :-



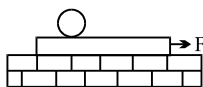
(D) cannot be interpreted

13. A cylinder is placed on a rough plank which in turn is placed on a smooth surface. The plank is pulled with a constant force F . The friction force can be given by which of the following diagrams :-



(D) cannot be interpreted

14. A plank with a uniform sphere placed on it rests on a smooth horizontal plane. Plank is pulled to right by a constant force F . If sphere does not slip over the plank. Which of the following is correct.



- (A) Acceleration of the centre of sphere is less than that of the plank.
(B) Work done by friction acting on the sphere is equal to its total kinetic energy.
(C) Total kinetic energy of the system is equal to work done by the force F
(D) None of the above

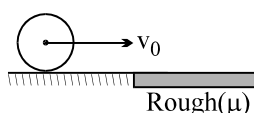
15. A block of mass m moves on a horizontal rough surface with initial velocity v . The height of the centre of mass of the block is h from the surface. Consider a point A on the surface.

- (A) angular momentum about A is mvh initially
(B) the velocity of the block decreases at time passes.
(C) torque of the forces acting on block is zero about A
(D) angular momentum is not conserved about A .

16. A man spinning in free space changes the shape of his body, eg. by spreading his arms or curling up. By doing this, he can change his
 (A) moment of inertia (B) angular momentum
 (C) angular velocity (D) rotational kinetic energy
17. A hollow sphere of radius R and mass m is fully filled with non viscous liquid of mass m . It is rolled down a horizontal plane such that its centre of mass moves with a velocity v . If it purely rolls
 (A) Kinetic energy of the sphere is $\frac{5}{6}mv^2$
 (B) Kinetic energy of the sphere is $\frac{4}{5}mv^2$
 (C) Angular momentum of the sphere about a fixed point on ground is $\frac{8}{3}mvR$
 (D) Angular momentum of the sphere about a fixed point on ground is $\frac{14}{5}mvR$

Question No. 18 to 21 (4 questions)

A ring of mass M and radius R sliding with a velocity v_0 suddenly enters into rough surface where the coefficient of friction is μ , as shown in figure.



18. Choose the correct statement(s)
 (A) As the ring enters on the rough surface, the limiting friction force acts on it
 (B) The direction of friction is opposite to the direction of motion
 (C) The friction force accelerates the ring in the clockwise sense about its centre of mass
 (D) As the ring enters on the rough surface it starts rolling
19. Choose the correct statement(s)
 (A) The momentum of the ring is conserved
 (B) The angular momentum of the ring is conserved about its centre of mass
 (C) The angular momentum of the ring conserved about any point on the horizontal surface
 (D) The mechanical energy of the ring is conserved
20. Choose the correct statement(s)
 (A) The ring starts its rolling motion when the centre of mass stationary
 (B) The ring starts rolling motion when the point of contact becomes stationary
 (C) The time after which the ring starts rolling is $\frac{v_0}{2\mu g}$
 (D) The rolling velocity is $\frac{v_0}{2}$

21. Choose the correct alternative(s)

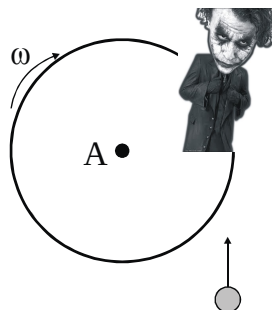
- (A) The linear distance moved by the centre of mass before the ring starts rolling is $\frac{3v_0^2}{8\mu g}$
- (B) The net work done by friction force is $-\frac{3}{8}mv_0^2$
- (C) The loss in kinetic energy of the ring is $\frac{mv_0^2}{4}$
- (D) The gain in rotational kinetic energy is $+\frac{mv_0^2}{8}$

MATRIX MATCH TYPE QUESTION

22. Column-I depicts various situations where some sudden events are taking place. Column-II describes changes in various parameters of systems immediately after the events taking place in column-I.

Column I

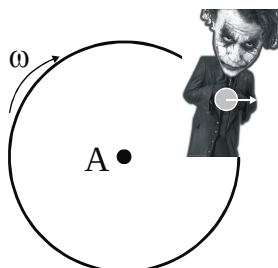
- (A) Joker is standing on revolving platform and batman throws the ball and joker catches the ball while it was moving horizontally.



Column II

- (P) Linear momentum remains conserved.

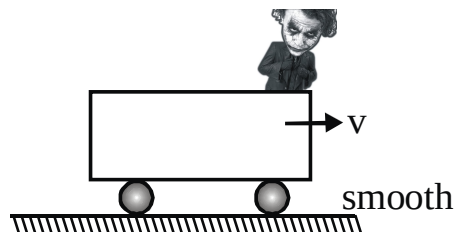
- (B) Joker, ball and platform is system. Joker throws the ball horizontally and perpendicular to his motion while standing on the revolving platform.



- (Q) Mechanical energy is conserved

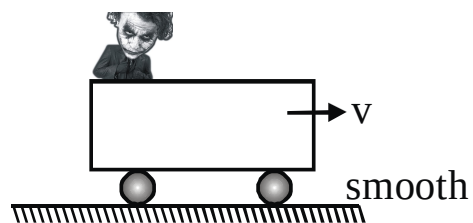
Joker, ball and platform is system.

- (C) Joker jumps horizontally towards right from the cart which is moving at speed v on smooth horizontal floor.



- (R) Mechanical energy increases.

- (D) Joker and cart is the system
 Joker drops himself vertically from the moving cart with no horizontal velocity relative to cart.



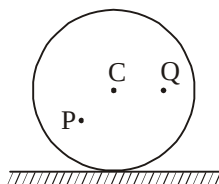
- (S) Mechanical energy decreases.

- (T) v or ω changes

Joker and cart is the system

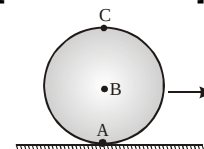
EXERCISE (J-A)

1. A disc is rolling without slipping with angular velocity ω . P and Q are two points equidistant from the centre C. The order of magnitude of velocity is **[IIT-JEE 2004]**



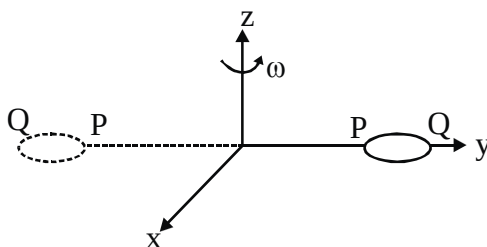
2. A sphere is rolling without slipping on a fixed horizontal plane surface. In the figure, A is the point of contact, B is the centre of the sphere and C is its topmost point. Then **[IIT-JEE 2009]**

- (A) $v_Q > v_C > v_P$ (B) $v_P > v_C > v_Q$ (C) $v_P = v_C, v_Q = v_C / 2$ (D) $v_P < v_C > v_Q$
(A) $\vec{v}_C - \vec{v}_A = 2(\vec{v}_B - \vec{v}_C)$ (B) $\vec{v}_C - \vec{v}_B = \vec{v}_B - \vec{v}_A$
(C) $|\vec{v}_C - \vec{v}_A| = 2|\vec{v}_B - \vec{v}_C|$ (D) $|\vec{v}_C - \vec{v}_A| = 4|\vec{v}_B|$

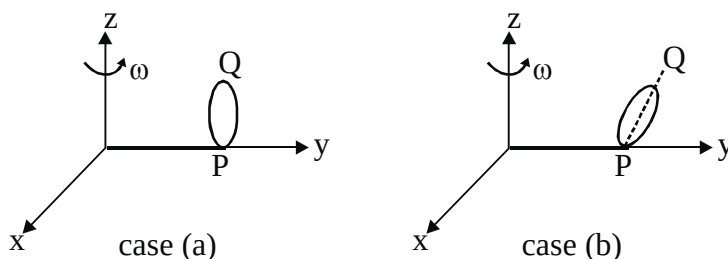


Paragraph for Questions 3 and 4

The general motion of a rigid body can be considered to be a combination of (i) a motion of its centre of mass about an axis, and (ii) its motion about an instantaneous axis passing through the centre of mass. These axes need not be stationary. Consider, for example, a thin uniform disc welded (rigidly fixed) horizontally at its rim to a massless stick, as shown in the figure. When the disc-stick system is rotated about the origin on a horizontal frictionless plane with angular speed ω , the motion at any instant can be taken as a combination of (i) a rotation of the centre of mass of the disc about the z-axis, and (ii) a rotation of the disc through an instantaneous vertical axis passing through its centre of mass (as is seen from the changed orientation of points P and Q). Both these motions have the same angular speed ω in this case.

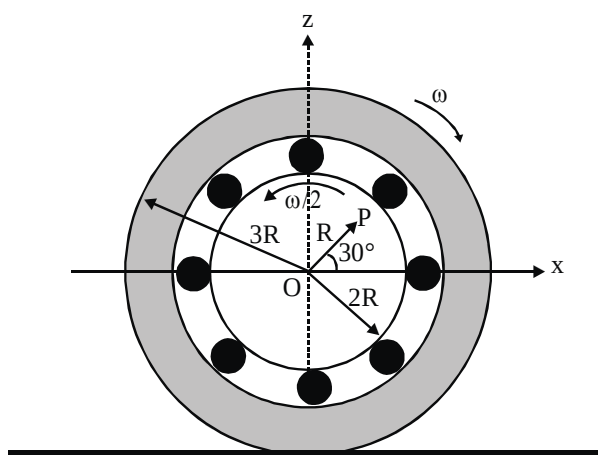


Now consider two similar systems as shown in the figure : case (A) the disc with its face vertical and parallel to x-z plane; Case (B) the disc with its face making an angle of 45° with x-y plane and its horizontal diameter parallel to x-axis. In both the cases, the disc is welded at point P, and the systems are rotated with constant angular speed ω about the z-axis. **[IIT-JEE 2012]**



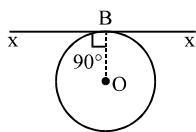
3. Which of the following statements regarding the angular speed about the instantaneous axis (passing through the centre of mass) is correct?
- (A) It is $\sqrt{2\omega}$ for both the cases. (B) It is ω for case (a); and $\frac{\omega}{\sqrt{2}}$ for case (b).
 (C) It is ω for case (a); and $\sqrt{2\omega}$ for case (b). (D) It is ω for both the cases.
4. Which of the following statements about the instantaneous axis (passing through the centre of mass) is correct?
- (A) It is vertical for both the cases (a) and (b).
 (B) It is vertical for case (a); and is at 45° to the x-z plane and lies in the plane of the disc for case (b).
 (C) It is horizontal for case (a); and is at 45° to the x-z plane and is normal to the plane of the disc for case (b).
 (D) It is vertical for case (a); and is at 45° to the x-z plane and is normal to the plane of the disc for case (b).
5. The figure shows a system consisting of (i) a ring of outer radius $3R$ rolling clockwise without slipping on a horizontal surface with angular speed ω and (ii) an inner disc of radius $2R$ rotating anti-clockwise with angular speed $\omega/2$. The ring and disc are separated by frictionless ball bearing. The system is in the x-z plane. The point P on the inner disc is at a distance R from the origin, where OP makes an angle of 30° with the horizontal. Then with respect to the horizontal surface,

[IIT-JEE 2012]



- (A) the point O has a linear velocity $3R\omega\hat{i}$
- (B) the point P has a linear velocity $\frac{11}{4}R\omega\hat{i} + \frac{\sqrt{3}}{4}R\omega\hat{k}$
- (C) the point P has a linear velocity $\frac{13}{4}R\omega\hat{i} - \frac{\sqrt{3}}{4}R\omega\hat{k}$
- (D) the point P has a linear velocity $\left(3 - \frac{\sqrt{3}}{4}\right)R\omega\hat{i} + \frac{1}{4}R\omega\hat{k}$

6. A thin wire of length L and uniform linear mass density ρ is bent into a circular loop with centre at O as shown. The moment of inertia of the loop about the axis XX' is: **[IIT-JEE'(Scr)'2000]**



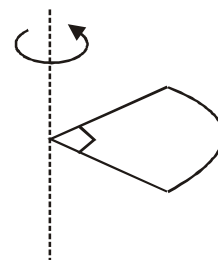
- (A) $\rho L^3/8\pi^2$ (B) $\rho L^3/16\pi^2$ (C) $5\rho L^3/16\pi^2$ (D) $3\rho L^3/8\pi^2$
7. One quarter sector is cut from a uniform circular disc of radius R . This sector has mass M . It is made to rotate about a line perpendicular to its plane and passing through the center of the original disc. Its moment of inertia about the axis of rotation is **[IIT-JEE 2000]**

(A) $\frac{1}{2} MR^2$

(B) $\frac{1}{4} MR^2$

(C) $\frac{1}{8} MR^2$

(D) $\sqrt{2} MR^2$



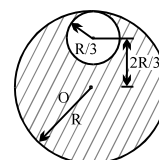
8. A disc has mass $9m$. A hole of radius $R/3$ is cut from it as shown in the figure. The moment of inertia of remaining part about an axis passing through the centre ' O ' of the disc and perpendicular to the plane of the disc is: **[IIT-JEE'2005 (Scr)]**

(A) $8 mR^2$

(B) $4 mR^2$

(C) $\frac{40}{9} mR^2$

(D) $\frac{37}{9} mR^2$



9. A solid sphere of mass M , radius R and having moment of inertia about an axis passing through the centre of mass as I , is recast into a disc of thickness t , whose moment of inertia about an axis passing through its edge and perpendicular to its plane remains I . Then, radius of the disc will be

(A) $2R/\sqrt{15}$

(B) $R\sqrt{2/15}$

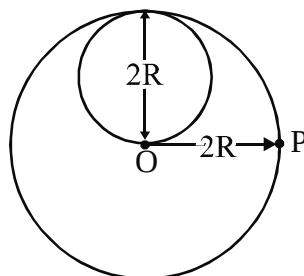
(C) $4R/\sqrt{15}$

(D) $R/4$ **[IIT-JEE 2006]**

10. Four solid spheres each of diameter $\sqrt{5}$ cm and mass 0.5 kg are placed with their centers at the corners of a square of side 4 cm. The moment of inertia of the system about the diagonal of the square is $N \times 10^{-4}$ kg-m², then N is **[IIT-JEE 2011]**

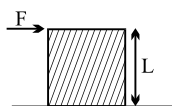
11. A lamina is made by removing a small disc of diameter $2R$ from a bigger disc of uniform mass density and radius $2R$, as shown in the figure. The moment of inertia of this lamina about axes passing through O and P is I_O and I_P respectively. Both these axes are perpendicular to the plane of the lamina. The ratio $\frac{I_P}{I_O}$ to the nearest integer is

[IIT-JEE 2012]



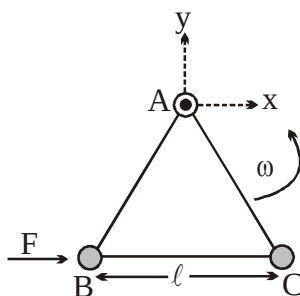
12. A cubical block of side L rests on a rough horizontal surface with coefficient of friction μ . A horizontal force F is applied on the block as shown. If the coefficient of friction is sufficiently high so that the block does not slide before toppling, the minimum force required to topple the block is:

[IIT-JEE'(Scr)'2000]

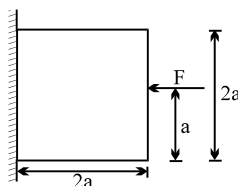


- (A) infinitesimal (B) $mg/4$ (C) $mg/2$ (D) $mg(1-\mu)$
13. A cylinder rolls up an inclined plane, reaches some height, and then rolls down (without slipping throughout these motions). The directions of the frictional force acting on the cylinder are
- (A) Up the incline while ascending and down the incline while descending [IIT-JEE 2002]
 (B) Up the incline while ascending as well as descending
 (C) down the incline while ascending and up the incline while descending
 (D) down the incline while ascending as well as descending
14. Three particles A, B, and C each of mass m , are connected to each other by three massless rigid rods to form a rigid, equilateral triangular body of side ℓ . This body is placed on a horizontal frictionless table (x - y plane) and is hinged to it at the point A, so that it can move without friction about the vertical axis through A (see figure). The body is set into rotational motion on the table about A with a constant angular velocity ω .
- (a) Find the magnitude of the horizontal force exerted by the hinge on the body.
 (b) At time T , when the side BC is parallel to the x -axis, a force F is applied on B along BC (as shown). Obtain the x -component and the y -component of the force exerted by the hinge on the body, immediately after time T .

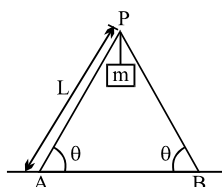
[IIT-JEE 2002]



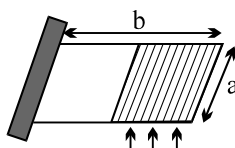
15. A block of mass m is held fixed against a wall by applying a horizontal force F . Which of the following option is incorrect: **[IIT-JEE 2005 (Scr)]**



- (A) friction force = mg (B) F will not produce torque
(C) normal will not produce torque (D) normal reaction = F
16. A cylinder of mass m and radius R rolls down an inclined plane of inclination θ . Calculate the linear acceleration of the axis of cylinder. **[IIT-JEE 2005]**
17. Two identical ladders, each of mass M and length L are resting on the rough horizontal surface as shown in the figure. A block of mass m hangs from P . If the system is in equilibrium, find the magnitude and the direction of frictional force at A and B . **[IIT-JEE 2005]**

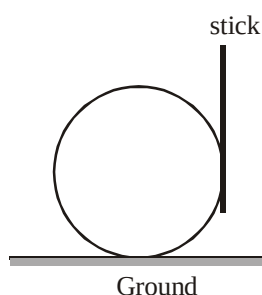


18. A solid cylinder is rolling down a rough inclined plane of inclination θ . Then **[IIT-JEE 2006]**
(A) The friction force is dissipative
(B) The friction force is necessarily changing
(C) The friction force will aid rotation but hinder translation
(D) The friction force is reduced if θ is reduced
19. There is a rectangular plate of mass M kg of dimensions $(a \times b)$. The plate is held in horizontal position by striking n small balls each of mass m per unit area per unit time. These are striking in the shaded half region of the plate. The balls are colliding elastically with velocity v . What is v ? It is given $n = 100$, $M = 3$ kg, $m = 0.01$ kg; $b = 2$ m; $a = 1$ m; $g = 10$ m/s². **[JEE 2006]**

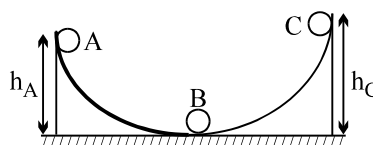


20. **Statement-1** : Two cylinders, one hollow (metal) and the other solid (wood) with the same mass and identical dimensions are simultaneously allowed to roll without slipping down an inclined plane from the same height. The hollow cylinder will reach the bottom of the inclined plane first.
and
Statement-2 : By the principle of conservation of energy, the total kinetic energies of both the cylinders are identical when they reach the bottom of the incline. **[JEE 2008]**
(A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1
(B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

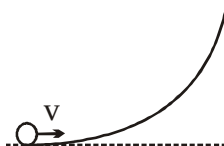
21. A block of base $10\text{ cm} \times 10\text{ cm}$ and height 15 cm is kept on an inclined plane. The coefficient of friction between them is $\sqrt{3}$. The inclination θ of this inclined plane from the horizontal plane is gradually increased from 0° . Then :- [IIT-JEE 2009]
- (A) at $\theta = 30^\circ$, the block will start sliding down the plane
 (B) the block will remain at rest on the plane up to certain θ and then it will topple
 (C) at $\theta = 60^\circ$, the block will start sliding down the plane and continue to do so at higher angles
 (D) at $\theta = 60^\circ$, the block will start sliding down the plane and on further increasing θ , it will topple at certain θ
22. A boy is pushing a ring of mass 2 kg and radius 0.5 m with a stick as shown in the figure. The stick applies a force of 2 N on the ring and rolls it without slipping with an acceleration of 0.3 m/s^2 . The coefficient of friction between the ground and the ring is large enough that rolling always occurs and the coefficient of friction between the stick and the ring is $(P/10)$. The value of P is [IIT-JEE 2011]



23. Two solid cylinders P and Q of same mass and same radius start rolling down a fixed inclined plane from the same height at the same time. Cylinder P has most of its mass concentrated near its surface, while Q has most of its mass concentrated near the axis. Which statement(s) is(are) correct? [IIT-JEE 2012]
- (A) Both cylinders P and Q reach the ground at the same time.
 (B) Cylinder P has larger acceleration than cylinder Q.
 (C) Both cylinders reach the ground with same translational kinetic energy.
 (D) Cylinder Q reaches the ground with larger angular speed.
24. A ball moves over a fixed track as shown in the figure. From A to B the ball rolls without slipping. Surface BC is frictionless. K_A , K_B and K_C are kinetic energies of the ball at A, B and C, respectively. Then [IIT-JEE 2006]



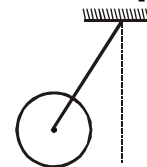
- (A) $h_A > h_C$; $K_B > K_C$
 (B) $h_A > h_C$; $K_C > K_A$
 (C) $h_A = h_C$; $K_B = K_C$
 (D) $h_A < h_C$; $K_B > K_C$
25. A small object of uniform density rolls up a curved surface with an initial velocity v . It reaches upto maximum height of $\frac{3v^2}{4g}$ with respect to the initial position. The object is [JEE 2007]



- (A) ring (B) solid sphere (C) hollow sphere (D) disc

26. A metal rod of length 'L' and mass 'm' is pivoted at one end. A thin disk of mass 'M' and radius 'R' ($< L$) is attached at its center to the free end of the rod. Consider two ways the disk is attached: (case A). The disk is not free to rotate about its center and (case B) the disk is free to rotate about its center. The rod-disk system performs SHM in vertical plane after being released from the same displaced position. Which of the following statement(s) is(are) true? **[IIT-JEE 2011]**

- (A) Restoring torque in case A = Restoring torque in case B
(B) Restoring torque in case A < Restoring torque in case B
(C) Angular frequency for case A > Angular frequency for case B
(D) Angular frequency for case A < Angular frequency for case B

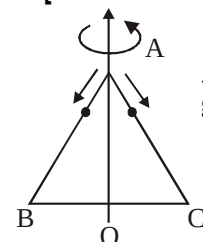


27. A rod AB of mass M and length L is lying on a horizontal frictionless surface. A particle of mass m travelling along the surface hits the end 'A' of the rod with a velocity v_0 in the direction perpendicular to AB. The collision is completely elastic. After the collision the particle comes to rest.

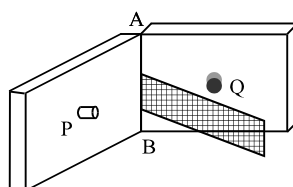
- (a) Find the ratio m/M . **[IIT-JEE' 2000]**
(b) A point P on the rod is at rest immediately after the collision. Find the distance AP.
(c) Find the linear speed of the point P at a time $\pi L/(3v_0)$ after the collision.

28. An equilateral triangle ABC formed from a uniform wire has two small identical beads initially located at A. The triangle is set rotating about the vertical axis AO. Then the beads are released from rest simultaneously and allowed to slide down, one along AB and the other along AC as shown. Neglecting frictional effects, the quantities that are conserved as the beads slide down, are **[IIT-JEE 2000]**

- (A) Angular velocity and total energy (kinetic and potential)
(B) Total angular momentum and total energy
(C) Angular velocity and moment of inertia about the axis of rotation.
(D) Total angular momentum and moment of inertia about the axis of rotation.

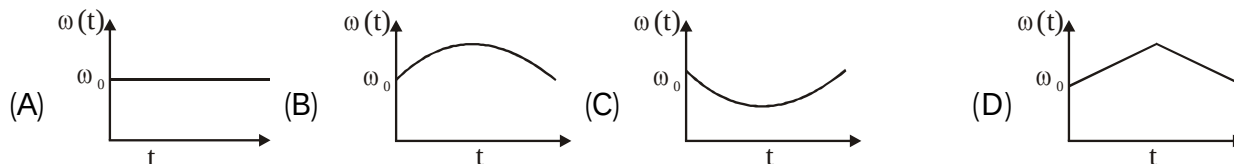


29. Two heavy metallic plates are joined together at 90° to each other. A laminar sheet of mass 30 Kg is hinged at the line AB joining the two heavy metallic plates. The hinges are frictionless. The moment of inertia of the laminar sheet about an axis parallel to AB and passing through its centre of mass is 1.2 Kg-m^2 . Two rubber obstacles P and Q are fixed, one on each metallic plate at a distance 0.5 m from the line AB. This distance is chosen so that the reaction due to the hinges on the laminar sheet is zero during the impact. Initially the laminar sheet hits one of the obstacles with an angular velocity 1 rad/s and turns back. If the impulse on the sheet due to each obstacle is 6 N-s. **[IIT-JEE' 2001]**



- (a) Find the location of the centre of mass of the laminar sheet from AB.
(b) At what angular velocity does the laminar sheet come back after the first impact ?
(c) After how many impacts, does the laminar sheet come to rest ?

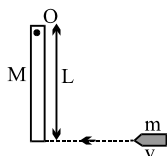
30. A circular platform is free to rotate in a horizontal plane about a vertical axis passing through its centre. A tortoise is sitting at the edge of the platform. Now, the platform is given an angular velocity ω_0 . When the tortoise move along a chord of the platform with a constant velocity (with respect to the platform), the angular velocity of the platform $\omega(t)$ will vary with time t as [IIT-JEE 2002]



31. A particle is moving in a horizontal uniform circular motion. The angular momentum of the particle is conserved about the point : [IIT-JEE'(Scr)2003]
 (A) Centre of the circle (B) Outside the circle
 (C) Inside the circle (D) Point on circumference
32. Two particles each of mass M are connected by a massless rod of length l . The rod is lying on the smooth surface. If one of the particle is given an impulse Mv as shown in the figure then angular velocity of the rod would be [IIT-JEE'(Scr)2003]



- (A) v/l (B) $2v/l$ (C) $v/2l$ (D) None
33. A child is standing with folded hands at the center of a platform rotating about its central axis. The kinetic energy of the system is K . The child now stretches his arms so that the moment of inertia of the system doubles. The kinetic energy of the system now is [JEE' 2004 (Scr)]
 (A) $2K$ (B) $K/2$ (C) $K/4$ (D) $4K$
34. A horizontal circular plate is rotating about a vertical axis passing through its centre with an angular velocity ω_0 . A man sitting at the centre having two blocks in his hands stretches out his hands so that the moment of inertia of the system doubles. If the kinetic energy of the system is K initially, its final kinetic energy will be [IIT-JEE 2004]
 (A) $2K$ (B) $K/2$ (C) K (D) $K/4$
35. A particle moves in circular path with decreasing speed. Which of the following is correct [IIT-JEE'2005 (Scr)]
 (A) $\vec{L}$ is constant
 (B) only direction of $\vec{L}$ is constant
 (C) acceleration $\vec{a}$ is towards the centre
 (D) it will move in a spiral and finally reach the centre
36. A wooden log of mass M and length L is hinged by a frictionless nail at O . A bullet of mass m strikes with velocity v and sticks to it. Find angular velocity of the system immediately after the collision about O . [IIT-JEE 2005]



Comprehension For Q. 37 to 39 (3 Questions)

Two discs A and B are mounted coaxially on a vertical axle. The discs have moments of inertia I and $2I$ respectively about the common axis. Disc A is imparted an initial angular velocity 2ω using the entire potential energy of a spring compressed by a distance x_1 . Disc B is imparted an angular velocity ω by a spring having the same spring constant and compressed by a distance x_2 . Both the discs rotate in the clockwise direction. **[JEE 2007]**

37. The value of ratio x_1/x_2 is / x_1/x_2 :-

- (A) 2 (B) $\frac{1}{2}$ (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

38. When disc B is brought in contact with disc A, they acquire a common angular velocity in time t . The average frictional torque on one disc by the other during this period is :

- (A) $\frac{2I\omega}{3t}$ (B) $\frac{9I\omega}{3t}$ (C) $\frac{9I\omega}{4t}$ (D) $\frac{3I\omega}{2t}$

39. The loss of kinetic energy the above process is :-

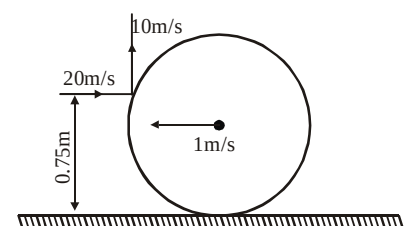
- (A) $\frac{I\omega^2}{2}$ (B) $\frac{I\omega^2}{3}$ (C) $\frac{I\omega^2}{4}$ (D) $\frac{I\omega^2}{6}$

40. If the resultant of the external forces acting on a system of particles is zero, then from an inertial frame, one can surely say that **[IIT-JEE 2009]**

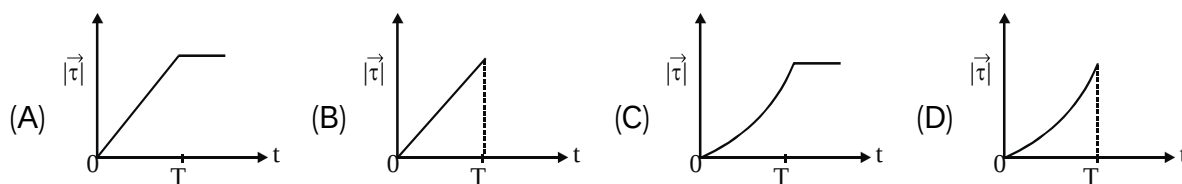
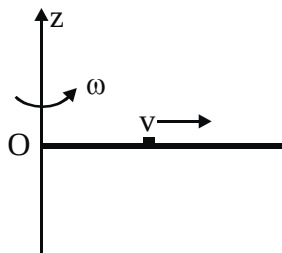
- (A) linear momentum of the system does not change in time
(B) kinetic energy of the system does not change in time
(C) angular momentum of the system does not change in time
(D) potential energy of the system does not change in time

41. A thin ring of mass 2 kg and radius 0.5 m is rolling without slipping on a horizontal plane with velocity 1 m/s. A small ball of mass 0.1 kg, moving with velocity 20 m/s in the opposite direction, hits the ring at a height of 0.75 m and goes vertically up with velocity 10 m/s. Immediately after the collision **[IIT-JEE 2011]**

- (A) the ring has pure rotation about its stationary CM
(B) the ring comes to a complete stop
(C) friction between the ring and the ground is to the left
(D) there is no friction between the ring and the ground

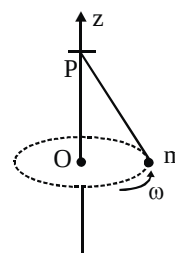


42. A thin uniform rod, pivoted at O , is rotating in the horizontal plane with constant angular speed ω , as shown in the figure. At time $t = 0$, a small insect starts from O and moves with constant speed v with respect to the rod towards the other end. It reaches the end of the rod at $t = T$ and stops. The angular speed of the system remains ω throughout. The magnitude of the torque ($|\vec{\tau}|$) on the system about O , as a function of time is best represented by which plot? [IIT-JEE 2012]



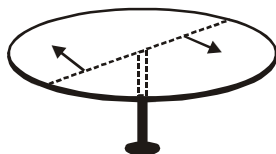
43. A small mass m is attached to a massless string whose other end is fixed at P as shown in the figure. The mass is undergoing circular motion in the x - y plane with centre at O and constant angular speed ω . If the angular momentum of the system, calculated about O and P are denoted by $\vec{L}_O$ and $\vec{L}_P$ respectively, then [IIT-JEE 2012]

- (A) $\vec{L}_O$ and $\vec{L}_P$ do not vary with time
 (B) $\vec{L}_O$ varies with time while $\vec{L}_P$ remains constant
 (C) $\vec{L}_O$ remains constant while $\vec{L}_P$ varies with time
 (D) $\vec{L}_O$ and $\vec{L}_P$ both vary with time

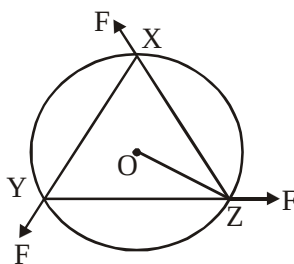


44. A uniform circular disc of mass 50 kg and radius 0.4 m is rotating with an angular velocity of 10 rad s^{-1} about its own axis, which is vertical. Two uniform circular rings, each of mass 6.25 kg and radius 0.2 m, are gently placed symmetrically on the disc in such a manner that they are touching each other along the axis of the disc and are horizontal. Assume that the friction is large enough such that the rings are at rest relative to the disc and the system rotates about the original axis. The new angular velocity (in rad s^{-1}) of the system is [IIT-JEE Advanced 2013]

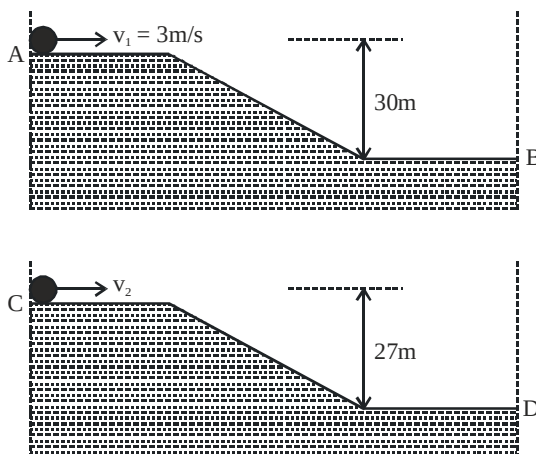
45. A horizontal circular platform of radius 0.5 m and mass 0.45 kg is free to rotate about its axis. Two massless spring toy-guns, each carrying a steel ball of mass 0.05 kg are attached to the platform at a distance 0.25 m from the centre on its either sides along its diameter (see figure). Each gun simultaneously fires the balls horizontally and perpendicular to the diameter in opposite directions. After leaving the platform, the balls have horizontal speed of 9 ms^{-1} with respect to the ground. The rotational speed of the platform in rad s^{-1} after the balls leave the platform is **[IIT-JEE Advanced 2014]**



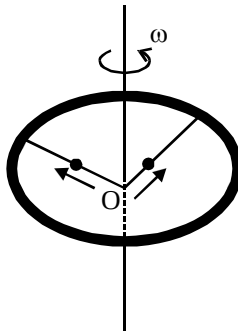
46. A uniform circular disc of mass 1.5 kg and radius 0.5 m is initially at rest on a horizontal frictionless surface. Three forces of equal magnitude $F = 0.5 \text{ N}$ are applied simultaneously along the three sides of an equilateral triangle XYZ with its vertices on the perimeter of the disc (see figure). One second after applying the forces, the angular speed of the disc in rad s^{-1} is **[IIT-JEE Advanced 2014]**



47. Two identical uniform discs roll without slipping on two different, surfaces AB and CD (see figure) starting at A and C with linear speeds v_1 and v_2 respectively, and always remain in contact with the surfaces. If they reach B and D with the same linear speed and $v_1 = 3 \text{ m/s}$, then v_2 in m/s is ($g = 10 \text{ m/s}^2$) **[JEE Advanced 2015]**



48. A ring of mass M and radius R is rotating with angular speed ω about a fixed vertical axis passing through its centre O with two point masses each of mass $\frac{M}{8}$ at rest at O . These masses can move radially outwards along two massless rods fixed on the ring as shown in the figure. At some instant the angular speed of the system is $\frac{8}{9}\omega$ and one of the masses is at a distance of $\frac{3}{5}R$ from O . At this instant the distance of the other mass from O is :
- [JEE Advanced 2015]**



- (A) $\frac{2}{3}R$ (B) $\frac{1}{3}R$ (C) $\frac{3}{5}R$ (D) $\frac{4}{5}R$
49. The densities of two solid spheres A and B of the same radii R vary with radial distance r as $\rho_A(r) = k\left(\frac{r}{R}\right)$ and $\rho_B(r) = k\left(\frac{r}{R}\right)^5$, respectively, where k is a constant. The moments of inertia of the individual spheres about axes passing through their centres are I_A and I_B , respectively. If $\frac{I_B}{I_A} = \frac{n}{10}$, the value of n is.
- [JEE Advanced 2015]**
50. A uniform wooden stick of mass 1.6 kg and length ℓ rests in an inclined manner on a smooth, vertical wall of height h ($< \ell$) such that a small portion of the stick extends beyond the wall. The reaction force of the wall on the stick is perpendicular to the stick. The stick makes an angle of 30° with the wall and the bottom of the stick is on a rough floor. The reaction of the wall on the stick is equal in magnitude to the reaction of the floor on the stick. The ratio h/ℓ and the frictional force f at the bottom of the stick are: ($g = 10 \text{ ms}^{-2}$)
- [JEE Advanced 2016]**

(A) $\frac{h}{\ell} = \frac{\sqrt{3}}{16}, f = \frac{16\sqrt{3}}{3} \text{ N}$

(B) $\frac{h}{\ell} = \frac{3}{16}, f = \frac{16\sqrt{3}}{3} \text{ N}$

(C) $\frac{h}{\ell} = \frac{3\sqrt{3}}{16}, f = \frac{8\sqrt{3}}{3} \text{ N}$

(D) $\frac{h}{\ell} = \frac{3\sqrt{3}}{16}, f = \frac{16\sqrt{3}}{3} \text{ N}$

51. The position vector $\vec{r}$ of a particle of mass m is given by the following equation $\vec{r}(t) = \alpha t^3 \hat{i} + \beta t^2 \hat{j}$, where

$\alpha = \frac{10}{3} \text{ ms}^{-3}$, $\beta = 5 \text{ ms}^{-2}$ and $m = 0.1 \text{ kg}$. At $t = 1 \text{ s}$, which of the following statement(s) is(are) true

about the particle?

[JEE Advanced 2016]

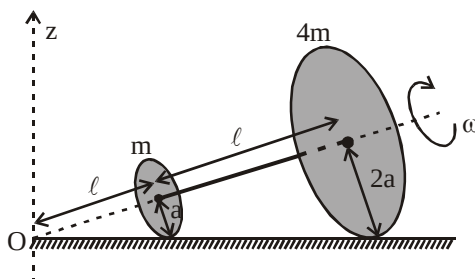
(A) The velocity $\vec{v}$ is given by $\vec{v} = (10\hat{i} + 10\hat{j})\text{ms}^{-1}$

(B) The angular momentum $\vec{L}$ with respect to the origin is given by $\vec{L} = -\left(\frac{5}{3}\right)\hat{k} \text{ Nms}$

(C) The force $\vec{F}$ is given by $\vec{F} = (\hat{i} + 2\hat{j})\text{N}$

(D) The torque $\vec{\tau}$ with respect to the origin is given by $\vec{\tau} = -\left(\frac{20}{3}\right)\hat{k} \text{ Nm}$

52. Two thin circular discs of mass m and $4m$, having radii of a and $2a$, respectively, are rigidly fixed by a massless, right rod of length $\ell = \sqrt{24}a$ through their center. This assembly is laid on a firm and flat surface, and set rolling without slipping on the surface so that the angular speed about the axis of the rod is ω . The angular momentum of the entire assembly about the point 'O' is $\vec{L}$ (see the figure). Which of the following statement(s) is(are) true ? [JEE Advanced 2016]



(A) The magnitude of angular momentum of the assembly about its center of mass is $17ma^2\omega/2$

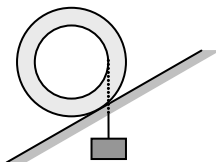
(B) The magnitude of the z-component of $\vec{L}$ is $55 ma^2\omega$

(C) The magnitude of angular momentum of center of mass of the assembly about the point O is $81 ma^2\omega$

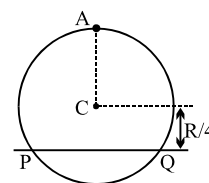
(D) The center of mass of the assembly rotates about the z-axis with an angular speed of $\omega/5$

Review Test

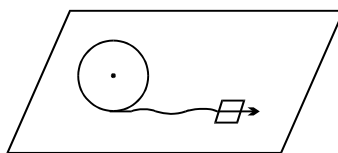
1. A 3.0 kg bobbin consists of a central cylinder of radius 5.0 cm and two end plates each of radius 6.0 cm. It is placed on a slotted incline, where friction is sufficient to prevent sliding. A block of mass 4.5 kg is suspended from a cord wound around the bobbin and passing through the slot under the incline. If the bobbin is in static equilibrium, what is the angle of tilt of the incline?



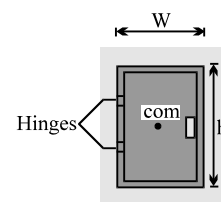
2. A uniform circular disc has radius R and mass m . A particle also of mass m is fixed at a point A on the wedge of the disc as in fig. The disc can rotate freely about a fixed horizontal chord PQ that is at a distance $R/4$ from the centre C of the disc. The line AC is perpendicular to PQ . Initially the disc is held vertical with the point A at its highest position. It is then allowed to fall so that it starts rotating about PQ . Find the linear speed of the particle at it reaches its lowest position.



3. A block of mass m is attached to a pulley disc of equal mass m , radius r by means of a slack string as shown. The pulley is hinged about its centre on a horizontal table and the block is projected with an initial velocity of 5 m/s. Find the velocity when the string becomes taut.

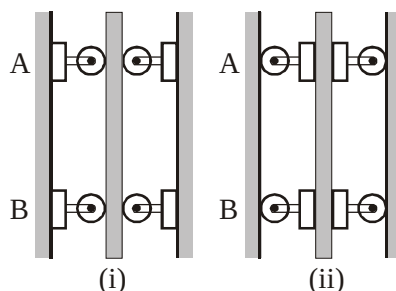


4. A uniform wooden door has mass m , height h , and width w . It is hanging from two hinges attached to one side; the hinges are located $h/3$ and $2h/3$ from the bottom of the door. Suppose that $m = 20.0$ kg, $h = 2.20$ m, and $w = 1.00$ m and the bottom smooth hinge is not screwed into the door frame. Find the forces acting on the door.

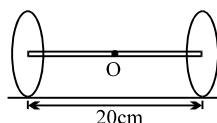


5. A circular disc of mass 300 gm and radius 20 cm can rotate freely about a vertical axis passing through its centre O . A small insect of mass 100 gm is initially at a point A on the disc (which is initially stationary) the insect starts walking from rest along the rim of the disc with such a time varying relative velocity that the disc rotates in the opposite direction with a constant angular acceleration $= 2\pi \text{ rad/s}^2$. After some time T , the insect is back at the point A . By what angle θ has the disc rotated till now ; as seen by a stationary earth observer? Also find the time T .

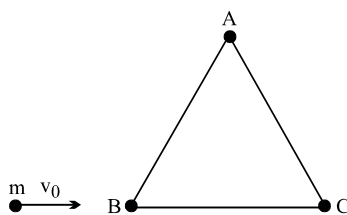
6. A bar of mass m is held as shown between 4 disks, each of mass m' & radius $r = 75$ mm. Determine the acceleration of the bar immediately after it has been released from rest, knowing that the normal forces exerted on the disks are sufficient to prevent any slipping and assuming that;



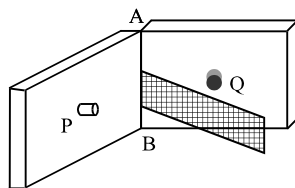
- (a) $m = 5$ kg and $m' = 2$ kg .
 (b) the mass of m' of the disks is negligible.
 (c) the mass of m of the bar is negligible .
7. Two thin circular disks of mass 2kg and radius 10cm each are joined by a rigid massless rod of length 20cm. The axis of the rod is along the perpendicular to the planes of the disks through their centre. The object is kept on a truck in such a way that the axis of the object is horizontal and perpendicular to the direction of motion of the truck. Its friction with the floor of the truck is large enough so that the object can roll on the truck without slipping. Take x-axis as the direction of motion of the truck and z -axis as the vertically upwards direction. If the truck has an acceleration of 9m/s^2 calculate :



- (a) the force of friction on each disk
 (b) The magnitude and the direction of the frictional torque acting on each disk about the centre of mass O of the object. Express the torque in the vector form of unit vectors in the x-y and z directions.
8. Three particles A, B, C of mass m each are joined to each other by massless rigid rods to form an equilateral triangle of side a . Another particle of mass m hits B with a velocity v_0 directed along BC as shown. The colliding particle stops immediately after impact .



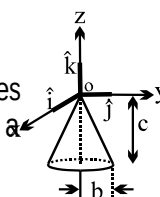
- (i) Calculate the time required by the triangle ABC to complete half-revolution in its subsequent motion.
 (ii) What is the net displacement of point B during this interval ?
9. Two heavy metallic plates are joined together at 90° to each other. A laminar sheet of mass 30 Kg is hinged at the line AB joining the two heavy metallic plates. The hinges are frictionless. The moment of inertia of the laminar sheet about an axis parallel to AB and passing through its centre of mass is 1.2 Kg-m^2 . Two rubber obstacles P and Q are fixed, one on each metallic plate at a distance 0.5 m from the line AB. This distance is chosen so that the reaction due to the hinges on the laminar sheet is zero during the impact. Initially the laminar sheet hits one of the obstacles with an angular velocity 1 rad/s and turns back. If the impulse on the sheet due to each obstacle is 6 N-s . [IIT-JEE' 2001]



- (a) Find the location of the centre of mass of the lamina sheet from AB.
 (b) At what angular velocity does the lamina sheet come back after the first impact ?
 (c) After how many impacts, does the lamina sheet come to rest ?

10. A solid cone hangs from a frictionless pivot at the origin O, as shown. If $\hat{i}$, $\hat{j}$ and

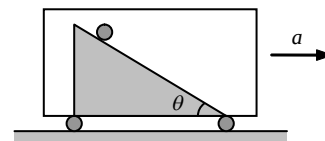
$\hat{k}$ are unit vectors, and a, b, and c are positive constants, which of the following forces F applied to the rim of the cone at a point P results in a torque τ on the cone with a negative component τ_z ?



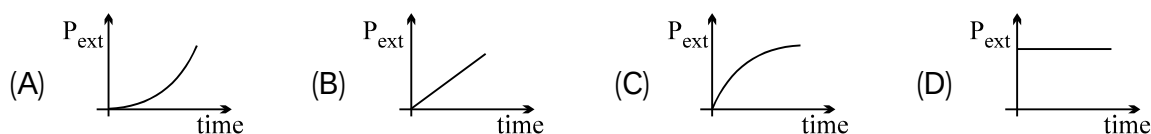
- (A) $F = a\hat{k}$, P is (0, b, -c) (B) $F = -a\hat{k}$, P is (0, -b, -c)
 (C) $F = a\hat{j}$, P is (-b, 0, -c) (D) None

11. Figure shows a smooth inclined plane fixed in a car accelerating on a horizontal road. The angle of incline θ is related to the acceleration a of the car as $a = g \tan \theta$. If the sphere is set in pure rolling on the incline.

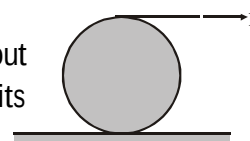
- (A) it will continue pure rolling.
 (B) it will slip down the plane.
 (C) its linear velocity will increase.
 (D) its linear velocity will slowly decrease.



12. A rod is hinged at its centre and rotated by applying a constant torque starting from rest. The power developed by the external torque as a function of time is :



13. A cylinder is pulled by constant force F on horizontal ground with the help of a light cord wrapped around it as shown in the figure. The cylinder rolls without slipping. When mass centre of the cylinder moves a distance x, what are its rotational and translational kinetic energies?



- (A) $\frac{Fx}{3}$ and $\frac{2Fx}{3}$ (B) $\frac{2Fx}{3}$ and $\frac{Fx}{3}$ (C) $\frac{2Fx}{3}$ and $\frac{4Fx}{3}$ (D) $\frac{4Fx}{3}$ and $\frac{2Fx}{3}$

14. Torque acting on a uniform circular wheel changes its angular momentum from $\vec{L}_0$ to $4\vec{L}_0$ in 4 s. Which of the following is necessarily correct?

- (A) Torque vector is $3\vec{L}_0 / 4$
 (B) Average torque vector is $3\vec{L}_0 / 4$
 (C) If torque is constant it equals to $3\vec{L}_0 / 4$
 (D) Torque vector cannot be decided from the given information.

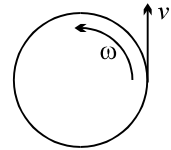
15. A figure skater is spinning at an angular frequency ω . She then pulls in her arms, reducing her moment of inertia by a factor of two. Compared to her initial rotational kinetic energy, her final rotational kinetic energy is
- (A) twice as large. (B) four times as large.
(C) one fourth as large. (D) one half as large.
16. A child with mass m is standing at the edge of a disc with moment of inertia I , radius R , and initial angular velocity ω . See figure given below. The child jumps off the edge of the disc with tangential velocity v with respect to the ground. The new angular velocity of the disc is

(A) $\sqrt{\frac{I\omega^2 - mv^2}{I}}$

(B) $\sqrt{\frac{(I + mR^2)\omega^2 - mv^2}{I}}$

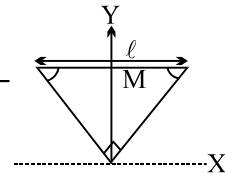
(C) $\frac{I\omega - mvR}{I}$

(D) $\frac{(I + mR^2)\omega - mvR}{I}$



Question No. 17 to 20 (4 questions)

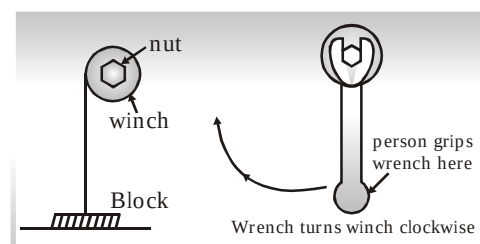
The figure shows an isosceles triangular plate of mass M and base L . The angle at the apex is 90° . The apex lies at the origin and the base is parallel to X-axis



17. The moment of inertia of the plate about the z-axis is
- (A) $\frac{ML^2}{12}$ (B) $\frac{ML^2}{24}$ (C) $\frac{ML^2}{6}$ (D) none of these
18. The moment of inertia of the plate about the x-axis is
- (A) $\frac{ML^2}{8}$ (B) $\frac{ML^2}{32}$ (C) $\frac{ML^2}{24}$ (D) $\frac{ML^2}{6}$
19. The moment of inertia of the plate about its base parallel to the x-axis is
- (A) $\frac{ML^2}{18}$ (B) $\frac{ML^2}{36}$ (C) $\frac{ML^2}{24}$ (D) none of these
20. The moment of inertia of the plate about the y-axis is
- (A) $\frac{ML^2}{6}$ (B) $\frac{ML^2}{8}$ (C) $\frac{ML^2}{24}$ (D) none of these
21. A block with a square base measuring a and height h , is placed on an inclined plane. The coefficient of friction is μ . The angle of inclination (θ) of the plane is gradually increased. The block will
- (A) topple before sliding if $\mu > \frac{a}{h}$ (B) topple before sliding if $\mu < \frac{a}{h}$
(C) slide before toppling if $\mu > \frac{a}{h}$ (D) slide before toppling if $\mu < \frac{a}{h}$

Question No. 22 to 24 (3 questions)

In figure, the winch is mounted on an axle, and the 6-sided nut is welded to the winch. By turning the nut with a wrench, a person can rotate the winch. For instance, turning the nut clockwise lifts the block off the ground, because more and more rope gets wrapped around the winch.



Three students agree that using a longer wrench makes it easier to turn the winch. But they disagree about why.

All three students are talking about the case where the winch is used, over a 10 s time interval, to lift the block one metre off the ground.

Student 1 By using a longer wrench, the person decreases the average force he must exert on the wrench, in order to lift the block one metre in 10 s.

Student 2 : Using a longer wrench reduces the work done by the person as he uses the winch to lift the block 1m in 10s.

Student 3 : Using a longer wrench reduces the power that the person must exert to lift the block 1m in 10s.

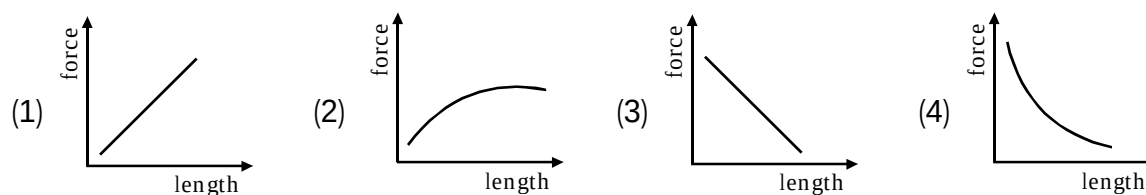
22. Student 1 is :-

- (A) correct, because the torque that the wrench must exert to lift the block doesn't depend on the wrench's length
- (B) correct, because using a longer wrench decreases the torque it must exert on the winch
- (C) incorrect, because the torque that the wrench must exert to lift the block doesn't depend on the wrench's length
- (D) Incorrect, because using a longer wrench decreases the torque it must exert on the winch.

23. Which of the following is true about student 2 and 3 :-

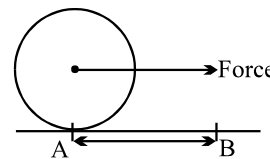
- (A) Student 2 and 3 are both correct
- (B) Student 2 is correct, but student 3 is incorrect
- (C) Student 3 is correct, but student 2 is incorrect
- (D) Student 2 and 3 are both incorrect

24. If several wrenches all apply the same torque to a nut, which graph best expresses the relationship between the force the person must apply to the wrench, and the length of the wrench :-

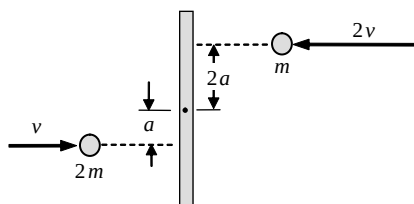


- (A) 1
- (B) 2
- (C) 3
- (D) 4

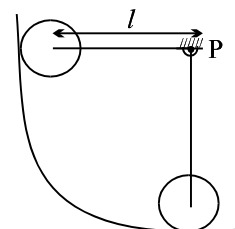
25. A disc of circumference s is at rest at a point A on a horizontal surface when a constant horizontal force begins to act on its centre. Between A and B there is sufficient friction to prevent slipping, and the surface is smooth to the right of B. $AB = s$. The disc moves from A to B in time T . To the right of B,



- (A) the angular acceleration of the disc will disappear, linear acceleration will remain unchanged
(B) linear acceleration of the disc will increase
(C) the disc will make one rotation in time $T/2$
(D) the disc will cover a distance greater than s in further time T .
26. The torque $\vec{\tau}$ on a body about a given point is found to be equal to $\vec{A} \times \vec{L}$ where $\vec{A}$ is a constant vector and $\vec{L}$ is the angular momentum of the body about that point. From this it follows that
- (A) $d\vec{L}/dt$ is perpendicular to $\vec{L}$ at all instants of time
(B) the components of $\vec{L}$ in the direction of $\vec{A}$ does not change with time
(C) the magnitude of $\vec{L}$ does not change with time
(D) $\vec{L}$ does not change with time
27. A uniform bar of length $6a$ and mass $8m$ lies on a smooth horizontal table as shown in figure. Two point masses m and $2m$ moving in the same horizontal plane with speed $2v$ and v respectively, strike the bar and stick to the bar after collision. After the collision, which of the following values are correct?



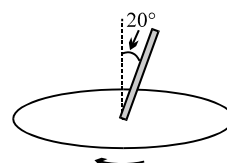
- (A) Speed of mass center = 0
(B) Angular speed = $0.2v/a$
(C) Angular speed = $0.6v/a$
(D) Kinetic energy = $3mv^2/5$
28. A sphere of mass M and radius R is attached by a light rod of length ℓ to a point P. The sphere rolls without slipping on a circular track as shown. It is released from the horizontal position. The angular momentum of the system about P when the rod becomes vertical is :



- (A) $M\sqrt{\frac{10}{7}}gl [l + R]$
(B) $M\sqrt{\frac{10}{7}}gl \left[l + \frac{2}{5}R \right]$
(C) $M\sqrt{\frac{10}{7}}gl \left[l + \frac{7}{5}R \right]$
(D) none of the above

Question No. 29 and 30 (2 questions)

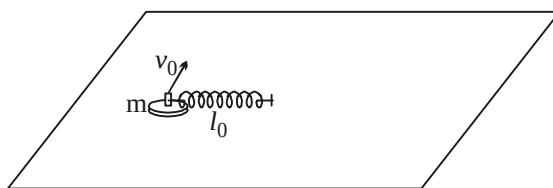
A uniform rod is fixed to a rotating turntable so that its lower end is on the axis of the turntable and it makes an angle of 20° to the vertical. (The rod is thus rotating with uniform angular velocity about a vertical axis passing through one end.) If the turntable is rotating clockwise as seen from above.



29. What is the direction of the rod's angular momentum vector (calculated about its lower end)?
 (A) vertically downwards (B) down at 20° to the horizontal
 (C) up at 20° to the horizontal (D) vertically upwards
30. Is there a torque acting on it, and if so in what direction?
 (A) yes, vertically (B) yes, horizontally
 (C) yes at 20° to the horizontal (D) no

Paragraph for Question No. 31 to 33.

A spring having initial unstretched length ℓ_0 is lying on a smooth table. Its one end is fixed and the other one is fastened to a small particle of mass m . The particle is imparted an initial speed v_0 horizontally in a direction perpendicular to the spring. In the course of the motion in horizontal plane, the maximum elongation of the spring is $\Delta\ell = \ell_0 / 10$. (Given : $m = 2.1$ kg, $\ell_0 = 21$ cm, $v_0 = 1.1$ m/s).



31. In the course of motion, which of the quantities relating to spring block system are conserved ?
 (A) kinetic energy (B) momentum
 (C) angular momentum (D) potential energy
32. Which of the following is correct about initial situation and situation at maximum elongation ?
 (A) the orientation of spring in both positions should be perpendicular to each other.
 (B) the velocity at maximum extension should be zero
 (C) the velocity at maximum extension as well as at initial position should be perpendicular to spring.
 (D) Acceleration should be zero at the maximum extension as well as at initial position.
33. **Student-A** : at maximum extension $mv'(\ell_0 + \Delta\ell) = mv_0\ell_0$.

Student-B : at maximum extension $k\Delta\ell = \frac{mv^2}{(\ell_0 + \Delta\ell)}$.

Student-C : $\frac{1}{2}mv_0^2 = \frac{1}{2}mv'^2 + \frac{1}{2}k\Delta\ell^2$.

where v' is velocity at instant of maximum extension :

- (A) Only Student-A and B are correct (B) Only Student-A and C are correct
 (C) Only Student-B and C are correct (D) All are correct

ANSWER KEY

EXERCISE (S-2)

1. Ans. $\frac{a\ell^4}{36}$

2. Ans. $(3/8) F_g$

3. Ans. $a = \left[\frac{b - \mu a}{3b + \mu a} \right] g$

4. Ans. (a) $5 \text{ m/s}^2 \rightarrow$, (b) $0.3 < h < 1.5 \text{ m}$

5. Ans. (a) $\sqrt{3} \text{ m}/\omega^2$ (b) $(F_{\text{net}})_x = -\frac{F}{4}$, $(F_{\text{net}})_y = \sqrt{3} \text{ m}/\omega^2$

6. Ans. $\sqrt{15} \text{ ft/sec}$

7. Ans. (a) $4\sqrt{\frac{3}{7}} \text{ m/s}$, (b) $\frac{200}{7} \text{ N}$

8. Ans. (i) (a) 3 g/L (clockwise)

(b) $\left(\frac{\sqrt{3}}{2} \hat{i} + \hat{j}\right) g = 1.323 \text{ g } \angle 49.1^\circ$

(c) $\left(\frac{\sqrt{3}}{2} \hat{i} - 2 \hat{j}\right) g = 2.18 \text{ g } \angle -66.6^\circ$

(ii) (a) g/L (clockwise)

(b) $-\left(\frac{\sqrt{3}}{2}\right) g \hat{i}$

(c) $-\left(\frac{\sqrt{3}}{2} \hat{i} + \hat{j}\right) g = 1.323 \text{ g } \angle -130.9^\circ$

9. Ans. (i) (a) $\frac{12g}{c}$ (cw)

(b) $-03.(\hat{i} + 2\hat{j}) g$

(ii) (a) $24 \text{ g}/17 \text{ c}$ (cw)

(b) $12 \text{ g}/17 \downarrow$

(iii) (a) 2.4 g/c (cw)

(b) $0.5 \text{ g } \downarrow$

10. Ans. 3

11. Ans. $\alpha_{\text{max}} = \frac{2g}{3r}$

12. Ans. (a) $a_c = \frac{4F}{3m_1 + 8m_2}$, $a_p = 2a_c$

(b) friction at the top of the cylinder = $3m_1 F / (3m_1 + 8m_2)$ towards right; friction at the bottom = $m_1 F / (3m_1 + 8m_2)$ towards right.

13. Ans. (a) pipe rolls without sliding

(b) pipe : $5/6 \text{ m/s } \rightarrow$, $125/12 \text{ rad/s}$ (anticlockwise); plate : $5/3 \text{ m/s } \rightarrow$

14. Ans. (a) $\theta_c = \cos^{-1}(4/7)$, (b) $v = \sqrt{4/7 g R}$, (c) $K_T / K_R = 6$

15. Ans. $v = \sqrt{\frac{14gR}{3}}$

16. Ans. (a) 6.68 m/s (b) 2.27 m

17. Ans. (i) $2v_0/3$, (ii) $t = v_0/3\mu g$, $W = \frac{1}{2} [3 \mu^2 m g^2 t^2 - 2 \mu m g t v_0]$ ($t < t_0$), $W = -\frac{1}{6} m v_0^2$ ($t > t_0$)

18. Ans. (a) $\omega/3$, (b) $\frac{\sqrt{37}}{3} m \omega R$, (c) $\frac{\sqrt{37}}{3} m \omega R$

19. Ans. 60°

EXERCISE (O-2)

- | | | | |
|--------------------|--------------------|--------------------|----------------------|
| 1. Ans. (A, C, D) | 2. Ans. (D) | 3. Ans. (A, B) | 4. Ans. (A, B, C, D) |
| 5. Ans. (B, C, D) | 6. Ans. (B, C) | 7. Ans. (B) | 8. Ans. (C) |
| 9. Ans. (A, C) | 10. Ans. (C) | 11. Ans. (B) | 12. Ans. (A) |
| 13. Ans. (B) | 14. Ans. (A, B, C) | 15. Ans. (A, B, D) | 16. Ans. (A, C, D) |
| 17. Ans. (C) | 18. Ans. (A, B, C) | 19. Ans. (C) | 20. Ans. (B, C, D) |
| 21. Ans. (A, C, D) | | | |

MATRIX MATCH TYPE QUESTION

22. Ans. (A) $\rightarrow$ (S,T) ; (B) $\rightarrow$ (R) ; (C) $\rightarrow$ (P,R,T) ; (D) $\rightarrow$ (P,Q)

EXERCISE (JA)

- | | | | |
|---|---|---|-----------------|
| 1. Ans. (A) | 2. Ans. (B, C) | 3. Ans. (D) | 4. Ans. (A) |
| 5. Ans. (A, B) | 6. Ans. (D) | 7. Ans. (A) | 8. Ans. (B) |
| 9. Ans. (A) | 10. Ans. 9 | 11. Ans. 3 | 12. Ans. (C) |
| 13. Ans. (B) | 14. Ans. (a) $\sqrt{3} \text{ m}/\omega^2$ (b) $(F_{\text{net}})_x = -\frac{F}{4}$, $(F_{\text{net}})_y = \sqrt{3} \text{ m}/\omega^2$ | | |
| 15. Ans. (C) | 16. Ans. $a_{\text{axis}} = \frac{2g \sin \theta}{3}$ | | |
| 17. Ans. $f = (M+m) g \frac{\cot \theta}{2}$ | 18. Ans. (C, D) | 19. Ans. 10 m/s | 20. Ans. (D) |
| 21. Ans. (B) | 22. Ans. 4 | 23. Ans. (D) | 24. Ans. (A, B) |
| 25. Ans. (D) | 26. Ans. (A, D) | 27. Ans. (a) $\frac{m}{M} = \frac{1}{4}$; (b) $x = \frac{2L}{3}$; (c) $\frac{v_0}{2\sqrt{2}}$ | |
| 28. Ans. (B) | | | |
| 29. Ans. (a) $\ell = 0.1\text{m}$; (b) $\omega' = 1\text{rad/s}$; (c) laminar sheet will never come to rest | | | |
| 30. Ans. (B) | 31. Ans. (A) | 32. Ans. (A) | 33. Ans. (B) |

34. Ans. (B) 35. Ans. (B) 36. Ans. $\omega = \frac{3mv}{(3m + M)L}$
37. Ans. (C) 38. Ans. (A) 39. Ans. (B) 40. Ans. (A)
41. Ans. (AC or C) 42. Ans. (B) 43. Ans. (C) 44. Ans. 8
45. Ans. 4 46. Ans. 2 47. Ans. 7
48. Ans. (C or D) or (C and D) 49. Ans. 6 50. Ans. (D)
51. Ans. (A, B, D) 52. Ans. (D)

REVIEW TEST

1. Ans. 30° 2. Ans. $v = \sqrt{5gR}$ 3. Ans. $10/3 \text{ m/s}$
4. Ans. $F_{dx} = \frac{3mgw}{2h}$ and $F_{dy} = mg$ 5. Ans. $t = 2/\sqrt{5} \text{ sec}$, $\theta = 4\pi/5 \text{ rad}$
6. Ans. (i) (a) $5g/9 \downarrow$ (b) $g \downarrow$ (c) 0 (ii) (a) $\frac{13g}{17} \downarrow$ (b) $g \downarrow$ (c) $\frac{2g}{3} \downarrow$
7. Ans. $6N, -0.6\hat{j} \pm 0.6\hat{k}$
8. Ans. (i) $t = \frac{6a\pi}{\sqrt{3}v_0}$; (ii) $s = \frac{a}{\sqrt{3}} \sqrt{1 + (2\pi + \sqrt{3})^2}$
9. Ans. (a) $\ell = 0.1\text{m}$; (b) $\omega' = 1\text{rad/s}$; (c) laminar sheet will never come to rest
10. Ans. (C) 11. Ans. (A) 12. Ans. (B) 13. Ans. (C)
14. Ans. (B, C) 15. Ans. (A) 16. Ans. (D) 17. Ans. (C)
18. Ans. (A) 19. Ans. (C) 20. Ans. (C) 21. Ans. (A, D)
22. Ans. (A) 23. Ans. (D) 24. Ans. (D) 25. Ans. (B, C, D)
26. Ans. (A, B, C) 27. Ans. (A, B, D) 28. Ans. (D) 29. Ans. (B)
30. Ans. (B) 31. Ans. (C) 32. Ans. (C) 33. Ans. (B)

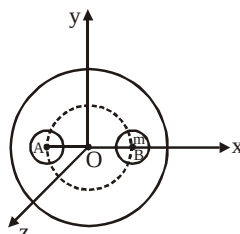
GRAVITATION

EXERCISE (S-2)

1. A small body of mass m is projected with a velocity just sufficient to make it reach from the surface of a planet (of radius $2R$ and mass $3M$) to the surface of another planet (of radius R and mass M). The distance between the centers of the two spherical planets is $6R$. The distance of the body from the center of bigger planet is ' x ' at any moment. During the journey, find the distance x where the speed of the body is (a) maximum (b) minimum. Assume motion of body along the line joining centres of planets.
2. A body is launched from the earth's surface at an angle $\alpha = 30^\circ$ to the horizontal at a speed $v_0 = \sqrt{\frac{1.5GM}{R}}$. Neglecting air resistance and earth's rotation, find (a) the height to which the body will rise. (b) the radius of curvature of trajectory at its top point.
3. A cord of length 64 m is used to connect a 100 kg astronaut to spaceship whose mass is much larger than that of the astronaut. Estimate the value of the tension in the cord. Assume that the spaceship is orbiting near earth surface. Assume that the spaceship and the astronaut fall on a straight line from the earth centre. The radius of the earth is 6400 km.
4. A remote sensing satellite is revolving in an orbit of radius x over the equator of earth. Find the area on earth surface in which satellite can not send message.
5. A pair of stars rotates about a common center of mass. One of the stars has a mass M which is twice as large as the mass m of the other. Their centres are at a distance d apart, d being large compared to the size of either star. (a) Derive an expression for the period of rotation of the stars about their common centre of mass in terms of d, m, G . (b) Compare the angular momentum of the two stars about their common centre of mass by calculating the ratio L_m / L_M . (c) Compare the kinetic energies of the two stars by calculating the ratio K_m / K_M .

EXERCISE (O-2)

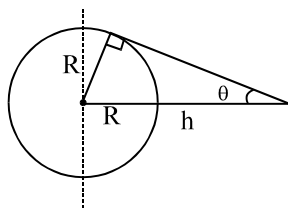
1. A solid sphere of uniform density and radius 4 units is located with its centre at the origin O of coordinates. Two spheres of equal radii 1 unit, with their centres at A (-2, 0, 0) and B (2, 0, 0) respectively, are taken out of the solid leaving behind spherical cavities as shown in figure. Then



- (A) The gravitational field due to this object at the origin is zero
 (B) The gravitational field at the point B (2, 0, 0) is zero
 (C) The gravitational potential is the same at all points of circle $y^2 + z^2 = 36$
 (D) The gravitational potential is the same at all points on the circle $y^2 + z^2 = 4$
2. A planet of mass m is in an elliptical orbit about the sun ($m \ll M_{\text{sun}}$) with an orbital period T . If A be the area of orbit, then its angular momentum would be :

- (A) $\frac{2mA}{T}$ (B) mAT (C) $\frac{mA}{2T}$ (D) $2mAT$

3. A planet revolves about the sun in elliptical orbit. The areal velocity $\left(\frac{dA}{dt}\right)$ of the planet is $4.0 \times 10^{16} \text{ m}^2/\text{s}$. The least distance between planet and the sun is $2 \times 10^{12} \text{ m}$. Then the maximum speed of the planet in km/s is :
- (A) 10 (B) 20 (C) 40 (D) None of these
4. A geostationary satellite is at a height h above the surface of earth. If earth radius is R

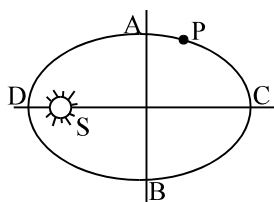


- (A) The minimum colatitude on earth upto which the satellite can be used for communication is $\sin^{-1} (R/R + h)$.
 (B) The maximum latitudes on earth upto which the satellite can be used for communication is $\cos^{-1} (R/R + h)$.
 (C) The area on earth escaped from this satellite is given as $2\pi R^2 (1 + \sin\theta)$

- (D) The area on earth escaped from this satellite is given as $2\pi R^2 (1 + \cos\theta)$
5. When a satellite in a circular orbit around the earth enters the atmospheric region, it encounters small air resistance to its motion. Then
- (A) its kinetic energy increases
 (B) its kinetic energy decreases
 (C) its angular momentum about the earth decreases
 (D) its period of revolution around the earth increases
6. Two satellites s_1 & s_2 of equal masses revolve in the same sense around a heavy planet in coplanar circular orbit of radii R & $4R$
- (A) the ratio of period of revolution of s_1 & s_2 is $1 : 8$.
 (B) their velocities are in the ratio $2 : 1$
 (C) their angular momentum about the planet are in the ratio $2 : 1$
 (D) the ratio of angular velocities of s_2 w.r.t. s_1 when all three are in the same line is $9 : 5$.
7. A double star is a system of two stars of masses m and $2m$, rotating about their centre of mass only under their mutual gravitational attraction. If r is the separation between these two stars then their time period of rotation about their centre of mass will be proportional to :
- (A) $r^{3/2}$ (B) r (C) $m^{1/2}$ (D) $m^{-1/2}$

Paragraph for Question No. 8 & 9

Figure shows the orbit of a planet P around the sun S. AB and CD are the minor and major axes of the ellipse.



8. If t_1 is the time taken by the planet to travel along ACB and t_2 the time along BDA, then
- (A) $t_1 = t_2$ (B) $t_1 > t_2$
 (C) $t_1 < t_2$ (D) nothing can be concluded
9. If U is the potential energy and K kinetic energy then $|U| > |K|$ at
- (A) Only D (B) Only C (C) both D & C (D) neither D nor C

EXERCISE (JA)

ALL ANSWERS OF THE QUESTIONS JEE-2007 ONWARDS ARE AS GIVEN ON IIT-JEE WEBSITE.

1. A double star system consists of two stars A and B which have time period T_A and T_B . Radius R_A and R_B and mass M_A and M_B . Choose the correct option :- **[IIT-JEE 2006]**

(A) If $T_A > T_B$ then $R_A > R_B$ (B) If $T_A > T_B$ then $M_A > M_B$

(C) $\left(\frac{T_A}{T_B}\right)^2 = \left(\frac{R_A}{R_B}\right)^3$ (D) $T_A = T_B$

2. Column-I describes some situations in which a small object moves. Column-II describes some characteristics of these motions. Match the situations in column I with the characteristics in column-II.

[IIT-JEE 2007]

Column-I

- (A) The object moves on the x-axis under a conservative force in such a way that its "speed" and "position" satisfy $v = c_1 \sqrt{c_2 - x^2}$, where c_1 and c_2 are positive constants.
- (B) The object moves on the x-axis in such a way that its velocity and its displacement from the origin satisfy $v = -kx$, where k is a positive constant.
- (C) The object is attached to one end of a mass-less spring of a given spring constant. The other end of the spring is attached to the ceiling of an elevator. Initially everything is at rest. The elevator starts going upwards with a constant acceleration a . The motion of the object is observed from the elevator during the period it maintains this acceleration.
- (D) The object is projected from the earth's surface vertically upwards with a speed

$$2\sqrt{\frac{GM_e}{R_e}}, \text{ where } M_e \text{ is the mass of the earth}$$

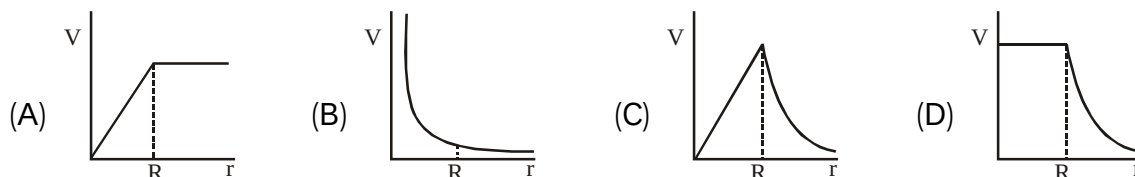
and R_e is the radius of the earth. Neglect forces from objects other than the earth.

Column-II

- (p) The object executes a simple harmonic motion.
- (q) The object does not change its direction.
- (r) The kinetic energy of the object keeps on decreasing
- (s) The object can change its direction only once.

3. A spherically symmetric gravitational system of particles has a mass density $\rho = \begin{cases} \rho_0 & \text{for } r \leq R \\ 0 & \text{for } r > R \end{cases}$

where ρ_0 is a constant. A test mass can undergo circular motion under the influence of the gravitational field of particles. Its speed V as a function of distance r ($0 < r < \infty$) from the centre of the system is represented by :- **[IIT-JEE 2008]**



4. **Statement – 1** : An astronaut in an orbiting space station above the Earth experiences weightlessness.
and **[IIT-JEE 2008]**

Statement – 2 : An object moving around the Earth under the influence of Earth's gravitational force is in a state of 'free-fall'.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

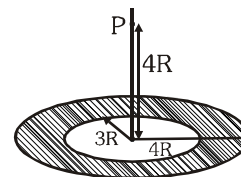
5. Gravitational acceleration on the surface of a planet is $\frac{\sqrt{6}}{11}g$, where g is the gravitational acceleration

on the surface of the earth. The average mass density of the planet is $\frac{2}{3}$ times that of the Earth. If the escape speed on the surface of the earth is taken to be 11 kms^{-1} , the escape speed on the surface of the planet in kms^{-1} will be **[IIT-JEE 2010]**

6. A binary star consists of two stars A (mass $2.2 M_s$) and B (mass $11 M_s$), where M_s is the mass of the sun. They are separated by distance d and are rotating about their centre of mass, which is stationary. The ratio of the total angular momentum of the binary star to the angular momentum of star B about the centre of mass is **[IIT-JEE 2010]**

7. A thin uniform annular disc (see figure) of mass M has outer radius $4R$ and inner radius $3R$. The work required to take a unit mass from point P on its axis to infinity is **[IIT-JEE 2010]**

- (A) $\frac{2GM}{7R}(4\sqrt{2}-5)$ (B) $-\frac{2GM}{7R}(4\sqrt{2}-5)$
 (C) $\frac{GM}{4R}$ (D) $\frac{2GM}{5R}(\sqrt{2}-1)$



8. A satellite is moving with a constant speed V in a circular orbit about the earth. An object of mass ' m ' is ejected from the satellite such that it just escapes from the gravitational pull of the earth. At the time of its ejection, the kinetic energy of the object is :- **[IIT-JEE 2011]**

(A) $\frac{1}{2}mV^2$ (B) mV^2 (C) $\frac{3}{2}mV^2$ (D) $2mV^2$

9. Two spherical planets P and Q have the same uniform density ρ , masses M_P and M_Q , and surface areas A and $4A$, respectively. A spherical planet R also has uniform density ρ and its mass is $(M_P + M_Q)$. The escape velocities from the planets P, Q and R, are V_P , V_Q and V_R , respectively. Then **[IIT-JEE 2012]**

(A) $V_Q > V_R > V_P$ (B) $V_R > V_Q > V_P$ (C) $V_R/V_P = 3$ (D) $V_P/V_Q = 1/2$

10. A planet of radius $R = \frac{1}{10} \times$ (radius of Earth) has the same mass density as Earth. Scientists dig a well

of depth $\frac{R}{5}$ on it and lower a wire of the same length and of linear mass density 10^{-3} kgm^{-1} into it. If

the wire is not touching anywhere, the force applied at the top of the wire by a person holding it in place is (take the radius of Earth = $6 \times 10^6 \text{ m}$ and the acceleration due to gravity on Earth is 10 ms^{-2})

[JEE-Advance 2014]

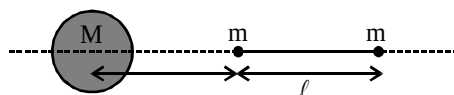
(A) 96 N (B) 108 N (C) 120 N (D) 150 N

11. A bullet is fired vertically upwards with velocity v from the surface of a spherical planet. When it reaches its maximum height, its acceleration due to the planet's gravity is $1/4^{\text{th}}$ of its value at the surface of the planet. If the escape velocity from the planet is $v_{\text{esc}} = v\sqrt{N}$, then the value of N is (ignore energy loss due to atmosphere) **[JEE-Advance 2015]**

12. A large spherical mass M is fixed at one position and two identical point masses m are kept on a line passing through the centre of M (see figure). The point masses are connected by a rigid massless rod of length ℓ and this assembly is free to move along the line connecting them. All three masses interact only through their mutual gravitational interaction. When the point mass nearer to M is at a distance

$r = 3\ell$ from M , the tension in the rod is zero for $m = k\left(\frac{M}{288}\right)$. The value of k is :

[JEE-Advance 2015]



13. A spherical body of radius R consists of a fluid of constant density and is in equilibrium under its own gravity. If $P(r)$ is the pressure at r ($r < R$), then the correct option(s) is(are) :- **[JEE-Advance 2015]**

(A) $P(r = 0) = 0$ (B) $\frac{P(r = 3R/4)}{P(r = 2R/3)} = \frac{63}{80}$ (C) $\frac{P(r = 3R/5)}{P(r = 2R/5)} = \frac{16}{21}$ (D) $\frac{P(r = R/2)}{P(r = R/3)} = \frac{20}{27}$

REVIEW TEST

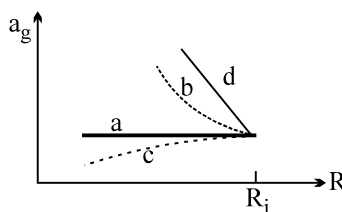
1. Let ω be the angular velocity of the earth's rotation about its axis. Assume that the acceleration due to gravity on the earth's surface has the same value at the equator and the poles in absence of rotation of earth. An object weighed at the equator gives the same reading as a reading taken at a depth d below earth's surface at a pole ($d \ll R$) The value of d is

(A) $\frac{\omega^2 R^2}{g}$ (B) $\frac{\omega^2 R^2}{2g}$ (C) $\frac{2\omega^2 R^2}{g}$ (D) $\frac{\sqrt{Rg}}{g}$

2. A spherical uniform planet is rotating about its axis. The velocity of a point on its equator is V . Due to the rotation of planet about its axis the acceleration due to gravity g at equator is $1/2$ of g at poles. The escape velocity of a particle on the pole of planet in terms of V is

(A) $V_e = 2V$ (B) $V_e = V$ (C) $V_e = V/2$ (D) $V_e = \sqrt{3} V$

3. A (nonrotating) star collapses onto itself from an initial radius R_i with its mass remaining unchanged. Which curve in figure best gives the gravitational acceleration a_g on the surface of the star as a function of the radius of the star during the collapse?



(A) a (B) b (C) c (D) d

4. A satellite can be in a geostationary orbit around earth at a distance r from the centre. If the angular velocity of earth about its axis doubles, a satellite can now be in a geostationary orbit around earth if its distance from the centre is :-

(A) $\frac{r}{2}$ (B) $\frac{r}{2\sqrt{2}}$ (C) $\frac{r}{(4)^{1/3}}$ (D) $\frac{r}{(2)^{1/3}}$

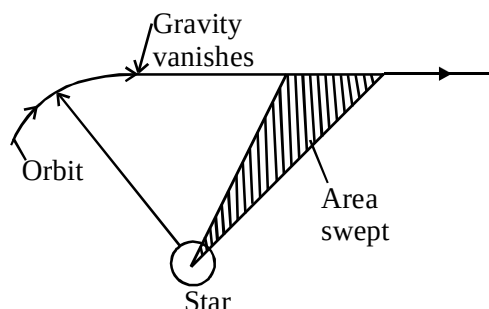
5. The fractional change in the value of free-fall acceleration g for a particle when it is lifted from the surface to an elevation h ($h \ll R$) is

(A) $\frac{h}{R}$ (B) $\frac{2h}{R}$ (C) $-\frac{2h}{R}$ (D) $-\frac{h}{R}$

6. The Sun travels in approximately circular orbit of radius R around the center of the galaxy and completes one revolution in time T . The Earth also revolves around the Sun in time t . Assume orbit of the Earth to be a circle of radius r ($r \ll R$) and whole mass of the galaxy centered on its center. By using only these given informations, find an expression for the ratio of the mass of the galaxy to that of the Sun.

(A) $\left(\frac{R}{r}\right)^3 \left(\frac{t}{T}\right)^2$ (B) $\left(\frac{R}{r}\right)^3 \left(\frac{T}{t}\right)^2$ (C) $\left(\frac{R}{r}\right)^2 \left(\frac{t}{T}\right)^3$ (D) $\left(\frac{R}{r}\right)^2 \left(\frac{T}{t}\right)^3$

7. A planet is orbiting a star when for no apparent reason the star's gravity suddenly vanishes. After which planet moves in a straight line. Mark the **CORRECT** statement(s) :



- (A) Newton's first law is obeyed on planet after gravity vanishes
 (B) Kepler's law of areas is obeyed only till the planet is in gravity of star
 (C) Kepler's law of areas is obeyed even after gravity vanishes
 (D) Angular momentum of planet about centre of star is conserved through out its motion
8. A body moving radially away from a planet of mass M , when at distance r from planet, explodes in such a way that two of its many fragments move in mutually perpendicular circular orbits around the planet. What will be (a) then velocity in circular orbits. (b) maximum distance between the two fragments before collision and (c) magnitude of their relative velocity just before they collide.
9. A hypothetical spherical planet of radius R and its density varies as $\rho = Kr$, where K is constant and r is the distance from the center. Determine the pressure caused by gravitational pull inside ($r < R$) the planet at a distance r measured from its center.
10. The Earth may be regarded as a spherically shaped uniform core of density ρ_1 and radius $R/2$ surrounded by a uniform shell of thickness $R/2$ and density ρ_2 . Find the ratio of $\frac{\rho_1}{\rho_2}$ if the value of acceleration due to gravity is the same at surface as at depth $R/2$ from the surface.
11. A binary star has a period (T) of 2 earth years while distance L between its components having masses M_1 and M_2 is four astronomical units. If $M_1 = M_s$ where M_s is the mass of sun, find the ratio $M_2/5M_s$.
12. Two uniform spherical stars made of same material have radii R and $2R$. Mass of the smaller planet is m . They start moving from rest towards each other from a large distance under mutual force of gravity. The collision between the stars is inelastic with coefficient of restitution $1/2$.
 (a) Find the kinetic energy of the system just after the collision.
 (b) Find the maximum separation between their centres after their first collision.

ANSWER KEY**EXERCISE (S-2)**

1. Ans. $2R, 3R[3 - \sqrt{3}]$ 2. Ans. (a) $h = \left(\frac{\sqrt{7}}{2} + 1\right)R$, (b) $1.125 R$
3. Ans. $T = 3 \times 10^{-2} \text{ N}$ 4. Ans. $\left(1 - \frac{\sqrt{x^2 - R^2}}{x}\right) 4\pi R^2$ 5. Ans. (a) $T = \frac{2\pi d^{3/2}}{\sqrt{3Gm}}$, (b) 2, (c) 2

EXERCISE (O-2)

1. Ans. (A,C,D) 2. Ans. (A) 3. Ans. (C) 4. Ans. (A,B,C) 5. Ans. (A,C)
6. Ans. (A,B,D) 7. Ans. (A,D) 8. Ans. (B) 9. Ans. (C)

EXERCISE (JA)

1. Ans. (D) 2. Ans. (A) -p, (B) -q,r (C) -p (D) -q,r 3. Ans. (C) 4. Ans. (A)
5. Ans. 3 6. Ans. 6 7. Ans. (A) 8. Ans. (B) 9. Ans. (B,D) 10. Ans. (B)
11. Ans. 2 12. Ans. 7 13. Ans. (B,C)

REVIEW TEST

1. Ans. (A) 2. Ans. (A) 3. Ans. (B) 4. Ans. (C) 5. Ans. (C)
6. Ans. (A) 7. Ans. (A,C,D) 8. Ans. (a) $\sqrt{\frac{GM}{r}}$; (b) $r\sqrt{2}$; (c) $\sqrt{\frac{2GM}{r}}$
9. Ans. $\frac{1}{4} K^2 G \pi (R^4 - r^4)$ 10. Ans. $7/3$ 11. Ans. 3 12. Ans. (a) $\frac{2Gm^2}{3R}$ (b) $4R$

FLUID MECHANICS

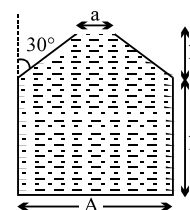
EXERCISE (O-2)

Fluid Statics & Surface Tension

1. A beaker is filled with water and is accelerated $a \text{ m/s}^2$ in $+x$ direction. The surface of water shall make an angle

(A) $\tan^{-1}(a/g)$ backwards (B) $\tan^{-1}(a/g)$ forwards
(C) $\cot^{-1}(g/a)$ backwards (D) $\cot^{-1}(g/a)$ forwards

2. The vessel shown in the figure has two sections. The lower part is a rectangular vessel with area of cross-section A and height h . The upper part is a conical vessel of height h with base area ' A ' and top area ' a ' and the walls of the vessel are inclined at an angle 30° with the vertical. A liquid of density ρ fills both the sections upto a height $2h$. Neglecting atmospheric pressure.



(A) The force F exerted by the liquid on the base of the vessel is $2h\rho g \frac{(A+a)}{2}$

(B) the pressure P at the base of the vessel is $2h\rho g \frac{A}{a}$

(C) the weight of the liquid W is greater than the force exerted by the liquid on the base

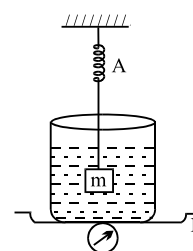
(D) the walls of the vessel exert a downward force $(F-W)$ on the liquid.

3. A cubical block of wood of edge 10cm and mass 0.92kg floats on a tank of water with oil of rel. density 0.6 to a depth of 4cm above water. When the block attains equilibrium with four of its sides edges vertical

(A) 1cm of it will be above the free surface of oil.
(B) 5cm of it will be under water.
(C) 2cm of it will be above the common surface of oil and water.
(D) 8cm of it will be under water.

4. The spring balance A reads 2 kg with a block m suspended from it. A balance B reads 5 kg when a beaker with liquid is put on the pan of the balance. The two balances are now so arranged that the hanging mass is inside the liquid in the beaker as shown in the figure in this situation:

(A) the balance A will read more than 2 kg
(B) the balance B will read more than 5 kg
(C) the balance A will read less than 2 kg and B will read more than 5 kg
(D) the balances A and B will read 2 kg and 5 kg respectively.



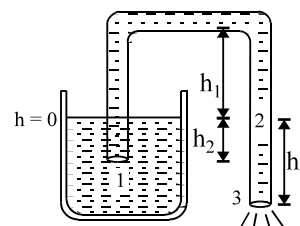
5. When an air bubble rises from the bottom of a deep lake to a point just below the water surface, the pressure of air inside the bubble

(A) is greater than the pressure outside it (B) is less than the pressure outside it
(C) increases as the bubble moves up (D) decreases as the bubble moves up

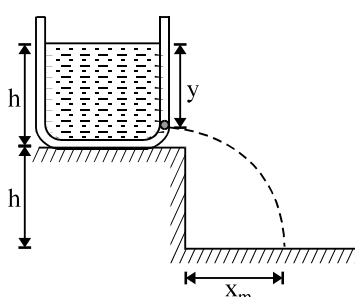
Fluid Dynamics & Viscosity

6. Figure shows a siphon. Choose the wrong statement:

- (A) Siphon works when $h_3 > 0$
 (B) Pressure at point 2 is $P_2 = P_0 - \rho gh_3$
 (C) Pressure at point 3 is P_0
 (D) None of the above
 (P_0 = atmospheric pressure)



7. A tank is filled upto a height h with a liquid and is placed on a platform of height h from the ground. To get maximum range x_m a small hole is punched at a distance of y from the free surface of the liquid. Then



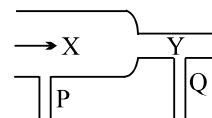
- (A) $x_m = 2h$ (B) $x_m = 1.5h$ (C) $y = h$ (D) $y = 0.75h$

8. Water coming out of a horizontal tube at a speed v strikes normally a vertically wall close to the mouth of the tube and falls down vertically after impact. When the speed of water is increased to $2v$.

- (A) the thrust exerted by the water on the wall will be doubled
 (B) the thrust exerted by the water on the wall will be four times
 (C) the energy lost per second by water striking the wall will also be four times
 (D) the energy lost per second by water striking the wall be increased eight times.

9. A steady flow of water passes along a horizontal tube from a wide section X to the narrower section Y, see figure. Manometers are placed at P and Q at the sections. Which of the statements A, B, C, D, is most correct?

- (A) water velocity at X is greater than at Y
 (B) the manometer at P shows lower pressure than at Q
 (C) kinetic energy per m^3 of water at X = kinetic energy per m^3 at Y
 (D) the manometer at P shows greater pressure than at Q



EXERCISE (J-A)

1. A large open tank has two holes in the wall. One is a square hole of side L at a depth y from the top and the other is a circular hole of radius R at a depth $4y$ from the top. When the tank is completely filled with water, the quantities of water flowing out per second from both holes are the same. Then, R is equal to:

[IIT-JEE 2000 (Scr.)]

- (A) $\frac{L}{\sqrt{2\pi}}$ (B) $2\pi L$ (C) L (D) $\frac{L}{2\pi}$

2. A hemispherical portion of radius R is removed from the bottom of a cylinder of radius R . The volume of the remaining cylinder is V and its mass is M . It is suspended by a string in a liquid of density ρ where it stays vertical. The upper surface of the cylinder is at a depth h below the liquid surface. The force on the bottom of the cylinder by the liquid is

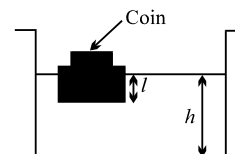
[IIT-JEE 2001 (Scr.)]

- (A) Mg (B) $Mg - V\rho g$ (C) $Mg + \pi R^2 h \rho g$ (D) $\rho g (V + \pi R^2 h)$

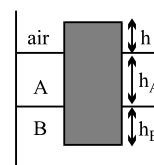
3. A wooden block, with a coin placed on its top, floats in water as shown in figure. The distances l and h are shown there. After some time the coin falls into the water. Then

[IIT-JEE 2002 (Scr.)]

- (A) l decreases and h increases
(B) l increases and h decreases
(C) both l and h increase
(D) both l and h decrease



4. A uniform solid cylinder of density 0.8 gm/cm^3 floats in equilibrium in a combination of two non mixing liquids A and B with its axis vertical. The densities of the liquids A and B are 0.7 gm/cm^3 and 1.2 gm/cm^3 , respectively. The height of liquid A is $h_A = 1.2 \text{ cm}$. The length of the part of the cylinder immersed in liquid B is $h_B = 0.8 \text{ cm}$.



(a) Find the total force exerted by liquid A on the cylinder.

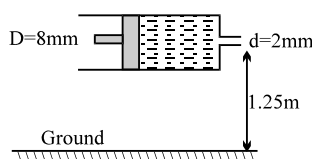
[IIT-JEE 2002]

(b) Find h , the length of the part of the cylinder in air.

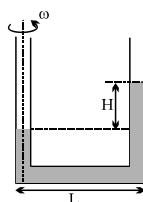
(c) The cylinder is depressed in such a way that its top surface is just below the upper surface of liquid A and is then released. Find the acceleration of the cylinder immediately after it is released.

5. Consider a horizontally oriented syringe containing water located at a height of 1.25 m above the ground. The diameter of the plunger is 8 mm and the diameter of the nozzle is 2 mm . The plunger is pushed with a constant speed of 0.25 m/s . Find the horizontal range of water stream on the ground. Take $g = 10 \text{ m/s}^2$.

[IIT-JEE 2004]

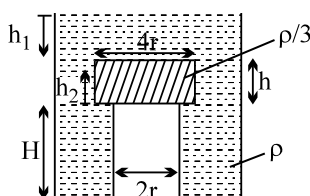


6. A solid sphere of radius R is floating in a liquid of density ρ with half of its volume submerged. If the sphere is slightly pushed and released, it starts performing simple harmonic motion. Find the frequency of these oscillations. [IIT-JEE 2004]
7. Water is filled in a container upto height 3m. A small hole of area 'a' is punched in the wall of the container at a height 52.5 cm from the bottom. The cross sectional area of the container is A . If $\frac{a}{A} = 0.1$ then v^2 is (where v is the velocity of water coming out of the hole) [IIT-JEE' 2005 (Scr)]
- (A) 48 (B) 51 (C) 50 (D) 51.5
8. A U tube is rotated about one of its limbs with an angular velocity ω . Find the difference in height H of the liquid (density ρ) level, where diameter of the tube $d \ll L$. [IIT-JEE 2005]



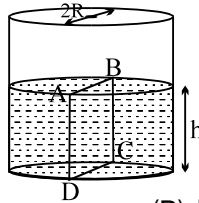
Paragraph for Q. 9 to 11

A wooden cylinder of diameter $4r$, height h and density $\rho/3$ is kept on a hole of diameter $2r$ of a tank, filled with water of density ρ as shown in the figure. The height of the base of cylinder from the base of tank is H . [IIT-JEE 2006]

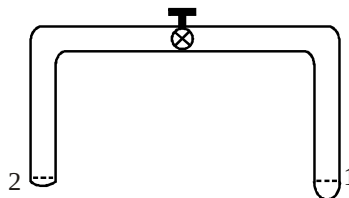


9. If level of liquid starts decreasing slowly when the level of liquid is at a height h_1 above the cylinder, the block just starts moving up. Then, value of h_1 is
- (A) $\frac{2h}{3}$ (B) $\frac{5h}{4}$ (C) $\frac{5h}{3}$ (D) $\frac{5h}{2}$
10. Let the cylinder is prevented from moving up, by applying a force and water level is further decreased. Then, height of water level (h_2 in figure) for which the cylinder remains in original position without application of force is
- (A) $\frac{h}{3}$ (B) $\frac{4h}{9}$ (C) $\frac{2h}{3}$ (D) h
11. If height h_2 of water level is further decreased, then
- (A) cylinder will not move up and remains at its original position.
 (B) for $h_2 = h/3$, cylinder again starts moving up
 (C) for $h_2 = h/4$, cylinder again starts moving up
 (D) for $h_2 = h/5$ cylinder again starts moving up

12. Water is filled up to a height h in a beaker of radius R as shown in the figure. The density of water is ρ , the surface tension of water is T and the atmospheric pressure is P_0 . Consider a vertical section ABCD of the water column through a diameter of the beaker. The force on water on one side of this section by water on the other side of this section has magnitude **[IIT-JEE 2007]**



- (A) $|2P_0Rh + \pi R^2\rho gh - 2RT|$ (B) $|2P_0Rh + R\rho gh^2 - 2RT|$
(C) $|P_0\pi R^2 + R\rho gh^2 - 2RT|$ (D) $|P_0\pi R^2 + R\rho gh^2 + 2RT|$
13. **Statement-1** : The stream of water flowing at high speed from a garden hose pipe tends to spread like a fountain when held vertically up, but tends to narrow down when held vertically down.
and
Statement-2 : In any steady flow of an incompressible fluid, the volume flow rate of the fluid remains constant. **[IIT-JEE 2008]**
- (A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1
(B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True
14. A glass tube of uniform internal radius (r) has a valve separating the two identical ends. Initially, the valve is in a tightly closed position. End 1 has a hemispherical soap bubble of radius r . End 2 has sub-hemispherical soap bubble as shown in figure. Just after opening the valve, **[IIT-JEE 2008]**



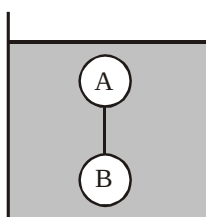
- (A) air from end 1 flows towards end 2. No change in the volume of the soap bubbles
(B) air from end 1 flows towards end 2. Volume of the soap bubble at end 1 decreases
(C) no change occurs
(D) air from end 2 flows towards end 1. Volume of the soap bubble at end 1 increases
15. A cylindrical vessel of height 500 mm has an orifice (small hole) at its bottom. The orifice is initially closed and water is filled in it up to height H . Now the top is completely sealed with a cap and the orifice at the bottom is opened. Some water comes out from the orifice and the water level in the vessel becomes steady with height of water column being 200 mm. Find the fall in height (in mm) of water level due to opening of the orifice. [Take atmospheric pressure = $1.0 \times 10^5 \text{ N/m}^2$, density of water = 1000 kg/m^3 and $g = 10 \text{ m/s}^2$. Neglect any effect of surface tension.] **[IIT-JEE-2009]**
16. Two soap bubbles A and B are kept in a closed chamber where the air is maintained at pressure 8 N/m^2 . The radii of bubbles A and B are 2 cm and 4 cm, respectively. Surface tension of the soap-water used to make bubbles is 0.04 N/m . Find the ratio n_B/n_A , where n_A and n_B are the number of moles of air in bubbles A and B, respectively. [Neglect the effect of gravity] **[IIT-JEE-2009]**

Paragraph for Q. 17 to 19

When liquid medicine of density ρ is to be put in the eye, it is done with the help of a dropper. As the bulb on the top of the dropper is pressed, a drop forms at the opening of the dropper. We wish to estimate the size of the drop.

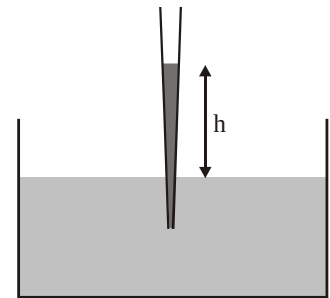
We first assume that the drop formed at the opening is spherical because that requires a minimum increase in its surface energy. To determine the size, we calculate the net vertical force due to the surface tension T when the radius of the drop is R . When this force becomes smaller than the weight of the drop, the drop gets detached from the dropper. **[IIT-JEE-2010]**

- 17.** If the radius of the opening of the dropper is r , the vertical force due to the surface tension on the drop of radius R (assuming $r \ll R$) is
- (A) $2\pi rT$ (B) $2\pi RT$ (C) $\frac{2\pi r^2 T}{R}$ (D) $\frac{2\pi R^2 T}{r}$
- 18.** If $r = 5 \times 10^{-4}$ m, $\rho = 10^3$ kgm $^{-3}$, $g = 10$ ms $^{-2}$, $T = 0.11$ Nm $^{-1}$, the radius of the drop when it detaches from the dropper is approximately
- (A) 1.4×10^{-3} m (B) 3.3×10^{-3} m (C) 2.0×10^{-3} m (D) 4.1×10^{-3} m
- 19.** After the drop detaches, its surface energy is
- (A) 1.4×10^{-6} J (B) 2.7×10^{-6} J (C) 5.4×10^{-6} J (D) 8.1×10^{-6} J
- 20.** Two solid spheres A and B of equal volumes but of different densities d_A and d_B are connected by a string. They are fully immersed in a fluid of density d_F . They get arranged into an equilibrium state as shown in the figure with a tension in the string. The arrangement is possible only if **[IIT-JEE-2011]**



- (A) $d_A < d_F$ (B) $d_B > d_F$ (C) $d_A > d_F$ (D) $d_A + d_B = 2d_F$
- 21.** A thin uniform cylindrical shell, closed at both ends, is partially filled with water. It is floating vertically in water in half-submerged state. If ρ_c is the relative density of the material of the shell with respect to water, then the correct statement is that the shell is **[IIT-JEE-2012]**
- (A) more than half-filled if ρ_c is less than 0.5 (B) more than half-filled if ρ_c is more than 1.0
- (C) half-filled if ρ_c is more than 0.5 (D) less than half-filled if ρ_c is less than 0.5
- 22.** A solid sphere of radius R and density ρ is attached to one end of a mass-less spring of force constant k . The other end of the spring is connected to another solid sphere of radius R and density 3ρ . The complete arrangement is placed in a liquid of density 2ρ and is allowed to reach equilibrium. The correct statement(s) is (are) **[JEE Advanced-2013]**
- (A) the net elongation of the spring is $\frac{4\pi R^3 \rho g}{3k}$
- (B) the net elongation of the spring is $\frac{8\pi R^3 \rho g}{3k}$
- (C) the light sphere is partially submerged.
- (D) the light sphere is completely submerged.

23. A glass capillary tube is of the shape of truncated cone with an apex angle α so that its two ends have cross sections of different radii. When dipped in water vertically, water rises in it to height h , where the radius of its cross section is b . If the surface tension of water is S , its density is ρ , and its contact angle with glass is θ , the value of h will be (g is the acceleration due to gravity)



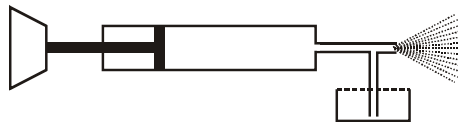
[JEE Advanced-2014]

- (A) $\frac{2S}{b\rho g} \cos(\theta - \alpha)$ (B) $\frac{2S}{b\rho g} \cos(\theta + \alpha)$ (C) $\frac{2S}{b\rho g} \cos(\theta - \alpha/2)$ (D) $\frac{2S}{b\rho g} \cos(\theta + \alpha/2)$

Paragraph for Questions 24 & 25

A spray gun is shown in the figure where a piston pushes air out of a nozzle. A thin tube of uniform cross section is connected to the nozzle. The other end of the tube is in a small liquid container. As the piston pushes air through the nozzle, the liquid from the container rises into the nozzle and is sprayed out. For the spray gun shown, the radii of the piston and the nozzle are 20 mm and 1 mm respectively. The upper end of the container is open to the atmosphere.

[JEE Advanced-2014]



24. If the piston is pushed at a speed of 5 mms^{-1} , the air comes out of the nozzle with a speed of
(A) 0.1 ms^{-1} (B) 1 ms^{-1} (C) 2 ms^{-1} (D) 8 ms^{-1}
25. If the density of air is ρ_a and that of the liquid ρ_ℓ , then for a given piston speed the rate (volume per unit time) at which the liquid is sprayed will be proportional to
- (A) $\sqrt{\frac{\rho_a}{\rho_\ell}}$ (B) $\sqrt{\rho_a \rho_\ell}$ (C) $\sqrt{\frac{\rho_\ell}{\rho_a}}$ (D) ρ_ℓ
26. A person in a lift is holding a water jar, which has a small hole at the lower end of its side. When the lift is at rest, the water jet coming out of the hole hits the floor of the lift at a distance d of 1.2 m from the person. In the following, state of the lift's motion is given in List I and the distance where the water jet hits the floor of the lift is given in List II. Match the statements from List I with those in List II and select the correct answer using the code given below the lists.

List - I

- (P) Lift is accelerating vertically up.
(Q) Lift is accelerating vertically down with an acceleration less than the gravitational acceleration.
(R) Lift is moving vertically up with constant speed.
(S) Lift is falling freely.

List-II

- (1) $d = 1.2 \text{ m}$
(2) $d > 1.2 \text{ m}$
(3) $d < 1.2 \text{ m}$
(4) No water leaks out of the jar

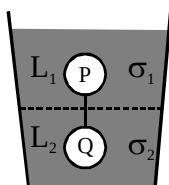
[JEE Advanced-2014]

Code :

- (A) P-2, Q-3, R-2, S-4 (B) P-2, Q-3, R-1, S-4
(C) P-1, Q-1, R-1, S-4 (D) P-2, Q-3, R-1, S-1

27. Two spheres P and Q of equal radii have densities ρ_1 and ρ_2 , respectively. The spheres are connected by a massless string and placed in liquids L_1 and L_2 of densities σ_1 and σ_2 and viscosities η_1 and η_2 , respectively. They float in equilibrium with the sphere P in L_1 and sphere Q in L_2 and the string being taut (see figure). If sphere P alone in L_2 has terminal velocity $\vec{V}_P$ and Q alone in L_1 has terminal velocity $\vec{V}_Q$, then

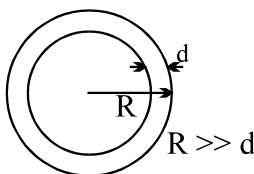
[JEE Advanced-2015]



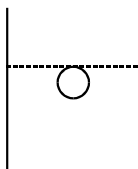
- (A) $\frac{|\vec{V}_P|}{|\vec{V}_Q|} = \frac{\eta_1}{\eta_2}$ (B) $\frac{|\vec{V}_P|}{|\vec{V}_Q|} = \frac{\eta_2}{\eta_1}$ (C) $\vec{V}_P \cdot \vec{V}_Q > 0$ (D) $\vec{V}_P \cdot \vec{V}_Q < 0$

Review Test

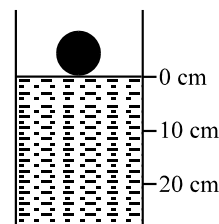
1. A test tube of thin walls has some lead shots in it at its bottom and the system floats vertically in water, sinking by a length $l_0 = 10 \text{ cm}$. A liquid of density less than that of water, is poured into the tube till the levels inside and outside the tube are even. If the tube now sinks to a length $l_0 = 40 \text{ cm}$, the specific gravity of the liquid is _____.
2. A vertical uniform U tube open at both ends contains mercury. Water is poured in one limb until the level of mercury is depressed 2 cm in that limb. What is the length of water column when this happens.
3. An expansible balloon filled with air floats on the surface of a lake with $2/3$ of its volume submerged. How deep must it be sunk in the water so that it is just in equilibrium neither sinking further nor rising? It is assumed that the temperature of the water is constant & that the height of the water barometer is 9 meters.
4. A soap bubble has radius R and thickness d ($d \ll R$) as shown. It collapses into a spherical drop. Find the ratio of excess pressure in the drop to the excess pressure inside the bubble.



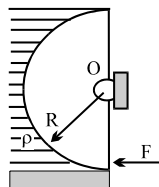
5. A ball is given velocity v_0 (greater than the terminal velocity v_T) in downward direction inside a highly viscous liquid placed inside a large container. The height of liquid in the container is H . The ball attains the terminal velocity just before striking at the bottom of the container. Draw graph between velocity of the ball and distance moved by the ball before getting terminal velocity.



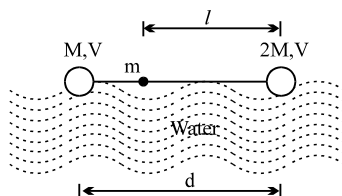
6. A spherical ball of radius $1 \times 10^{-4} \text{ m}$ and density 10^4 kg/m^3 falls freely under gravity through a distance h before entering a tank of water. If after entering the water the velocity of the ball does not change, find h . The viscosity of water is $9.8 \times 10^{-6} \text{ N-s/m}^2$.
7. A spherical ball of density ρ and radius 0.003 m is dropped into a tube containing a viscous fluid filled up to the 0 cm mark as shown in the figure. Viscosity of the fluid = 1.260 N.m^{-2} and its density $\rho_L = \rho/2 = 1260 \text{ kg.m}^{-3}$. Assume the ball reaches a terminal speed by the 10 cm mark. Find the time taken by the ball to traverse the distance between the 10 cm and 20 cm mark.
($g = \text{acceleration due to gravity} = 10 \text{ ms}^{-2}$)



8. A light semi cylindrical gate of radius R is pivoted at its mid point O , of the diameter as shown in the figure holding liquid of density ρ . The force F required to prevent the rotation of the gate is equal to

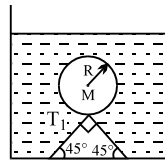


- (A) $2\pi R^3 \rho g$ (B) $2\rho g R^3 l$ (C) $\frac{2R^2 l \rho g}{3}$ (D) none of these
9. A heavy hollow cone of radius R and height h is placed on a horizontal table surface, with its flat base on the table. The whole volume inside the cone is filled with water of density ρ . The circular rim of the cone's base has a watertight seal with the table's surface and the top apex of the cone has a small hole. Neglecting atmospheric pressure find the total upward force exerted by water on the cone is
- (A) $(2/3)\pi R^2 h \rho g$ (B) $(1/3)\pi R^2 h \rho g$ (C) $\pi R^2 h \rho g$ (D) None
10. A dumbbell is placed in water of density ρ . It is observed that by attaching a mass m to the rod, the dumbbell floats with the rod horizontal on the surface of water and each sphere exactly half submerged as shown in the figure. The volume of the mass m is negligible. The value of length l is

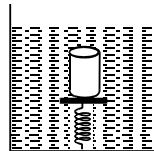


- (A) $\frac{d(V_p - 3M)}{2(V_p - 2M)}$ (B) $\frac{d(V_p - 2M)}{2(V_p - 3M)}$ (C) $\frac{d(V_p + 2M)}{2(V_p - 3M)}$ (D) $\frac{d(V_p - 2M)}{2(V_p + 3M)}$
11. Two cylinders of same cross-section and length L but made of two material of densities d_1 and d_2 are connected together to form a cylinder of length $2L$. The combination floats in a liquid of density d with a length $L/2$ above the surface of the liquid. If $d_1 > d_2$ then:
- (A) $d_1 > \frac{3}{4}d$ (B) $\frac{d}{2} > d_1$ (C) $\frac{d}{4} > d_1$ (D) $d < d_1$
12. A small ball of relative density 0.8 falls into water from a height of 2m. The depth to which the ball will sink is (neglect viscous forces):
- (A) 8 m (B) 2 m (C) 6 m (D) 4 m
13. A small wooden ball of density ρ is immersed in water of density σ to depth h and then released. The height H above the surface of water up to which the ball will jump out of water is
- (A) $\frac{\sigma h}{\rho}$ (B) $\left(\frac{\sigma}{\rho} - 1\right)h$ (C) h (D) zero

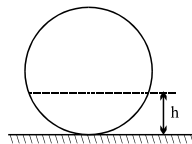
14. A hollow sphere of mass M and radius r is immersed in a tank of water (density ρ_w). The sphere would float if it were set free. The sphere is tied to the bottom of the tank by two wires which makes angle 45° with the horizontal as shown in the figure. The tension T_1 in the wire is :



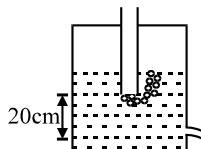
- (A) $\frac{\frac{4}{3}\pi R^3 \rho_w g - Mg}{\sqrt{2}}$ (B) $\frac{2}{3}\pi R^3 \rho_w g - Mg$ (C) $\frac{\frac{4}{3}\pi R^3 \rho_w g - Mg}{2}$ (D) $\frac{4}{3}\pi R^3 \rho_w g + Mg$
15. A cylindrical block of area of cross-section A and of material of density ρ is placed in a liquid of density one-third of density of block. The block compresses a spring and compression in the spring is one-third of the length of the block. If acceleration due to gravity is g , the spring constant of the spring is:



- (A) ρAg (B) $2\rho Ag$ (C) $2\rho Ag/3$ (D) $\rho Ag/3$
16. A liquid is filled in a spherical container of radius R till a height h . At this position the liquid surface at the edges is also horizontal. The contact angle is



- (A) 0 (B) $\cos^{-1}\left(\frac{R-h}{R}\right)$ (C) $\cos^{-1}\left(\frac{h-R}{R}\right)$ (D) $\sin^{-1}\left(\frac{R-h}{R}\right)$
17. A tube is attached as shown in closed vessel containing water. The velocity of water coming out from a small hole is :



- (A) $\sqrt{2}$ m/s (B) 2 m/s
(C) depends on pressure of air inside vessel (D) None of these
18. Fountains usually seen in gardens are generated by a wide pipe with an enclosure at one end having many small holes. Consider one such fountain which is produced by a pipe of internal diameter 2 cm in which water flows at a rate 3 ms^{-1} . The enclosure has 100 holes each of diameter 0.05 cm. The velocity of water coming out of the holes is (in ms^{-1}):
- (A) 0.48 (B) 96 (C) 24 (D) 48

19. A cylindrical vessel filled with water upto height of H stands on a horizontal plane. The side wall of the vessel has a plugged circular hole touching the bottom. The coefficient of friction between the bottom of vessel and plane is μ and total mass of water plus vessel is M . What should be minimum diameter of hole so that the vessel begins to move on the floor if plug is removed (here density of water is ρ)

(A) $\sqrt{\frac{2\mu M}{\pi\rho H}}$ (B) $\sqrt{\frac{\mu M}{2\pi\rho H}}$ (C) $\sqrt{\frac{\mu M}{\rho H}}$ (D) none

20. A Newtonian fluid fills the clearance between a shaft and a sleeve. When a force of 800N is applied to the shaft, parallel to the sleeve, the shaft attains a speed of 1.5 cm/sec. If a force of 2.4 kN is applied instead, the shaft would move with a speed of

(A) 1.5 cm/sec (B) 13.5 cm/sec (C) 4.5 cm/sec (D) None

ANSWER KEY

EXERCISE (O-2)

1. A,C 2. D 3. C,D 4. B,C 5. A,D 6. D 7. A,C 8. B,D 9. D

EXERCISE (J-A)

1. A 2. D 3. D 4. (a) 0, (b) $h = 0.25$ cm, (c) $a = g/6$ (upward) 5. $x = 2$ m

6. $f = \frac{1}{2\pi} \sqrt{\frac{3g}{2R}}$ 7. C 8. $H = \frac{L^2 \omega^2}{2g}$ 9. C 10. B 11. A 12. B 13. A 14. B 15. 6

16. 6 17. C 18. A 19. B 20. ABD 21. A 22. A, D 23. D

24. C 25. A 26. C 27. A, D

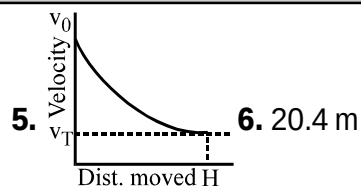
REVIEW TEST

1. (0.75)

2. 54.4 cm

3. 4.5m

4. $\left(\frac{R}{24d}\right)^{\frac{1}{3}}$



7. 0.5 s

8. (D)

9. (A)

10. (B)

11. (A)

12. (A)

13. (B)

14. (A)

15. (B)

16. (B)

17. (B)

18. (D)

19. (A)

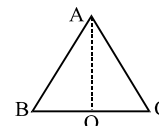
20. (C)

[illegible]

ELASTICITY, THERMAL EXPANSION, CALORIMETRY & HEAT TRANSFER

EXERCISE (S-2)

1. Three aluminium rods of equal length form an equilateral triangle ABC. Taking O (mid point of rod BC) as the origin. Find the increase in Y-coordinate of center of mass per unit change in temperature of the system. Assume the length of the each rod is 2m, and $\alpha_{al} = 4\sqrt{3} \times 10^{-6} / ^\circ\text{C}$



2. A metal rod A of 25cm lengths expands by 0.050 cm, when its temperature is raised from 0°C to 100°C . Another rod B of a different metal of length 40cm expands by 0.040 cm for the same rise in temperature. A third rod C of 50cm length is made up of pieces of rods A and B placed end to end expands by 0.03 cm on heating from 0°C to 50°C . Find the lengths of each portion of the composite rod.
3. A wire of cross-sectional area $4 \times 10^{-4} \text{ m}^2$, modulus of elasticity $2 \times 10^{11} \text{ N/m}^2$ and length 1 m is stretched between two vertical rigid poles. A mass of 1 kg is suspended at its middle. Calculate the angle it makes with the horizontal.
4. A copper calorimeter of mass 100 gm contains 200 gm of a mixture of ice and water. Steam at 100°C under normal pressure is passed into the calorimeter and the temperature of the mixture is allowed to rise to 50°C . If the mass of the calorimeter and its contents is now 330 gm, what was the ratio of ice and water in the beginning? Neglect heat losses.

Given : Specific heat capacity of copper = $0.42 \times 10^3 \text{ J kg}^{-1}\text{K}^{-1}$,

Specific heat capacity of water = $4.2 \times 10^3 \text{ J kg}^{-1}\text{K}^{-1}$,

Specific heat of fusion of ice = $3.36 \times 10^5 \text{ J kg}^{-1}$

Latent heat of condensation of steam = $22.5 \times 10^5 \text{ J kg}^{-1}$

5. An isosceles triangle is formed with a rod of length l_1 and coefficient of linear expansion α_1 for the base and two thin rods each of length l_2 and coefficient of linear expansion α_2 for the two pieces, if the distance between the apex and the midpoint of the base remain unchanged as the temperatures varied show that

$$\frac{l_1}{l_2} = 2 \sqrt{\frac{\alpha_2}{\alpha_1}}.$$

6. A steel drill making 180 rpm is used to drill a hole in a block of steel. The mass of the steel block and the drill is 180 gm. If the entire mechanical work is used up in producing heat and the rate of raise in temperature of the block and the drill is 0.5°C/s . Find

(a) the rate of working of the drill in watts, and (b) the torque required to drive the drill.

Specific heat of steel = 0.1 and $J = 4.2 \text{ J/cal}$. Use : $P = \tau \omega$

7. A liquid takes 5 minutes to cool from 80°C to 50°C . How much time will it take to cool from 60°C to 30°C ? The temperature of surrounding is 20°C . Use exact method.

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

- A cylindrical wire of radius 1 mm, length 1 m, Young's modulus = 2×10^{11} N/m², poisson's ratio $\mu = \pi/10$ is stretched by a force of 100 N. Its radius will become

(A) 0.99998 mm (B) 0.99999 mm (C) 0.99997 mm (D) 0.99995 mm
- The coefficient of linear expansion of copper is $17 \times 10^{-6} (^{\circ}\text{C})^{-1}$. A copper statue is 93 m tall on the summer morning of temperature 25°C . What is maximum order of increase in magnitude of the height in statue (maximum temperature of day is 45°C)

(A) 0.1 mm (B) 1 mm (C) 10 mm (D) 100 mm
- The coefficients of thermal expansion of steel and a metal X are respectively 12×10^{-6} and 2×10^{-6} per $^{\circ}\text{C}$. At 40°C , the side of a cube of metal X was measured using a steel vernier callipers. The reading was 100 mm. Assuming that the calibration of the vernier was done at 0°C , then the actual length of the side of the cube at 0°C will be

(A) > 100 mm (B) < 100 mm
 (C) $= 100$ mm (D) data insufficient to conclude
- The volume of the bulb of a mercury thermometer at 0°C is V_0 and cross section of the capillary is A_0 . The coefficient of linear expansion of glass is α_g per $^{\circ}\text{C}$ and the cubical expansion of mercury γ_m per $^{\circ}\text{C}$. If the mercury just fills the bulb at 0°C , what is the length of mercury column in capillary at $T^{\circ}\text{C}$.

(A) $\frac{V_0 T(\gamma_m + 3\alpha_g)}{A_0(1 + 2\alpha_g T)}$ (B) $\frac{V_0 T(\gamma_m - 3\alpha_g)}{A_0(1 + 2\alpha_g T)}$ (C) $\frac{V_0 T(\gamma_m + 2\alpha_g)}{A_0(1 + 3\alpha_g T)}$ (D) $\frac{V_0 T(\gamma_m - 2\alpha_g)}{A_0(1 + 3\alpha_g T)}$
- Four rods of same material with different radii r and length ℓ are used to connect two reservoirs of heat at different temperatures. Which one will conduct most heat ?

(A) $r = 2\text{cm}, \ell = 0.5\text{m}$ (B) $r = 2\text{cm}, \ell = 2\text{m}$ (C) $r = 0.5\text{cm}, \ell = 0.5\text{m}$ (D) $r = 1\text{cm}, \ell = 1\text{m}$
- An experiment is performed to measure the specific heat of copper. A lump of copper is heated in an oven, then dropped into a beaker of water. To calculate the specific heat of copper, the experimenter must know or measure the value of all of the quantities below EXCEPT the

(A) heat capacity of water and beaker
 (B) original temperature of the copper and the water
 (C) final (equilibrium) temperature of the copper and the water
 (D) time taken to achieve equilibrium, after the copper is dropped into the water

7. One end of a conducting rod is maintained at temperature 50°C and at the other end, ice is melting at 0°C . The rate of melting of ice is doubled if:
- (A) the temperature is made 200°C and the area of cross-section of the rod is doubled
 (B) the temperature is made 100°C and length of rod is made four times
 (C) area of cross-section of rod is halved and length is doubled
 (D) the temperature is made 100°C and the area of cross-section of rod and length both are doubled.

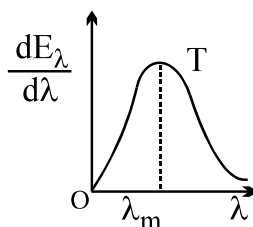
MULTIPLE CORRECT TYPE QUESTIONS

8. A bimetallic strip is formed out of two identical strips one of copper and the other of brass. The coefficient of linear expansion of the two metals are α_C and α_B . On heating, the temperature of the strip goes up by ΔT and the strip bends to form an arc of radius of curvature R . Then R is
- (A) proportional to ΔT (B) inversely proportional to ΔT
 (C) proportional to $|\alpha_B - \alpha_C|$ (D) inversely proportional to $|\alpha_B - \alpha_C|$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 9 and 10.

The figure shows a radiant energy spectrum graph for a black body at a temperature T .



9. Choose the **CORRECT** statement(s)
- (A) The radiant energy is not equally distributed among all the possible wavelengths
 (B) For a particular wavelength the spectral intensity is maximum
 (C) The area under the curve is equal to the total rate at which heat is radiated by the body at that temperature
 (D) None of these
10. If the temperature of the body is raised to a higher temperature T' , then choose the correct statement(s)
- (A) The intensity of radiation for every wavelength increases
 (B) The maximum intensity occurs at a shorter wavelength
 (C) The area under the graph increases
 (D) The area under the graph is proportional to the fourth power of temperature

EXERCISE (JA)

ALL ANSWERS OF THE QUESTIONS JEE-2007 ONWARDS ARE AS GIVEN ON IIT-JEE WEBSITE.

1. 2 litre water at 27°C is heated by a 1 kW heater in an open container. On an average heat is lost to surroundings at the rate 160 J/s. The time required for the temperature to reach 77°C is

[JEE' 2005 (Scr)]

- (A) 8 min 20 sec (B) 10 min (C) 7 min (D) 14 min

2. 1 calorie is the heat required to increased the temperature of 1 gm of water by 1°C from

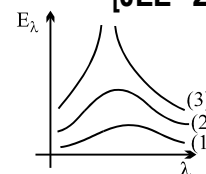
[JEE' 2005 (Scr)]

- (A) 13.5°C to 14.5°C at 76 mm of Hg (B) 14.5°C to 15.5°C at 760 mm of Hg
 (C) 0°C to 1°C at 760 mm of Hg (D) 3°C to 4°C to 760 mm of Hg

3. Three graphs marked as 1, 2, 3 representing the variation of maximum emissive power and wavelength of radiation of the sun, a welding arc and a tungsten filament. Which of the following combination is correct

[JEE' 2005 (Scr)]

- (A) 1-bulb, 2 $\rightarrow$ welding arc, 3 $\rightarrow$ sun
 (B) 2-bulb, 3 $\rightarrow$ welding arc, 1 $\rightarrow$ sun
 (C) 3-bulb, 1 $\rightarrow$ welding arc, 2 $\rightarrow$ sun
 (D) 2-bulb, 1 $\rightarrow$ welding arc, 3 $\rightarrow$ sun



4. In which of the following phenomenon heat convection does not take place

[JEE' 2005 (Scr)]

- (A) land and sea breeze
 (B) boiling of water
 (C) heating of glass surface due to filament of the bulb
 (D) air around the furnace

5. A spherical body of area A , and emissivity $e = 0.6$ is kept inside a black body. What is the rate at which energy is radiated per second at temperature T

[JEE' 2005 (Scr)]

- (A) $0.6 \sigma AT^4$ (B) $0.4 \sigma AT^4$ (C) $0.8 \sigma AT^4$ (D) $1.0 \sigma AT^4$

6. In an insulated vessel, 0.05 kg steam at 373 K and 0.45 kg of ice at 253 K are mixed. Then, find the final temperature of the mixture.

Given, $L_{\text{fusion}} = 80 \text{ cal/g} = 336 \text{ J/g}$, $L_{\text{vaporization}} = 540 \text{ cal/g} = 2268 \text{ J/g}$,
 $S_{\text{ice}} = 2100 \text{ J/kg K} = 0.5 \text{ cal/gK}$ and $S_{\text{water}} = 4200 \text{ J/kg K} = 1 \text{ cal/gK}$

[JEE 2006]

7. In a dark room with ambient temperature T_0 , a black body is kept at a temperature T . Keeping the temperature of the black body constant (at T), Sun rays are allowed to fall on the black body through a hole in the roof of the dark room. Assuming that there is no change in the ambient temperature of the room, which of the following statement(s) is/are correct?

[JEE 2006]

- (A) The quantity of radiation absorbed by the black body in unit time will increase.
 (B) Since emissivity = absorptivity, hence the quantity of radiation emitted by black body in unit time will increase.
 (C) Black body radiates more energy in unit time in the visible spectrum.
 (D) The reflected energy in unit time by the black body remains same.

8. Column I gives some devices and column II gives some processes on which the functioning of these devices depend. Match the devices in column-I with the processes in column-II and indicate your answer by darkening appropriate bubbles in the 4×4 matrix given in the ORS. [JEE 2007]

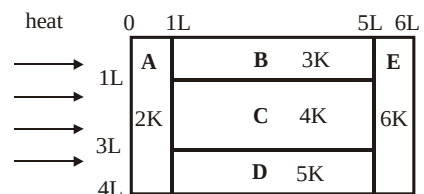
Column-I

- (A) Bimetallic strip
(B) Steam engine
(C) Incandescent lamp
(D) Electric fuse

Column-II

- (P) Radiation from a hot body
(Q) Energy conversion
(R) Melting
(S) Thermal expansion of solids.

9. A metal rod AB of length $10x$ has its one end A in ice at 0°C and the other end B in water at 100°C . If a point P on the rod is maintained at 40°C , then it is found that equal amounts of water and ice evaporate and melt per unit time. The latent heat of evaporation of water is 540 cal/g and latent heat of melting of ice is 80 cal/g . If the point P is at a distance of λx from the ice end A, find the value of λ . [Neglect any heat loss to the surrounding] [JEE 2009]
10. Two spherical bodies A (radius 6 cm) and B (radius 18 cm) are at temperatures T_1 and T_2 , respectively. The maximum intensity in the emission spectrum of A is at 500 nm and in that of B is at 1500 nm. Considering them to be black bodies, what will be the ratio of the rate of total energy radiated by A to that of B? [JEE 2010]
11. A piece of ice (heat capacity = $2100 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$ and latent heat = $3.36 \times 10^5 \text{ J kg}^{-1}$) of mass m grams is at -5°C at atmospheric pressure. It is given 420 J of heat so that the ice starts melting. Finally when the ice–water mixture is in equilibrium, it is found that 1 gm of ice has melted. Assuming there is no other heat exchange in the process, the value of m is [JEE 2010]
12. A 0.1 kg mass is suspended from a wire of negligible mass. The length of the wire is 1m and its cross-sectional area is $4.9 \times 10^{-7} \text{ m}^2$. If the mass is pulled a little in the vertically downward direction and released, it performs simple harmonic motion of angular frequency 140 rad s^{-1} . If the Young's modulus of the material of the wire is $n \times 10^9 \text{ Nm}^{-2}$, the value of n is [IIT-JEE 2010]
13. A composite block is made of slabs A, B, C, D and E of different thermal conductivities (given in terms of a constant K) and sizes (given in terms of length, L) as shown in the figure. All slabs are of same width. Heat 'Q' flows only from left to right through the blocks. Then in steady state [JEE 2011]
- (A) heat flow through A and E slabs are same
(B) heat flow through slab E is maximum
(C) temperature difference across slab E is smallest
(D) heat flow through C = heat flow through B + Heat flow through D
14. Steel wire of length 'L' at 40°C is suspended from the ceiling and then a mass 'm' is hung from its free end. The wire is cooled down from 40° to 30°C to regain its original length 'L'. The coefficient of linear thermal expansion of the steel is $10^{-5}/^\circ\text{C}$, Young's modulus of steel is 10^{11} N/m^2 and radius of the wire is 1 mm. Assume that $L \gg$ diameter of the wire. Then the value of 'm' in kg is nearly



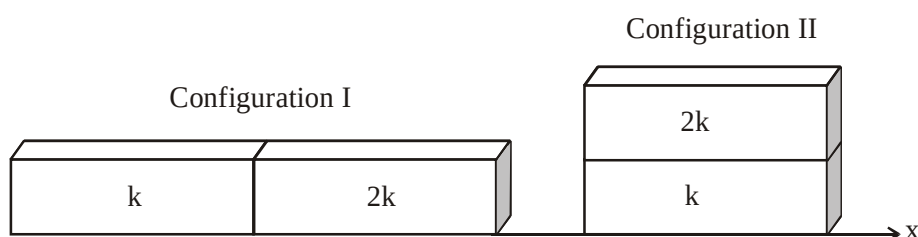
[JEE 2011]

15. Three very large plates of same area are kept parallel and close to each other. They are considered as ideal black surfaces and have very high thermal conductivity. The first and third plates are maintained at temperatures $2T$ and $3T$ respectively. The temperature of the middle (i.e. second) plate under steady state condition is

[JEE 2012]

(A) $\left(\frac{65}{2}\right)^{1/4} T$ (B) $\left(\frac{97}{4}\right)^{1/4} T$ (C) $\left(\frac{97}{2}\right)^{1/4} T$ (D) $(97)^{1/4} T$

16. Two rectangular blocks, having identical dimensions, can be arranged either in configuration I or in configuration II as shown in the figure. One of the blocks has thermal conductivity k and the other $2k$. The temperature difference between the ends along the x -axis is the same in both the configurations. It takes $9s$ to transport a certain amount of heat from the hot end to the cold end in the configuration I. The time to transport the same amount of heat in the configuration II is :- [JEE-Advance-2013]



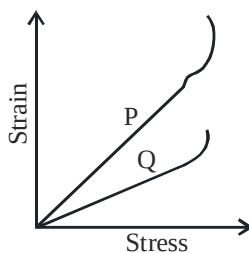
- (A) $2.0 s$ (B) $3.0 s$ (C) $4.5 s$ (D) $6.0 s$
17. One end of a horizontal thick copper wire of length $2L$ and radius $2R$ is welded to an end of another horizontal thin copper wire of length L and radius R . When the arrangement is stretched by applying forces at two ends, the ratio of the elongation in the thin wire to that in the thick wire is :- [JEE-Advance-2013]
- (A) 0.25 (B) 0.50 (C) 2.00 (D) 4.00
18. Parallel rays of light of intensity $I = 912 \text{ Wm}^{-2}$ are incident on a spherical black body kept in surroundings of temperature 300 K . Take Stefan-Boltzmann constant $\sigma = 5.7 \times 10^{-8} \text{ Wm}^{-2} \text{ K}^{-4}$ and assume that the energy exchange with the surroundings is only through radiation. The final steady state temperature of the black body is close to :- [JEE-Advance-2014]
- (A) 330 K (B) 660 K (C) 990 K (D) 1550 K
19. Two spherical stars A and B emit blackbody radiation. The radius of A is 400 times that of B and A

emits 10^4 times the power emitted from B. The ratio $\left(\frac{\lambda_A}{\lambda_B}\right)$ of their wavelengths λ_A and λ_B at which the

peaks occur in their respective radiation curves is.

20. In plotting stress versus strain curves for two materials P and Q, a student by mistake puts strain on the y-axis and stress on the x-axis as shown in the figure. Then the correct statement(s) is (are):-

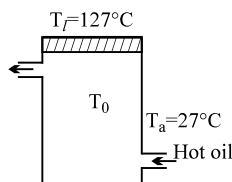
[JEE-Advance-2015]



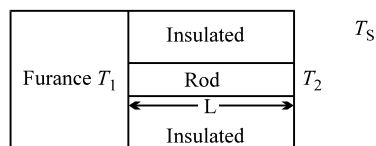
- (A) P has more tensile strength than Q
- (B) P is more ductile than Q
- (C) P is more brittle than Q
- (D) The Young's modulus of P is more than that of Q

Review Test

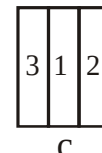
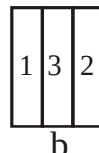
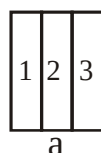
1. A flow calorimeter is used to measure the specific heat of a liquid. Heat is added at a known rate to a stream of the liquid as it passes through the calorimeter at a known rate. Then a measurement of the resulting temperature difference between the inflow and the outflow points of the liquid stream enables us to compute the specific heat of the liquid. A liquid of density 0.2 g/cm^3 flows through a calorimeter at the rate of $10 \text{ cm}^3/\text{s}$. Heat is added by means of a 250-W electric heating coil, and a temperature difference of 25°C is established in steady-state conditions between the inflow and the outflow points. Find the specific heat of the liquid.
2. A solid receives heat by radiation over its surface at the rate of 4 kW . The heat convection rate from the surface of solid to the surrounding is 5.2 kW , and heat is generated at a rate of 1.7 kW over the volume of the solid. The rate of change of the average temperature of the solid is 0.5°C s^{-1} . Find the heat capacity of the solid.
3. Ice at -20°C is filled upto height $h = 10 \text{ cm}$ in a uniform cylindrical vessel. Water at temperature $\theta^\circ\text{C}$ is filled in another identical vessel upto the same height $h = 10 \text{ cm}$. Now, water from second vessel is poured into first vessel and it is found that level of upper surface falls through $\Delta h = 0.5 \text{ cm}$ when thermal equilibrium is reached. Neglecting thermal capacity of vessels, change in density of water due to change in temperature and loss of heat due to radiation, calculate initial temperature θ of water. Given,
 Density of water: $\rho_w = 1 \text{ gm cm}^{-3}$ Density of ice: $\rho_i = 0.9 \text{ gm/cm}^3$
 Specific heat of water: $s_w = 1 \text{ cal/gm } ^\circ\text{C}$ Specific heat of ice: $s_i = 0.5 \text{ cal/gm } ^\circ\text{C}$
 Specific latent heat of ice: $L = 80 \text{ cal/gm}$
4. A highly conducting solid cylinder of radius a and length l is surrounded by a co-axial layer of a material having thermal conductivity K and negligible heat capacity. Temperature of surrounding space (out side the layer) is T_0 , which is higher than temperature of the cylinder. If heat capacity per unit volume of cylinder material is s and outer radius of the layer is b , calculate time required to increase temperature of the cylinder from T_1 to T_2 . Assume end faces to be thermally insulated.
5. Hot oil is circulated through an insulated container with a wooden lid at the top whose conductivity $K = 0.149 \text{ J/(m}^\circ\text{C-sec)}$, thickness $t = 5 \text{ mm}$, emissivity $= 0.6$. Temperature of the top of the lid in steady state is at $T_l = 127^\circ$. If the ambient temperature $T_a = 27^\circ\text{C}$. Calculate **[JEE' 2003]**
 (a) rate of heat loss per unit area due to radiation from the lid.
 (b) temperature of the oil. (Given $\sigma = \frac{17}{3} \times 10^{-8}$)



6. One end of a rod of length L and cross-sectional area A is kept in a furnace of temperature T_1 . The other end of the rod is kept at a temperature T_2 . The thermal conductivity of the material of the rod is K and emissivity of the rod is e . It is given that $T_2 = T_s + \Delta T$ where $\Delta T \ll T_s$, T_s being the temperature of the surroundings. If $\Delta T \propto (T_1 - T_s)$, find the proportionality constant. Consider that heat is lost only by radiation at the end where the temperature of the rod is T_2 . **[JEE 2004]**

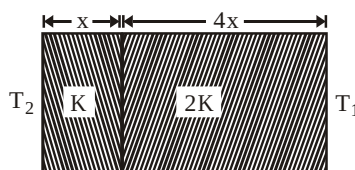


7. Steel ruptures when a shear of $3.5 \times 10^8 \text{ Nm}^{-2}$ is applied. The force needed to punch a 1 cm diameter hole in a steel sheet 0.3 cm thick is nearly :- **[JEE-MAINS ONLINE]**
 (A) $3.3 \times 10^4 \text{ N}$ (B) $1.4 \times 10^4 \text{ N}$ (C) $1.1 \times 10^4 \text{ N}$ (D) $2.7 \times 10^4 \text{ N}$
8. The bulk modulus of copper is $1.4 \times 10^{11} \text{ Pa}$ and the coefficient of linear expansion is $1.7 \times 10^{-5} (\text{C}^\circ)^{-1}$. What hydrostatic pressure is necessary to prevent a copper block from expanding when its temperature is increased from 20°C to 30°C ?
 (A) $6.0 \times 10^5 \text{ Pa}$ (B) $7.1 \times 10^7 \text{ Pa}$ (C) $5.2 \times 10^6 \text{ Pa}$ (D) 40 atm
9. The coefficient of apparent expansion of a liquid in a copper vessel is C and in a silver vessel is S. The coefficient of volume expansion of copper is γ_c . What is the coefficient of linear expansion of silver?
 (A) $\frac{(C + \gamma_c + S)}{3}$ (B) $\frac{(C - \gamma_c + S)}{3}$ (C) $\frac{(C + \gamma_c - S)}{3}$ (D) $\frac{(C - \gamma_c - S)}{3}$
10. Two identical conducting rods are first connected independently to two vessels, one containing water at 100°C and the other containing ice at 0°C . In the second case, the rods are joined end to end and connected to the same vessels. Let q_1 and $q_2 \text{ g/s}$ be the rate of melting of ice in the two cases respectively. The ratio q_2/q_1 is **[JEE' 2004 (Scr.)]**
 (A) 1/2 (B) 2/1 (C) 4/1 (D) 1/4
11. Figure shows three different arrangements of materials 1, 2 and 3 to form a wall. Thermal conductivities are $k_1 > k_2 > k_3$. The left side of the wall is 20°C higher than the right side. Temperature difference ΔT across the material 1 has following relation in three cases :



- (A) $\Delta T_a > \Delta T_b > \Delta T_c$ (B) $\Delta T_a = \Delta T_b = \Delta T_c$ (C) $\Delta T_a = \Delta T_b > \Delta T_c$ (D) $\Delta T_a = \Delta T_b < \Delta T_c$
12. The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and 2K and thickness x and 4x, respectively are T_2 and T_1 ($T_2 > T_1$).

The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2 - T_1)K}{x} \right) f$, with f equals to-

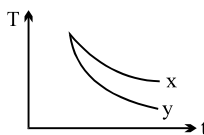


[AIEEE - 2004]

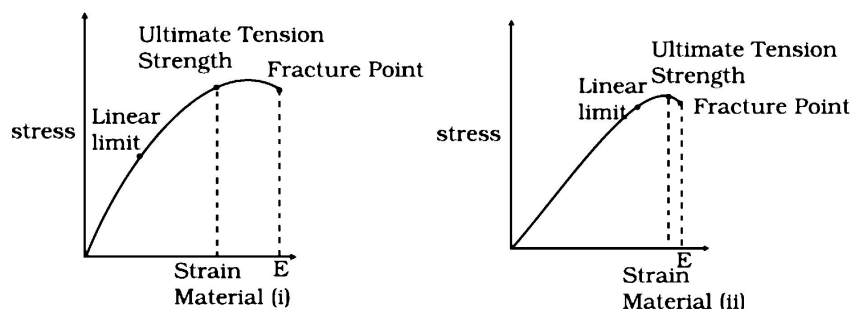
- (A) 1 (B) 1/2 (C) 2/3 (D) 1/3

13. If emissivity of bodies X and Y are e_x and e_y and absorptive power are A_x and A_y then

[JEE' (Scr) 2003]



- (A) $e_y > e_x$; $A_y > A_x$ (B) $e_y < e_x$; $A_y < A_x$ (C) $e_y > e_x$; $A_y < A_x$ (D) $e_y = e_x$; $A_y = A_x$
14. Three discs A, B, and C having radii 2 m, 4 m and 6 m respectively are coated with carbon black on their outer surfaces. The wavelengths corresponding to maximum intensity are 300 nm, 400 nm and 500 nm respectively. The power radiated by them are Q_A , Q_B and Q_C respectively. [JEE' 2004 (Scr.)]
- (A) Q_A is maximum (B) Q_B is maximum (C) Q_C is maximum (D) $Q_A = Q_B = Q_C$
15. If the temperature of the sun were to increase from T to $2T$ and its radius from R to $2R$, then the ratio of the radiant energy received on earth to what it was previously, will be- [AIEEE - 2004]
- (A) 4 (B) 16 (C) 32 (D) 64
16. The stress-strain graphs for two materials are shown in figure (assume same scale).



- (A) Material (ii) is more elastic than material (i) and hence material (ii) is more brittle.
 (B) Material (i) and (ii) have the same elasticity and the same brittleness.
 (C) Material (ii) is elastic over a larger region of strain as compared to (i).
 (D) Material (ii) is more brittle than material (i).
17. A thermally insulated vessel contains some water at 0°C . The vessel is connected to a vacuum pump to pump out water vapour. This results in some water getting frozen. It is given Latent heat of vaporization of water at $0^\circ\text{C} = 21 \times 10^5 \text{ J/kg}$ and latent heat of freezing of water = $3.36 \times 10^5 \text{ J/kg}$. The maximum percentage amount of water that will be solidified in this manner will be :-
- (A) 86.2% (B) 33.6% (C) 21% (D) 24.36%
18. Ice at 0°C is added to 200 g of water initially at 70°C in a vacuum flask. When 50 g of ice has been added and has all melted the temperature of the flask and contents is 40°C . When a further 80g of ice has been added and has all melted, the temperature of the whole is 10°C . Calculate the specific latent heat of fusion of ice. [Take $S_w = 1 \text{ cal / gm } ^\circ\text{C}$.]
- (A) $3.8 \times 10^5 \text{ J/ kg}$ (B) $1.2 \times 10^5 \text{ J/ kg}$ (C) $2.4 \times 10^5 \text{ J/ kg}$ (D) $3.0 \times 10^5 \text{ J/ kg}$

19. A rod of length 2m at 0°C and having expansion coefficient $\alpha = (3x + 2) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ where x is the distance (in cm) from one end of rod. The length of rod at 20°C is :

(A) 2.124 m (B) 3.24 m (C) 2.0120 m (D) 3.124 m

20. A liquid is given some heat.

Statement A : Some liquid evaporates.

Statement B : The liquid starts boiling.

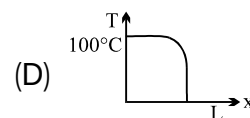
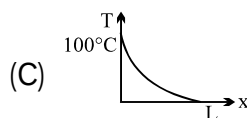
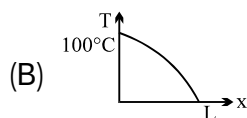
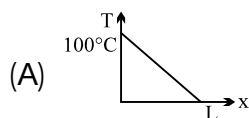
(A) A implies B and B implies A

(B) B implies A but, A does not imply B

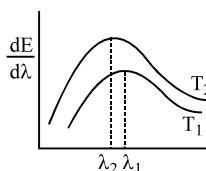
(C) A implies B but B does not imply A

(D) Neither A implies B nor B implies A

21. A rod of length L and uniform cross-sectional area has varying thermal conductivity which changes linearly from $2K$ at end A to K at the other end B. The ends A and B of the rod are maintained at constant temperature 100°C and 0°C , respectively. At steady state, the graph of temperature : $T = T(x)$ where x = distance from end A will be



22. The spectral emissive power E_λ for a body at temperature T_1 is plotted against the wavelength and area under the curve is found to be A . At a different temperature T_2 the area is found to be $9A$. Then $\lambda_1/\lambda_2 =$



(A) 3 (B) $1/3$ (C) $1/\sqrt{3}$ (D) $\sqrt{3}$

23. A black body is at a temperature of 2880 K . The energy of radiation emitted by this object with wavelength between 499 nm and 500 nm is U_1 , between 999 nm and 1000 nm is U_2 and between 1499 nm and 1500 nm is U_3 . The Wien constant $b = 2.88 \times 10^6 \text{ nm K}$. Then

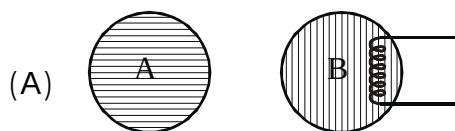
(A) $U_1 = 0$ (B) $U_3 = 0$ (C) $U_1 > U_2$ (D) $U_2 > U_1$

MATRIX MATCH TYPE QUESTION

24. A & B are two black bodies of radii r_A and r_B respectively, placed in surrounding of temperature T_0 . At steady state the temperature of A & B is T_A & T_B respectively.

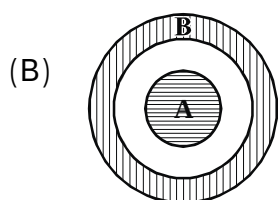
Column I

Column II



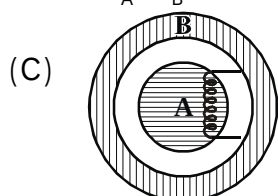
(P) $T_A = T_B$

- A & B are solid sphere
- $r_A = r_B$
- Body 'B' is being heated by a heater of constant power 'P'



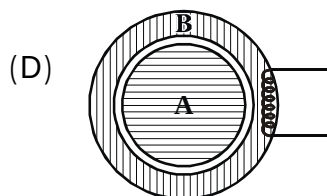
(Q) $T_A < T_B$

- B is thin spherical shell
- A is a solid sphere
- $r_A < r_B$



- (R) Heat received by A is more than heat radiated by it at steady state.

- B is thin spherical shell
- A is a solid sphere
- $r_A < r_B$
- Body A is being heated by a heater of constant power 'P'



- (S) Radiation spectrum of A & B is distinguishable

- B is thin spherical shell
- A is a solid sphere
- $r_A \approx r_B$
- Body B is being heated by a heater of constant power 'P'

- (T) Steady state can't be achieved

25. A sample 'A' of liquid water and a sample B of ice of equal mass are kept in 2 nearby containers so that they can exchange heat with each other but are thermally insulated from the surroundings. The graphs in column-II show the sketch of temperature T of samples versus time t . Match with appropriate description in column-I.

Column I

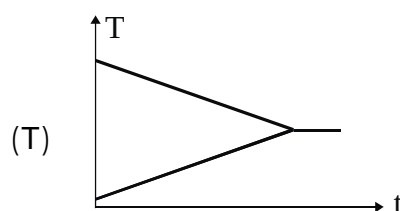
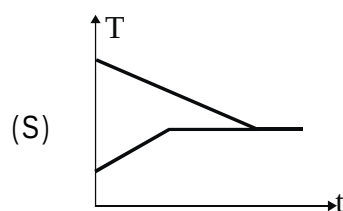
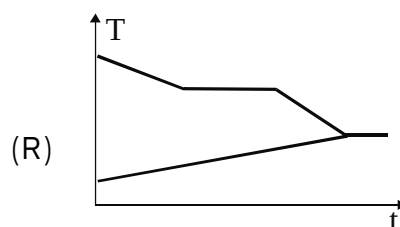
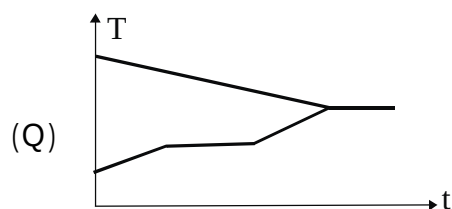
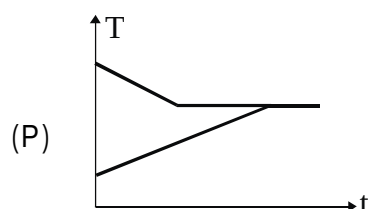
(A) Equilibrium temperature is above melting point of ice.

(B) At least some of water freezes.

(C) At least some of ice melts.

(D) Equilibrium temperature is below freezing point of water

Column II



EXERCISE (S-2)

1. Ans. $4 \times 10^{-6} \text{ m/}^\circ\text{C}$ 2. Ans. 10cm, 40cm 3. Ans. $1/200 \text{ rad}$ 4. Ans. 1 : 1.26
6. Ans. (a) 37.8 J/s (Watts), (b) 2.005 N-m 7. Ans. 10 min.

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. Ans. (D) 2. Ans. (C) 3. Ans. (A) 4. Ans. (B)
5. Ans. (A) 6. Ans. (D) 7. Ans. (D)

MULTIPLE CORRECT TYPE QUESTIONS

8. Ans. (B,D)

COMPREHENSION TYPE QUESTIONS

9. Ans. (A,B) 10. Ans. (A,B,C,D)

EXERCISE (J-A)

1. Ans. (A) 2. Ans. (B) 3. Ans. (A) 4. Ans. (C)
5. Ans. (A) 6. Ans. 273 K 7. Ans. (A, D)
8. Ans. A - Q,S ; B - Q ; C - P,Q ; D - Q,R or A - S ; B - Q ; C - P ; D - R
9. Ans. 9 10. Ans. 9 11. Ans. 8 12. Ans. 4
13. Ans. ACD or ABCD 14. Ans. 3 15. Ans. (C) 16. Ans. (A)
17. Ans. (C) 18. Ans. (A) 19. Ans. 2 20. Ans. (A,B)

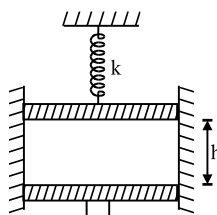
REVIEW TEST

1. Ans. $5000 \text{ J/}^\circ\text{C kg}$ 2. Ans. $1000 \text{ J (}^\circ\text{C)}^{-1}$ 3. Ans. 45°C
4. Ans. $\frac{a^2 s}{2K} \log_e \left(\frac{b}{a} \right) \log_e \left(\frac{T_0 - T_1}{T_0 - T_2} \right)$ 5. Ans. (a) 595 watt/m²; (b) $T_0 \approx 420 \text{ K}$
6. Ans. $\frac{K}{4e\sigma L T_s^3 + K}$ 7. Ans. (A) 8. Ans. (B)
9. Ans. (C) 10. Ans. (D) 11. Ans. (B) 12. Ans. (D)
13. Ans. (A) 14. Ans. (B) 15. Ans. (D) 16. Ans. (C,D)
17. Ans. (A) 18. Ans. (A) 19. Ans. (C) 20. Ans. (B)
21. Ans. (B) 22. Ans. (D) 23. Ans. (D)
24. Ans. (A - Q,S ; B - P ; C - S ; D - Q,S) 25. Ans. (A - Q ; B - P,R ; C - Q,S ; D - R)

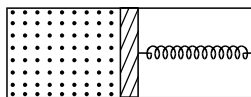
KINETIC THEORY OF GASES

EXERCISE (S-2)

1. A barometer is faulty. When the true barometer reading are 73 and 75 cm of Hg, the faulty barometer reads 69 cm and 70 cm respectively.
 - (i) What is the total length of the barometer tube?
 - (ii) What is the true reading when the faulty barometer reads 69.5 cm ?
 - (iii) What is the faulty barometer reading when the true barometer reads 74 cm?
2. An ideal gas at NTP is enclosed in a adiabatic vertical cylinder having area of cross section $A = 27 \text{ cm}^2$, between two light movable pistons as shown in the figure. Spring with force constant $k = 3700 \text{ N/m}$ is in a relaxed state initially. Now the lower piston is moved upwards a height $h/2$, h being the initial length of gas column. It is observed that the upper piston moves up by a distance $h/16$. Find h taking γ for the gas to be 1.5. Also find the final temperature of the gas.

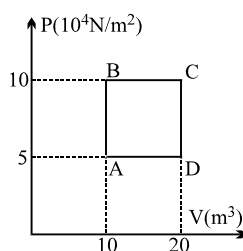


3. A thermally insulated vessel is divided into two parts by a heat-insulating piston which can move in the vessel without the friction. The left part of the vessel contains one mole of an ideal monatomic gas, & the right part is empty. The piston is connected to the right wall of the vessel through a spring whose length in free state is equal to the length of the vessel as shown in the figure. Determine the heat capacity C of the system, neglecting the heat capacities of the vessel, piston and spring.



4. A weightless piston divides a thermally insulated cylinder into two parts of volumes V and $3V$. 2 moles of an ideal gas at pressure $P = 2$ atmosphere are confined to the part with volume $V = 1$ litre. The remainder of the cylinder is evacuated. The piston is now released and the gas expands to fill the entire space of the cylinder. The piston is then pressed back to the initial position. Find the increase of internal energy in the process and final temperature of the gas. The ratio of the specific heat of the gas $\gamma = 1.5$.

5. Two moles of an ideal monoatomic gas are confined within a cylinder by a massless & frictionless spring loaded piston of cross-sectional area $4 \times 10^{-3} \text{ m}^2$. The spring is, initially in its relaxed state. Now the gas is heated by an electric heater, placed inside the cylinder, for some time. During this time, the gas expands and does 50 J of work in moving the piston through a distance 0.10 m. The temperature of the gas increases by 50 K. Calculate the spring constant & the heat supplied by the heater. [$P_0 = 10^5 \text{ Pa}$]
6. A sample of 2 kg of monoatomic Helium (assumed ideal) is taken through the process ABC and another sample of 2 kg of the same gas is taken through the process ADC as in figure. Given, molecular mass of Helium = 4

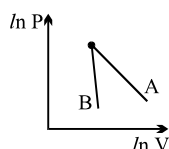


- What is the temperature of Helium in each of the states A, B, C & D ?
- Is there any way of telling afterwards which sample of Helium went through the process ABC and which went through the process ADC? Write Yes or No.
- How much is the heat involved in each of the processes ABC ADC.

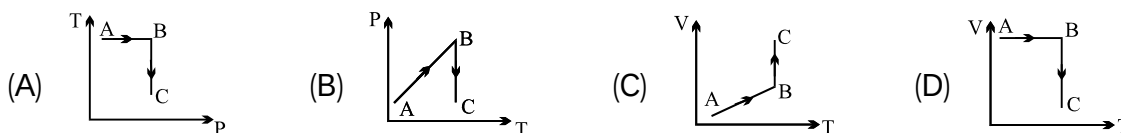
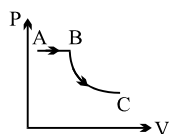
EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

- An open and wide glass tube is immersed vertically in mercury in such a way that length 0.05 m extends above mercury level. The open end of the tube is closed and the tube is raised further by 0.43 m. The length of air column above mercury level in the tube will be : Take $P_{\text{atm}} = 76$ cm of mercury
(A) 0.215 m (B) 0.2 m (C) 0.1 m (D) 0.4 m
- The figure, shows the graph of logarithmic reading of pressure and volume for two ideal gases A and B undergoing adiabatic process. From figure it can be concluded that



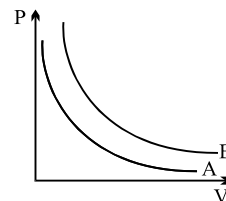
- (A) gas B is diatomic (B) gas A and B both are diatomic
(C) gas A is monoatomic (D) gas B is monoatomic & gas A is diatomic
- 1 kg of a gas does 20 kJ of work and receives 16 kJ of heat when it is expanded between two states. A second kind of expansion can be found between the initial and final state which requires a heat input of 9 kJ. The work done by the gas in the second expansion is :
(A) 32 kJ (B) 5 kJ (C) - 4 kJ (D) 13 kJ
 - A process is shown in the diagram. Which of the following curves may represent the same process?



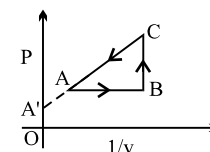
- An ideal gas expands in such a way that $PV^2 = \text{constant}$ throughout the process.
(A) The graph of the process of T-V diagram is a parabola.
(B) The graph of the process of T-V diagram is a straight line.
(C) Such an expansion is possible only with heating.
(D) Such an expansion is possible only with cooling.

6. Figure shows the pressure P versus volume V graphs for two different gas sample at a given temperature. M_A and M_B are masses of two samples, n_A and n_B are numbers of moles. Which of the following **must be incorrect**.

- (A) $M_A > M_B$
 (B) $M_A < M_B$
 (C) $n_A > n_B$
 (D) $n_A < n_B$

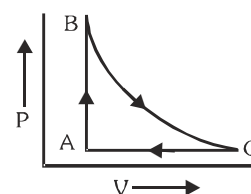


7. An enclosed ideal gas is taken through a cycle as shown in the figure. Then
 (A) Along AB, temperature decreases while along BC temperature increases
 (B) Along AB, temperature increases while along BC the temperature decreases.
 (C) Along CA work is done by the gas and the internal energy remains constant.
 (D) Along CA work is done on the gas and internal energy of the gas increases.



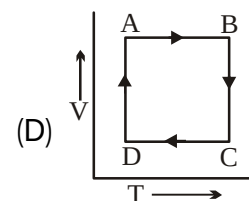
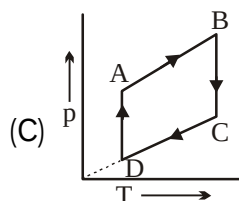
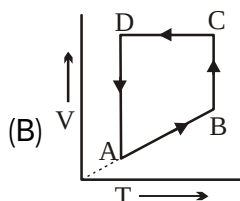
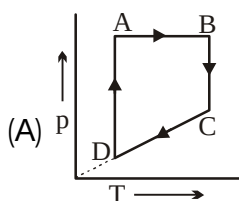
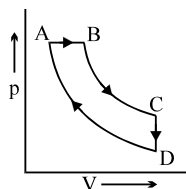
8. One mole of monoatomic ideal gas undergoes a cyclic process ABCA as shown in figure. Process BC is adiabatic. The temperatures at A, B and C are 300, 600 and 450K respectively. Choose the correct statement(s).

- (A) In process CA change in internal energy is $225R$.
 (B) In process AB change in internal energy is $-150R$.
 (C) In process BC change in internal energy is $-225R$.
 (D) Change in internal energy during the whole cyclic process is $+150R$.

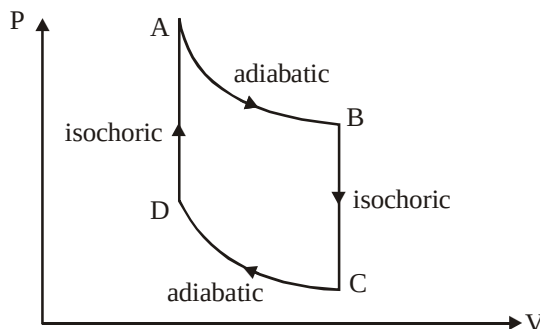


MULTIPLE CORRECT TYPE QUESTIONS

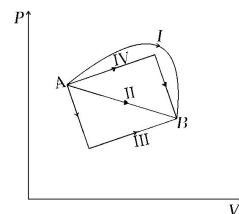
9. A gas expands such that its initial and final temperature are equal. Also, the process followed by the gas traces a straight line on the P - V diagram :
 (A) The temperature of the gas remains constant throughout.
 (B) The temperature of the gas first increases and then decreases
 (C) The temperature of the gas first decreases and then increases
 (D) The straight line has a negative slope.
10. A cyclic process ABCD is shown in the p - V diagram. Which of the following curves represents the same process if BC & DA are isothermal processes



11. Two moles of helium gas is taken through the cycle ABCDA as shown in the figure. If $T_A = 1000\text{ K}$, $2P_A = 3P_B = 6P_C$.



- (A) work done by the gas in the process A to B is 3741 J.
 (B) heat lost by the gas in the process B to C is 10600 J.
 (C) temperature T_D is 2000 K.
 (D) none of these
12. Figure shows the P - V diagram of an ideal gas undergoing a change of state from A to B. Four different parts I, II, III and IV as shown in the figure may lead to the same change of state.
- (A) Change in internal energy is same in IV and III cases, but not in I and II.
 (B) Change in internal energy is same in all the four cases.
 (C) Work done is maximum in case I
 (D) Work done is minimum in case II.

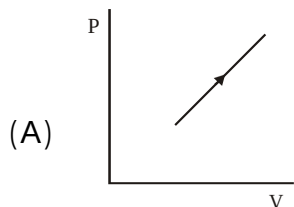


MATRIX MATCH TYPE QUESTION

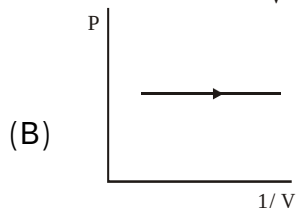
13.

Column I
(Pressure volume graph)

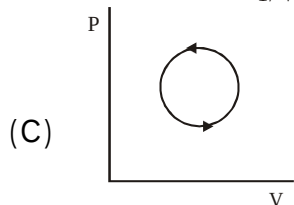
Column II
(W is work done by gas,
Q heat supply to gas)



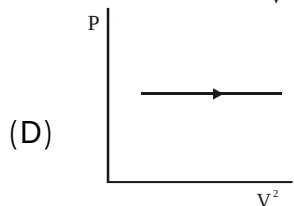
(P) $W > 0$



(Q) $W < 0$



(R) $Q > 0$



(S) $Q < 0$

SUPPLEMENT FOR JEE-MAINS

1. Two engines A and B have their sources at 400 K and 350 K and sinks at 350 K and 300 K respectively. Which engine is more efficient and by how much ?
2. A carnot engine working between 400 K and 800 K has a work output of 1200 J per cycle. What is the amount of heat energy supplied to the engine from source per cycle?
3. A carnot engine works as a refrigerator between 250 K and 300 K. If it receives 750 cal of heat from the reservoir at the lower temperature. Calculate the amount of heat rejected at the higher temperature.
4. The temperature insides & outside of refrigerator are 260 K and 315 K respectively. Assuming that the refrigerator cycle is reversible, calculate the heat delivered to surroundings for every joule of work done.
5. A refrigerator takes heat from water at 0°C and transfer it to room at 27°C. If 100 kg of water is converted in ice at 0°C then calculate the workdone. (Latent heat of ice is 3.4×10^5 J/kg)
6. The coefficient of performance of a refrigerator working between 10° C and 20° C is :-
 (A) 1 (B) 28.3 (C) 29.3 (D) 3.53
7. A refrigerator freezes 5 kg of water at 0°C into ice at 0°C in a time interval of 20 minutes. Assume that room temperature is 20°C. Calculate the minimum power needed to accomplish it ?
 (A) 24.4 Watt (B) 0.1025 Watt (C) 0.0244 Watt (D) 102.5 Watt
8. A carnot engine whose source temperature is at 400 K takes 100 kcal of heat at this temperature in each cycle and gives 70 kcal to the sink. Calculate (i) the temperature of the sink and (ii) the efficiency of the engine.
 (A) 280 K, 30 % (B) 270 K, 30% (C) 280 K, 10 % (D) None of these
9. According to the second law of thermodynamics:
 (A) Heat energy cannot be completely converted to work
 (B) Work cannot be completely converted to heat energy
 (C) For all cyclic processes we have $dQ/T < 0$
 (D) The reason all heat engine efficiencies are less than 100% is friction, which is unavoidable
 (E) All of the above are true
10. A reversible refrigerator operates between a low temperature reservoir at T_C and a high temperature reservoir at T_H . Its coefficient of performance is given by :
 (A) $(T_H - T_C)/T_C$ (B) $T_C/(T_H - T_C)$ (C) $(T_H - T_C)/T_H$ (D) $T_H/(T_H - T_C)$
 (E) $T_H(T_H + T_C)$
11. An inventor claims to have a heat engine that has an efficiency of 40% when it operates between a high temperature reservoir of 150°C and a low temperature reservoir of 30°C. This engine:
 (A) Must violate the zeroth law of thermodynamics
 (B) Must violate the first law of thermodynamics
 (C) Must violate the second law of thermodynamics
 (D) Must violate the third law of thermodynamics
 (E) Does not necessarily violate any of the laws of thermodynamics
12. A carnot heat engine runs between a cold reservoir at temperature T_C and a hot reservoir at temperature T_H . You want to increase its efficiency. Of the following, which change results in the greatest increase in efficiency? The value of ΔT is the same for all changes.
 (A) Raise the temperature of the hot reservoir by ΔT
 (B) Raise the temperature of the cold reservoir by ΔT
 (C) Lower the temperature of the hot reservoir by ΔT
 (D) Lower the temperature of the cold reservoir by ΔT
 (E) Lower the temperature of the hot reservoir by $1/2\Delta T$ and raise the temperature of the cold reservoir by $1/2\Delta T$.
13. A Carnot engine takes 3×10^6 cal of heat from reservoir at 627° and gives it to a sink at 27°C. Then work done by the engine is :-
 (A) 4.2×10^6 J (B) 8.4×10^6 J (C) 16.8×10^6 J (D) zero

5. While the piston is at a distance $2L$ from the top, the hole at the top is sealed. The piston is then released, to a position where it can stay in equilibrium. In this condition, the distance of the piston from the top is :- [JEE 2007]

(A) $\left(\frac{2P_0\pi R^2}{\pi R^2 P_0 + Mg} \right) (2L)$

(B) $\left(\frac{P_0\pi R^2 - Mg}{\pi R^2 P_0} \right) (2L)$

(C) $\left(\frac{P_0\pi R^2 + Mg}{\pi R^2 P_0} \right) (2L)$

(D) $\left(\frac{P_0\pi R^2}{\pi R^2 P_0 - Mg} \right) (2L)$

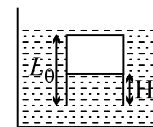
6. The piston is taken completely out of the cylinder. The hole at the top is sealed. A water tank is brought below the cylinder and put in a position so that the water surface in the tank is at the same level as the top of the cylinder as shown in the figure. The density of the water is ρ . In equilibrium, the height H of the water column in the cylinder satisfies [JEE 2007]

(A) $\rho g(L_0 - H)^2 + P_0(L_0 - H) + L_0 P_0 = 0$

(B) $\rho g(L_0 - H)^2 - P_0(L_0 - H) - L_0 P_0 = 0$

(C) $\rho g(L_0 - H)^2 + P_0(L_0 - H) - L_0 P_0 = 0$

(D) $\rho g(L_0 - H)^2 - P_0(L_0 - H) + L_0 P_0 = 0$



7. **Statement-1** : The total translational kinetic energy of all the molecules of a given mass of an ideal gas is 1.5 times the product of its pressure and its volume. [JEE 2007]

and

Statement-2 : The molecules of a gas collide with each other and the velocities of the molecules change due to the collision.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

8. An ideal gas is expanding such that $PT^2 = \text{constant}$. The coefficient of volume expansion of the gas is : [JEE 2008]

(A) $\frac{1}{T}$

(B) $\frac{2}{T}$

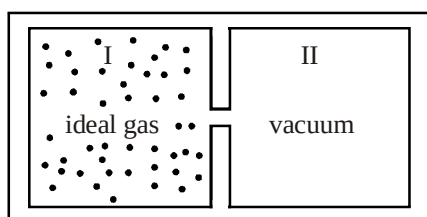
(C) $\frac{3}{T}$

(D) $\frac{4}{T}$

9. **Column I** contains a list of processes involving expansion of an ideal gas. Match this with **Column II**, describing the thermodynamic change during this process. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. **[JEE 2008]**

Column I

- (A) An insulated container has two chambers separated by a valve. Chamber I contains an ideal gas and the chamber II has vacuum. The valve is opened..

**Column II**

- (P) The temperature of the gas decreases

- (B) An ideal monoatomic gas expands to twice its original volume such that pressure

- (Q) The temperature of the gas increases or remains constant

$$P \propto \frac{1}{V^2}, \text{ where } V \text{ is the volume of the gas.}$$

- (C) An ideal monoatomic gas expands to twice its

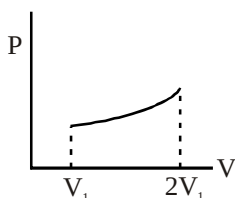
- (R) The gas loses heat

$$\text{original volume such that its pressure } P \propto \frac{1}{V^{4/3}},$$

where V is its volume.

- (D) An ideal monoatomic gas expands such that its pressure P and volume V follows the behaviour shown in the graph.

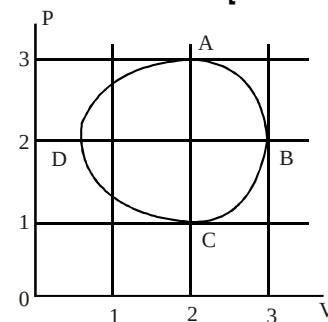
- (S) The gas gains heat



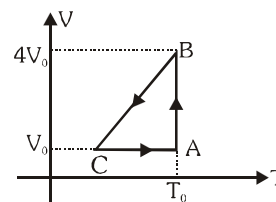
10. C_v and C_p denote the molar specific heat capacities of a gas at constant volume and constant pressure, respectively. Then **[JEE-2009]**

- (A) $C_p - C_v$ is larger for a diatomic ideal gas than for a monoatomic ideal gas
 (B) $C_p + C_v$ is larger for a diatomic ideal gas than for a monoatomic ideal gas
 (C) C_p / C_v is larger for a diatomic ideal gas than for a monoatomic ideal gas
 (D) $C_p \cdot C_v$ is larger for a diatomic ideal gas than for a monoatomic ideal gas

11. The figure shows the P-V plot of an ideal gas taken through a cycle ABCDA. The part ABC is a semicircle and CDA is half of an ellipse. Then, [JEE-2009]



- (A) the process during the path $A \rightarrow B$ is isothermal
(B) heat flows out of the gas during the path $B \rightarrow C \rightarrow D$
(C) work done during the path $A \rightarrow B \rightarrow C$ is zero
(D) positive work is done by the gas in the cycle ABCDA
12. A real gas behaves like an ideal gas if its
(A) pressure and temperature are both high
(B) pressure and temperature are both low
(C) pressure is high and temperature is low
(D) pressure is low and temperature is high
13. One mole of an ideal gas in initial state A undergoes a cyclic process ABCA, as shown in the figure. its pressure at A is P_0 . Choose the correct option(s) from the following. [JEE-2010]



- (A) Internal energies at A and B are the same
(B) Work done by the gas in process AB is $P_0 V_0 \ln 4$
(C) Pressure at C is $\frac{P_0}{4}$
(D) Temperature at C is $\frac{T_0}{4}$
14. A diatomic ideal gas is compressed adiabatically to $1/32$ of its initial volume. In the initial temperature of the gas is T_i (in Kelvin) and the final temperature is aT_i , the value of a is [JEE-2010]
15. 5.6 liter of helium gas at STP is adiabatically compressed to 0.7 liter. Taking the initial temperature to be T_1 , the work done in the process is [JEE-2011]

- (A) $\frac{9}{8} RT_1$ (B) $\frac{3}{2} RT_1$ (C) $\frac{15}{8} RT_1$ (D) $\frac{9}{2} RT_1$

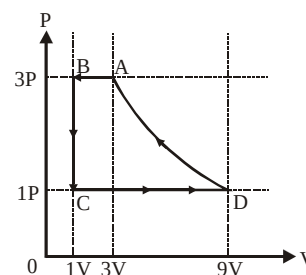
16. One mole of a monatomic ideal gas is taken through a cycle ABCDA as shown in P-V diagram. Column II gives the characteristics involved in the cycle. Match them with each of the processes given in Column I. [JEE-2011]

Column I

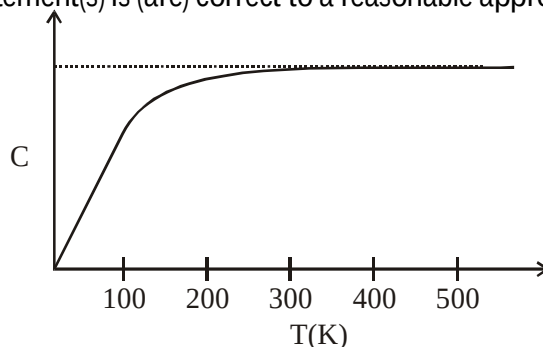
- (A) Process $A \rightarrow B$
(B) Process $B \rightarrow C$
(C) Process $C \rightarrow D$
(D) Process $D \rightarrow A$

Column II

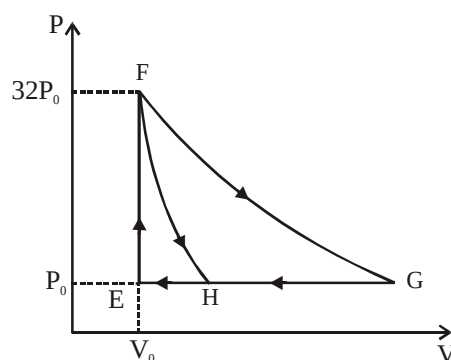
- (P) Internal energy decreases.
(Q) Internal energy increases.
(R) Heat is lost.
(S) Heat is gained.
(T) Work is done on the gas.



17. A mixture of 2 moles of helium gas (atomic mass = 4 amu) and 1 mole of argon gas (atomic mass = 40 amu) is kept at 300 K in a container. The ratio of the rms speeds $\left(\frac{v_{rms}(\text{helium})}{v_{rms}(\text{argon})} \right)$ is
 (A) 0.32 (B) 0.45 (C) 2.24 (D) 3.16 [JEE-2012]
18. Two moles of ideal helium gas are in a rubber balloon at 30°C. The balloon is fully expandable and can be assumed to require no energy in its expansion. The temperature of the gas in the balloon is slowly changed to 35°C. The amount of heat required in raising the temperature is nearly
 (Take : $R = 8.31 \text{ J/mol.K}$) [JEE-2012]
 (A) 62 J (B) 104 J (C) 124 J (D) 208 J
19. Two non-reactive monoatomic ideal gases have their atomic masses in the ratio 2 : 3. The ratio of their partial pressures, when enclosed in a vessel kept at a constant temperature, is 4 : 3. The ratio of their densities is :- [JEE-2013]
 (A) 1 : 4 (B) 1 : 2 (C) 6 : 9 (D) 8 : 9
20. The figure below shows the variation of specific heat capacity (C) of a solid as a function of temperature (T). The temperature is increased continuously from 0 to 500 K at a constant rate. Ignoring any volume change, the following statement(s) is (are) correct to a reasonable approximation :- [JEE-2013]



- (A) the rate at which heat is absorbed in the range 0–100 K varies linearly with temperature T.
 (B) heat absorbed in increasing the temperature from 0–100 K is less than the heat required for increasing the temperature from 400–500 K.
 (C) there is no change in the rate of heat absorption in the range 400–500 K
 (D) the rate of heat absorption increases in the range 200–300 K
21. One mole of a monatomic ideal gas is taken along two cyclic processes $E \rightarrow F \rightarrow G \rightarrow E$ and $E \rightarrow F \rightarrow H \rightarrow E$ as shown in the PV diagram. The processes involved are purely isochoric, isobaric, isothermal or adiabatic. [JEE-2013]



Match the paths in List I with the magnitudes of the work done in the List II and select the correct answer using the codes given below the lists.

List I

- P. $G \rightarrow E$
Q. $G \rightarrow H$
R. $F \rightarrow H$
S. $F \rightarrow G$

Codes :

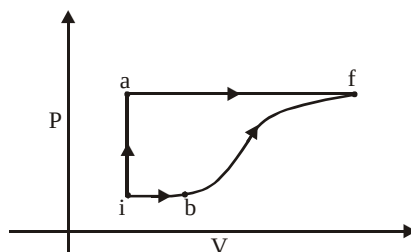
	P	Q	R	S
(A)	4	3	2	1
(B)	4	3	1	2
(C)	3	1	2	4
(D)	1	3	2	4

List II

1. $160 P_0 V_0 \ln 2$
2. $36 P_0 V_0$
3. $24 P_0 V_0$
4. $31 P_0 V_0$

22. A thermodynamic system is taken from an initial state i with internal energy $U_i = 100$ J to the final state f along two different paths iaf and ibf, as schematically shown in the figure. The work done by the system along the paths af, ib and bf are $W_{af} = 200$ J, $W_{ib} = 50$ J and $W_{bf} = 100$ J respectively. The heat supplied to the system along the path iaf, ib and bf are Q_{iaf} , Q_{ib} and Q_{bf} respectively. If the internal energy of the system in the state b is $U_b = 200$ J and $Q_{iaf} = 500$ J, the ratio Q_{bf}/Q_{ib} is.

[JEE-Advance-2014]



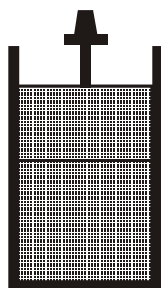
Paragraph for Questions 23 and 24

In the figure a container is shown to have a movable (without friction) piston on top. The container and the piston are all made of perfectly insulating material allowing no heat transfer between outside and inside the container. The container is divided into two compartments by a rigid partition made of a thermally conducting material that allows slow transfer of heat. The lower compartment of the container is filled with 2 moles of an ideal monatomic gas at 700 K and the upper compartment is filled with 2 moles of an ideal diatomic gas

at 400 K. The heat capacities per mole of an ideal monatomic gas are $C_V = \frac{3}{2}R$, $C_P = \frac{5}{2}R$, and those for

an ideal diatomic gas are $C_V = \frac{5}{2}R$, $C_P = \frac{7}{2}R$.

[JEE-Advance-2014]



23. Consider the partition to be rigidly fixed so that it does not move. When equilibrium is achieved, the final temperature of the gases will be

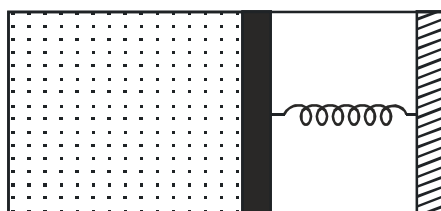
- (A) 550 K (B) 525 K (C) 513 K (D) 490 K

24. Now consider the partition to be free to move without friction so that the pressure of gases in both compartments is the same. Then total work done by the gases till the time they achieve equilibrium will be
 (A) 250 R (B) 200 R (C) 100 R (D) -100 R
25. A container of fixed volume has a mixture of one mole of hydrogen and one mole of helium in equilibrium at temperature T. Assuming the gases are ideal, the correct statement(s) is (are) :-

[JEE-Advance-2015]

- (A) The average energy per mole of the gas mixture is $2RT$.
 (B) The ratio of speed of sound in the gas mixture to that in helium gas is $\sqrt{6/5}$.
 (C) The ratio of the rms speed of helium atoms to that of hydrogen molecules is $1/2$.
 (D) The ratio of the rms speed of helium atoms to that of hydrogen molecules is $1/\sqrt{2}$.
26. An ideal monoatomic gas is confined in a horizontal cylinder by a spring loaded piston (as shown in the figure). Initially the gas is at temperature T_1 , pressure P_1 and volume V_1 and the spring is in its relaxed state. The gas is then heated very slowly to temperature T_2 , pressure P_2 and volume V_2 . During this process the piston moves out by a distance x . Ignoring the friction between the piston and the cylinder, the correct statement(s) is(are) :-

[JEE-Advance-2015]



- (A) If $V_2 = 2V_1$ and $T_2 = 3T_1$, then the energy stored in the spring is $\frac{1}{4} P_1 V_1$
 (B) If $V_2 = 2V_1$ and $T_2 = 3T_1$, then the change in internal energy is $3 P_1 V_1$
 (C) If $V_2 = 3V_1$ and $T_2 = 4T_1$, then the work done by the gas is $\frac{7}{3} P_1 V_1$
 (D) If $V_2 = 3V_1$ and $T_2 = 4T_1$, then the heat supplied to the gas is $\frac{17}{6} P_1 V_1$
27. A gas is enclosed in a cylinder with a movable frictionless piston. Its initial thermodynamic state at pressure $P_i = 10^5$ Pa and volume $V_i = 10^{-3}$ m³ changes to a final state at $P_f = \left(\frac{1}{32}\right) \times 10^5$ Pa and $V_f = 8 \times 10^{-3}$ m³ in an adiabatic quasi-static process, such that $P^3 V^5 = \text{constant}$. Consider another thermodynamic process that brings the system from the same initial state to the same final state in two steps: an isobaric expansion at P_i followed by an isochoric (isovolumetric) process at volumes V_f . The amount of heat supplied to the system in the two step process is approximately

[JEE-Advance-2016]

- (A) 112 J (B) 294 J (C) 588 J (D) 813 J

REVIEW TEST

1. A cycle followed by an engine (made of one mole of perfect gas in a cylinder with a piston) is shown in figure.

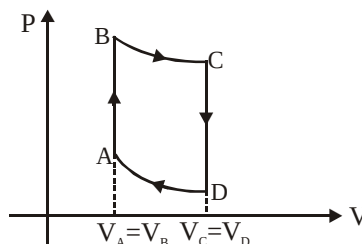
A to B : volume constant

B to C : adiabatic

C to D : volume constant

D to A : adiabatic

$$V_C = V_D = 2V_A = 2V_B$$



(a) In which part of the cycle heat is supplied to the engine from outside?

(b) In which part of the cycle heat is being given to the surrounding by the engine?

(c) What is the work done by the engine in one cycle? Write your answer in term of P_A , P_B , V_A

(d) What is the efficiency of the engine?

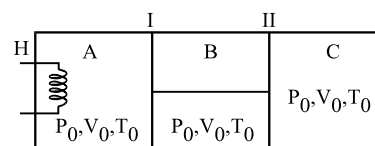
$$[\gamma = \frac{5}{3} \text{ for the gas}], (C_V = \frac{3}{2}R \text{ for one mole})$$

2. A vessel of volume $V = 30\text{ l}$ is separated into three equal parts by stationary semipermeable thin membranes as shown in the Figure. The left, middle and right parts are filled with $m_{H_2} = 30\text{ g}$ of hydrogen, $m_{O_2} = 160\text{ g}$ of oxygen, and $m_{N_2} = 70\text{ g}$ of nitrogen respectively. The left partition lets through only hydrogen, while the right partition lets through hydrogen and nitrogen. What will be the pressure in each part of the vessel after the equilibrium has been set in if the vessel is kept at a constant temperature $T = 300\text{ K}$?

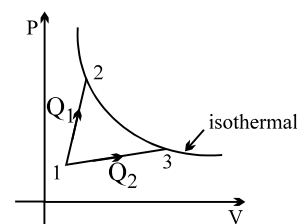
H_2	O_2	N_2
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3. The figure shows an insulated cylinder divided into three parts A, B and C. Pistons I and II are connected by a rigid rod and can move without friction inside the cylinder. Piston I is perfectly conducting while piston II is perfectly insulating. The initial state of the gas ($\gamma = 1.5$) present in each compartment A, B and C is as shown. Now, compartment A is slowly given heat through a heater H such that the final volume of C becomes $\frac{4V_0}{9}$. Assume the gas to be ideal and find.

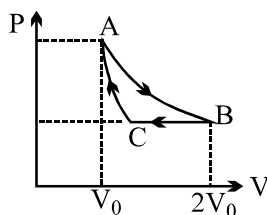
- (a) Final pressures in each compartment A, B and C
(b) Final temperatures in each compartment A, B and C
(c) Heat supplied by the heater
(d) Work done by gas in A and B.
(e) Heat flowing across piston I.



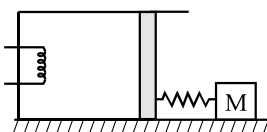
4. A gas takes part in two processes in which it is heated from the same initial state 1 to the same final temperature. The processes are shown on the P-V diagram by the straight line 1-2 and 1-3. 2 and 3 are the points on the same isothermal curve. Q_1 and Q_2 are the heat transfer along the two processes. Then in which case will the heat transfer be more.



5. In a cyclic process ABCA consisting of isothermal expansion AB, isobaric compression BC and adiabatic compression CA, find the efficiency of cycle (Given : $T_A = T_B = 400\text{ K}$, $\gamma = 1.5$)



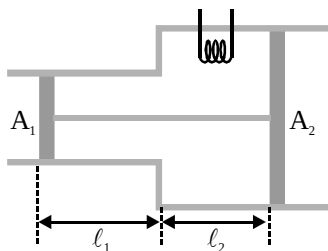
6. An adiabatic cylinder has 8 gram of helium. A light smooth adiabatic piston is connected to a light spring of force constant 300 N/m . The other end of the spring is connected with a block of mass 1 kg kept on a rough horizontal surface of coefficient of friction $\mu = 0.3$. Area of cross-section of cylinder is $A = 25\text{ cm}^2$. Initially the spring is in a relaxed position and the temperature of the gas is 400 K . The gas is heated slowly for some time by means of an electric heater so as to bring the block M on the verge of motion. Take $P_{\text{atm}} = 10^5\text{ N/m}^2$. Find



- (a) the work done by the gas
 (b) the final temperature
 (c) heat supplied by the heater
7. Monoatomic, diatomic and triatomic gases whose initial volume and pressure are same, are compressed till their volume becomes half the initial volume.
 (A) If the compression is adiabatic then monoatomic gas will have maximum final pressure.
 (B) If the compression is adiabatic then triatomic gas will have maximum final pressure.
 (C) If the compression is adiabatic then their final pressure will be same.
 (D) If the compression is isothermal then their final pressure will be different.
8. Water is heated in an open pan where the air pressure is 10^5 Pa . The water remains a liquid, which expands by a small amount as it is heated. Determine the ratio of the heat absorbed by the water to the work done by water. (γ for water = $10^{-3}/^\circ\text{C}$, $S = 1\text{ cal/gm}^\circ\text{C}$)
 (A) 4.2×10^3 (B) 4.2×10^5 (C) 4.2×10^2 (D) 4.2×10^4
9. An ideal gas expands from volume V_1 to V_2 . This may be achieved by either of the three processes: isobaric, isothermal and adiabatic. Let ΔU be the change in internal energy of the gas, Q be the quantity of heat added to the system and W be the work done by the gas. Identify which of the following statements is false for ΔU ?
 (A) ΔU is least under adiabatic process.
 (B) ΔU is greatest under adiabatic process.
 (C) ΔU is greatest under the isobaric process.
 (D) ΔU in isothermal process lies in-between the values obtained under isobaric and adiabatic processes.

Paragraph for Question Nos. 10 to 12

The walls of the two connecting cylinders shown in the figure are adiabatic (thermally insulating). The cross-sectional areas of the parts are A_1 and A_2 . There is a well-fitting but freely moving, thermally insulating piston in each cylinder, at a distance $\ell_1 = \ell_2 = \ell$ from the point where the cross-sectional area changes. The pistons are connected to each other by a thin and rigid rod. The enclosed volume contains air. The temperature and air pressure are T_0 and p_0 both inside and outside. The heater filament inside is operated for time t at a power of P .



10. Mark the **CORRECT** statement :-

- (A) Process performed is adiabatic (B) Piston shifts towards left
(C) Process performed is polytropic (D) Piston shifts towards right

11. What is change in temperature (Q is heat energy supplied, v_0 is initial volume) :-

- (A) $\Delta T = \frac{2}{5} \frac{QT_0}{p_0 V_0}$ (B) $\Delta T = \frac{2}{7} \frac{QT_0}{p_0 V_0}$ (C) $\Delta T = \frac{5}{7} \frac{QT_0}{p_0 V_0}$ (D) $\Delta T = \frac{QT_0}{p_0 V_0}$

12. What is displacement (x) of piston till equilibrium is reached :-

- (A) $x = \frac{2Pt}{(A_2 - A_1)7p_0}$ (B) $x = \frac{2Pt}{(A_2 - A_1)5p_0}$ (C) $x = \frac{2Pt}{(A_2 - A_1)3p_0}$ (D) $x = \frac{2Pt}{(A_2 - A_1)9p_0}$

13. A barometer tube, containing mercury, is lowered in a vessel containing mercury until only 50 cm of the tube is above the level of mercury in the vessel. If the atmospheric pressure is 75 cm of mercury, what is the pressure at the top of the tube ?

- (A) 33.3 kPa (B) 66.7 kPa (C) 3.33 MPa (D) 6.67 MPa

14. At temperature T , N molecules of gas A each having mass m and at the same temperature $2N$ molecules of gas B each having mass $2m$ are filled in a container. The mean square velocity of molecules of gas B is v^2 and mean square of x component of velocity of molecules of gas A is w^2 . The ratio of w^2/v^2 is :

- (A) 1 (B) 2 (C) 1/3 (D) 2/3

15. When unit mass of water boils to become steam at 100°C , it absorbs Q amount of heat. The densities of water and steam at 100°C are ρ_1 and ρ_2 respectively and the atmospheric pressure is p_0 . The increase in internal energy of the water is

- (A) Q (B) $Q + p_0 \left(\frac{1}{\rho_1} - \frac{1}{\rho_2} \right)$ (C) $Q + p_0 \left(\frac{1}{\rho_2} - \frac{1}{\rho_1} \right)$ (D) $Q - p_0 \left(\frac{1}{\rho_1} + \frac{1}{\rho_2} \right)$

16. A cylinder made of perfectly non conducting material closed at both ends is divided into two equal parts by a heat proof piston. Both parts of the cylinder contain the same masses of a gas at a temperature $t_0 = 27^\circ\text{C}$ and pressure $P_0 = 1$ atm. Now if the gas in one of the parts is slowly heated to $t = 57^\circ\text{C}$ while the temperature of first part is maintained at t_0 the distance moved by the piston from the middle of the cylinder will be (length of the cylinder = 84 cm)

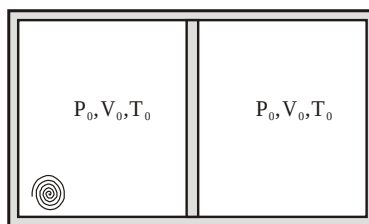
- (A) 3 cm (B) 5 cm (C) 2 cm (D) 1 cm

17. A vertical cylinder with heat-conducting walls is closed at the bottom and is fitted with a smooth light piston. It contains one mole of an ideal gas. The temperature of the gas is always equal to the surrounding's temperature, T_0 . The piston is moved up slowly to increase the volume of the gas to η times. Which of the following is incorrect?
- (A) Work done by the gas is $RT_0 \ln \eta$.
 (B) Work done against the atmosphere is $RT_0(\eta - 1)$.
 (C) There is no change in the internal energy of the gas.
 (D) The final pressure of the gas is $\frac{1}{(\eta - 1)}$ times its initial pressure.
18. A gas is enclosed in a vessel at a constant temperature at a pressure of 5 atmosphere and volume 4 litre. Due to a leakage in the vessel, after some time, the pressure is reduced to 4 atmosphere. As a result, the
- (A) volume of the gas decreased by 20%
 (B) average K.E. of gas molecule decreases by 20%
 (C) 20% of the gas escaped due to the leakage
 (D) 25% of the gas escaped due to the leakage
19. Let v_{av} , v_{rms} and v_p respectively denote mean speed, root mean square speed and the most probable speed of the molecule in an ideal monoatomic gas at absolute temperature T . The mass of a molecule is m then :
- (A) no molecule can have speed greater than $\sqrt{2} v_{rms}$
 (B) no molecule can have speed less than $v_p / \sqrt{2}$
 (C) $v_p < v_{av} < v_{rms}$
 (D) the average kinetic energy of a molecule is $3/4 mv_p^2$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 20 to 22

One mole of a monoatomic ideal gas occupies two chambers of a cylinder partitioned by means of a movable piston. The walls of the cylinder as well as the piston are thermal insulators. Initially equal amounts of gas fill both the chambers at (P_0, V_0, T_0) . A coil is burnt in the left chamber which absorbs heat and expands, pushing the piston to the right. The gas on the right chamber is compressed until its pressure becomes $32 P_0$.



20. The final volume of left chamber is :-
- (A) $\frac{V_0}{8}$ (B) $\frac{15}{8} V_0$ (C) $\frac{7}{8} V_0$ (D) $\frac{9}{8} V_0$
21. The work done on the gas in the right chamber is :-
- (A) $\frac{9}{2} P_0 V_0$ (B) $-\frac{9}{2} P_0 V_0$ (C) $\frac{13}{2} P_0 V_0$ (D) $\frac{17}{2} P_0 V_0$
22. The change in internal energy of the gas in the left chamber is :-
- (A) $\frac{186}{4} RT_0$ (B) $\frac{177}{4} RT_0$ (C) $\frac{59}{2} RT_0$ (D) $\frac{131}{4} RT_0$

Answer key

EXERCISE (S-2)

1. Ans. (i) 74 cm, (ii) 73.94 cm, (iii) 69.52 cm 2. Ans. 1.6 m, 364 K
3. Ans. $C = 2R$ 4. Ans. 400 J, $2T_0$ 5. Ans. 2000 N/m, 1295 J
6. Ans. (i) $T_A = 120.33 \text{ K}$, $T_B = 240.66 \text{ K}$, $T_C = 481.32 \text{ K}$, $T_D = 240.66 \text{ K}$,
 (ii) No
 (iii) $\Delta Q_{ABC} = 3.25 \times 10^6 \text{ J}$; $\Delta Q_{ADC} = 2.75 \times 10^6 \text{ J}$

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. Ans. (C) 2. Ans. (D) 3. Ans. (D) 4. Ans. (C)
 5. Ans. (D) 6. Ans. (C) 7. Ans. (A) 8. Ans. (C)

MULTIPLE CORRECT TYPE QUESTIONS

9. Ans. (B,D) 10. Ans. (A,B) 11. Ans. (A,B) 12. Ans. (B,C)

MATRIX MATCH TYPE QUESTION

13. Ans. (A) $\rightarrow$ (P,R) ; (B) $\rightarrow$ (Q, S) ; (C) $\rightarrow$ (Q, S) ; (D) $\rightarrow$ (P, R)

SUPPLEMENT FOR JEE-MAINS

1. Ans. B & 1.8% more 2. Ans. 2400 joule per cycle 3. Ans. 900 cal
 4. Ans. 5.7 J 5. Ans. $3.36 \times 10^6 \text{ J}$ 6. Ans. (B)
 7. Ans. (D) 8. Ans. (A) 9. Ans. (A) 10. Ans. (B)
 11. Ans. (C) 12. Ans. (D) 13. Ans. (B)

EXERCISE (JA)

1. Ans. (A) 2. Ans. (a) $T_{\text{final}} = 70^\circ\text{C}$, (b) 0.05 J, (c) 19999.95 J
 3. Ans. (A) $\rightarrow$ S ; (B) $\rightarrow$ P and R ; (C) $\rightarrow$ R; (D) $\rightarrow$ Q and S
 4. Ans. (A) 5. Ans. (D) 6. Ans. (C) 7. Ans. (B)

8. Ans. (C) 9. Ans. (A) q (b) p,r (c) p,s (d) q,s 10. Ans. (B,D)
 11. Ans. (B,D) 12. Ans. (D) 13. Ans. (A,B,C,D) 14. Ans. 4
 15. Ans. (A) 16. Ans. (A) $\rightarrow$ (P,R,T) ; (B) $\rightarrow$ (P,R) ; (C) $\rightarrow$ (Q,S) ; (D) $\rightarrow$ (R,T)
 17. Ans. (D) 18. Ans. (D) 19. Ans. (D)
 20. Ans. (A, B, C, D) or (B, C, D) 21. Ans. (A) 22. Ans. 2
 23. Ans. (D) 24. Ans. (D) 25. Ans. (A,B,D) 26. Ans. (A,B,C)
 27. Ans. (C)

REVIEW TEST

1. Ans. (a) A to B (b) C to D

$$(c) W_{AB} = \frac{1}{1-\gamma} (2^{1-\gamma} - 1) (P_B - P_A) V_A = \frac{3}{2} \left(1 - \left(\frac{1}{2} \right)^{2/3} \right) (P_B - P_A) V_A \quad (d) \left[1 - \left(\frac{1}{2} \right)^{2/3} \right]$$

2. Ans. $p_1 = p_{H_2} \simeq 1.25 \times 10^6 \text{ Pa}$; $p_2 = p_{H_2} + p_{O_2} + p_{N_2} \simeq 2.8125 \times 10^6 \text{ Pa}$; $p_3 = p_{H_2} + p_{N_2} \simeq 1.5625 \times 10^6 \text{ Pa}$

3. Ans. (a) Final pressure in A = $\frac{27}{8} P_0$ = Final pressure in C, Final pressure in B = $\frac{21}{4} P_0$

(b) Final temperature in A (and B) = $\frac{21}{4} T_0$, Final temperature in C = $\frac{3}{2} T_0$,

(c) $18 P_0 V_0$

(d) work done by gas in A = $+ P_0 V_0$, work done by gas in B = 0,

(e) $\frac{17}{2} P_0 V_0$

4. Ans. $Q_1 < Q_2$ 5. Ans. $1 - \frac{3 \left(1 - \frac{1}{2^{1/3}} \right)}{\ln 2}$

6. Ans. 2.515, 404.8 K,

7. Ans. (A) 8. Ans. (D) 9. Ans. (B) 10. Ans. (D)
 11. Ans. (B) 12. Ans. (A) 13. Ans. (A) 14. Ans. (D)
 15. Ans. (B) 16. Ans. (C) 17. Ans. (D) 18. Ans. (C)
 19. Ans. (C,D)

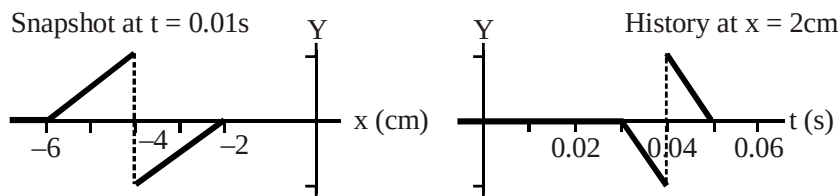
COMPREHENSION TYPE QUESTIONS

20. Ans. (B) 21. Ans. (A) 22. Ans. (B)

MECHANICAL WAVES

EXERCISE (O-2)

1. Figure shows a snapshot graph and a history graph for a wave pulse on a stretched string. They describe the same wave from two perspectives.



- (A) the wave is travelling in positive x-direction (B) the wave is travelling in negative x-direction
(C) the speed of the wave is 2 m/s. (D) the peak is located at $x = -6$ cm at $t = 0$.
2. $Y(x,t) = 0.05 / [(4x + 2t)^2 + 5]$ represents a moving wave pulse, where x and y are in meters and t is in seconds. Then which statement(s) are **CORRECT**:
(A) Pulse is moving in $-x$ direction (B) Wave speed is 0.5 m/s
(C) Maximum particle displacement is 1 cm (D) It is a symmetric pulse
3. A transverse wave travelling on a taut string is represented by:
 $Y = 0.01 \sin 2\pi (10 t - x) \rightarrow Y$ and x are in meters and t in seconds. Then
(A) The speed of the wave is 10 m/s.
(B) Closest points on the string which differ in phase by 60° are $(1/6)$ m apart.
(C) Maximum particle velocity is $\pi/4$ m/s.
(D) The phase of a certain point on the string changes by 120° in $(1/20)$ seconds.
4. $y(x, t) = 0.8 / [(4x + 5t)^2 + 5]$ represents a moving pulse, where x & y are in meter and t in second. Then:
[JEE '99, 3]
(A) pulse is moving in $+x$ direction (B) in 2s it will travel a distance of 2.5 m
(C) its maximum displacement is 0.16 m (D) it is a symmetric pulse.
5. At a certain moment, the photograph of a string on which a harmonic wave is travelling to the right is shown. Then, which of the following is true regarding the velocities of the points P, Q and R on the string.
-
- (A) v_P is upwards (B) $v_Q = -v_R$
(C) $|v_P| > |v_Q| = |v_R|$ (D) $v_Q = v_R$
6. Which of the following combinations can give standing wave .
(A) $y_1 = A \sin^2 (\omega t - kx)$; $y_2 = -A \sin^2 (\omega t + kx)$
(B) $y_1 = A \sin (kx - \omega t)$; $y_2 = A \cos (\omega t + kx)$
(C) $y_1 = 2A \cos^2 (\omega t - kx + \pi)$; $y_2 = A [\sin 2 (\omega t + kx) - 1]$
(D) $y_1 = A \sin (kx - \omega t + 30^\circ)$; $y_2 = A \cos (\omega t + kx - 60^\circ)$.

7. If the tension in a stretched string fixed at both ends is changed by 20%, the fundamental frequency is found to increase by 15Hz. then the
- (A) original frequency is 157 Hz
- (B) velocity of propagation of the transverse wave along the string changes by 5%
- (C) velocity of propagation of the transverse wave along the string changes by 10%.
- (D) fundamental wave length on the string does not change.
8. String of length L whose one end is $x=0$ vibrates according to the relations given by different equations. Choose the **CORRECT** statements.

(A) $y = A \sin \frac{\pi x}{L} \sin \omega t$ has 1 antinodes, 2 nodes

(B) $y = A \cos \frac{\pi x}{L} \sin \omega t$ has 2 antinodes, 1 nodes

(C) $y = A \sin \frac{2\pi x}{L} \sin \omega t$ has 3 nodes, 2 antinodes

(D) $y = A \cos \frac{2\pi x}{L} \sin \omega t$ has 3 antinodes, 2 nodes

9. The vibration of a string fixed at both ends are described by $Y = 2 \sin(\pi x) \sin(100\pi t)$ [where Y is in mm, x is in cm and t in second] then
- (A) Maximum displacement of the particle at $x = 1/6$ cm would be 1 mm.
- (B) Velocity of the particle at $x = 1/6$ cm at time $t = 1/600$ sec will be $157\sqrt{3}$ mm/s
- (C) If the length of the string be 10 cm, number of loop in it would be 5
- (D) None of these
10. The (x, y) co-ordinates of the corners of a square plate are $(0, 0)$ $(L, 0)$ (L, L) & $(0, L)$. The edges of the plate are clamped & transverse standing waves are set up in it. If $u(x, y)$ denotes the displacement of the plate at the point (x, y) at some instant of time, the possible expression(s) for u is/are : (a = positive constant)

(A) $a \cos\left(\frac{\pi x}{2L}\right) \cos\left(\frac{\pi y}{2L}\right)$

(B) $a \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi y}{L}\right)$

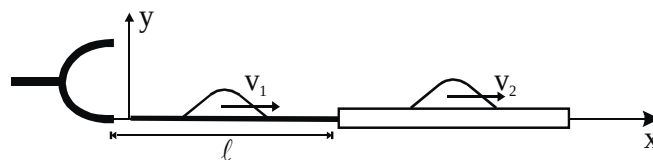
(C) $a \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi y}{L}\right)$

(D) $a \cos\left(\frac{2\pi x}{L}\right) \sin\left(\frac{\pi y}{L}\right)$

11. In a travelling one dimensional mechanical sinusoidal, wave
- (A) potential energy and kinetic energy of an element become maximum simultaneously.
 - (B) all particles oscillate with the same frequency and the same amplitude
 - (C) all particles may come to rest simultaneously
 - (D) we can find two particles, in a length equal to half of a wavelength, which have the same non zero acceleration simultaneously.

Paragraph for Question Nos. 12 to 14

A harmonic oscillator at $x=0$, oscillates with a frequency $\frac{\omega}{2\pi}$ and amplitude a . It is generating waves at end of a thin string in which velocity of wave is v_1 and which is connected to another heavier string in which velocity of wave is v_2 as shown, length of first string is ℓ .



12. If harmonic oscillator oscillates by an equation $y = a \sin \omega t$. The equation of incident wave in first string is

(A) $y = a \sin \omega \left(t - \frac{x}{v_1} \right)$ (B) $y = a \sin \omega \left(t + \frac{x}{v_1} \right)$

(C) $y = a \sin \left[\omega \left(t - \frac{x}{v_1} \right) + \pi \right]$ (D) $y = a \sin \left[\omega \left(t + \frac{x}{v_1} \right) + \pi \right]$

13. Equation of transmitted wave in second string if its amplitude is a_t is

(A) $y = a_t \sin \omega \left(t - \frac{x}{v_2} \right)$ (B) $y = a_t \sin \omega \left(t - \frac{\ell}{v_1} \right)$

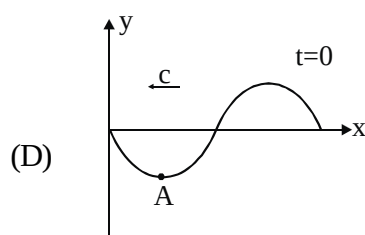
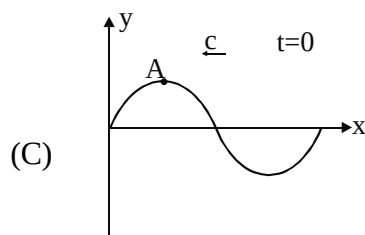
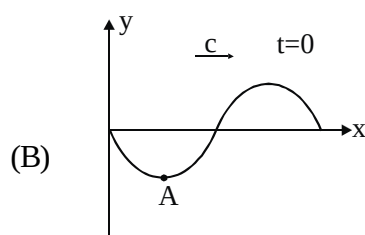
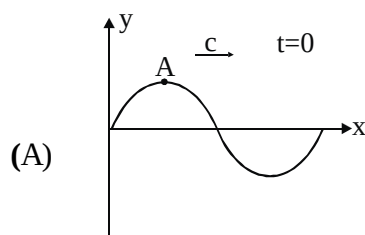
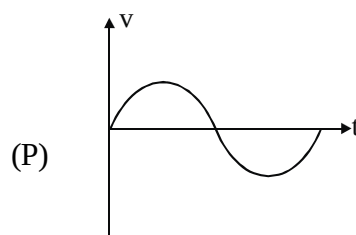
(C) $y = a_t \sin \omega \left(t - \frac{\ell}{v_1} - \frac{x - \ell}{v_2} \right)$ (D) $y = a_t \sin \omega \left(t - \frac{x}{v_2} \right)$

14. Equation of reflected wave, if it is reflecting at the joint and amplitude of reflected wave is a_r

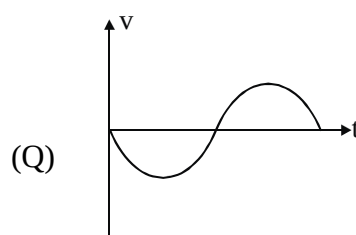
(A) $y = a_r \sin \omega \left(t - \frac{x}{v_2} \right)$ (B) $y = a_r \sin \left[\omega \left(t - \frac{\ell}{v_1} - \frac{\ell - x}{v_1} \right) + \pi \right]$

(C) $y = a_r \sin \left[\omega \left(t + \frac{x}{v_1} \right) + \pi \right]$ (D) $y = a_r \sin \left[\omega \left(t + \frac{2\ell + x}{v_1} \right) + \pi \right]$

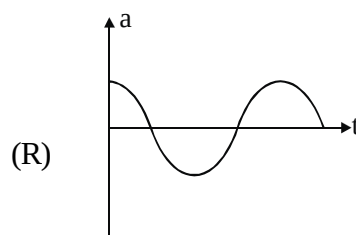
15. In column-I transverse waves travelling on a string at $t = 0$ is shown. Wave velocity is indicated by 'c'. Column-II describes variation of different parameters for particle A or for all the particles.

Column-I**Column-II**

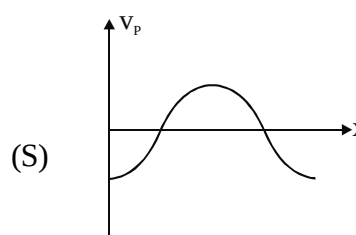
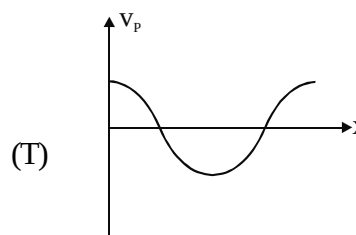
For particle A



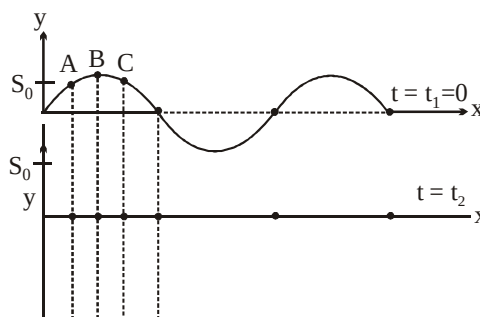
For particle A



For particle A

At $t = 0$ for all the particlesAt $t = 0$ for all the particles

16. The graphs show the standing wave on a string at two successive instants of time t_1, t_2 . A, B, C are points on the string (so is the maximum displacement amplitude of the standing wave) table II gives observations about net mechanical energy for the time interval between t_1 & t_2 .



Column-I

Column-II

- | | |
|-------|---|
| (A) A | (P) Kinetic energy at this element is increasing. |
| (B) B | (Q) Energy is flowing towards right through this point at t_1 . |
| (C) C | (R) Energy is flowing left through this point at t_1 . |
| | (S) No net energy ever crosses this point |
17. In case of mechanical wave a particle oscillates and during oscillation its kinetic energy and potential energy changes.

Column I

Column II

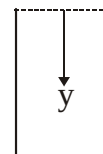
- | | |
|---|---------------------------------|
| (A) When particle of travelling wave is passing through mean position. | (P) Kinetic energy is maximum |
| (B) When particle of travelling wave is at extreme position. | (Q) Potential energy is maximum |
| (C) When particle between node and antinode in standing wave is passing through mean position | (R) Kinetic energy is minimum |
| (D) When particle between node and antinode in standing wave is at extreme position | (S) Potential energy is minimum |
18. Which of the following statements are wrong about the velocity of sound in air:
- | | |
|---|--|
| (A) decreases with increases in temperature | (B) increases with decrease in temperature |
| (C) decreases as humidity increases | (D) independent of density of air. |
19. A person hears a single-frequency sound from a speaker that emits sound uniformly in all directions. The person records the decibel level reading. Which of the following processes would increase the decibel reading the most ?
- | |
|---|
| (A) The power output of the speaker is doubled. |
| (B) The frequency of the sound is doubled keeping amplitude constant. |
| (C) The temperature of the air is doubled on the Kelvin scale. |
| (D) The area over which the sound is transmitted is halved. |

20. A car moves towards a hill with speed v_c . It blows a horn of frequency f which is heard by an observer following the car with speed v_o . The speed of sound in air is v .
- (A) the wavelength of sound reaching the hill is $\frac{v}{f}$
- (B) the wavelength of sound reaching the hill is $\frac{v - v_c}{f}$
- (C) the beat frequency observed by the observer is $\left(\frac{v + v_o}{v - v_c} \right) f$
- (D) the beat frequency observed by the observer is $\frac{2v_c (v + v_o) f}{v^2 - v_c^2}$
21. Two tuning forks of frequency 250 Hz and 256 Hz produce beats. If a maximum of intensity is observed just now, after how much time the minimum is observed at the same place ?
- (A) $\frac{1}{18}$ sec (B) $\frac{1}{4}$ sec. (C) $\frac{1}{3}$ sec. (D) $\frac{1}{12}$ sec.

Paragraph for Question No. 22 to 24

In an organ pipe (may be closed or open) of length 1m standing wave is setup, whose equation for longitudinal displacement is given by $\xi = (0.1 \text{ mm})$

$\cos \frac{2\pi}{0.8} (y) \cos (400) t$ where y is measured from the top of the tube in meters and t in second.



22. The upper end and the lower ends of the tube are respectively:
 (A) open – closed (B) closed – open (C) open – open (D) closed – closed
23. The air column is vibrating in
 (A) First overtone (B) Second overtone (C) Third harmonic (D) Fundamental mode
24. Equation of the standing wave in terms of excess pressure is (Bulk modulus of air $B = 5 \times 10^5 \text{ N/m}^2$)

(A) $P_{\text{ex}} = (125 \pi \text{ N/m}^2) \sin \frac{2\pi}{0.8} (y) \cos (400t)$

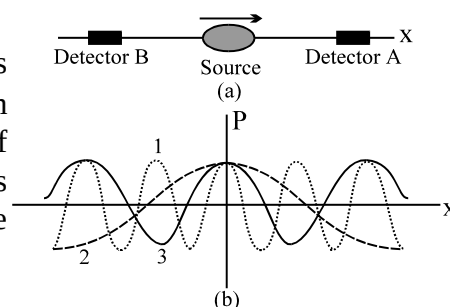
(B) $P_{\text{ex}} = (125 \pi \text{ N/m}^2) \cos \frac{2\pi}{0.8} (y) \sin (400t)$

(C) $P_{\text{ex}} = (225 \pi \text{ N/m}^2) \sin \frac{2\pi}{0.8} (y) \cos (200t)$

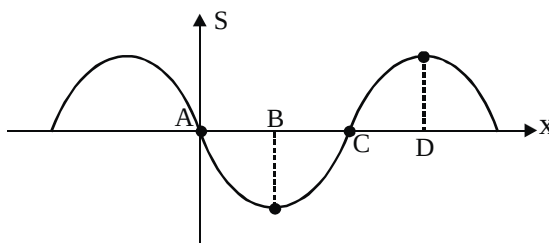
(D) $P_{\text{ex}} = (225 \pi \text{ N/m}^2) \cos \frac{2\pi}{0.8} (y) \sin (200t)$

Paragraph for Question No. 25 to 28

A source emitting a sound wave at a certain frequency moves with constant speed along an x-axis figure (a). The source moves directly towards a stationary detector A and directly away from another stationary detector B. The superimposed three plots of figure (b) indicate the pressure function $P(x)$ of the sound wave as measured by detector A, by detector B, and by someone (c) in the rest frame of the source.



25. Which of the following plot corresponds to the measurement done by detector A?
(A) 1 (B) 2 (C) 3 (D) These plots are not possible
26. The plot corresponding to the measurement done by detector B is
(A) 1 (B) 2 (C) 3 (D) These plots are not possible
27. The plot corresponding to the measurement done by the detector C is
(A) 1 (B) 2 (C) 3 (D) These plots are not possible
28. Now the source stops and begins to move along y-axis with same speed, the plot which corresponds to the measurement of B now is
(A) 1 (B) 2 (C) 3 (D) none of these
29. Figure shows a graph of particle displacement function of x at $t=0$ for a longitudinal wave travelling in positive x-direction in a gas. A, B, C, D denote position of particles in space.



Column-I

- (A) point A
- (B) point B
- (C) point C
- (D) point D

Column II

- (P) Particle velocity is in direction of wave propagation
- (Q) Maximum magnitude of strain
- (R) Excess pressure is zero
- (S) Maximum density
- (T) Maximum magnitude of excess pressure

17. **Statement-1:** Two longitudinal waves given by equations : $y_1(x, t) = 2a \sin(\omega t - kx)$ and $y_2(x, t) = a \sin(2\omega t - 2kx)$ will have equal intensity.

Statement-1: Intensity of waves of given frequency in same medium is proportional to square of amplitude only. [AIEEE - 2011]

- (1) Statement-1 is false, statement-2 is true
 (2) Statement-1 is true, statement-2 is false
 (3) Statement-1 is true, statement-2 true; statement-2 is the correct explanation of statement-1
 (4) Statement-1 is true, statement-2 is true; statement -2 is not correct explanation of statement-1.
18. A travelling wave represented by $y = A \sin(\omega t - kx)$ is superimposed on another wave represented by $y = A \sin(\omega t + kx)$. The resultant is :- [AIEEE-2011]

(1) A standing wave having nodes at $x = \left(n + \frac{1}{2}\right) \frac{\lambda}{2}$, $n = 0, 1, 2$

(2) A wave travelling along + x direction

(3) A wave travelling along -x direction

(4) A standing wave having nodes at $x = \frac{n\lambda}{2}$; $n = 0, 1, 2$

19. A sonometer wire of length 1.5m is made of steel. The tension in it produces an elastic strain of 1%. What is the fundamental frequency of steel if density and elasticity of steel are $7.7 \times 10^3 \text{ kg/m}^3$ and $2.2 \times 10^{11} \text{ N/m}^2$ respectively ? [JEE Mains-2013]

- (1) 188.5 Hz (2) 178.2 Hz (3) 200.5 Hz (4) 770 Hz

20. A pipe of length 85 cm is closed from one end. Find the number of possible natural oscillations of air column in the pipe whose frequencies lie below 1250 Hz. The velocity of sound in air is 340 m/s. [JEE Mains - 2014]

- (1) 6 (2) 4 (3) 12 (4) 8

21. A train is moving on a straight track with speed 20 ms^{-1} . It is blowing its whistle at the frequency of 1000 Hz. The percentage change in the frequency heard by a person standing near the track as the train passes him is (speed of sound = 320 ms^{-1}) close to :- [JEE Mains-2015]

- (1) 18% (2) 24% (3) 6% (4) 12%

22. A uniform string of length 20m is suspended from a rigid support. A short wave pulse is introduced at its lowest end. It starts moving up the string. The time taken to reach the support is :- (take $g = 10 \text{ ms}^{-2}$) [JEE Mains-2016]

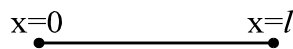
- (1) $\sqrt{2} \text{ s}$ (2) $2\pi\sqrt{2} \text{ s}$ (3) 2s (4) $2\sqrt{2} \text{ s}$

23. A pipe open at both ends has a fundamental frequency f in air. The pipe is dipped vertically in water so that half of it is in water. The fundamental frequency of the air column is now :- [JEE Mains-2016]

- (1) $f(2) \frac{f}{2}$ (3) $\frac{3f}{4}$ (4) 2f

EXERCISE - JA

1. A string between $x = 0$ and $x = l$ vibrates in fundamental mode. The amplitude A , tension T and mass per unit length μ is given. Find the total energy of the string. **[IIT-JEE 2003(Scr)]**

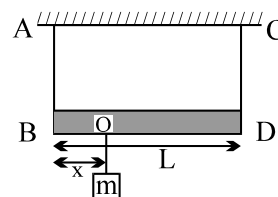


2. A string fixed at both ends is in resonance in its 2nd harmonic with a tuning fork of frequency f_1 . Now its one end becomes free. If the frequency of the tuning fork is increased slowly from f_1 then again a resonance is obtained when the frequency is f_2 . If in this case the string vibrates in n th harmonic then **[IIT-JEE 2005(Scr)]**

(A) $n = 3, f_2 = \frac{3}{4} f_1$ (B) $n = 3, f_2 = \frac{5}{4} f_1$ (C) $n = 5, f_2 = \frac{5}{4} f_1$ (D) $n = 5, f_2 = \frac{3}{4} f_1$

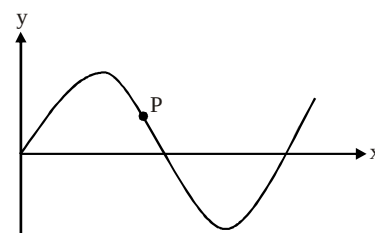
3. A transverse harmonic disturbance is produced in a string. The maximum transverse velocity is 3 m/s and maximum transverse acceleration is 90 m/s^2 . If the wave velocity is 20 m/s then find the waveform. **[IIT-JEE 2005]**

4. A massless rod is suspended by two identical strings AB and CD of equal length. A block of mass m is suspended from point O such that BO is equal to ' x '. Further, it is observed that the frequency of 1st harmonic (fundamental frequency) in AB is equal to 2nd harmonic frequency in CD. Then, length of BO is **[IIT-JEE 2006]**



(A) $\frac{L}{5}$ (B) $\frac{L}{4}$ (C) $\frac{4L}{5}$ (D) $\frac{3L}{4}$

5. A transverse sinusoidal wave moves along a string in the positive x -direction at a speed of 10 cm/s. The wavelength of the wave is 0.5 m and its amplitude is 10 cm. At a particular time t , the snap-shot of the wave is shown in figure. The velocity of point P when its displacement is 5 cm is : **[IIT-JEE 2008]**



(A) $\frac{\sqrt{3}\pi}{50} \hat{j} \text{ m/s}$ (B) $-\frac{\sqrt{3}\pi}{50} \hat{j} \text{ m/s}$ (C) $\frac{\sqrt{3}\pi}{50} \hat{i} \text{ m/s}$ (D) $-\frac{\sqrt{3}\pi}{50} \hat{i} \text{ m/s}$

6. A vibrating string of certain length ℓ under a tension T resonates with a mode corresponding to the first overtone (third harmonic) of an air column of length 75 cm inside a tube closed at one end. The string also generates 4 beats per second when excited along with a tuning fork of frequency n . Now when the tension of the string is slightly increased the number of beats reduces to 2 per second. Assuming the velocity of sound in air to be 340 m/s, the frequency n of the tuning fork in Hz is. **[IIT-JEE 2008]**

(A) 344 (B) 336 (C) 117.3 (D) 109.3

7. A 20 cm long string having a mass of 1.0 g, is fixed at both the ends. The tension in the string is 0.5 N. The string is set into vibrations using an external vibrator of frequency 100 Hz. Find the separation (**in cm**) between the successive nodes on the string. **[IIT-JEE 2009]**
8. When two progressive waves $y_1 = 4 \sin(2x - 6t)$ and $y_2 = 3 \sin\left(2x - 6t - \frac{\pi}{2}\right)$ are superimposed, the amplitude of the resultant wave is **[IIT-JEE 2010]**
9. A police van moving with velocity 22 m/s and emitting sound of frequency 176 Hz, follows a motor cycle in turn is moving towards a stationary car and away from the police van. The stationary car is emitting frequency 165 Hz. If motorcyclist does not hear any beats then his velocity is ($v_s = 330$ m/s) **[JEE 2003 (Scr)]**
- (A) 22 m/s (B) 24 m/s (C) 20 m/s (D) 18 m/s
10. A tuning fork of frequency 480 Hz resonates with a tube closed at one end of length, 16 cm and diameter 5 cm in fundamental mode. Calculate velocity of sound in air. **[JEE 2003]**
11. A cylindrical tube when sounded with a tuning fork gives, first resonance when length of air column is 0.1 m and gives second resonance when the length of air column is 0.35 m. Then end correction is **[JEE 2003 (Scr)]**
- (A) 0.025 m (B) 0.020 m (C) 0.018 m (D) 0.012 m
12. A closed organ pipe of length L and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases are equal in both the pipes. Both the pipes are vibrating in their first overtone with same frequency. The length of the open organ pipe is **[JEE' 2004 (Scr)]**
- (A) $\frac{L}{3}$ (B) $\frac{4L}{3}$ (C) $\frac{4L}{3} \sqrt{\frac{\rho_1}{\rho_2}}$ (D) $\frac{4L}{3} \sqrt{\frac{\rho_2}{\rho_1}}$
13. A source of sound of frequency 600 Hz is placed inside water. The speed of sound in water is 1500 m/s and in air it is 300 m/s. The frequency of sound recorded by an observer who is standing in air is **[JEE' 2004 (Scr)]**
- (A) 200 Hz (B) 3000 Hz (C) 120 Hz (D) 600 Hz
14. A whistling train approaches a junction. An observer standing at junction observes the frequency to be 2.2 KHz and 1.8 KHz of the approaching and the receding train. Find the speed of the train (speed sound = 300 m/s). **[JEE 2005]**

Paragraph for Question Nos. 15 to 17 (3 questions)

Two waves $y_1 = A \cos(0.5\pi x - 100\pi t)$ and $y_2 = A \cos(0.46\pi x - 92\pi t)$ are travelling in a pipe placed along x-axis. **[JEE 2006]**

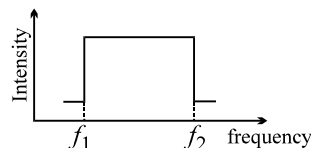
15. Find the number of times intensity is maximum in time interval of 1 sec.
- (A) 4 (B) 6 (C) 8 (D) 10
16. Find wave velocity of louder sound
- (A) 100 m/s (B) 192 m/s (C) 200 m/s (D) 96 m/s

17. At $x = 0$ how many times the value of $y_1 + y_2$ is zero in one second?
(A) 100 (B) 46 (C) 192 (D) 96
18. In the experiment to determine the speed of sound using a resonance column, [JEE 2007]
(A) prongs of the tuning fork are kept in a vertical plane
(B) prongs of the tuning fork are kept in a horizontal plane
(C) in one of the two resonances observed, the length of the resonating air column is close to the wavelength of sound in air
(D) in one of the two resonances observed, the length of the resonating air column is close to half of the wavelength of sound in air

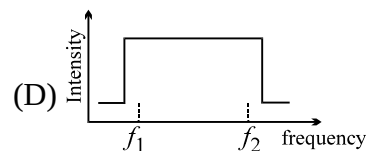
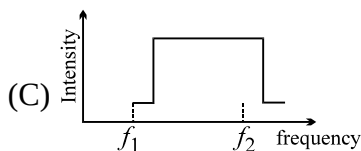
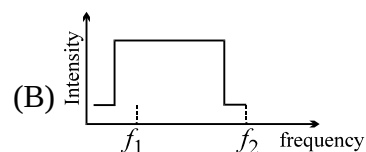
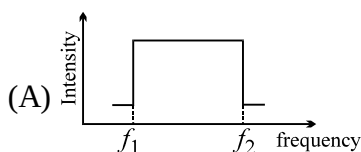
Paragraph for Question Nos. 19 to 21 (3 questions)

[JEE 2007]

Two trains A and B are moving with speeds 20 m/s and 30 m/s respectively in the same direction on the same straight track, with B ahead of A. The engines are at the front ends. The engine of train A blows a long whistle. Assume that the sound of the whistle is composed to components varying in frequency from $f_1 = 800$ Hz to $f_2 = 1120$ Hz, as shown in the figure. The spread in the frequency (highest frequency – lowest frequency) is thus 320 Hz. The speed of sound in still air is 340 m/s



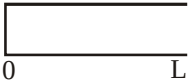
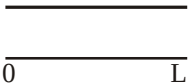
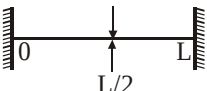
19. The speed of sound of the whistle is
(A) 340 m/s for passengers in A and 310 m/s for passengers in B
(B) 360 m/s for passengers in A and 310 m/s for passengers in B
(C) 310 m/s for passengers in A and 360 m/s for passengers in B
(D) 340 m/s for passengers in both the trains
20. The distribution of the sound intensity of the whistle as observed by the passengers in train A is best represented by



21. The spread of frequency as observed by the passengers in train B is
(A) 310 Hz (B) 330 Hz (C) 350 Hz (D) 290 Hz
22. A police car with a siren of frequency 8 kHz is moving with uniform velocity 36 km/hr towards a tall building which reflects the sound waves. The speed of sound in air is 320 m/s. The frequency of the siren heard by the car driver is [JEE 2011]
(A) 8.50 kHz (B) 8.25 kHz (C) 7.75 kHz (D) 7.50 kHz

23. **Column I** shows four systems, each of the same length L , for producing standing waves. The lowest possible natural frequency of a system is called its fundamental frequency, whose wavelength is denoted as λ_f . Match each system with statements given in **Column II** describing the nature and wavelength of the standing waves.

[JEE 2011]

Column I	Column II
(A) Pipe closed at one end 	(P) Longitudinal waves
(B) Pipe open at both ends 	(Q) Transverse Waves
(C) Stretched wire clamped at both ends 	(R) $\lambda_f = L$
(D) Stretched wire clamped at both ends and at mid-point 	(S) $\lambda_f = 2L$ (T) $\lambda_f = 4L$

24. A person blows into open-end of a long pipe. As a result, a high-pressure pulse of air travels down the pipe. When this pulse reaches the other end of the pipe. [JEE 2012]

- (A) a high-pressure pulse starts travelling up the pipe, if the other end of the pipe is open
 (B) a low -pressure pulse starts travelling up the pipe, if the other end of the pipe is open
 (C) a low pressure pulse starts travelling up the pipe, if the other end of the pipe is closed
 (D) a high-pressure pulse starts travelling up the pipe, if the other end of the pipe is closed

25. A student is performing the experiment of Resonance Column. The diameter of the column tube is 4 cm. The frequency of the tuning fork is 512 Hz. The air temperature is 38°C in which the speed of sound is 336 m/s. The zero of the meter scale coincides with the top end of the Resonance column tube. When the first resonance occurs, the reading of the water level in the column is [JEE 2012]

- (A) 14.0 cm (B) 15.2 cm (C) 16.4 cm (D) 17.6 cm

26. A horizontal stretched string, fixed at two ends, is vibrating in its fifth harmonic according to the equation, $y(x, t) = (0.01\text{m}) \sin [(62.8 \text{ m}^{-1})x] \cos [(628 \text{ s}^{-1})t]$. Assuming $\pi = 3.14$, the correct statement(s) is (are) [JEE Advanced-2013]

- (A) The number of nodes is 5.
 (B) The length of the string is 0.25 m.
 (C) The maximum displacement of the midpoint of the string, from its equilibrium position is 0.01m.
 (D) The fundamental frequency is 100 Hz.

27. Two vehicles, each moving with speed u on the same horizontal straight road, are approaching each other. Wind blows along the road with velocity w . One of these vehicles blows a whistle of frequency f_1 . An observer in the other vehicle hears the frequency of the whistle to be f_2 . The speed of sound in still air is V . The correct statement(s) is (are) :-

[JEE Advanced-2013]

- (A) If the wind blows from the observer to the source, $f_2 > f_1$
 (B) If the wind blows from the source to the observer, $f_2 > f_1$
 (C) If the wind blows from observer to the source, $f_2 < f_1$
 (D) If the wind blows from the source to the observer, $f_2 < f_1$

28. One end of a taut string of length 3m along the x -axis is fixed at $x = 0$. The speed of the waves in the string is 100 ms^{-1} . The other end of the string is vibrating in the y direction so that stationary waves are set up in the string. The possible waveform (s) of these stationary waves is(are) :-

[JEE-Advance-2014]

(A) $y(t) = A \sin \frac{\pi x}{6} \cos \frac{50\pi t}{3}$

(B) $y(t) = A \sin \frac{\pi x}{3} \cos \frac{100\pi t}{3}$

(C) $y(t) = A \sin \frac{5\pi x}{6} \cos \frac{250\pi t}{3}$

(D) $y(t) = A \sin \frac{5\pi x}{2} \cos 250\pi t$

29. A student is performing an experiment using a resonance column and a tuning fork of frequency 244 s^{-1} . He is told that the air in the tube has been replaced by another gas (assume that the column remains filled with the gas). If the minimum height at which resonance occurs is $(0.350 \pm 0.005) \text{ m}$, the gas in the tube is

(Useful information : $\sqrt{167RT} = 640 \text{ J}^{1/2} \text{ mole}^{-1/2}$; $\sqrt{140RT} = 590 \text{ J}^{1/2} \text{ mole}^{-1/2}$. The molar masses M

in grams are given in the options. Take the values of $\sqrt{\frac{10}{M}}$ for each gas as given there.)

[JEE Advanced-2014]

(A) Neon $\left(M = 20, \sqrt{\frac{10}{20}} = \frac{7}{10} \right)$

(B) Nitrogen $\left(M = 28, \sqrt{\frac{10}{28}} = \frac{3}{5} \right)$

(C) Oxygen $\left(M = 32, \sqrt{\frac{10}{32}} = \frac{9}{16} \right)$

(D) Argon $\left(M = 36, \sqrt{\frac{10}{36}} = \frac{17}{32} \right)$

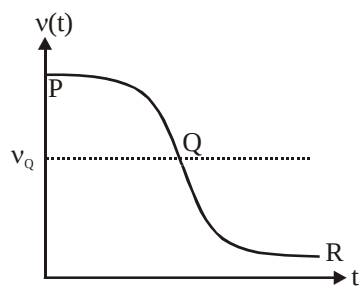
30. Four harmonic waves of equal frequencies and equal intensities I_0 have phase angles $0, \pi/3, 2\pi/3$ and π . When they are superposed, the intensity of the resulting wave is nI_0 . The value of n is.

[JEE Advanced-2015]

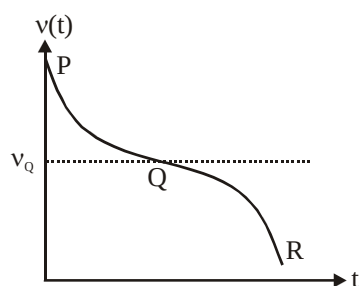
31. Two loudspeakers M and N are located 20m apart and emit sound at frequencies 118 Hz and 121 Hz, respectively. A car is initially at a point P, 1800 m away from the midpoint Q of the line MN and moves towards Q constantly at 60 km/hr along the perpendicular bisector of MN. It crosses Q and eventually reaches a point R, 1800 m away from Q. Let $v(t)$ represent the beat frequency measured by a person sitting in the car at time t . Let v_P, v_Q and v_R be the beat frequencies measured at locations P, Q and R, respectively. The speed of sound in air is 330 ms^{-1} . Which of the following statement(s) is(are) true regarding the sound heard by the person ?

[JEE Advanced-2016]

(A) The plot below represents schematically the variation of beat frequency with time



(B) The plot below represents schematically the variations of beat frequency with time



(C) The rate of change in beat frequency is maximum when the car passes through Q

(D) $v_P + v_R = 2v_Q$

REVIEW TEST

1. A steel wire has a mass of 5g/m and is under tension 450 N. Find the maximum average power that can be carried by the transverse wave in the wire if the amplitude is not to exceed 20% of the wavelength.
2. A 40 cm long wire having a mass 3.2 gm and area of cross-section 1 mm^2 is stretched between the supports 40.05 cm apart. In its fundamental mode it vibrates with a frequency $1000/64 \text{ Hz}$. Find the young's modulus of the wire.
3. In a mixture of gases, the average number of degrees of freedom per molecule is 6. The rms speed of the molecules of the gas is c . Find the velocity of sound in the gas.
4. A, B and C are three tuning forks. Frequency of A is 350 Hz. Beats produced by A and B are 5 per second and by B and C are 4 per second. When a wax is put on A beat frequency between A and B is 2 Hz and between A and C is 6 Hz. Then, find the frequency of B and C respectively.
5. The air column in a pipe closed at one end is made to vibrate in its second overtone by a tuning fork of frequency 440 Hz. The speed of sound in air is 330 ms^{-1} . End corrections may be neglected. Let P_0 denote the mean pressure at any point in the pipe & ΔP_0 the maximum amplitude of pressure variation.

[JEE '98]

(i) Find the length L of the air column.

(ii) What is the amplitude of pressure variation at the middle of the column?

(iii) What are the maximum & minimum pressures at the open end of the pipe.

(iv) What are the maximum & minimum pressures at the closed end of the pipe?

6. Two narrow cylindrical pipes A and B have the same length. Pipe A is open at both ends and is filled with a monoatomic gas of molar mass M_A . Pipe B is open at one end and closed at the other end, and is filled with a diatomic gas of molar mass M_B . Both gases are at the same temperature.

[JEE 2002]

- (a) If the frequency of the second harmonic of the fundamental mode in pipe A is equal to the frequency of the third harmonic of the fundamental mode in pipe B, determine the value of M_A/M_B .
- (b) Now the open end of pipe B is also closed (so that the pipe is closed at both ends). Find the ratio of the fundamental frequency in pipe A to that in pipe B.]

7. A plane sound wave of frequency f_0 and wavelength λ_0 travels horizontally toward the right. It strikes and is reflected from a large, rigid, vertical plane surface, perpendicular to the direction of propagation of the wave and moving towards the left with a speed v .

(a) How many positive wave crests strike the surface in a time interval t ?

(b) At the end of this time interval, how far to the left of the surface is the wave that was reflected at the beginning of the time interval?

(c) What is the wavelength of the reflected waves, in terms of λ_0 ?

(d) What is the frequency, in terms of f_0 ?

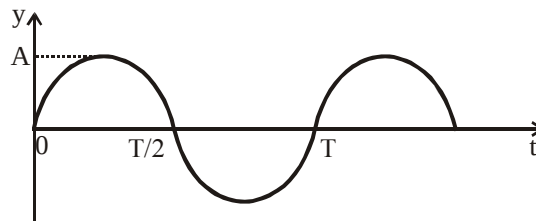
(e) A listener is at rest at the left of the moving surface. Describe the sensation of sound that he hears as a result of the combined effect of the incident and reflected wave trains.

8. Here given graph shows the motion of particle at $x = \frac{\lambda}{2}$ of a progressive wave. Then the equation of the wave if wave is going in +ve x-direction and if wave is going in -ve x-direction will be respectively.

(Here, $T = \frac{2\pi}{\omega}$)

(A) $y = A \sin(\omega t - kx + \pi)$, $y = A \sin(kx + \omega t + \pi)$

(B) $y = A \cos(\omega t - kx + \pi)$, $y = A \cos(\omega t + kx + \pi)$



(C) $y = A \sin(kx - \omega t + \pi)$, $y = A \sin(kx + \omega t - \pi)$

(D) $y = A \sin(kx - \omega t - \pi)$, $y = A \sin(kx + \omega t - \pi)$

9. A sinusoidal wave travelling in the positive direction of x on a stretched string has amplitude 2.0 cm, wavelength 1 m and wave velocity 5.0 m/s. At $x = 0$ and $t = 0$, it is given that displacement $y = 0$ and

$\frac{\partial y}{\partial t} < 0$. Express the wave function correctly in the form $y = f(x, t)$:-

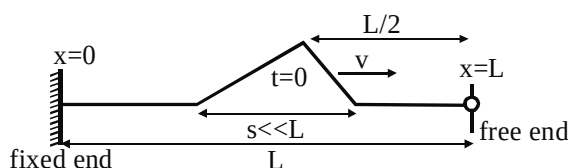
(A) $y = (0.02 \text{ m}) \sin 2\pi(x - 5t)$

(B) $y = (0.02 \text{ cm}) \cos 2\pi(x - 5t)$

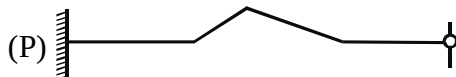
(C) $y = (0.02 \text{ m}) \sin 2\pi\left(x - 5t + \frac{1}{4}\right)$

(D) $y = (0.02 \text{ cm}) \cos 2\pi\left(x - 5t + \frac{1}{4}\right)$

10. A small pulse travelling with speed v in a string is shown at $t = 0$, moving towards free end. Which of these is not **CORRECTLY** matched.



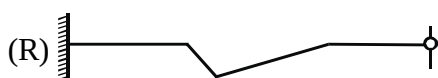
(i) $t = \frac{L}{v}$



(ii) $t = \frac{2L}{v}$



(iii) $t = \frac{3L}{v}$



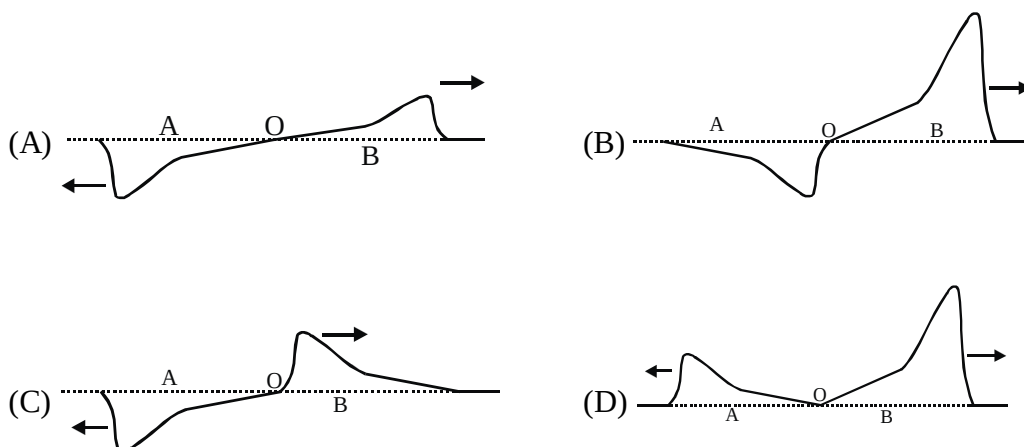
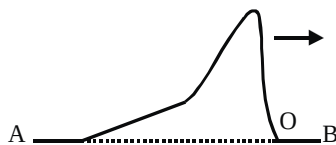
(A) (i)

(B) (ii)

(C) (iii)

(D) None of these

11. Two strings, A and B, of lengths $4L$ and L respectively and same mass M each, are tied together to form a knot 'O' and stretched under the same tension. A transverse wave pulse is sent along the composite string from the side A, as shown to the right. Which of the following diagrams correctly shows the reflected and transmitted wave pulses near the knot 'O'?



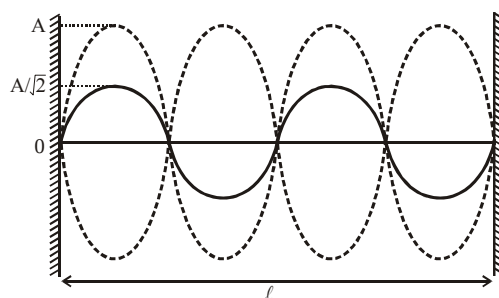
12. Here given snap shot at $t = \frac{T}{2}$ of a standing wave. Then the equations of the wave will be when particles are moving towards their extreme and when particles are moving towards the mean position respectively
(Here, $T = \frac{2\pi}{\omega}$).

(A) $y = A \sin \left(kx - \frac{\pi}{4} \right) \sin \left(\omega t - \frac{\pi}{4} \right)$, $y = A \sin kx \sin \left(\omega t + \frac{\pi}{4} \right)$

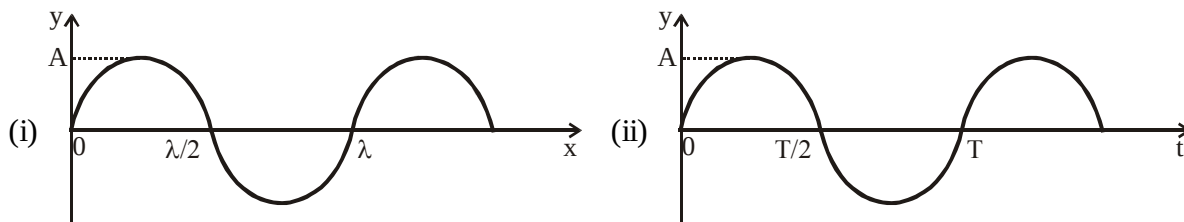
(B) $y = A \sin kx \sin \left(\omega t + \frac{\pi}{4} \right)$, $y = A \sin kx \sin \omega t$

(C) $y = A \sin kx \sin \left(\omega t - \frac{\pi}{4} \right)$, $y = A \sin kx \sin \left(\omega t + \frac{\pi}{4} \right)$

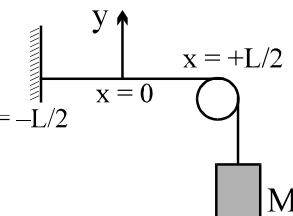
(D) $y = A \sin kx \cos \left(\omega t - \frac{\pi}{4} \right)$, $y = A \sin kx \sin \left(\omega t + \frac{\pi}{4} \right)$



13. Here given figure (i) shows snap shot at $t = T/4$ and figure (ii) shows motion of particle at $x = \lambda/4$. Then the possible equations of the wave will be (Here, $T = \frac{2\pi}{\omega}$) :-



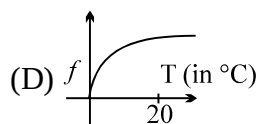
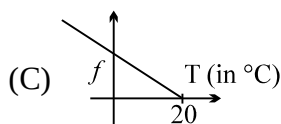
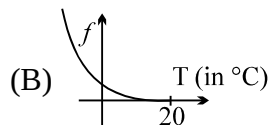
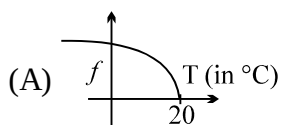
- (A) $y = A \sin \left(\omega t + kx - \frac{\pi}{2} \right)$
- (B) $y = A \sin \left(\omega t - kx + \frac{\pi}{2} \right)$
- (C) Both $y = A \cos \left(\omega t + kx - \frac{\pi}{2} \right)$ and $y = A \cos \left(\omega t - kx + \frac{\pi}{2} \right)$
- (D) Both $y = A \sin \left(\omega t + kx - \frac{\pi}{2} \right)$ and $y = A \sin \left(\omega t - kx + \frac{\pi}{2} \right)$
14. A string of length L is fixed at one end and under tension due to a mass $M = 25 \text{ kg}$ hanging from the other end, as shown in the sketch. Coordinate axes are chosen so that the string runs from $x = -L/2$ to $x = L/2$, and y is the transverse displacement. The string is vibrating at one of its resonant frequencies with displacement :



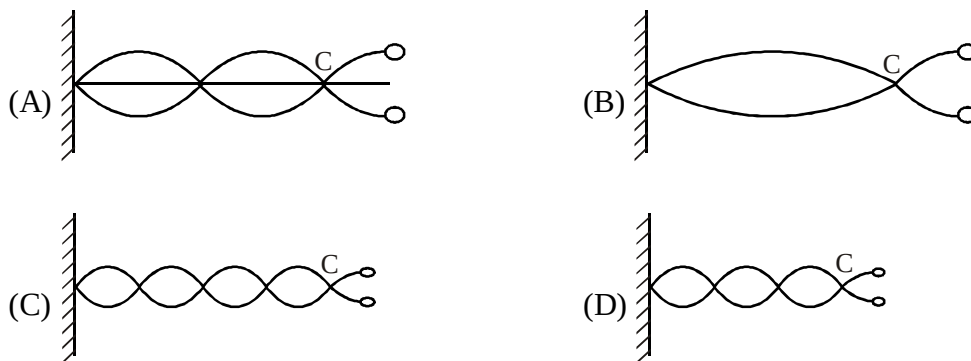
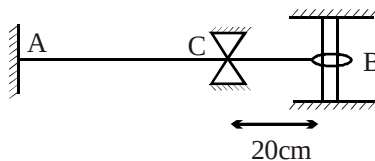
$$y(x,t) = 0.05 \cos(12.0x) \sin(360t) \quad \dots(1)$$

with x, y in meters and t in seconds. What are the three smallest possible values of L consistent with equation (1)?

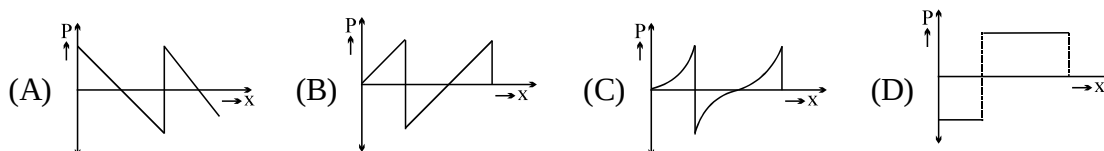
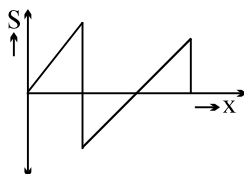
- (A) $\lambda, 2\lambda, 3\lambda$ (B) $\lambda/2, 3\lambda/2, 5\lambda/2$ (C) $\lambda/2, \lambda, 3\lambda/2$ (D) $\lambda/4, \lambda/2, 3\lambda/4$
15. A metal wire is clamped between two vertical walls. At 20°C the unstrained length of the wire is exactly equal to the separation between walls. If the temperature of the wire is decreased the graph between fundamental frequency (f) and temperature (T) of the wire is



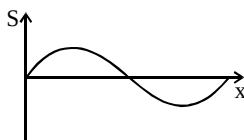
16. A 1m long wire having tension T is fixed at A and free at B. The point C, 20 cm from B is constrained to be stationary. What is shape of string for fundamental mode?



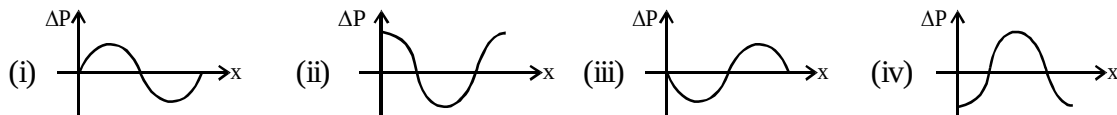
17. A sound wave is propagating through a medium whose displacement (s) vs x graph shown in the figure. Its pressure (P) vs x graph is



18. If a sound wave is travelling and snap shot at $t = 0$ is as shown in figure.

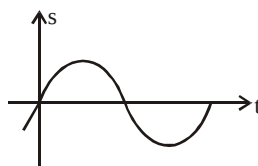


Choose snapshot of pressure variation.

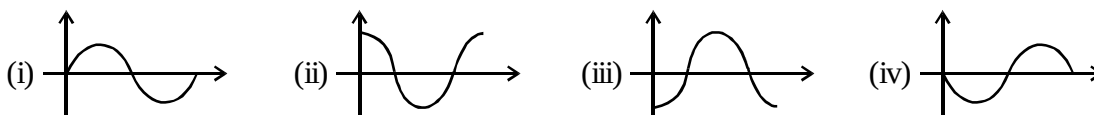


- (A) For wave travelling towards right or left (i) is correct.
(B) For wave travelling towards right graph (iv) and for wave travelling towards left graph (ii) is correct.
(C) For wave travelling towards right graph (i) and for wave travelling towards left graph (iii) is correct
(D) For wave travelling towards right or left (ii) is correct.

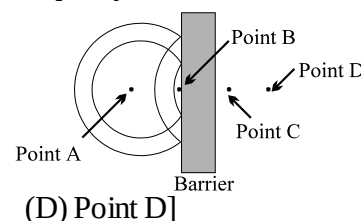
19. A sound wave is travelling towards left and its s-t graph is as shown for $x = 0$.



Choose **CORRECT** matching of the graphs :-



- (A) Graph (i) shows $P-t$ for $x = 0$ and graph (iv) shows $P-x$ at $t = 0$
 (B) Graph (iii) shows $P-t$ for $x = 0$ and graph (iii) shows $P-x$ at $t = 0$
 (C) Graph (iv) shows $P-t$ for $x = 0$ and graph (i) shows $P-x$ at $t = 0$
 (D) Graph (ii) shows $P-t$ for $x = 0$ and graph (ii) shows $P-x$ at $t = 0$
20. A plane transverse wave is propagating in a direction making an angle of 30° with positive x -axis in the x - y plane. Find phase difference between points $(0, 0, 0)$ and $(1, 1, 1)$. Wavelength of the wave is 1m :-
 (A) 2π rad (B) $(\sqrt{3}+1)\pi$ rad (C) $(\sqrt{2}+1)\pi$ rad (D) None
21. Spherical sound waves are emitted uniformly in all directions from a point source. The variation in sound level SL as a function of distance $r(> 0)$ from the source can be written as :-
 (A) $SL = -b \log r^a$ (B) $SL = a - b \log r$ (C) $SL = a - b(\log r)^2$ (D) $SL = a - \frac{b}{r^2}$
 where a and b are positive constant.
22. Two sound waves, having sinusoidal waveform but different wavelength and different amplitude, are said to have :-
 (A) the same pitch, but different quality (B) the same pitch, but different intensity
 (C) the same quantity, but different intensity (D) different intensity and quality
23. The diagram shows two water pulses which were originally produced at point A and are beginning to be reflected from a wide barrier as shown. Which of the points in the diagram would represent a virtual source for the reflected waves ?
 (A) Point A (B) Point B (C) Point C (D) Point D



24. In a resonance tube experiment, an 80 cm air column is in resonance with a tuning fork in first overtone. Which equation can represent correct pressure variation in the air column ($x = 0$ is the top point of the tube, neglect end correction, speed of sound = 320 m/sec) :-

- (A) $A \sin \frac{15\pi}{8} x \cos 600\pi t$ (B) $A \cos \frac{15\pi}{8} x \sin 600\pi t$
 (C) $A \cos \frac{15\pi}{8} x \sin 300\pi t$ (D) $A \sin \frac{15\pi}{8} x \sin 300\pi t$

25. If l_1 and l_2 are the lengths of air column for the first and second resonance when a tuning fork of frequency n is sounded on a resonance tube, then the distance of the displacement antinode from the top end of the resonance tube is:

(A) $2(l_2 - l_1)$ (B) $\frac{1}{2}(2l_1 - l_2)$ (C) $\frac{l_2 - 3l_1}{2}$ (D) $\frac{l_2 - l_1}{2}$

26. The superposition of two waves

$$y_1 = (0.006 \text{ cm}) \cos \left[2\pi \left(\frac{156}{s} t \right) + kx \right] \text{ and } y_2 = (0.004 \text{ cm}) \cos \left[2\pi \left(\frac{150}{s} t \right) + k'x \right]$$

at the location $x = 0$ in space results in

- (A) a pure tone of 156 Hz
(B) a tone of frequency 153 Hz as well as beats at beat frequency of 6 Hz in a 156 Hz tone.
(C) beats at a beat frequency of 6 Hz in a 153 Hz tone.
(D) a tone at a frequency of 156 Hz, as well as beats at a beat frequency of 6 Hz in a 153 Hz tone.
27. Consider a hypothetical wave pulse (at time $t = 0$) given by the following (y, x in meter)

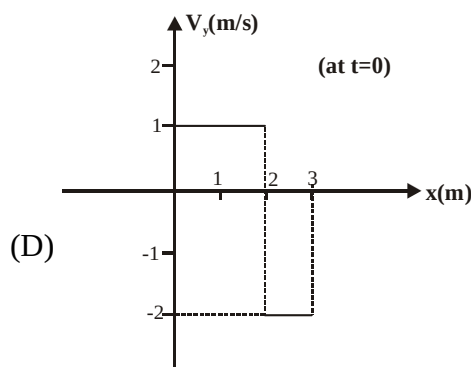
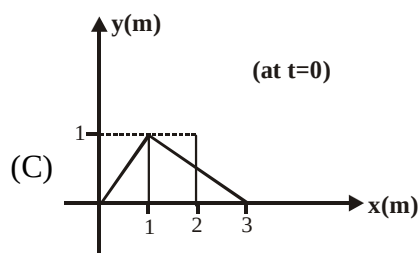
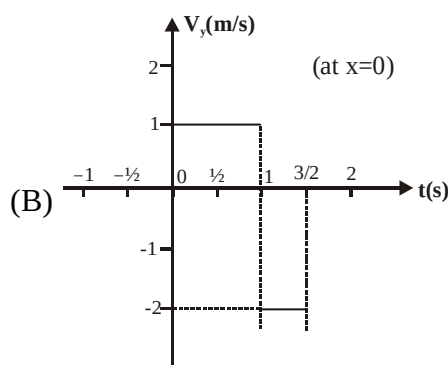
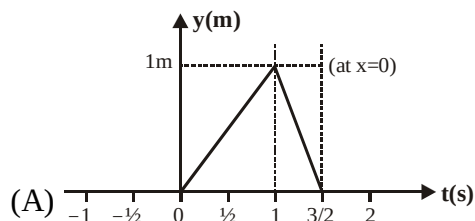
$$y = 0, x < 0$$

$$y = x/2, 2 > x \geq 0$$

$$y = 3 - x, 3 \geq x \geq 2$$

$$y = 0, x > 3$$

The pulse travels leftwards (negative x direction) at speed $v = 2$ m/s. Which of the following plots are correct? [V_y is the velocity of particle]



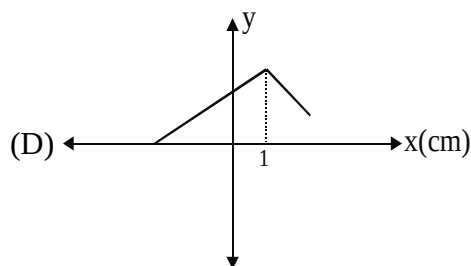
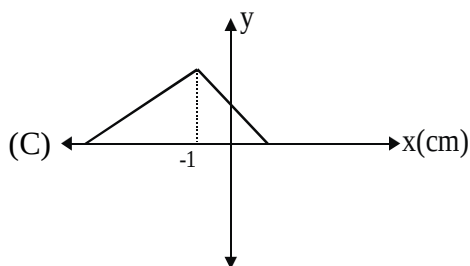
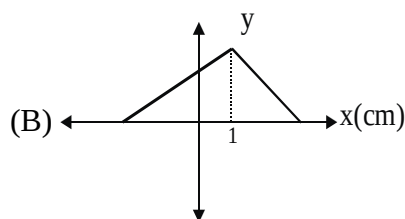
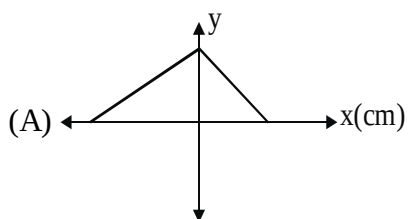
28. An string has resonant frequencies given by 1001 Hz and 2639 Hz.
 (A) If the string is fixed at one end only, 910 Hz can be a resonance frequency.
 (B) If the string is fixed at one end only, 1911 Hz can be a resonance frequency.
 (C) If the string is fixed at both the ends, 364 Hz can be one of the resonant frequency.
 (D) 1001 Hz is definitely not the fundamental frequency of the string.
29. A narrow steel rod of length 'L' is rigidly clamped at its mid-point and transverse standing waves of frequency 'f' are set up in it. The speed of transverse waves in the rod is 'c'. Then
 (A) The free ends of the rod must be Antinodes
 (B) The fundamental frequency 'f' of the rod is $c/(L)$
 (C) The second overtone frequency of the rod is $5c/(2L)$
 (D) 'f' can be any integral multiple of the fundamental frequency
30. A clamped string is oscillating in nth harmonic, then
 (A) total energy of oscillations will be proportional to n^2 times that of fundamental frequency
 (B) total energy of oscillations will be proportional to $(n-1)^2$ times that of fundamental frequency
 (C) average kinetic energy of the string over a complete oscillations is half of that of the total energy of the string.
 (D) none of these

Paragraph for Question Nos. 31 to 33

At a time $t=0$ a pulse moving in the +ve x-direction with velocity v in the shape given by the function between

$$\text{with } y = \begin{cases} \frac{x}{4} + 1 & -4 < x < 0 \\ -x + 1 & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}, \text{ where } x \text{ and } y \text{ are in cm.}$$

31. If velocity is 10 m/s, which of the graphs shows the shape and position of the string at a time $t = 1$ millise



32. The wave equation that explicitly gives the displacement of any point at any time is

$$(A) y = \begin{cases} \frac{1}{4}(x - Vt) + 1 & -4 < (x - Vt) < 0 \\ -(x - Vt) + 1 & 0 < (x - Vt) < 1 \\ 0 & \text{otherwise} \end{cases} \quad (B) y = \begin{cases} \frac{1}{4}(x + Vt) + 1 & -4 < (x + Vt) < 0 \\ -(x + Vt) + 1 & 0 < (x + Vt) < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$(C) y = \begin{cases} -\frac{1}{4}(x + Vt) + 1 & -4 < (x + Vt) < 0 \\ (x + Vt) + 1 & 0 < (x + Vt) < 1 \\ 0 & \text{otherwise} \end{cases} \quad (D) y = \begin{cases} -\frac{1}{4}(x - Vt) + 1 & -4 < (x - Vt) < 0 \\ (x - Vt) + 1 & 0 < (x - Vt) < 1 \\ 0 & \text{otherwise} \end{cases}$$

33. Velocity of a point at $x = 20$ m as a function of time is given as

$$(A) v = \begin{cases} \frac{1}{4} \text{ m/s} & 1.999 < t < 2 \text{ s} \\ -1 \text{ m/s} & 2 < t < 2.004 \text{ s} \\ 0 & \text{otherwise} \end{cases} \quad (B) v = \begin{cases} 1 \text{ m/s} & 1.999 < t < 2 \text{ s} \\ -\frac{1}{4} \text{ m/s} & 2 < t < 2.004 \text{ s} \\ 0 & \text{otherwise} \end{cases}$$

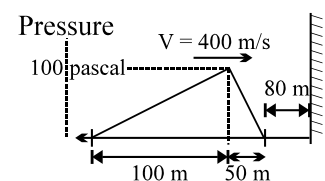
$$(C) v = \begin{cases} 10 \text{ m/s} & 1.999 < t < 2 \text{ s} \\ -\frac{5}{2} \text{ m/s} & 2 < t < 2.004 \text{ s} \\ 0 & \text{otherwise} \end{cases} \quad (D) v = \begin{cases} \frac{5}{2} \text{ m/s} & 1.999 < t < 2 \text{ s} \\ -10 \text{ m/s} & 2 < t < 2.004 \text{ s} \\ 0 & \text{otherwise} \end{cases}$$

34. The particle displacement of a travelling longitudinal wave is represented by $S = S(x, t)$. The midpoints of a compression zone and an adjacent rarefaction zone are represented by the letter 'C' and 'R'. Which of the following is true?

- (A) $|\partial S / \partial x|_C = |\partial S / \partial x|_R$
 (B) $|\partial S / \partial t|_C = |\partial S / \partial t|_R = 0$
 (C) $(\text{pressure})_C - (\text{pressure})_R = 2 |\partial S / \partial x|_C \times \text{Bulk modulus of air.}$
 (D) Particles of air are stationary mid-way between 'C' and 'R'.

Paragraph for Question No. 35 to 38

A plane pressure pulse triangular in shape approaches a rigid wall along normal at a speed of 400 m/s. At time $t = 0$, situation is shown in the figure. The peak pressure is 100 P. By the wall pulse gets reflected and pressure near the wall gets doubled. Height of the wall is 2m and width is also 2m. A detector on the wall records a minimum excess pressure of 16 pascal.



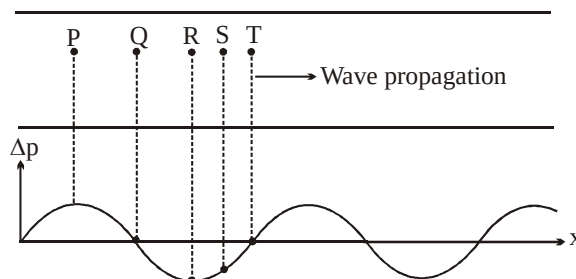
35. When for the first time detector will record the pressure pulse
 (A) 22×10^{-2} s (B) 21×10^{-2} s (C) 31.5×10^{-2} s (D) None of these
36. For how much time the detector will record the pulse
 (A) 37.5×10^{-2} s (B) 34.5×10^{-2} s (C) 31.5×10^{-2} s (D) None of these
37. What is the maximum force applied by the pulse on the wall
 (A) 400 N (B) 800 N (C) 200 N (D) None of these
38. Total impulse imparted by the pulse on the wall will be
 (A) 300 Ns (B) 150 Ns (C) 750 Ns (D) None of these

Paragraph for Question No. 39 to 41

A metallic rod of length 1m has one end free and other end rigidly clamped. Longitudinal stationary waves are set up in the rod in such a way that there are total six antinodes present along the rod. The amplitude of an antinode is 4×10^{-6} m. Young's modulus and density of the rod are 6.4×10^{10} N/m² and 4×10^3 Kg/m³ respectively. Consider the free end to be at origin and at $t=0$ particles at free end are at positive extreme.

39. The equation describing displacements of particles about their mean positions is
 (A) $s = 4 \times 10^{-6} \cos\left(\frac{11\pi}{2}x\right) \cos(22\pi \times 10^3 t)$ (B) $s = 4 \times 10^{-6} \cos\left(\frac{11\pi}{2}x\right) \sin(22\pi \times 10^3 t)$
 (C) $s = 4 \times 10^{-6} \cos(5\pi x) \cos(20\pi \times 10^3 t)$ (D) $s = 4 \times 10^{-6} \cos(5\pi x) \sin(20\pi \times 10^3 t)$
40. The equation describing stress developed in the rod is
 (A) $140.8\pi \times 10^4 \cos\left(\frac{11}{2}\pi x + \pi\right) \cos(22\pi \times 10^3 t)$
 (B) $140.8\pi \times 10^4 \sin\left(\frac{11}{2}\pi x + \pi\right) \cos(22\pi \times 10^3 t)$
 (C) $128\pi \times 10^4 \cos(5\pi x + \pi) \cos(20\pi \times 10^3 t)$
 (D) $128\pi \times 10^4 \sin(5\pi x + \pi) \cos(20\pi \times 10^3 t)$
41. The magnitude of strain at midpoint of the rod at $t = 1$ sec is
 (A) $11\sqrt{3}\pi \times 10^{-6}$ (B) $11\sqrt{2}\pi \times 10^{-6}$
 (C) $10\sqrt{3}\pi \times 10^{-6}$ (D) $10\sqrt{2}\pi \times 10^{-6}$

42. Sound is travelling in a long tube towards right and the graph of excess pressure variation versus position (at some instant) is given below.



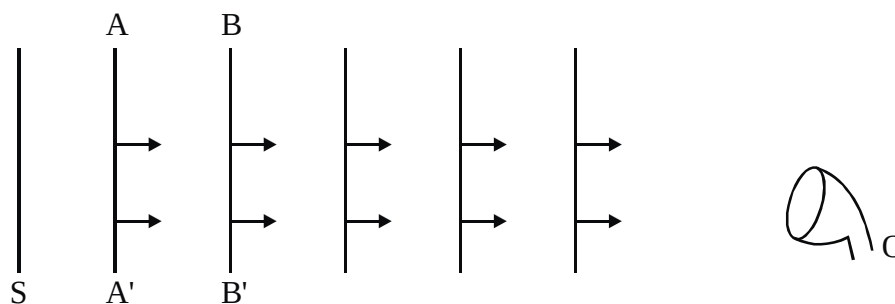
Match velocities in column-I with column-II. P, Q, R, S, T are medium particles inside the tube.

Column-I

Column-II

- | | |
|-------------------------------|-------|
| (A) velocity is towards right | (P) P |
| (B) velocity is towards left | (Q) Q |
| (C) velocity is zero | (R) R |
| (D) Speed is maximum | (S) S |
| | (T) T |

43. Consider a large plane diaphragm 'S' emitting sound and a detector 'O'. The diagram shows plane wavefronts for the sound wave travelling in air towards right when source, observer and medium are at rest. AA' and BB' are fixed imaginary planes. Column-I describes about the motion of source, observer or medium and column-II describes various effects. Match them correctly.



Column I

Column II

- | | |
|--|---|
| (A) Source starts moving towards right | (P) Distance between any two wavefronts will increase. |
| (B) Air starts moving towards right | (Q) Distance between any two wavefronts will decrease. |
| (C) Observer and source both move towards left with same speed. | (R) The time needed by sound to move from plane AA' to BB' will increase. |
| (D) Source and medium (air) both move towards right with same speed. | (S) The time needed by sound to move from plane AA' to BB' will decrease. |
| | (T) Frequency received by observer increases. |

ANSWER KEY**EXERCISE (O-2)**

- | | | | | | |
|---------------------------------------|--|---|----------|-----------|------------|
| 1. A,C,D | 2. A,B,C,D | 3. A,B | 4. B,C,D | 5. C,D | 6. A,B,C,D |
| 7. A, C, D | 8. A,B,C,D | 9. A,B | 10. B, C | 11. A,B,D | 12. A |
| 13. C | 14. B | 15. (A) (Q, S) ; (B) (P,R,T) ; (C) (Q, T) ; (D) (P,R,S) | | | |
| 16. (A) (P,Q) ; (B) (P,S) ; (C) (P,R) | | 17. (A) (P,Q) ; (B) (R,S) ; (C) (P,S) ; (D) (Q,R) | | | |
| 18. A,B,C,D | 19. B | 20. B,D | 21. B | 22. A | 23. B |
| 24. A | 25. A | 26. B | 27. C | | |
| 28. D | 29. (A) (P,Q,S,T) ; (B) (R) ; (C) (Q, T) ; (D) (R) | | | | |

EXERCISE (J-A)

- | | | | |
|-------------------------------------|------------------|---|--------------|
| 1. $E = \frac{A^2 \pi^2 T}{4l}$ | 2. C | 3. $y = (10 \text{ cm}) \sin \left(30 t \pm \frac{3}{2} x + f \right)$ | |
| 4. A | 5. A | 6. A | 7. 5 |
| 8. 5 | 9. (A) | 10. 336 m/s | 11. (A) |
| 12. (C) | 13. (D) | 14. $V_s = 30 \text{ m/s}$ | 15. (A) |
| 16. (C) | 17. (A) | 18. (A) | 19. (B) |
| 20. (A) | 21. (A) | 22. (A) | |
| 23. (A) P,T (B) P,S (C) Q,S (D) Q,R | | 24. (B,D) | 25. (B) |
| 26. B,C | 27. A,B | 28. Ans. (A,C,D) | 29. Ans. (D) |
| 30. Ans. 3 | 31. Ans. (A,C,D) | | |

REVIEW TEST

- | | | | |
|---------------------------------|---------------------------------|---------|-----------------------|
| 1. $10.8 \times 10^4 \text{ W}$ | 2. $1 \times 10^9 \text{ Nm}^2$ | 3. 2c/3 | 4. 345, 341 or 349 Hz |
|---------------------------------|---------------------------------|---------|-----------------------|
5. (i) $L = \frac{15}{16} \text{ m}$, (ii) $\frac{\Delta P_0}{\sqrt{2}}$, (iii) $P_{\max} = P_{\min} = P_0$ (iv) $P_{\max} = P_0 + \Delta P_0$, $P_{\min} = P_0 - \Delta P_0$
- | | | |
|---------------------------------|-------|-------------|
| 6. (a) 2.116, (b) $\frac{3}{4}$ | 21. 5 | 22. $11f/9$ |
|---------------------------------|-------|-------------|
7. (a) $\left(\frac{v + \lambda_0 f_0}{\lambda_0} \right) t$ (b) $(\lambda_0 f_0 - v) t$ (c) $\lambda_0 \left(\frac{\lambda_0 f_0 - v}{\lambda_0 f_0 + v} \right)$ (d) $f_0 \left(\frac{\lambda_0 f_0 + v}{\lambda_0 f_0 - v} \right)$ (e) $\frac{2v f_0}{\lambda_0 f_0 - v}$

8. (A) 9. (A) 10. (B) 11. (A) 12. (C) 13. (D) 14. (B) 15. (A)
 16. (A) 17. (D) 18. (B) 19. (B) 20. (B) 21. (B) 22. (C) 23. (C)
 24. (A) 25. (C) 26. (D) 27. (A,B,D) 28. (B,C,D) 29. (A,C) 30. (A,C) 31. (B)
 32. (A) 33. (C) 34. (A,C,D) 35. (B) 36. (B) 37. (B) 38. (B) 39. (A)
 40. (B) 41. (B) 42. (A) (P) ; (B) (R, S) ; (C) (Q,T) ; (D) (P,R)
 43. (A) (QT) ; (B) (PS) ; (C) (P) ; (D) (ST)

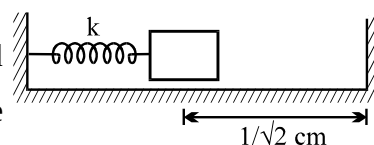
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SIMPLE HARMONIC MOTION

EXERCISE (S-2)

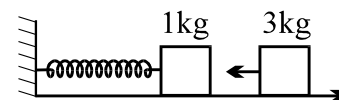
1. A body undergoing SHM about the origin has its equation is given by $x = 0.2 \cos 5\pi t$. Find its average speed from $t = 0$ to $t = 0.7$ sec.
2. A body of mass 1 kg is suspended from a weightless spring having force constant 600 N/m. Another body of mass 0.5 kg moving vertically upwards hits the suspended body with a velocity of 3.0 m/s and get embedded in it. Find the frequency of oscillations and amplitude of motion.

3. A block of mass 0.9 kg attached to a spring of force constant k is lying on a frictionless floor. The spring is compressed to $\sqrt{2}$ cm and the block is at a distance $1/\sqrt{2}$ cm from the wall as shown in the figure. When the block is released, it makes elastic collision with the wall and its period of motion is 0.2 sec. Find the approximate value of k .



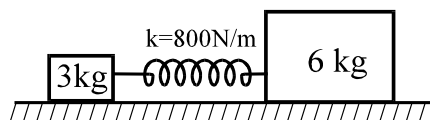
4. One end of an ideal spring is fixed to a wall at origin O and the axis of spring is parallel to x -axis. A block of mass $m = 1$ kg is attached to free end of the spring and it is performing SHM. Equation of position of block in coordinate system shown is $x = 10 + 3\sin 10t$, t is in second and x in cm. Another block of mass $M = 3$ kg, moving towards the origin with velocity 30 cm/s collides with the block performing SHM at $t = 0$ and gets struck to it, calculate :

- (i) new amplitude of oscillations.
- (ii) new equation for position of the combined body.
- (iii) loss of energy during collision. Neglect friction.



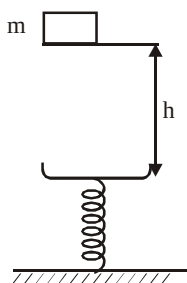
5. The system shown in the figure can move on a smooth surface. The spring is initially compressed by 6 cm and then released. Find

- (a) time period;
- (b) amplitude of 3 kg block and
- (c) maximum momentum of 6 kg block

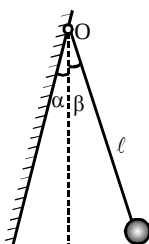


6. A mass m , which is attached to a spring with spring constant k , oscillates on a horizontal table, with amplitude A . At an instant when the spring is stretched by $\sqrt{3}A/2$, a second mass m is dropped vertically onto the original mass and immediately sticks to it. What is the amplitude of the resulting motion?

7. A body of mass m fell from a height h onto the pan of a spring balance. The masses of the pan and the spring are negligible, the stiffness of the latter is x . Having stuck to the pan, the body starts performing harmonic oscillations in the vertical direction. Find the amplitude and the energy of these oscillations.



8. A ball is suspended by a thread of length ℓ at the point O on the wall, forming a small angle α with the vertical as shown in figure. Then the thread with the ball was deviated through a small angle β ($\beta > \alpha$) and set free. Assuming the collision of the ball against the wall to be perfectly elastic, find the oscillation period of such a pendulum.



EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. A particle executes SHM on a straight line path. The amplitude of oscillation is 2 cm. When the displacement of the particle from the mean position is 1 cm, the numerical value of magnitude of acceleration is equal to the numerical value of magnitude of velocity. The frequency of SHM (in second^{-1}) is:

(A) $2\pi\sqrt{3}$ (B) $\frac{2\pi}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2\pi}$ (D) $\frac{1}{2\pi\sqrt{3}}$

2. A particle executes SHM with time period T and amplitude A. The maximum possible average velocity in time $\frac{T}{4}$ is

(A) $\frac{2A}{T}$ (B) $\frac{4A}{T}$ (C) $\frac{8A}{T}$ (D) $\frac{4\sqrt{2}A}{T}$

3. A particle moves along the x-axis according to : $x = A.[1 + \sin \omega t]$. What distance does it travel between $t = 0$ and $t = 2.5\pi/\omega$?

(A) 4A (B) 6A (C) 5A (D) None

4. Two particles are in SHM in a straight line about same equilibrium position. Amplitude A and time period T of both the particles are equal. At time $t=0$, one particle is at displacement $y_1 = +A$ and the other at $y_2 = -A/2$, and they are approaching towards each other. After what time they cross each other ?

(A) $T/3$ (B) $T/4$ (C) $5T/6$ (D) $T/6$

5. Vertical displacement of a plank with a body of mass 'm' on it is varying according to law $y = \sin \omega t + \sqrt{3} \cos \omega t$. The minimum value of ω for which the mass just breaks off the plank and the moment it occurs first after $t = 0$ are given by: (y is positive vertically upwards)

(A) $\sqrt{\frac{g}{2}}, \frac{\sqrt{2}}{6} \frac{\pi}{\sqrt{g}}$ (B) $\frac{g}{\sqrt{2}}, \frac{2}{3} \sqrt{\frac{\pi}{g}}$ (C) $\sqrt{\frac{g}{2}}, \frac{\pi}{3} \sqrt{\frac{2}{g}}$ (D) $\sqrt{2g}, \sqrt{\frac{2\pi}{3g}}$

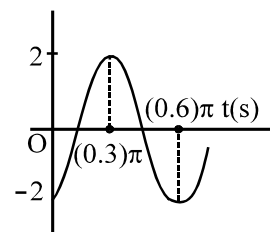
6. Part of a simple harmonic motion is graphed in the figure, where y is the displacement from the mean position. The correct equation describing this S.H.M is

(A) $y = 4 \cos (0.6t)$

(B) $y = 2 \sin \left(\frac{10}{3}t - \frac{\pi}{2} \right)$

(C) $y = 4 \sin \left(\frac{10}{3}t + \frac{\pi}{2} \right)$

(D) $y = 2 \cos \left(\frac{10}{3}t + \frac{\pi}{2} \right)$

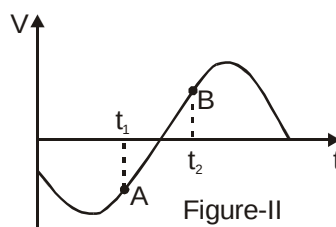
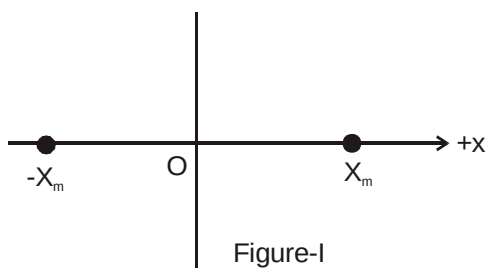


7. A particle executing a simple harmonic motion of period 2s. When it is at its extreme displacement from its mean position, it receives an additional energy equal to what it had in its mean position. Due to this, in its subsequent motion,
- (A) its amplitude will change and become equal to $\sqrt{2}$ times its previous amplitude
 (B) its periodic time will become doubled i.e. 4s
 (C) its potential energy will be decreased
 (D) it will continue to execute simple harmonic motion of the same amplitude and period as before receiving the additional energy.
8. A particle is executing SHM of amplitude A, about the mean position $x = 0$. Which of the following cannot be a possible phase difference between the positions of the particle at $x = +A/2$ and $x = -A/\sqrt{2}$.

- (A) 75° (B) 165° (C) 135° (D) 195°

MULTIPLE CORRECT TYPE QUESTIONS

1. A necessary and sufficient condition for SHM is
- (A) constant period.
 (B) constant acceleration.
 (C) proportionality between acceleration and displacement from equilibrium position.
 (D) proportionality between the restoring force and displacement from equilibrium position.
2. For a SHM with given angular frequency, two arbitrary initial conditions are necessary and sufficient to determine the motion completely. These initial conditions may be-
- (A) Amplitude and initial phase
 (B) Amplitude and total energy of oscillation
 (C) Initial phase and total energy of oscillation
 (D) Initial position and initial velocity
3. A particle is executing SHM between points $-X_m$ and X_m , as shown in figure-I. The velocity $V(t)$ of the particle is partially graphed and shown in figure-II. Two points A and B corresponding to time t_1 and time t_2 respectively are marked on the $V(t)$ curve.



- (A) At time t_1 , it is going towards X_m .
 (B) At time t_1 , its speed is decreasing.
 (C) At time t_2 , its position lies in between $-X_m$ and O.
 (D) The phase difference $\Delta\phi$ between points A and B must be expressed as $90^\circ < \Delta\phi < 180^\circ$.

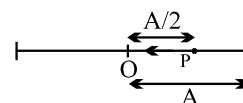
4. The amplitude of a particle executing SHM about O is 10 cm. Then:
- (A) When the K.E. is 0.64 of its max. K.E. its displacement is 6cm from O.
- (B) When the displacement is 5 cm from O its K.E. is 0.75 of its max.P.E.
- (C) Its total energy at any point is equal to its maximum K.E.
- (D) Its velocity is half the maximum velocity when its displacement is half the maximum displacement.
5. A particle starts from a point P at a distance of $A/2$ from the mean position O & travels towards left as shown in the figure. If the time period of SHM, executed about O is T and amplitude A then the equation of motion of particle is :

(A) $x = A \sin \left(\frac{2\pi}{T} t + \frac{\pi}{6} \right)$

(B) $x = A \sin \left(\frac{2\pi}{T} t + \frac{5\pi}{6} \right)$

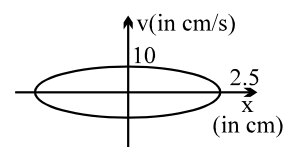
(C) $x = A \cos \left(\frac{2\pi}{T} t + \frac{\pi}{6} \right)$

(D) $x = A \cos \left(\frac{2\pi}{T} t + \frac{\pi}{3} \right)$



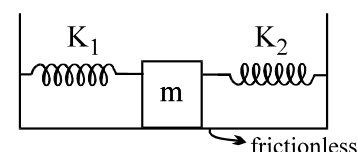
6. A spring has natural length 40 cm and spring constant 500 N/m. A block of mass 1 kg is attached at one end of the spring and other end of the spring is attached to ceiling. The block released from the position, where the spring has length 45 cm.
- (A) the block will perform SHM of amplitude 5 cm.
- (B) the block will have maximum velocity $30\sqrt{5}$ cm/sec.
- (C) the block will have maximum acceleration 15 m/s^2
- (D) the minimum potential energy of the spring will be zero.
7. The figure shows a graph between velocity and displacement (from mean position) of a particle performing SHM:

- (A) the time period of the particle is 1.57s
- (B) the maximum acceleration will be 40 cm/s^2
- (C) the velocity of particle is $2\sqrt{21}$ cm/s when it is at a distance 1 cm from the mean position.
- (D) none of these



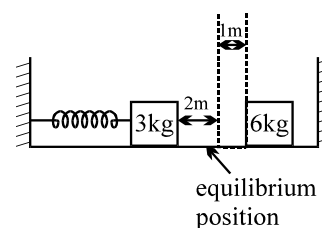
8. Two springs with negligible masses and force constant of $K_1 = 200 \text{ Nm}^{-1}$ and $K_2 = 160 \text{ Nm}^{-1}$ are attached to the block of mass $m = 10 \text{ kg}$ as shown in the figure. Initially the block is at rest, at the equilibrium position in which both springs are neither stretched nor compressed. At time $t = 0$, a sharp impulse of 50 Ns is given to the block with a hammer.

- (A) Period of oscillations for the mass m is $\frac{\pi}{3}$ s.
- (B) Maximum velocity of the mass m during its oscillation is 5 ms^{-1} .
- (C) Data are insufficient to determine maximum velocity.
- (D) Amplitude of oscillation is 0.42 m.



9. A mass of 0.2 kg is attached to the lower end of a massless spring of force-constant 200 N/m, the upper end of which is fixed to a rigid support. Which of the following statements is/are true?
- (A) In equilibrium, the spring will be stretched by 1 cm.
- (B) If the mass is raised till the spring is unstretched state and then released, it will go down by 2 cm before moving upwards.
- (C) The frequency of oscillation will be nearly 5 Hz.
- (D) If the system is taken to the moon, the frequency of oscillation will be the same as on the earth.
10. The potential energy of a particle of mass 0.1 kg, moving along x-axis, is given by $U = 5x(x-4)$ J where x is in metres. It can be concluded that
- (A) the particle is acted upon by a constant force.
- (B) the speed of the particle is maximum at $x = 2$ m
- (C) the particle executes simple harmonic motion
- (D) the period of oscillation of the particle is $\pi/5$ s.
11. Two blocks of masses 3 kg and 6 kg rest on a horizontal smooth surface.

The 3 kg block is attached to a spring with a force constant $k = 900 \text{ Nm}^{-1}$ which is compressed 2 m from beyond the equilibrium position. The 6 kg mass is at rest at 1 m from mean position. 3 kg mass strikes the 6 kg mass and the two stick together.



- (A) velocity of the combined masses immediately after the collision is 10 ms^{-1}
- (B) velocity of the combined masses immediately after the collision is 5 ms^{-1}
- (C) Amplitude of the resulting oscillation is $\sqrt{2}$ m
- (D) Amplitude of the resulting oscillation is $\sqrt{5/2}$ m.
12. The equation of motion for an oscillating particle is given by $x = 3\sin(4\pi t) + 4\cos(4\pi t)$, where x is in mm and t is in second
- (A) The motion is simple harmonic
- (B) The period of oscillation is 0.5 s
- (C) The amplitude of oscillation is 5 mm
- (D) The particle starts its motion from the equilibrium

EXERCISE (JA)

ALL ANSWERS OF THE QUESTIONS JEE-2007 ONWARDS ARE AS GIVEN ON IIT-JEE WEBSITE.

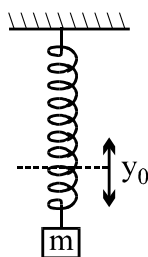
1. A simple pendulum has time period T_1 . When the point of suspension moves vertically up according to the equation $y = kt^2$ where $k = 1 \text{ m/s}^2$ and t is time then the time period of the pendulum is T_2 then $\left(\frac{T_1}{T_2}\right)^2$ is :-

[IIT JEE (Scr) 2005]

- (A) $\frac{5}{6}$ (B) $\frac{11}{10}$ (C) $\frac{6}{5}$ (D) $\frac{5}{4}$

2. A small body attached to one end of a vertically hanging spring is performing SHM about its mean position with angular frequency ω and amplitude a . If at a height y from the mean position the body gets detached from the spring, calculate the value of y so that the height H attained by the mass is maximum. The body does not interact with the spring during its subsequent motion after detachment. ($a\omega^2 > g$).

[IIT JEE 2005]



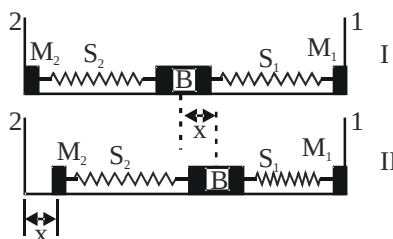
3. Function $x = A \sin^2 \omega t + B \cos^2 \omega t + C \sin \omega t \cos \omega t$ represents SHM :- [IIT JEE 2006]

- (A) for any value of A , B and C (except $C = 0$) (B) if $A = -B$; $C = 2B$, amplitude = $|B\sqrt{2}|$
(C) if $A = B$; $C = 0$ (D) if $A = B$; $C = 2B$, amplitude = $|B|$

4. A block (B) is attached to two unstretched springs S_1 and S_2 with spring constants k and $4k$, respectively (see figure 1). The other ends are attached to identical supports M_1 and M_2 not attached to the walls. The springs and supports have negligible mass. There is no friction anywhere. The block B is displaced towards wall 1 by a small distance x (figure II) and released. The block returns and moves a maximum distance y towards wall 2. Displacements x and y are measured with respect to the equilibrium position of the block

B. The ratio $\frac{y}{x}$ is :-

[IIT JEE 2008]

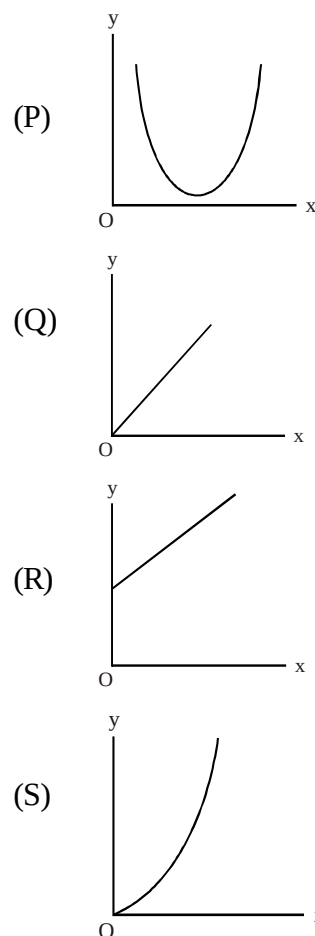


- (A) 4 (B) 2 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

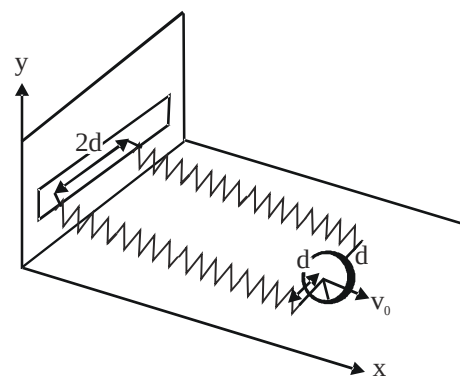
5. **Column-I** gives a list of possible set of parameters measured in some experiments. The variations of the parameters in the form of graphs are shown in **Column-II**. Match the set of parameters given in **Column-I** with the graphs given in **Column-II**. [IIT JEE 2008]

Column-I

- (A) Potential energy of a simple pendulum (y axis) as a function of displacement (x axis)
- (B) Displacement (y axis) as a function of time (x axis) for an one dimensional motion at zero or constant acceleration when the body is moving along the positive x-direction.
- (C) Range of a projectile (y axis) as a function of its velocity (x axis) when projected at a fixed angle.
- (D) The square of the time period (y axis) of a simple pendulum as a function of its length (x axis)

Column-II**Comprehension (6 to 8)**

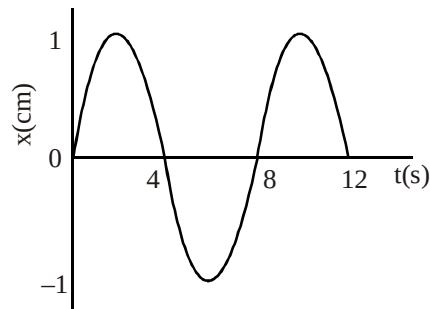
A uniform thin cylindrical disk of mass M and radius R is attached to two identical massless springs of spring constant k which are fixed to the wall as shown in the figure. The springs are attached to the axle of the disk symmetrically on either side at a distance d from its centre. The axle is massless and both the springs and the axle are in a horizontal plane. The unstretched length of each spring is L . The disk is initially at its equilibrium position with its centre of mass (CM) at a distance L from the wall. The disk rolls without slipping with velocity $\vec{v}_0 = v_0 \hat{i}$. The coefficient of friction is μ . [IIT JEE 2008]



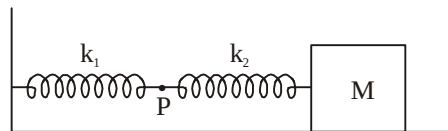
6. The net external force acting on the disk when its centre of mass is at displacement x with respect to its equilibrium position, is :

- (A) $-kx$ (B) $-2kx$ (C) $-\frac{2kx}{3}$ (D) $-\frac{4kx}{3}$

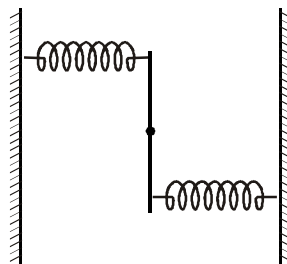
7. The centre of mass of the disk undergoes simple harmonic motion with angular frequency ω , equal to
 (A) $\sqrt{\frac{k}{M}}$ (B) $\sqrt{\frac{2k}{M}}$ (C) $\sqrt{\frac{2k}{3M}}$ (D) $\sqrt{\frac{4k}{3M}}$
8. The maximum value of v_0 for which the disk will roll without slipping, is :
 (A) $\mu g \sqrt{\frac{M}{k}}$ (B) $\mu g \sqrt{\frac{M}{2k}}$ (C) $\mu g \sqrt{\frac{3M}{k}}$ (D) $\mu g \sqrt{\frac{5M}{2k}}$
9. The x-t graph of a particle undergoing simple harmonic motion is shown below. The acceleration of the particle at $t = 4/3$ s is
[IIT JEE 2009]



- (A) $\frac{\sqrt{3}}{32} \pi^2 \text{ cm/s}^2$ (B) $-\frac{\pi^2}{32} \text{ cm/s}^2$ (C) $\frac{\pi^2}{32} \text{ cm/s}^2$ (D) $-\frac{\sqrt{3}}{32} \pi^2 \text{ cm/s}^2$
10. The mass M shown in the figure oscillates in simple harmonic motion with amplitude A . The amplitude of the point P is
[IIT JEE 2009]



- (A) $\frac{k_1 A}{k_2}$ (B) $\frac{k_2 A}{k_1}$ (C) $\frac{k_1 A}{k_1 + k_2}$ (D) $\frac{k_2 A}{k_1 + k_2}$
11. A uniform rod of length L and mass M is pivoted at the centre. Its two ends are attached to two springs of equal spring constants k . The springs are fixed to rigid supports as shown in the figure, and the rod is free to oscillate in the horizontal plane. The rod is gently pushed through a small angle θ in one direction and released. The frequency of oscillation is :-
[IIT JEE 2009]

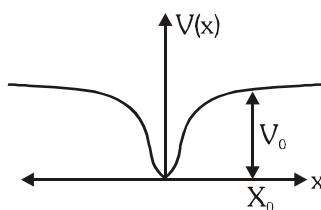


- (A) $\frac{1}{2\pi} \sqrt{\frac{2k}{M}}$ (B) $\frac{1}{2\pi} \sqrt{\frac{k}{M}}$ (C) $\frac{1}{2\pi} \sqrt{\frac{6k}{M}}$ (D) $\frac{1}{2\pi} \sqrt{\frac{24k}{M}}$

Comprehension (12 to 14)

When a particle of mass m moves on the x -axis in a potential of the form $V(x) = kx^2$, it performs simple harmonic motion. The corresponding time period is proportional to $\sqrt{\frac{m}{k}}$, as can be seen easily using dimensional analysis. However, the motion of a particle can be periodic even when its potential energy increases on both sides of $x = 0$ in a way different from kx^2 and its total energy is such that the particle does not escape to infinity. Consider a particle of mass m moving on the x -axis. Its potential energy is $V(x) = \alpha x^4$ ($\alpha > 0$) for $|x|$ near the origin and becomes a constant equal to V_0 for $|x| \geq X_0$ (see figure)

[IIT-JEE 2010]



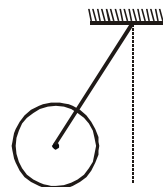
12. If the total energy of the particle is E , it will perform periodic motion only if :-
 (A) $E < 0$ (B) $E > 0$ (C) $V_0 > E > 0$ (D) $E > V_0$
13. For periodic motion of small amplitude A , the time period T of this particle is proportional to :-
 (A) $A\sqrt{\frac{m}{\alpha}}$ (B) $\frac{1}{A}\sqrt{\frac{m}{\alpha}}$ (C) $A\sqrt{\frac{\alpha}{m}}$ (D) $\frac{1}{A}\sqrt{\frac{\alpha}{m}}$
14. The acceleration of this particle for $|x| > X_0$ is :-
 (A) proportional to V_0 (B) proportional to $\frac{V_0}{mX_0}$
 (C) proportional to $\sqrt{\frac{V_0}{mX_0}}$ (D) Zero
15. A point mass is subjected to two simultaneous sinusoidal displacements in x -direction, $x_1(t) = A \sin \omega t$ and $x_2(t) = A \sin\left(\omega t + \frac{2\pi}{3}\right)$. Adding a third sinusoidal displacement $x_3(t) = B \sin(\omega t + \phi)$ brings the mass to a complete rest. The values of B and ϕ are :-

[IIT-JEE 2011]

- (A) $\sqrt{2}A, \frac{3\pi}{4}$ (B) $A, \frac{4\pi}{3}$ (C) $\sqrt{3}A, \frac{5\pi}{6}$ (D) $A, \frac{\pi}{3}$

16. A metal rod of length 'L' and mass 'm' is pivoted at one end. A thin disk of mass 'M' and radius 'R' (< L) is attached at its center to the free end of the rod. Consider two ways the disk is attached: (case A). The disk is not free to rotate about its center and (case B) the disk is free to rotate about its center. The rod-disk system performs SHM in vertical plane after being released from the same displaced position. Which of the following statement(s) is(are) true? [IIT-JEE 2011]

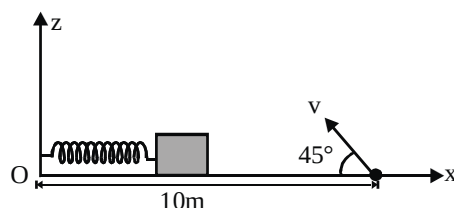
- (A) Restoring torque in case A = Restoring torque in case B
(B) Restoring torque in case A < Restoring torque in case B
(C) Angular frequency for case A > Angular frequency for case B
(D) Angular frequency for case A < Angular frequency for case B



17. A small block is connected to one end of a massless spring of un-stretched length 4.9 m. The other end of the spring (see the figure) is fixed. The system lies on a horizontal frictionless surface. The block is stretched by 0.2 m and released from rest at $t = 0$. It then executes simple harmonic motion with

angular frequency $\omega = \frac{\pi}{3}$ rad/s. Simultaneously at $t = 0$, a small pebble is projected with speed v from

point P at an angle of 45° as shown in the figure. Point P is at a horizontal distance of 10m from O. If the pebble hits the block at $t = 1$ s, the value of v is:- (take $g = 10 \text{ m/s}^2$) [IIT-JEE 2012]



- (A) $\sqrt{50}$ m/s (B) $\sqrt{51}$ m/s (C) $\sqrt{52}$ m/s (D) $\sqrt{53}$ m/s

18. A particle of mass m is attached to one end of a mass-less spring of force constant k , lying on a frictionless horizontal plane. The other end of the spring is fixed. The particle starts moving horizontally from its equilibrium position at time $t = 0$ with an initial velocity u_0 . When the speed of the particle is $0.5 u_0$, it collides elastically with a rigid wall. After this collision :- [JEE-Advanced-2013]

- (A) the speed of the particle when it returns to its equilibrium position is u_0
(B) the time at which the particle passes through the equilibrium position for the first time is

$$t = \pi \sqrt{\frac{m}{k}}$$

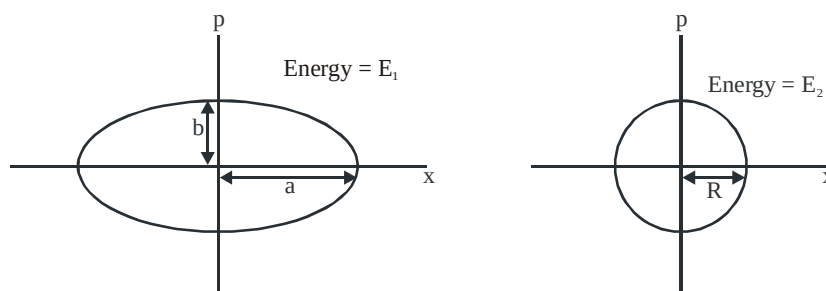
- (C) the time at which the maximum compression of the spring occurs is $t = \frac{4\pi}{3} \sqrt{\frac{m}{k}}$

- (D) the time at which the particle passes through the equilibrium position for the second time is

$$t = \frac{5\pi}{3} \sqrt{\frac{m}{k}}$$

19. Two independent harmonic oscillators of equal mass are oscillating about the origin with angular frequencies ω_1 and ω_2 and have total energies E_1 and E_2 , respectively. The variations of their momenta p with positions x are shown in the figures. If $\frac{a}{b} = n^2$ and $\frac{a}{R} = n$, then the correct equation (s) is (are)

[JEE-Advanced-2015]



- (A) $E_1 \omega_1 = E_2 \omega_2$ (B) $\frac{\omega_2}{\omega_1} = n^2$ (C) $\omega_1 \omega_2 = n^2$ (D) $\frac{E_1}{\omega_1} = \frac{E_2}{\omega_2}$
20. A particle of unit mass is moving along the x -axis under the influence of a force and its total energy is conserved. Four possible forms of the potential energy of the particle are given in column I (a and U_0 are constants). Match the potential energies in column I to the corresponding statement(s) in column-II

[JEE-Advanced-2015]

Column-I**Column-II**

(A) $U_1(x) = \frac{U_0}{2} \left[1 - \left(\frac{x}{a} \right)^2 \right]^2$

(P) The force acting on the particle is zero at $x = a$.

(B) $U_2(x) = \frac{U_0}{2} \left(\frac{x}{a} \right)^2$

(Q) The force acting on the particle is zero at $x = 0$.

(C) $U_3(x) = \frac{U_0}{2} \left(\frac{x}{a} \right)^2 \exp \left[- \left(\frac{x}{a} \right)^2 \right]$

(R) The force acting on the particle is zero at $x = -a$.

(D) $U_4(x) = \frac{U_0}{2} \left[\frac{x}{a} - \frac{1}{3} \left(\frac{x}{a} \right)^3 \right]$

(S) The particle experiences an attractive force towards $x = 0$ in the region $|x| < a$.

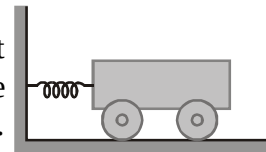
(T) The particle with total energy $\frac{U_0}{4}$ can oscillate about the point $x = -a$.

21. A block with mass M is connected by a massless spring with stiffness constant k to a rigid wall and moves without friction on a horizontal surface. The block oscillates with small amplitude A about an equilibrium position x_0 . Consider two cases : (i) when the block is at x_0 ; and (ii) when the block is at $x = x_0 + A$. In both the cases, a particle with mass m ($< M$) is softly placed on the block after which they stick to each other. Which of the following statement(s) is(are) true about the motion after the mass m is placed on the mass M ? **[JEE-Advanced-2016]**

- (A) The amplitude of oscillation in the first case changes by a factor of $\sqrt{\frac{M}{m+M}}$, whereas in the second case it remains unchanged
- (B) The final time period of oscillation in both the cases is same
- (C) The total energy decreases in both the cases
- (D) The instantaneous speed at x_0 of the combined masses decreases in both the cases.

REVIEW TEST

1. A small body of mass m is fixed to the middle of a stretched string of length 2ℓ . In the equilibrium position the string tension is equal to T_0 . Find the angular frequency of small oscillations of the body in the transverse direction. The mass of the string is negligible, the gravitational field is absent.
2. A cart consists of a body and four wheels on frictionless axles. The body has a mass m . The wheels are uniform disks of mass M and radius R . The cart rolls, without slipping, back and forth on a horizontal plane under the influence of a spring attached to one end of the car (figure). The spring constant is k . Taking into account the moment of inertia of the wheels, find a formula for the frequency of the back and forth motion of the cart.



3. The motion of a simple pendulum is given by

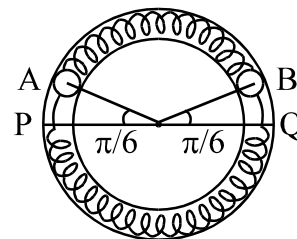
$$\theta = A \cos \left(\sqrt{\frac{g}{\ell}} t \right) \quad (\text{symbols have their usual meaning})$$

- (a) Find the tension in the string of this pendulum as a function of time assume that $\theta \ll 1$.
 - (b) At what time is the tension maximum? What is the value of this maximum tension?
4. A physical pendulum has the shape of a disk of radius R . the pendulum swings about an axis perpendicular to the plane of the disk and at distance ℓ from the center of the disk.

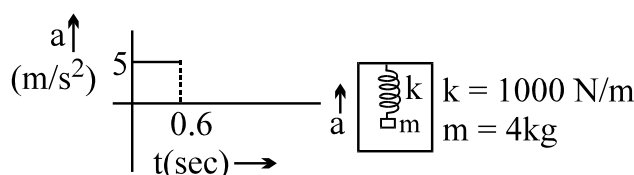
- (a) Show that the frequency of the oscillations of this pendulum is $\omega = \sqrt{\frac{g\ell}{\frac{1}{2}R^2 + \ell^2}}$

- (b) For what value of ℓ is this frequency at a maximum?

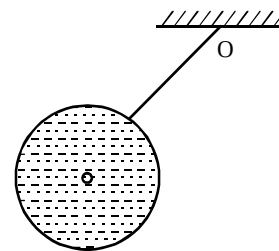
5. Two identical balls A and B each of mass 0.1 kg are attached to two identical massless springs. The spring mass system is constrained to move inside a rigid smooth pipe in the form of a circle as in fig. The pipe is fixed in a horizontal plane. The centres of the ball can move in a circle of radius 0.06 m . Each spring has a natural length $0.06\pi \text{ m}$ and force constant 0.1 N/m . Initially both the balls are displaced by an angle of $\theta = \pi/6$ radian with respect to diameter PQ of the circle and released from rest



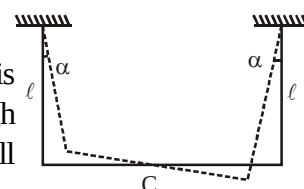
- (a) Calculate the frequency of oscillation of the ball B.
 - (b) What is the total energy of the system.
 - (c) Find the speed of the ball A when A and B are at the two ends of the diameter PQ.
6. A spring block (force constant $k = 1000 \text{ N/m}$ and mass $m = 4 \text{ kg}$) system is suspended from the ceiling of an elevator such that block is initially at rest. The elevator begins to move upwards at $t = 0$. Acceleration time graph of the elevator is shown in the figure. Draw the displacement x (from its initial position taking upwards as positive) vs time graph of the block with respect to the elevator starting from $t = 0$ to $t = 1 \text{ sec}$. Take $\pi^2 = 10$.



7. A pendulum is constructed as a light thin-walled sphere of radius R filled up with water and suspended at the point O from a light rigid rod. The distance between the point O and the centre of the sphere is equal to ℓ . How many times will the small oscillations of such a pendulum change after the water freezes? The viscosity of water and the change of its volume on freezing are to be neglected.

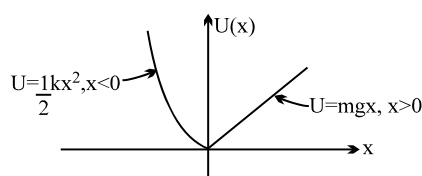


8. A uniform rod of mass $m = 1.5$ kg suspended by two identical threads $\ell = 90$ cm in length was turned through a small angle about the vertical axis passing through its middle point C . The threads deviated in the process through an angle $\alpha = 5.0^\circ$. Then the rod was released to start performing small oscillations. Find:



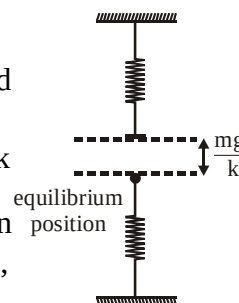
- (a) the oscillation period;
(b) the rod's oscillation energy.

9. Two physical pendulums perform small oscillations about the same horizontal axis with frequencies ω_1 and ω_2 . Their moments of inertia relative to the given axis are equal to I_1 and I_2 respectively. In a state of stable equilibrium the pendulums were fastened rigidly together. What will be the frequency of small oscillations of the compound pendulum?
10. A particle of mass m moves in the potential energy U shown above. Find the period of the motion when the particle has total energy E .



11. A particle of mass 4 kg moves between two points A and B on a smooth horizontal surface under the action of two forces such that when it is at a point P , the forces are $2\vec{PA}$ N and $2\vec{PB}$ N. If the particle is released from rest at A , find the time it takes to travel a quarter of the way from A to B .

12. Figure shows a small block of mass m attached to a spring of force constant k and an identical spring hangs from ceiling. Initially lower spring is in compressed state with compression equal to $\frac{3mg}{k}$ from natural length of spring. When block is released it strikes upper spring and sticks to it. What is the amplitude of oscillation (in cm) after sticking given $mg = 10$ N, $k = 100\sqrt{7}$ N/m.



13. Two masses m_1 and m_2 connected by a light spring of natural length ℓ_0 is compressed completely and tied by a string. This system while moving with a velocity v_0 along +ve x -axis pass through the origin at $t = 0$. At this position the string snaps. Position of mass m_1 at time t is given by the equation

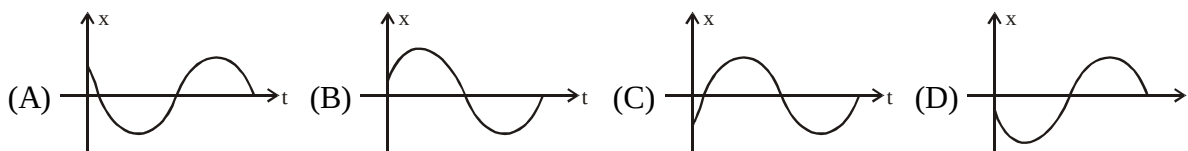
$$x_1(t) = v_0 t - A(1 - \cos \omega t)$$

Calculate :

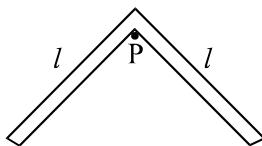
- (i) Position of the particle m_2 as a function of time. (ii) ℓ_0 in terms of A .

14. A particle is performing S.H.M. and at $t = \frac{3T}{4}$, is at position $= \frac{A}{\sqrt{2}}$ and moving towards the origin.

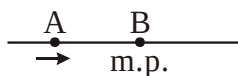
Equilibrium position of the particle is at $x = 0$. After $t = \frac{3T}{2}$ what will be the graph of the particle according to person B.



15. The time taken by a particle performing SHM to pass from point A to B where its velocities are same is 2 seconds. After another 2 seconds it returns to B. The time period of oscillation is (in seconds)
 (A) 2 (B) 8 (C) 6 (D) 4
16. Two pendulums have time periods T and $5T/4$. They start SHM at the same time from the mean position. After how many oscillations of the smaller pendulum they will be again in the same phase:
 (A) 5 (B) 4 (C) 11 (D) 9
17. A rod whose ends are A & B of length 25 cm is hanged in vertical plane. When hanged from point A and point B the time periods calculated are 3 sec & 4 sec respectively. Given the moment of inertia of rod about axis perpendicular to the rod is in ratio 9 : 4 at points A and B. Find the distance of the centre of mass from point A.
 (A) 9 cm (B) 5 cm (C) 25 cm (D) 20 cm
18. A system of two identical rods (L-shaped) of mass m and length l are resting on a peg P as shown in the figure. If the system is displaced in its plane by a small angle θ , find the period of oscillations:

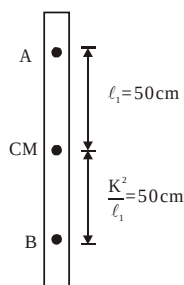


- (A) $2\pi\sqrt{\frac{\sqrt{2}l}{3g}}$ (B) $2\pi\sqrt{\frac{2\sqrt{2}l}{3g}}$ (C) $2\pi\sqrt{\frac{2l}{3g}}$ (D) $3\pi\sqrt{\frac{l}{3g}}$
19. The rotation of earth about its axis is :-
 (A) periodic motion. (B) simple harmonic motion.
 (C) periodic but not simple harmonic motion. (D) non-periodic motion.
20. A particle is oscillating simple harmonically with angular frequency ω and amplitude A . It is at a point (A) at certain instant (shown in figure). At this instant it is moving towards mean position (B). It takes time t to reach mean position (B). If time period of oscillation is T , the average speed between A and B is :-



- (A) $\frac{A \sin \omega t}{t}$ (B) $\frac{A \cos \omega t}{t}$ (C) $\frac{A \sin \omega t}{T}$ (D) $\frac{A \cos \omega t}{T}$

21. The time taken by a particle performing SHM to pass from point A to B where its velocities are same is 2 seconds. After another 2 seconds it returns to B. The time period of oscillation is (in seconds)
(A) 2 (B) 8 (C) 6 (D) 4
22. Time period of a particle executing SHM is 8 sec. At $t = 0$ it is at the mean position. The ratio of the distance covered by the particle in the 1st second to the 2nd second is :
(A) $\frac{1}{\sqrt{2}+1}$ (B) $\sqrt{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}+1$
23. Two particles A and B perform SHM along the same straight line with the same amplitude 'a', same frequency 'f' and same equilibrium position 'O'. The greatest distance between them is found to be $3a/2$. At some instant of time they have the same displacement from mean position. What is the displacement?
(A) $a/2$ (B) $a\sqrt{7}/4$ (C) $\sqrt{3}a/2$ (D) $3a/4$
24. A wire frame in the shape of an equilateral triangle is hinged at one vertex so that it can swing freely in a vertical plane, with the plane of the Δ always remaining vertical. The side of the frame is $1/\sqrt{3}$ m. The time period in seconds of small oscillations of the frame will be
(A) $\frac{\pi}{\sqrt{2}}$ (B) $\pi\sqrt{2}$ (C) $\frac{\pi}{\sqrt{6}}$ (D) $\frac{\pi}{\sqrt{5}}$
25. For a particle acceleration is defined as $\vec{a} = \frac{-5x\hat{i}}{|x|}$ for $x \neq 0$ and $\vec{a} = 0$ for $x = 0$. If the particle is initially at rest at $(a, 0)$ what is period of motion of the particle.
(A) $4\sqrt{2a/5}$ sec. (B) $8\sqrt{2a/5}$ sec.
(C) $2\sqrt{2a/5}$ sec. (D) cannot be determined
26. Two particles are performing SHM with same angular frequency and amplitudes A and 2A respectively along same straight line with same mean position. They cross each other at position A/2 distance from mean position in opposite direction. The phase between them is :
(A) $\frac{5\pi}{6} - \sin^{-1}\left(\frac{1}{4}\right)$ (B) $\frac{\pi}{6} - \sin^{-1}\left(\frac{1}{4}\right)$
(C) $\frac{5\pi}{6} - \cos^{-1}\left(\frac{1}{4}\right)$ (D) $\frac{\pi}{6} - \cos^{-1}\left(\frac{1}{4}\right)$
27. The time period of a bar pendulum when suspended at distances 30 cm and 50 cm from its centre of gravity comes out to be the same. If the mass of the body is 2kg. Find out its moment of inertia about an axis passing through first point.



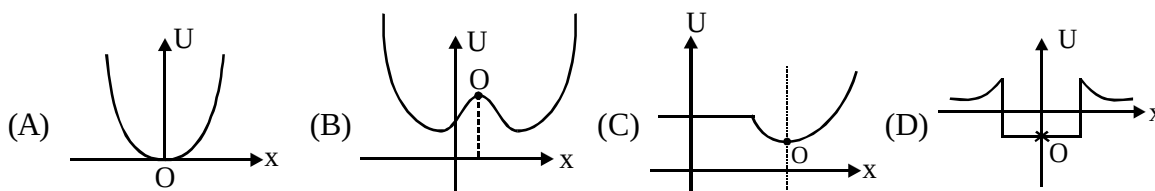
- (A) 0.24 kg-m^2 (B) 0.72 kg-m^2 (C) 0.48 kg-m^2 (D) Data insufficient

28. The position vector of a particle that is moving in space is given by

$$\vec{r} = (1 + 2 \cos 2\omega t)\hat{i} + (3 \sin^2 \omega t)\hat{j} + (3)\hat{k}$$

in the ground frame. All units are in SI. Choose the correct statement (s) :

- (A) The particle executes SHM in the ground frame about the mean position $\left(1, \frac{3}{2}, 3\right)$
- (B) The particle executes SHM in a frame moving along the z-axis with a velocity of 3 m/s.
- (C) The amplitude of the SHM of the particle is $\frac{5}{2}$ m.
- (D) The direction of the SHM of the particle is given by the vector $\left(\frac{4}{5}\hat{i} - \frac{3}{5}\hat{j}\right)$
29. A system is oscillating with undamped simple harmonic motion. Then the
- (A) average total energy per cycle of the motion is its maximum kinetic energy.
- (B) average total energy per cycle of the motion is $\frac{1}{\sqrt{2}}$ times its maximum kinetic energy.
- (C) root mean square velocity is $\frac{1}{\sqrt{2}}$ times its maximum velocity
- (D) mean velocity is 1/2 of maximum velocity.
30. Two particles execute SHM with amplitude A and 2A and angular frequency ω and 2ω respectively. At $t = 0$ they start with some initial phase difference. At $t = \frac{2\pi}{3\omega}$ they are in same phase. Their initial phase difference is :
- (A) $\frac{\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{4\pi}{3}$ (D) π
31. Potential energy vs displacement graph is given for different situations, then for a small displacement about position 'O' in which of the following motion is S.H.M. :-



ANSWER KEY
EXERCISE (S-2)

- 1. Ans.** 2 m/s **2. Ans.** $10/\pi$ Hz, $\frac{5\sqrt{37}}{6}$ cm **3. Ans.** 100 Nm^{-1}
4. Ans. 3 cm, $x = 10 - 3\sin 5t$; $\Delta E = 0.135 \text{ J}$ **5. Ans.** (a) $\frac{\pi}{10}$ sec, (b) 4 cm, (c) 2.40 kg m/s.
6. Ans. $\sqrt{\frac{7}{8}} \text{ A}$ **7. Ans.** $a = (mg/x) \sqrt{1 + 2hx/mg}$, $\frac{1}{2}x \left(a + \frac{mg}{x} \right)^2$
8. Ans. $T = 2\sqrt{\ell/g} [\pi/2 + \sin^{-1}(\alpha/\beta)]$

EXERCISE (O-2)
SINGLE CORRECT TYPE QUESTIONS

- 1. Ans.** (C) **2. Ans.** (D)
3. Ans. (C) **4. Ans.** (D)
5. Ans. (A) **6. Ans.** (B) **7. Ans.** (A)
8. Ans. (C)

MULTIPLE CORRECT TYPE QUESTIONS

- 1. Ans.** (D) **2. Ans.** (A,D) **3. Ans.** (B, C)
4. Ans. (A,B,C) **5. Ans.** (B,D)
6. Ans. (B,C,D) **7. Ans.** (A,B,C)
8. Ans. (A,B) **9. Ans.** (A,B,C,D) **10. Ans.** (B,C,D) **11. Ans.** (A,C)
12. Ans. (A,B,C)

EXERCISE (JA)

- 1. Ans.** (C) **2. Ans.** $y = \frac{mg}{k} = \frac{g}{\omega^2} < a$ **3. Ans.** (A,B,D) **4. Ans.** (C)
5. Ans. (A) P,S (B) Q,R,S (C) S (D) Q **6. Ans.** (D) **7. Ans.** (D) **8. Ans.** (C)
9. Ans. (D) **10. Ans.** (D) **11. Ans.** (C) **12. Ans.** (C) **13. Ans.** (B)
14. Ans. (D) **15. Ans.** (B) **16. Ans.** (A,D) **17. Ans.** (A) **18. Ans.** (A,D)
19. Ans. (B), (D) **20. Ans.** (A)-P,Q,R,T; (B)-Q,S; (C)-P,Q,R,S; (D)-P,R,T **21. Ans.** (A,B,D)

REVIEW TEST

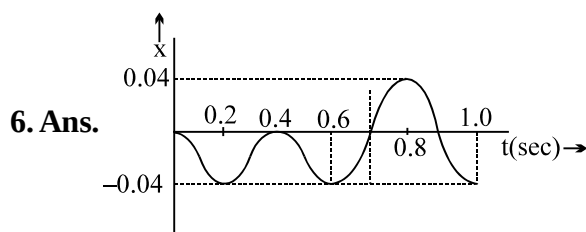
1. Ans. $\omega = \sqrt{\frac{2T_0}{m\ell}}$

2. Ans. $\omega = \sqrt{\frac{k}{(m+6M)}}$

3. Ans. (a) $T = mg + mgA^2 \sin^2 \sqrt{\frac{g}{\ell}} t$ (b) $t = (2n+1) \frac{\pi}{2} \sqrt{\frac{\ell}{g}}$; $n \in \mathbb{I}$ $T_{\max} = mg + mgA^2$

4. Ans. (b) $\frac{R}{\sqrt{2}}$

5. Ans. $f = \frac{1}{\pi}$; $E = 4\pi^2 \times 10^{-5} \text{ J}$; $v = 2\pi \times 10^{-2} \text{ m/s}$



7. Ans. Will increase $\sqrt{1 + \frac{2}{5} \left(\frac{R}{\ell} \right)^2}$ times. It is taken into account here that the water (when in liquid phase) moves translationwise, and the system behaves as a mathematical pendulum.

8. Ans. (a) $T = 2\pi \sqrt{\ell/3g} = 1.1 \text{ s}$; (b) $E = \frac{1}{2} mg\ell\alpha^2 = 0.05 \text{ J}$

9. Ans. $\omega = \sqrt{(I_1\omega_1^2 + I_2\omega_2^2)/(I_1 + I_2)}$

10. Ans. $\pi\sqrt{m/k} + 2\sqrt{2E/mg^2}$

11. Ans. $\frac{\pi}{3} \text{ s}$

12. Ans. 5

13. Ans. (i) $x_2 = v_0 t + \frac{m_1}{m_2} A(1 - \cos \omega t)$ (ii) $\ell_0 = \left(\frac{m_1}{m_2} + 1 \right) A$

14. Ans. (B)

15. Ans. (B)

16. Ans. (A)

17. Ans. (D)

18. Ans. (B)

19. Ans. (A,C,D)

20. Ans. (A)

21. Ans. (B)

22. Ans. (D)

23. Ans. (B)

24. Ans. (D)

25. Ans. (A)

26. Ans. (A)

27. Ans. (C)

28. Ans. (A,C,D)

29. Ans. (A,C)

30. Ans. (B,C)

31. Ans. (A,C)

This image shows a full page of blank white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for writing or drawing. There are no margins, text, or other markings present.

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